FHIR-STU3 — Conformance Module (sections 5 – 5.9.9)

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5 Conformance Module

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This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Work Group FHIR Infrastructure | Ballot Status: Informative

5.0 Conformance Module

The Conformance Module represents metadata about the datatypes, resources and API features of the FHIR specification and can be used to create derived specifications.

CapabilityStatement StructureDefinition OperationDefinition SearchParameter | CompartmentDefinition DataElement ImplementationGuide ElementDefinition (datatype) | Full profiling details Detailed conformance rules

5.0.1 Introduction

The core FHIR specification describes a set of resources, frameworks and APIs that are used in many different contexts in healthcare. However, there is wide variability between jurisdictions and across the healthcare ecosystem around practices, requirements, regulations, education and what actions are feasible and/or beneficial.

For this reason, the FHIR specification is a "platform specification" - it creates a common platform or foundation on which a variety of different solutions are implemented. As a consequence, this specification usually requires further adaptation to particular contexts of use.

Typically, these adaptations specify:

* Rules about which resource elements are or are not used, and what additional elements are added that are not part of the base specification

* Rules about which of FHIR's RESTful API, messaging and document features are used, and how

* Rules about which terminologies are used in particular elements

* Descriptions of how the Resource elements and API features map to local requirements and/or implementations

Note that because of the nature of the healthcare ecosystem, there may be multiple overlapping sets of adaptations - by healthcare domain, by country, by institution, and/or by vendor/implementation.

FHIR provides a set of resources that can be used to represent and share the adaptations listed above in a computable fashion. These resources are collectively called the conformance resources. Although these conformance resources can be used in isolation they are typically used in the context of an Implementation Guide or a Capability Statement:

* Implementation Guides are documents published by a domain, institution or vendor that describe how FHIR is adapted to support a certain use case (or set of use cases). An implementation guide combines a set of conformance resources and supporting narrative into a document for use by implementers.

* A capability statement which uses the conformance resources to document how a client or server has implemented FHIR, i.e. which aspects of the specification and API are implemented and how.

5.0.2 Contents

The content of an Implementation Guide is described using the ImplementationGuide resource, while the capability statement is represented by the CapabilityStatement resource. These two resources make use of the complete set of conformance resources to fully capture the set of adaptations they represent. Note that the CapabilityStatement resource is one of the conformance resources, the first just describing the capabilities of a system, while the latter is the set of all conformance resources, including:

* The defines how a particular structure (Resource, Extension or Data Type) is used to:

StructureDefinition

* Describe how existing elements in resources and/or data types are used and identify existing elements that are not used. Exact rules on how to do this can be found in the section on conformance rules

* Define extensions that can be used in resources or data types.

* Reference the Value Sets that specify the content of coded elements, see the terminology module for more details.

* The MessageDefinition describes messages that can be sent and received, including the driving event, contents to be exchanged and responsibilities on receipt.

* The OperationDefinition describes additional operations in addition to those in the base specification.

* The SearchParameter describes additional search capabilities in addition to those in the base specification

* The CompartmentDefinition describes a logical grouping for resources, used in access control or search

* The DataElement resource is a formal description of a single piece of information that can be gathered and reported.

Conformance resources may be used independently, not just within the context of an ImplementationGuide resource or capability statement. See the section Common use cases for examples of such uses.

The conformance resources and their relationships are shown below:

Resources shown with a dotted box are described in other sections of the specification: ValueSet, ConceptMap and StructureMap are from the section on terminology, TestScript is part of the section on Implementer Support.

5.0.3 Security and Privacy

The conformance resources do not represent patient-related data, and as such are less susceptible to data breaching. Some caution is required however:

* StructureDefinitions may contain invariants formulated as structured expressions that are evaluated by external engines (i.e. xpath), which -if unproperly sandboxed- could provide low-level access to the system

* A server may be implemented in such a way that it will automatically support additional search parameters or compartments definitions uploaded to that system. This may result in searches that use large amounts of processing power, expose sensitive patient data, or both.

* If a server acts as a repository of conformance resources, it should tightly control modifications to its data: updates to conformance resources are usually managed by a review process to ensure backwards compatibility and stability. Worst case, changes to these resources may result in dependent systems running the risk of being misconfigured.

5.0.4 Common use Cases

Conformance resources are commonly used as part of an Implementation Guide or CapabilityStatement resource. There are many ways to use the resources independently however, including:

* An instance of a resource may reference one or more StructureDefinitions to claim conformance to the rules laid out in those StructureDefinitions

* A server may serve CapabilityStatement resources, effectively functioning as a discovery endpoint for services within an organization

* A server may hold a set of DataElement resources to become a Data Dictionary of reusable datamodels as per ISO 11179.

* A code-generation tool may use StructureDefinitions and OperationDefinitions to generate code that represents the structures as classes and operations as remotely callable functions to provide an easier programming model to a software developer.

* A server may act as a repository of NameSpace resources, so vendors and implementers can quickly look up the URLs or oids for a given terminology or identifier system.

5.0.5 Developmental Roadmap

A subset of the conformance resources has been tested and used in production tooling and as such have reached a maturity level where changes become less likely. These are CapabilityStatement, StructureDefinition, ValueSet, OperationDefinition and SearchParameter.

Other resources are still under development:

* ImplementationGuide: used in the HL7 production tooling, but has not received much use outside of these tools yet.

* CompartmentDefinition: new in STU3, and as such has not undergone much production use

* DataElement: the exact role of "Data Element Definitions" is still being studied

Over the past two years, these resources have been mainly used in the tools used to build the FHIR publication, early-adopter implementation guides and the FHIR Foundation conformance resource registry. In the 2016-2018 timeframe, we expect to see more widespread use of these resources in validation tooling, code-generators and more extensive model-based guide authoring tools.

5.1 Profiling FHIR

URL: https://hl7.org/fhir/STU3/profiling.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.1.0 Profiling FHIR

FHIR Infrastructure Work Group | Maturity Level: 4 | Ballot Status: Trial Use

The base FHIR specification (this specification) describes a set of base resources, frameworks and APIs that are used in many different contexts in healthcare. However, there is wide variability between jurisdictions and across the healthcare ecosystem around practices, requirements, regulations, education and what actions are feasible and/or beneficial.

For this reason, the FHIR specification is a "platform specification" - it creates a common platform or foundation on which a variety of different solutions are implemented. As a consequence, this specification usually requires further adaptation to particular contexts of use. Typically, these adaptations specify:

* Rules about which resource elements are or are not used, and what additional elements are added that are not part of the base specification

* Rules about which API features are used, and how

* Rules about which terminologies are used in particular elements

* Descriptions of how the Resource elements and API features map to local requirements and/or implementations

Note that because of the nature of the healthcare ecosystem, there may be multiple overlapping sets of adaptations - by healthcare domain, by country, by institution, and/or by vendor/implementation.

5.1.0.1 Glossary

FHIR defines a cascade of artifacts for this purpose:

Artifact | Description | DAF example
Implementation Guide (IG) | A coherent and bounded set of adaptations that are published as a single unit. Validation occurs within the context of the Implementation Guide | DAF IG
Package | A group of related adaptations that are published as a group within an Implementation Guide | DAF Medication Usage
Conformance Resource | A single resource in a package that makes rules about how an implementation works. These are described below | DAF Prescription

The term "profile" is a general one that is used about either a "package" or an "item". "Profiling" is a general term that describes the process of creating an implementation guide, or any of the conformance resources found in one.

5.1.0.2 Conformance Resources

Typically, Implementation Guides both restrict and extend APIs, resources and terminologies. FHIR provides a set of resources that can be used to represent and share the decisions that have been made, and allows implementers to build useful services from them. These resources are known as the conformance resources. These conformance resources allow implementers to:

* Indicate that some API calls are not used for a particular situation, and provide additional details about how API calls are used (Capability Statement)

* Add additional operations or search parameters not in the base specification (using the OperationDefinition resource or the SearchParameter Resource

* Define how a particular structure (Resource, Extension or Data Type) is used (Resource):

StructureDefinition

* Describe how existing elements in resources are used

* Identify existing elements that are not used

* Define extensions that can be used in resources or data types

* Mix custom and standard terminologies and choose which codes from these to use for a particular coded element(Value Set and StructureDefinition Resources)

* Map between local and standard terminologies or content models (Concept Map Resource)

* Register system namespaces for identifiers and terminologies (NamingSystem Resource)

* Describe and register in a Data Dictionary specific Data Elements that are used across systems (DataElement Resource)

These resources need to be used as discussed below, and also following the basic concepts for extension that are described in "Extensibility". For implementer convenience, the specification itself publishes its base definitions using these same resources.

5.1.0.3 Two uses of Profiles

The CapabilityStatement resource describes two different uses for profiles on resources: Resource Profiles and System Profiles. Resource Profiles are specified using the CapabilityStatement.rest.resource.profile element and System Profiles are specified using the CapabilityStatement.profile element.

5.1.0.3.1 CapabilityStatement.rest.resource.profile

These profiles describe the general features that are supported by the system for each kind of resource. Typically, this is the superset of all the different use-cases implemented by the system. This is a resource-level perspective of a system's functionality.

5.1.0.3.2 CapabilityStatement.profile

These profiles describe the information handled/produced by the system on a per use case basis. Some examples of the uses for these kind of profiles:

* A Laboratory service producing a set of different reports - general chemistry, blood count, etc. Typical labs would support several hundred different reports

* A care manager which handles a set of different types of care plans and associated clinical resources

* A medications formulary that handles several different levels of sophistication in its medication representations

Typically, these profiles are a series of variations on the same set of resources - different use cases leading to handling the resources that represent them differently. These use cases described above all pertain to system that produce and publish data, but the same concept applies to systems that consume data. For instance:

* An expert service that provides analysis on several different sets of data conforming to a particular pattern - tests x,y and z with particular codes and units

For a producer system and a consumer system to exchange data successfully based on one of these supported system profiles, it's not enough to know that the systems happen to have system profiles that overlap for the use case of interest; the consumer must be able to filter the total set of resources made available by the producer system and deal only with the ones relevant to the use case.

As an example, consider a laboratory system generating thousands of reports a day. 1% of those reports are a particular endocrine report that a decision support system knows how to process. Both systems declare that they support the particular endocrine report profile, but how does the expert system actually find the endocrine reports that it knows how to process?

One possible option is for the expert system to receive every single report coming from the lab system, check whether it conforms to the profile or not, and then decide whether to process it. Checking whether a resource conforms to a particular profile or not is a straight forward operation (one option is to use the provided tools for this), but this is very inefficient way - the expert system has to receive and process 100 times many resources as it uses. To help a consumer find the correct set of reports for a use-case, a producer of resources also SHALL, for any profile declared in Conformance.profile:

* Mark resources with profile assertions documenting the profile(s) they conform to (this enables indexing by the profile)

* (if a server) support searching by the _profile parameter for the declared profiles

Beyond these requirements, a producer of resources SHOULD ensure that any resource instance that would reasonably be expected to conform to the declared profiles SHOULD be published in this form.

STU Note: there are many uninvestigated issues associated with this use of profiles. HL7 is actively seeking feedback from users who experiment in this area, and users should be prepared for changes to features and obligations in this area in the future.

Feedback is welcome here .

5.1.0.4 Extending and Restricting the API

A CapabilityStatement resource lists the REST interactions (read, update, search, etc.) that a server provides or that a client uses, along with some supporting information for each. It can also be used to define a set of desired behavior (e.g. as part of a specification or a Request for Proposal). The only interaction that servers are required to support is the capabilities interaction itself - to retrieve the server's CapabilityStatement. Beyond that, servers and clients support and use whichever API calls are relevant to their use case.

In addition to the operations that FHIR provides, servers may provide additional operations that are not part of the FHIR specification. Implementers can safely do this by appending a custom operation name prefixed with '$' to an existing FHIR URL, as the Operations framework does. The Conformance resource supports defining what OperationDefinitions make use of particular names on an end-point. If services are defined that are not declared using OperationDefinition, it may be appropriate to use longer names, reducing the chance of collision (and confusion) with services declared by other interfaces. The base specification will never define operation names with a "." in them, so implementers are recommended to use some appropriate prefix for their names (such as "ihe.someService") to reduce the likelihood of name conflicts.

Implementations are encouraged, but not required, to define operations using the standard FHIR operations framework - that is, to declare the operations using the OperationDefinition resource, but some operations may involve formats that can't be described that way.

Implementations are also able to extend the FHIR API using additional content types. For instance, it might be useful to read or update the appointment resources using a vCard based format. vCard defines its own mime type, and these additional mime types can safely be used in addition to those defined in this specification.

5.1.0.5 Extending and Restricting Resources

Extending and restricting resources (collectively known as 'profiling a resource') is done with a "StructureDefinition" resource, which is a statement of rules about how the elements in a resource are used, and where extensions are used in a resource.

5.1.0.6 Changing Cardinality

One key function of profiles is to change the cardinality of an element. A profile can restrict the cardinality of an element within the limits of the base structure it is constraining. This table summarizes what types of restrictions are allowed:

derived (across) base (down)	0..0 (Not used)	0..1 (optional)	0..n (optional, many)	1..1 (required)	1..n (at least 1)
0..1	yes	yes	no	yes	no
0..*	yes	yes	yes	yes	yes
1..1	no	no	no	yes	no
1..*	no	no	no	yes	yes

When a profile is constraining another profile where there are more cardinality options (e.g. low is not just 0 or 1, and high is not just 1 or *), the same principles still apply: the constraining profile can only allow what the base profile allows.

5.1.0.7 Limitations of Use

What Structure Definitions can do when they are constraining existing resources and datatypes is limited in some respects:

* Profiles cannot break the rules established in the base specification (e.g. cardinality as described above)

* Profiles cannot specify default values or meanings for elements

* Profiles cannot give more specific names to elements

* It must be safe to process a resource without knowing the profile

The consequence of this is that if a profile mandates extended behavior that cannot be ignored, it must also mandate the use of a modifier extension. Another way of saying this is that knowledge must be explicit in the instance, not implicit in the profile.

As an example, if a profile wished to describe that a Procedure resource was being negated (e.g. asserting that it never happened), it could not simply say in the profile itself that this is what the resource means; instead, the profile must say that the resource must have an extension that represents this knowledge.

There is a facility to mark resources to indicate that they can only be safely understood by a process that is aware of and understands a set of published rules. For more information, see Restricted Understanding of Resources.

5.1.0.8 Using Structure Definitions

A "constraint" Structure Definition specifies a set of restrictions on the content of a FHIR resource or data type, or an additional set of constraints on an existing profile. A given structure definition is identified by its canonical URL, which SHOULD be the URL at which it is published. The following kinds of statements can be made about how an element is used, using a series of Element Definitions:

* Restricting the cardinality of the element; e.g. the base might allow 0..*, and a particular application might support 1..2

* Ruling out use of an element by setting its maximum cardinality to 0

* Restricting the contents of an element to a single fixed value

* Making additional constraints on the content of nested elements within the resource (expressed as XPath statements)

* Restricting the types for an element that allows multiple types

* Requiring a typed element or the target of a resource reference to conform to another structure profile (declared in the same profile, or elsewhere)

* Specifying a binding to a different terminology value set (see below)

* Providing refined definitions, comments/usage notes and examples for the elements defined in a Resource to reflect the usage of the element within the context of the Profile

* Providing more specific or additional mappings (e.g. to HL7 v2 or HL7 v3) for the resource when used in a particular context

* Declaring that one or more elements in the structure must be 'supported' (see below)

Any changed definitions SHALL be restrictions that are consistent with the rules defined in the resource in the FHIR Specification from which the profile is derived. Note that some of these restrictions can be enforced by tooling (and are by the FHIR tooling), but others (e.g. alignment of changes to descriptive text) cannot be automatically enforced.

Note that structure definitions cannot 'remove' mappings and constraints that are defined in the base structure, but for purposes of clarity, they can refrain from repeating them.

A structure definition contains a linear list of element definitions. The inherent nested structure of the elements is derived from the path value of each element. For instance, a sequence of the element paths like this:

* Root

* Root.childA

* Root.childA.grandchild1

* Root.childB

defines the following structure:

```
<Root>
  <childA>
    <grandChild1/>
  </childA>
  <childB/>
</Root>
```

or its JSON equivalent. The structure is coherent - children are never implied, and the path statements are always in order. The element list is a linear list rather than being explicitly nested because element definitions are frequently re-used in multiple places within a single definition, and this re-use is easier with a flat structure.

5.1.0.9 Differential vs Snapshot

Structure Definitions may contain a differential statement, a snapshot statement or both.

Differential statements describe only the differences that they make relative to another structure definition (which is most often the base FHIR resource or data type). For example, a profile may make a single element mandatory (cardinality 1..1). In the case of a differential structure, it will contain a single element with the path of the element being made mandatory, and a cardinality statement. Nothing else is stated - all the rest of the structural information is implied (note: this means that a differential profile can be sparse and only mention the elements that are changed, without having to list the full structure. This rule includes the root element - it is not needed in a sparse differential).

Note that a differential can choose not to constrain elements. Doing so means that the profile will be more flexible in terms of compatibility with other profiles, but will require more work to support from implementing systems. Alternatively, a profile can constrain all optional elements to be not present (max cardinality = 0) - this closes the content, which makes implementation easier, but also reduces its usefulness.

In order to properly understand a differential structure, it must be applied to the structure definition on which it is based. In order to save tools from needing to support this operation (which is computationally intensive - and impossible if the base structure is not available), a StructureDefinition can also carry a "snapshot" - a fully calculated form of the structure that is not dependent on any other structure. The FHIR project provides tools for the common platforms that can populate a snapshot from a differential (note that the tools generate complete verbose snapshots; they do not support suppressing mappings or constraints).

StructureDefinitions can contain both a differential and a snapshot view. In fact, this is the most useful form - the differential form serves the authoring process, while the snapshot serves the implementation tooling. StructureDefinition resources used in operational systems should always have the snapshot view populated.

5.1.0.10 Slicing

One common feature of constraining Structure Definitions is to take an element that may occur more than once (e.g. in a list), and split the list into a series of sublists, each with different restrictions on the elements in the sublist with associated additional meaning. In FHIR, this operation is known as "Slicing" a list. It is common to "slice" a list into sub-lists each containing just one element, effectively putting constraints on each element in the list. This technique can also be used on elements that do not repeat, but that have a choice of data types.

Here is an example to illustrate the process:

In this example, the base structure definition for the resource Observation defines the "component" element which contains a nested code and a value for observations that have multiple values. A classic example of this kind of observation is a blood pressure measurement - it contains 2 values, one for systolic, and one for diastolic (example).

This diagram shows the conceptual process of 'slicing' the component list into systolic and diastolic slices (note: to avoid clutter, the "name" attribute of Observation is shown as just a code not a full CodeableConcept).

The structure definition for Blood Pressure splits the component list into two sublists of one element each: a systolic element, and a diastolic element. Each of these elements has a fixed value for the code element (a fixed LOINC code for the name), and both have a value of type Quantity. This process is known as "slicing" and the Systolic and Diastolic elements are called "slices".

Note that when the resource is exchanged, the serialization format that is exchanged is not altered by the constraining definition. This means that the item profile names defined in the structure definition ("systolic", etc. in this example) are never exchanged. A resource instance looks like this:

```
<Observation>
  ...
  <component>
    <code {LOINC="8480-6"}/>
    <value ...>
  </component>
  <component>
    <code {LOINC="8462-4"}/>
    <value ...>
  </component>
</Observation>
```

In order to determine that the first related item corresponds to "Systolic" in the structure definition, so that it can determine to which additional constraints for a sub-list the item conforms, the system checks the values of the elements. In this case, the "code" element in the target resource can be used to determine which slice that target refers to. This element is called the "discriminator".

5.1.0.11 Discriminator

In the general case, systems processing resources using a structure definition that slices a list can determine the slice corresponding to an item in the list by checking whether the item's content meets the rules specified for the slice. This would require a processor to be able to check all the rules applied in the slice and to do so speculatively in a depth-first fashion. Both of these requirements are inappropriately difficult for an operational system, and particularly for generated code (e.g. software that is automatically produced based on the StructureDefinition). Thus, to provide a better way to distinguish slices, a sliced element can designate a field or set of fields that act as a "discriminator" - they are used to tell the slices apart.

When a discriminator is provided, the composite of the values of the elements designated in the discriminator is unique and distinct for each possible slice and applications can easily determine which slice an item in a list is. The intention is that this can be done in generated code, e.g. using a switch/case statement.

When a constraining structure designates one or more discriminators, it SHALL ensure that the possible values for each slice are different and non-overlapping, so that the slices can easily be distinguished.

Each discriminator is a pair of values: a type that indicates how the field is processed when evaluating the discriminator, and a FHIRPath expression that identifies the element in which the discriminator is found. There are five different processing types for discriminators:

value | The slices have different values in the nominated element

exists | The slices are differentiated by the presence or absence of the nominated element

pattern | The slices have different values in the nominated element, as determined by testing them against the applicable ElementDefinition.pattern[x]

type | The slices are differentiated by type of the nominated element to a specifed profile

profile | The slices are differentiated by conformance of the nominated element to a specifed profile

The FHIRPath statement that allows for the selection of the element on which the discriminator is based is a restricted FHIRPath statement that is allowed to include:

* Element selections (e.g. FHIRPath statements without "()" such as component.value)

* The function extension(url) to allow selection of a particular extension

* The function resolve() to allow slicing to across resource boundaries

Each slice must use the element definition for the element in the discriminator(s) to ensure that the slices are clearly differentiated (by assigning a fixed value, a specific type, or a profile, depending on the discriminator type. If the type is 'value', then the element definition must use either ElementDefinition.fixed[x] or, if the element has a terminology binding, a required binding with a Value Set that enumerates the list of possible codes in the value set ("extensional definition").

It is the composite (combined) values of the discriminators that are unique, not each discriminator alone. For example, a slice on a list of items that are references to other resources could designate fields from different resources, where each resource only has one of the designated elements, as long as they are distinct across slices.

A structure definition is not required to designate any discriminator at all for a slice, but those that don't identify discriminators are describing content that is very difficult to process, and so this is discouraged.

Within a structure definition, a slice is defined using multiple element entries that share a path but have distinct names. These entries together form a "slice group" that is:

* Initiated by a "slicing entry" That is, the first element in a slice group must contain a slicing property that defines the discriminator for all members of the group. It also contains the unconstrained definition of the element that is sliced, potentially including children of the unconstrained element, if there are any

* Mutually exclusive. This means that each element in a slice group SHALL describe a distinct set of values for the group's discriminators. Because of this constraint, an element in a resource instance will never match more than one element in a given slice group. If no discriminators are named, it SHOULD still be possible to differentiate the slices based on their properties, though it may be substantially harder to do so.

* Serialized as a group. The entries in a slice group must be adjacent in a serialized structure definition, or, if there are any intervening elements, those elements must be "compatible with" the group. Concretely, this means that any intervening elements must have a path that starts with the slice groups's path. For example, an element with a path of Observation.name.extension would be compatible with (and thus, would not "break up") a slice group whose path was Observation.name

Some examples of descriminators:

Context | Discriminator Type | Discriminator Path | Interpretation
List.entry | value | item.reference.resolve().name | Entries are differentiated by the name element on the target resource - probably an observation, which could be determined by other information in the profile
List.entry | type | item.reference.resolve() | Entries are differentiated by the type of the target element that the reference points to
List.entry | profile | item.reference.resolve() | Entries are differentiated by a profile tag on the target of the reference, as specified by a structure definition (todo: how to do that?)
List.entry | value | item.extension("http://acme.org/extensions/test").code | Entries are differentiated by the value of the code element in the extension with the designated URL
List.entry.extension | value | url | Extensions are differentiated by the value of their url property (usually how extensions are sliced)
List.entry | type, value | item.reference.resolve(), item.reference.resolve().code | Extensions are differentiated by the combination of the type of the referenced resource, and, if it has one, the code element of that resource. This would be appropriate for where a List might be composed of a Condition, and set of observations, each differentiated by its name - the condition has no name, so that is evaluated as a null in the discriminator set
Observation.value[x] | type | $this | Different constraints (e.g. "must support", usage notes, vocabulary bindings, etc.) are asserted for different supported types for the multi-typed element Observation.value[x]

Note that discriminator types of type and profile can also be used where a repeating element contains a resource directly (e.g. DomainResource.contained, Bundle.entry, Parameters.parameter.resource).

The examples of slicing and discriminators show exactly how this and other typical uses of slicing are represented in profiles.

Note that extensions are always sliced by the url element, though they may be resliced on additional elements where required.

5.1.0.12 Default Slice

There is a special slice, called the default slice. This allows a profile to describe a set if specific slices, and then make a set of rules that apply to all the remaining content that is not in one of the defined slices. Some rules about the default slice:

* It is identified because the name of the slice is @default. The sliceName '@default' is reserved and cannot be used in any other context

* Default slices are only allowed when the slicing rule = closed

* Default slices must not fix the value of the discriminator elements

* Default slices can be re-sliced in dependent profiles

One use of a default slice would be the case where the profile slices an identifier element to require a set of known identifiers, where the type element is prohibited (since they are known identifiers) but requires type on all other identifiers if any are present. In this case, the default slice makes no rules about the identiifer.system (which is the slicing discriminator), but fixes the cardinality of type to 1..1 in the @default slice.

5.1.0.13 Re-profiling and Re-slicing

Profiles can be based on other profiles, and apply further constraints to those already specified. This is a useful technique, but implementers should be wary of over-use - humans have understanding the implications of deep stacks of constraining profiles.

When a profile constrains another profile, it can make additional constrainta, including extending the discriminator, adding new slices (if the slices are not already closed), and slicing inside the existing slices.

The rules for changing the slicing are as follows:

* Rule = open can be changed to rule = closed, and unordered can be changed to ordered)

* If a discriminator for an element is declared in a parent profile, child profiles referencing that element must either declare the same discriminator or must declare a new discriminator that includes the parent discriminator content. I.e. additional discriminator paths may be added, but none of the existing paths can be removed.

It's sometimes necessary to slice data that has already been sliced in the base profile - that is, create new slices within the existing slices. This is called "Re-slicing". The rules for re-slicing are as follows:

When you slice, you define a name for each new slice. The name has to be unique across the set of slices in the profile. So if profile A defines an element X with cardinality 0..*, and profile B is derived from profile A, then profile B can either:

* make a constraint on X with no name - in which case the profile is adding constraints to all slices of X; or

* make a constraint on X with a name - in which case the profile is describing a specific slice on X, and the constraints only apply to that slice; or

* it can do both

Then, profile C derives from profile B. Profile C can do the following:

* make a constraint on X with no name - in which case the profile is constraining all appearances of X; or

* make a constraint on X with a different name from that used in profile B - in which case the profile is describing a specific new slice on X, and the constraints only apply to that slice; or

* make a constraint on X with the same name as that used in profile B - in which case the profile is making new constraints on the slice defined in profile B; or

* some combination of the above options

Note: it is possible for Profile C to make rules that are incompatible with profile B, in which case there is no set of instances that can be valid against profile C

In addition to the above, there are times when Profile C will need to further slice a slice defined in B. In this case, there's a need to reference both the name of the original slice from Profile B as well as to define a name for the slice defined within Profile C. This is done by separating the names using "/". For example, if Profile B defines the slice "example", and profile C defines the slice "example/example1", then this is deemed to be "example1" slice of the example slice. This process can continue indefinitely by separating each layer of slicing names with the "/" character. This pattern applies to @default too: @default/@default.

5.1.0.14 Extension Definitions

An extension definition defines the URL that identifies the extension and which is used to refer to the extension definition when it is used in a resource.

The extension definition also defines the context where the extension can be used (usually a particular path or a data type) and then defines the extension element using the same details used to profile the structural elements that are part of resources. This means that a single extension can be defined once and used on different Resource and/or datatypes, e.g. one would only have to define an extension for "hair color" once, and then specify that it can be used on both Patient and Practitioner.

For further discussion of defining and using extensions, along with some examples, see Extensibility.

5.1.0.14.1 Using Extensions in Profiles

Once defined, an extension can be used in an instance of a resource without any Profile declaring that it can, should or must be, but Profiles can be used to describe how an extension is used.

To actually prescribe the use of an extension in an instance, the extension list on the resource needs to be sliced. This is shown in the extensibility examples

Note that the minimum cardinality of an extension SHALL be a valid restriction on the minimum cardinality in the definition of the extension. if the minimum cardinality of the extension is 1 when it is defined, it can only be mandatory when it is added to a profile. This is not recommended - the minimum cardinality of an extension should usually be 0.

5.1.0.15 Binding Definitions

Coded elements have bindings that link from the element to a definition of the set of possible codes the element may contain. The binding identifies the definition of the set of possible codes and controls how tightly the set of the possible codes is interpreted.

The set of possible codes is either a formal reference to a ValueSet resource, which may be version specific, or a general reference to some web content that defines a set of codes. The second is most appropriate where a set of values is defined by some external standard (such as mime types). Alternatively, where the binding is incomplete (e.g. under development) just a text description of the possible codes can be provided.

Bindings have a property that defines how the strongly implementations are required to use the set of codes. See Binding Strength.

5.1.0.16 Mixing Custom and Standard Terminologies

CodeSystem resources can be used to carry definitions of local codes (Example) and ValueSets can mix a combination of local codes and standard codes (e.g. LOINC, SNOMED), or just to choose a particular set of standard codes (examples: LOINC, SNOMED, RxNorm). Profiles can bind to these value sets instead of the ones defined in the base specification, following these rules:

Binding Strength in base specification | Customization Rules in Profiles
required | The value set can only contain codes contained in the value set specified by the FHIR specification
extensible | The value set can contain codes not found in the base value set. These additional codes SHOULD NOT have the same meaning as existing codes in the base value set
preferred | The value set can contain whatever is appropriate for local use
example | The value set can contain whatever is appropriate for local use

Note that local codes are not as interoperable as standard published code systems (e.g. LOINC, SNOMED CT, so it is preferable to use standard code systems.

5.1.0.17 Changing Binding Strength in Profiles

A profile can change the terminology binding of an element - both strength and value set - within the limits of the base structure it is constraining. This table summarizes the changes that can be made to the binding strength:

derived (across) base (down) | required | extensible | preferred | example
required | yes | no | no | no
extensible | yes | yes | no | no
preferred | yes | yes | yes | no
example | yes | yes | yes | yes

Note that a constraining profile may leave the binding strength the same and change the value set instead. Whatever the constraining profile does, it cannot make codes valid that are invalid in the base structure/profile.

5.1.0.18 Must Support

One property that can be declared on profiles that is not declared on the resource or data type definitions is "Must Support". This is a boolean property. If true, it means that systems claiming to conform to a given profile must "support" the element. This is distinct from cardinality. It is possible to have an element with a minimum cardinality of "0", but still expect systems to support the element.

The meaning of "support" is not defined by the base FHIR specification, but can be set to true in a profile. When a profile does this, it SHALL also make clear exactly what kind of "support" is required. Examples might include:

* The system must be able to store and retrieve the element

* The system must display the element to the user and/or allow the user to capture the element via the UI

* The element must appear in an output report

* The element must be taken into account when performing decision support, calculations or other processing

* etc.

The specific meaning of "Must Support" for the purposes of a particular profile SHALL be described in the element.definition, the general StructureDefinition.description or in other documentation for the implementation guide the profile is part of.

If creating a profile based on another profile, Must Support can be changed from false to true, but cannot be changed from true to false. Note that an element that has the property IsModifier is not necessarily a "key" element (e.g. one of the important elements to make use of the resource), nor is it automatically mustSupport - however both of these things are more likely to be true for IsModifier elements than for other elements.

5.1.0.19 Search Criteria

The final thing implementations can do is to define search criteria in addition to those defined in the specification itself. Search criteria fall into one of four categories:

* Enabling search on core elements that don't have standard search criteria defined (e.g. searching Observation by normal range)

* Enabling search on elements that already have a standard search criteria defined, but with custom matching rules; e.g. a sounds-like search on Practitioner name

* Enabling search on a particular extension

* Enabling search that doesn't correspond to a single element but rather a combination of elements or computation on an element; e.g. searching for patients by age

Additional Search Parameters can be defined using the SearchParameter resource.

5.1.0.20 Presenting Profiles

When this specification describes a profile, the profile is presented in 5 different forms:

* Text Summary

* Differential Table

* Snapshot Table

* XML Template

* JSON Template

5.1.0.20.1 Text Summary

This presents a short summary of the profile - a combination of the author's summary, and some automatically generated summary content

5.1.0.20.2 Differential Table

This is a view of the differential statement (see above). For context, additional information not in the differential is also show partially transparent

5.1.0.20.3 Snapshot Table

This is a view of the snapshot produced by the profile (see above). The information is a comprehensive view of what the profile means

5.1.0.20.4 XML Template

Not one yet

5.1.0.20.5 JSON Template

Not one yet

5.1.0.21 Supporting Multiple Profiles

Applications may be required to support more than one profile at a time. A typical example might be an EHR application that is required to support a general purpose data sharing profile (such as DAF), and also must support specific profiles for decision support using the same interface.

The impact of supporting two sets of profiles depends on whether resources are being created or consumed. When an application is creating content, it must create content that conforms to both sets of profiles - that is, the intersection of the profiles. When an application is consuming information, then it must be able to consume content that conforms to either set of profiles - that is, the union of the profiles.

Since applications generally consume and produce resources at the same time, conforming to more than one profile may not be possible, unless the profiles are designed to make statements at different levels - and the case above is one such case, where one profile is focused on data access, provenance, and availability, the other profile is focused on clinical content.

Accordingly, profiles can relate to each other in four different ways. Each profile can be thought of in terms of the set of instances that conform to the profile:

* non-overlapping: there no instances that conform to profiles A & B (technically, the intersection of profiles A & B is an empty set)

* partly overlapping: some instances conform to both A & B, but others only conform to A or B

* one set contained in the other: all resources that conform to A conform to B, but only some of the ones that conform to B conform to A (or vice versa)

* identical sets: all resources that conform to A also conform to B and any resources that don't conform to B, don't conform to A

Profiles can be compared to determine their compatibility. One such comparison can be found (no - todo: bring this into the build) between DAF and QICore . Note that this comparison is generated by tooling under ongoing development, and is purely draft content to demonstrate the idea of profile comparison.

5.1.1 Slicing and Discriminator Examples

URL: https://hl7.org/fhir/STU3/profiling-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.1.1 Slicing and Discriminator Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

5.1.1.1 Slicing Patient Contact Details

One common use of slicing is to describe different constraints on different kinds of patient contact details. In this example, as Patient.telecom is defined as: ContactPoint [0..*] where the ContactPoint has system, value and use.

Consider the case where the profile should say:

* There must be a single home phone number

* The patient may have a work phone number

* The patient may have an email address

* No other types of contact are allowed

An example of a patient resource that meets these rules:

```
<Patient>
... snip ...
  <telecom>
    <system value="phone" />
    <use value="home" />
    <value value="5551234567" />
  </telecom>
  <telecom>
    <system value="email" />
    <value value="someone@acme.org" />
  </telecom>
... snip ...
</Patient>
```

To do this, the profile that implements these rules needs to do the following:

* On the base Patient.telecom element: define that slicing is discriminated by system and use, and that the slices are closed ("No other types of contact are allowed"). Order is left unfixed

* Then define 3 slices:

* home phone: fixed values for system and use, and cardinality 1..1

* work phone: fixed values for system and use, and cardinality 0..1

* email: fixed value for system, cardinality 0..0 for use, and cardinality 0..1

In a StructureDefinition, this will look like:

```
  <!-- setting up the slicing -->
  <element>
    <path value="Patient.telecom"/>
    <slicing>
      <discriminator value="system"/>
      <discriminator value="use"/>
      <rules value="closed"/>
    </slicing>
    <!-- net cardinality rules -->
    <min value="1"/>
    <max value="3"/>
  </element>

  <!-- first slice: home phone -->
  <element>
    <path value="Patient.telecom"/>
    <name value="HomePhone"/> <!-- mandatory - gives the slice a name -->
    <min value="1"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.system"/>
    <min value="1"/>
    <fixedCode value="phone"/>
  </element>
  <element>
    <path value="Patient.telecom.value"/>
    <min value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.use"/>
    <min value="1"/>
    <fixedCode value="home"/>
  </element>

  <!-- second slice: work phone -->
  <element>
    <path value="Patient.telecom"/>
    <name value="WorkPhone"/> <!-- mandatory - gives the slice a name -->
    <min value="0"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.system"/>
    <min value="1"/>
    <fixedCode value="phone"/>
  </element>
  <element>
    <path value="Patient.telecom.value"/>
    <min value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.use"/>
    <min value="1"/>
    <fixedCode value="work"/>
  </element>

  <!-- third slice: email -->
  <element>
```

```
    <path value="Patient.telecom"/>
    <name value="Email"/> <!-- mandatory - gives the slice a name -->
    <min value="0"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.system"/>
    <min value="1"/>
    <fixedCode value="email"/>
  </element>
  <element>
    <path value="Patient.telecom.value"/>
    <min value="1"/>
  </element>
  <element>
    <path value="Patient.telecom.use"/>
    <min value="0"/>
  </element>
```

Note: lots of definition detail has been left out, and only the parts relevant to the pattern are shown. Also, providing a fixed value makes the minimum cardinality irrelevant, but it is shown here for completeness.

This table illustrates the relationship between the instance and the ElementDefinitions:

Path	Name	Min	Max	Fixed
<Patient>	Patient			
Patient.telecom		1	3	
<telecom>	Patient.telecom	HomePhone	1	1
<system value="phone" />	Patient.telecom.system		1	1	phone
<value value="5551234567" />	Patient.telecom.value		1	1	
<use value="home" />	Patient.telecom.use		1	1	home
</telecom>				
Patient.telecom	WorkPhone	0	1	
<telecom>	Patient.telecom	Email	0	1
<system value="email" />	Patient.telecom.system		1	1	email
<value value="someone@acme.org" />	Patient.telecom.value		1	1
</telecom>	Patient.telecom.use			

5.1.1.2 Blood Pressure Example

Another use of Slicing is for Blood Pressure Measurements, where the profile says:

* There must be two components

* The first has LOINC code 8480-6, and a quantity

* The second has LOINC code 8462-4, and a quantity

* Other components are allowed (posture, etc., but not profiled in the base blood pressure profile)

An example of an Observation resource that meets these rules:

```
<Observation>
... snip ...
  <component>
    <code>
      <coding>
        <system value="http://loinc.org" />
        <code value="8480-6" />
        <display value="Systolic blood pressure" />
      </coding>
    </code>
    <valueQuantity>
      <value value="120" />
      <unit value="mmHg" />
      <system value="http://unitsofmeasure.org" />
      <code value="mm[Hg]" />
    </valueQuantity>
  </component>
  <component>
    <code>
      <coding>
        <system value="http://loinc.org" />
        <code value="8462-4" />
        <display value="Diastolic blood pressure" />
      </coding>
    </code>
    <valueQuantity>
      <value value="80" />
      <unit value="mmHg" />
      <system value="http://unitsofmeasure.org" />
      <code value="mm[Hg]" />
    </valueQuantity>
  </component>
</Patient>
```

To do this, the profile that implements these rules needs to do the following:

* On the base Observation.component element: define that slicing is discriminated by code. Order is left unfixed, and rules left open.

* Then define 2 slices:

* systolic: fixed values for code, cardinality 1..1, value is a Quantity

* diastolic: fixed values for code, cardinality 1..1, value is a Quantity

In a StructureDefinition, this will look like:

```
  <!-- setting up the slicing -->
  <element>
    <path value="Observation.component"/>
    <slicing>
      <discriminator value="code"/>
    </slicing>
    <!-- net cardinality rules -->
    <min value="2"/>
    <max value="*"/>
  </element>

  <!-- first slice: systolic -->
  <element>
    <path value="Observation.component"/>
    <name value="systolic"/> <!-- mandatory - gives the slice a name -->
    <min value="1"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Observation.component.code"/>
    <min value="1"/>
    <fixedCodeableConcept>
      <coding>
        <system value="http://loinc.org" />
        <code value="8480-6" />
        <display value="Systolic blood pressure" />
      </coding>
    </fixedCodeableConcept>
  </element>
  <element>
    <path value="Observation.component.valueQuantity"/>
    <min value="1"/>
  </element>

  <!-- second slice: diastolic -->
  <element>
    <path value="Observation.component"/>
    <name value="diastolic"/> <!-- mandatory - gives the slice a name -->
    <min value="1"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Observation.component.code"/>
    <min value="1"/>
    <fixedCodeableConcept>
      <coding>
        <system value="http://loinc.org" />
        <code value="8462-4" />
        <display value="Diastolic blood pressure" />
      </coding>
    </fixedCodeableConcept>
  </element>
  <element>
    <path value="Observation.component.valueQuantity"/>
    <min value="1"/>
  </element>
```

Note: lots of definition detail has been left out, and only the parts relevant to the pattern are shown. E.g. A real blood pressure profile would fix unit, an overall Observation code etc.

5.1.1.3 Extensions

For another example, consider slicing extensions. The base extension on every element is defined as a list (0..*) of extensions, and each extension has a url that identifies it, and a value. Consider an example where a profile defines that for a particular element (named Patient), there are two extensions, with URLs http://acme.com/a and http://acme.com/b. In addition, the profile allows other extensions to be used.

Technically, the profile achieves this by "slicing" the extension list, into two slices, and saying that the slicing is "open" - that there can be other slices introduced. Here are the relevant parts of the Profile on patient:

```
<StructureDefinition xmlns="http://hl7.org/fhir">
  <!-- snip -->
    <baseType value="Patient" />
    <baseDefinition value="http://hl7.org.fhir/StructureDefinition/Patient" />
    <derivation value="constraint" />
    <snapshot>
      <element>
        <path value="Patient"/>
        <!-- snip -->
```

```
      </element>
      <element>
        <path value="Patient.extension"/>
        <!-- this first element defines the slicing, and carries the base definition forward -->
        <slicing>
          <discriminator value="url"/> <!-- Extensions are always discriminated by URL -->
          <ordered value="false"/>     <!-- we don't care what order they appear in -->
          <rules value="open"/>        <!-- other extensions can be used -->
        </slicing>
        <!-- -- snip definition -->
      </element>
      <!-- first extension -->
      <element>
        <path value="Patient.extension"/>
        <!-- snip most of definition -->
        <type>
         <code value="Extension"/>
         <!-- the profile for an extension is a reference to the extension definition itself -
           this implies a profile, and happens to fix the @url value to the desired URL -->
         <profile value="http://acme.com/a"/>
        </type>
      </element>
      <!-- second extension -->
      <element>
        <path value="Patient.extension"/>
        <!-- snip most of definition -->
        <type>
         <code value="Extension"/>
         <!-- the profile for an extension is a reference to the extension definition itself -
           this implies a profile, and happens to fix the @url value to the desired URL -->
         <profile value="http://acme.com/b"/>
        </type>
      </element>
      <!-- snip rest of profile -->
    </snapshot>
</StructureDefinition>
```

Here's a patient example that conforms to this profile:

```
<Patient xmlns="http://hl7.org/fhir">
  <!-- two extensions, the order doesn't matter -->
  <extension url="http://acme.com/b">
    <!-- this has the right url, and so matches the second slice -->
    <!-- snip whatever value extension would have -->
  </extension>
  <extension url="http://acme.com/a">
    <!-- this has the right url, and so matches the first slice -->
    <!-- snip whatever value extension would have -->
  </extension>
  <!-- the rest of patient -->
</Patient>
```

5.1.1.4 Diagnostic Report & Observation

In this example, a profile on an diagnostic report says that it must have 4 observations, each with a different LOINC code (e.g. a classic lab panel). In this case (taken from the Example Lipid Profile), the structure that applies to DiagnosticReport will say that there are 4 slices on DiagnosticReport.result, each conforming to a different structure, which are also contained in the same profile. Each of those structures will constrain the LOINC code in the observation.

```
<!-- first structure, the DiagnosticReport -->
<StructureDefinition xmlns="http://hl7.org/fhir">
  <!-- snip -->
  <url value="http://acme.org/fhir/StructureDefinition/lipid-report"/>
  <name value="LipidProfile"/>
  <baseType value="DiagnosticReport"/>
  <baseDefinition value="http://hl7.org/fhir/StructureDefinition/DiagnosticReport"/>
  <derivation value="constraint" />

  <!-- snip -->
  <snapshot>
    <!-- snip elements -->
    <element>
      <!-- first definition for result -->
      <path value="DiagnosticReport.result"/>
      <slicing>
        <!-- this is sliced by the code value of the target of the reference -->
        <discriminator value="reference.code"/>
        <!-- have to be in the specified order -->
        <ordered value="true"/>
        <!-- this profile says, no other observations allowed -->
        <rules value="closed"/>
      </slicing>
      <!-- snip definition -->
    </element>
    <!-- first slice: Cholesterol -->
    <element>
      <path value="DiagnosticReport.result"/>
      <name value="Cholesterol"/>
      <!-- snip definition parts -->
      <type>
```

```
        <code value="Reference"/>
        <!-- this element must conform to the "Cholesterol" structure -->
        <profile value="http://acme.org/fhir/StructureDefinition/Cholesterol"/>
      </type>
    </element>
    <!-- next 3 slices all the same, but different names for profile -->
    <element>
      <path value="DiagnosticReport.result"/>
      <name value="Triglyceride"/>
      <!-- snip definition parts -->
      <type>
        <code value="Reference"/>
        <!-- this element must conform to the "Triglyceride" structure -->
        <profile value="http://acme.org/fhir/StructureDefinition/Triglyceride"/>
      </type>
    </element>
    <element>
      <path value="DiagnosticReport.result"/>
      <name value="LDLCholesterol"/>
      <!-- snip definition parts -->
      <type>
        <code value="Reference"/>
        <!-- this element must conform to the "LDLCholesterol" structure -->
        <profile value="http://acme.org/fhir/StructureDefinition/LDLCholesterol"/>
      </type>
    </element>
    <element>
      <path value="DiagnosticReport.result"/>
      <name value="HDLCholesterol"/>
      <!-- snip definition parts -->
      <type>
        <code value="Reference"/>
        <!-- this element must conform to the "HDLCholesterol" structure -->
        <profile value="http://acme.org/fhir/StructureDefinition/HDLCholesterol"/>
      </type>
    </element>
    <!-- snip elements -->
  </snapshot>
</StructureDefinition>

<!-- now, the second structure, for the Cholesterol profile -->
<StructureDefinition>
  <url value="http://acme.org/fhir/StructureDefinition/Cholesterol"/>
  <name value="Cholesterol"/>
  <baseType value="Observation"/>
  <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Observation"/>
  <derivation value="constraint" />
  <snapshot>
    <!-- snip elements -->
    <element>
      <!-- this the element definition for name. Because of the
        slicing / discriminator rules in the LipidReport profile
        that references it, it is required to fix the value of
        the name element -->
      <path value="Observation.code"/>
      <definition>
         <!-- there are actually 3 ways to fix a CodeableConcept
         to a single fixed value. Here, we used the simplest one -->
         <valueCodeableConcept>
           <!-- just fix the value to the right code -->
           <coding>
             <system value="http://loinc.org"/>
             <code value="35200-5"/>
             <display value="Cholesterol"/>
           </coding>
         </valueCodeableConcept>
      </definition>
    </element>
    <!-- snip elements -->
  </snapshot>
</StructureDefinition>

<!-- Triglyceride profile -->
<StructureDefinition>
  <url value="http://acme.org/fhir/StructureDefinition/Triglyceride"/>
  <name value="Triglyceride"/>
  <baseType value="Observation"/>
  <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Observation"/>
  <derivation value="constraint" />
  <snapshot>
    <!-- snip elements -->
    <element>
      <!-- this the element definition for name. Because of the
        slicing / discriminator rules in the LipidReport profile
        that references it, it is required to fix the value of
        the name element -->
      <path value="Observation.code"/>
      <definition>
         <!-- there's actually 3 ways to fix a CodeableConcept
```

```
          to a single fixed value. Here, we used the simplest one -->
          <valueCodeableConcept>
            <!-- just fix the value to the right code -->
            <coding>
              <system value="http://loinc.org"/>
              <code value="35217-9"/>
              <display value="Triglyceride"/>
            </coding>
          </valueCodeableConcept>
        </definition>
      </element>
      <!-- snip elements -->
    </snapshot>
  </StructureDefinition>

  <!-- LDLCholesterol profile -->
  <StructureDefinition>
    <url value="http://acme.org/fhir/StructureDefinition/LDLCholesterol"/>
    <name value="LDLCholesterol"/>
    <baseType value="Observation"/>
    <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Observation"/>
    <derivation value="constraint" />
    <snapshot>
      <!-- snip elements -->
      <element>
        <!-- this the element definition for name. Because of the
          slicing / discriminator rules in the LipidReport profile
          that references it, it is required to fix the value of
          the name element -->
        <path value="Observation.code"/>
        <definition>
           <!-- because of the way that LDL cholesterol measurements works
             (well, in this context- it varies), there's 2 different LOINC
             codes for either measured or calculated. So here, we bind to
             a value set -->
         <binding>
          <conformance value="required"/> <!-- must be required if this is a discriminator -->
          <!-- snip the actual value set reference, but it refers to a value
               set with two LOINC codes, one for each kind of LDL, which in
               this case are LOINC codes 18262-6 and 13457-7 -->
         </binding>
       </definition>
      </element>
      <!-- snip elements -->
    </snapshot>
  </StructureDefinition>

  <!-- HDLCholesterol profile -->
  <StructureDefinition>
    <url value="http://acme.org/fhir/StructureDefinition/HDLCholesterol"/>
    <name value="HDLCholesterol"/>
    <baseType value="Observation"/>
    <baseDefinition value="http://hl7.org/fhir/StructureDefinition/Observation"/>
    <derivation value="constraint" />
    <snapshot>
      <!-- snip elements -->
      <element>
        <!-- this the element definition for name. Because of the
          slicing / discriminator rules in the LipidReport profile
          that references it, it is required to fix the value of
          the name element -->
        <path value="Observation.code"/>
        <definition>
           <!-- there's actually 3 ways to fix a CodeableConcept
           to a single fixed value. Here, we used the simplest one -->
           <valueCodeableConcept>
             <!-- just fix the value to the right code -->
             <coding>
               <system value="http://loinc.org"/>
               <code value="2085-9"/>
               <display value="LDL Cholesterol"/>
             </coding>
           </valueCodeableConcept>
        </definition>
      </element>
      <!-- snip elements -->
    </snapshot>
  </StructureDefinition>
```

Here is an instance that meets the rules for this profile:

```
  <!-- first, the diagnostic report -->
  <DiagnosticReport xmlns="http://hl7.org/fhir">
    <!-- snip -->
    <!-- here's the set of results. We don't know what
      slices they are or anything until we go off, find
      the references, and look in them -->
    <result>
      <reference value="Observation/cholesterol"/>
    </result>
```

```
  <result>
    <reference value="Observation/triglyceride"/>
  </result>
  <result>
    <reference value="Observation/ldlcholesterol"/>
  </result>
  <result>
    <reference value="Observation/hdlcholesterol"/>
  </result>
  <!-- snip -->
</DiagnosticReport>

<!-- Observation, id = cholesterol -->
<Observation xmlns="http://hl7.org/fhir">
  <!-- the observation starts with the name, as specified
    by the profile for the first slice  -->
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="35200-5"/>
      <display value="Cholesterol"/>
    </coding>
  </code>
  <!-- snip -->
</Observation>

<!-- Observation, id = triglyceride -->
<!-- this code matches the second slice. good -->
<Observation xmlns="http://hl7.org/fhir">
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="35217-9"/>
      <display value="Triglyceride"/>
    </coding>
  </code>
  <!-- snip -->
</Observation>

<!-- Observation, id = hdlcholesterol -->
<!-- this code matches the fourth slice. good -->
<Observation xmlns="http://hl7.org/fhir">
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="2085-9"/>
      <display value="HDL Cholesterol"/>
    </coding>
  </code>
  <!-- snip -->
</Observation>

<!-- Observation, id = ldlcholesterol -->
<!-- this code matches the third slice. good -->
<Observation id="ldlcholesterol">
  <code>
    <coding>
      <system value="http://loinc.org"/>
      <code value="13457-7"/>
      <display value="LDL Chol. (Calc)"/>
    </coding>
  </code>
  <!-- snip -->
</Observation>
```

Note that this version isn't valid, because the slices are not in the correct order:

```
<!-- first, the diagnostic report -->
<DiagnosticReport xmlns="http://hl7.org/fhir">
  <!-- snip -->
  <!-- here's the set of results. We don't know what
    slices they are or anything until we go off, find
    the references, and look in them -->
  <result>
    <reference value="Observation/cholesterol"/>
  </result>
  <result>
    <reference value="Observation/triglyceride"/>
  </result>
    <!-- Oops the last two are out of order -->
  <result>
    <reference value="Observation/hdlcholesterol"/>
  </result>
  <result>
    <reference value="Observation/ldlcholesterol"/>
  </result>
  <!-- snip -->
</DiagnosticReport>
```

5.1.1.5 Composition Sections

Most uses of Composition involve conformance to a profile that specifies which sections will exist, and what their contents will be. This is yet another example of slicing. A typical document content profile might specify a section structure something like this:

* Reason for visit Narrative, LOINC Code 29299-5

* Medications, LOINC Code 46057-6

* Prescribed Medications, LOINC Code 66149-6

* OTC medications, 66150-4 (optional)

* Vital Signs, LOINC Code 8716-3

Real profiles will contain lots of detail about the sections, but these are omitted here in the interests of clarity.

An example of a Composition that meets these rules:

```
<Composition>
... snip ...
  <section>
    <code>
      <coding>
        <system value="http://loinc.org" />
        <use value="29299-5" />
        <value value="Reason for visit Narrative" />
      </coding>
    </code>
    ... snip ...
  </section>
  <section>
    <code>
      <coding>
        <system value="http://loinc.org" />
        <use value="46057-6" />
        <value value="Medications section" />
      </coding>
    </code>
    ... snip ...
    <section>
      <code>
        <coding>
          <system value="http://loinc.org" />
          <use value="66149-6" />
          <value value="Prescribed medications" />
        </coding>
      </code>
      ... snip ...
    </section>
    <section>
      <code>
        <coding>
          <system value="http://loinc.org" />
          <use value="66150-4" />
          <value value="O medications" />
        </coding>
      </code>
      ... snip ...
    </section>
  </section>
  <section>
    <code>
      <coding>
        <system value="http://loinc.org" />
        <use value="8716-3" />
        <value value="Vital signs" />
      </coding>
    </code>
    ... snip ...
  </section>
</Composition>
```

To do this, the profile that implements these rules needs to do the following:

* On the base Composition.section element: define that slicing is discriminated by code, and that the slices are closed and ordered

* Then define 3 slices on Composition.section:

* reason-for-visit: fixed code, cardinality 1..1

* medication: fixed code, cardinality 1..1

* vital-signs: fixed code, cardinality 1..1

* Then, in the medication slice, slice the Composition.section.section: define that slicing is discriminated by code, and that the slices are closed and ordered

* Then define 2 slices on the Composition.section.section in medication:

* prescribed: fixed code, cardinality 1..1

* otc: fixed code, cardinality 0..1

In a StructureDefinition, this will look like:

```
<!-- setting up the slicing on Composition.section -->
<element>
  <path value="Composition.section"/>
  <slicing>
    <discriminator value="code"/>
    <ordered value="true"/>
    <rules value="closed"/>
  </slicing>
  <!-- net cardinality rules -->
  <min value="3"/>
  <max value="3"/>
</element>

<!-- first slice: reason for visit -->
<element>
  <path value="Composition.section"/>
  <name value="reason-for-visit"/>
  <min value="1"/>
  <max value="1"/>
</element>
<element>
  <path value="Composition.section.code"/>
  <min value="1"/>
  <fixedCodeableConcept>
    <coding>
      <system value="http://loinc.org" />
      <use value="29299-5" />
      <value value="Reason for visit Narrative" />
    </coding>
  </fixedCodeableConcept>
</element>

<!-- second slice: medications -->
<element>
  <path value="Composition.section"/>
  <name value="medications"/>
  <min value="1"/>
  <max value="1"/>
</element>
<element>
  <path value="Composition.section.code"/>
  <min value="1"/>
  <fixedCodeableConcept>
    <coding>
      <system value="http://loinc.org" />
      <use value="46057-6" />
      <value value="Medications section" />
    </coding>
  </fixedCodeableConcept>
</element>

<!-- setting up the inner slicing on medication Composition.section.section -->
<element>
  <path value="Composition.section.section"/>
  <slicing>
    <discriminator value="code"/>
    <ordered value="true"/>
    <rules value="closed"/>
  </slicing>
  <!-- net cardinality rules -->
  <min value="1"/>
  <max value="2"/>
</element>

<!-- first inner slice: prescribed medications -->
<element>
  <path value="Composition.section.section"/>
  <name value="prescribed"/>
  <min value="1"/>
  <max value="1"/>
</element>
<element>
  <path value="Composition.section.section.code"/>
  <min value="1"/>
  <fixedCodeableConcept>
    <coding>
      <system value="http://loinc.org" />
      <use value="66149-6" />
      <value value="Prescribed medications" />
    </coding>
  </fixedCodeableConcept>
</element>

<!-- second inner slice: over the counter medications -->
<element>
  <path value="Composition.section.section"/>
  <name value="otc"/>
```

```
    <min value="0"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Composition.section.section.code"/>
    <min value="1"/>
    <fixedCodeableConcept>
      <coding>
        <system value="http://loinc.org" />
        <use value="66150-4" />
        <value value="Over the counter medications" />
      </coding>
    </fixedCodeableConcept>
  </element>

  <!-- third slice: Vital Signs -->
  <element>
    <path value="Composition.section"/>
    <name value="vital-signs"/>
    <min value="1"/>
    <max value="1"/>
  </element>
  <element>
    <path value="Composition.section.code"/>
    <min value="1"/>
    <fixedCodeableConcept>
      <coding>
        <system value="http://loinc.org" />
        <use value="8716-3" />
        <value value="Vital signs" />
      </coding>
    </fixedCodeableConcept>
  </element>
```

On This Page:

Patient Contact Details

Blood Pressure

Extensions

Diagnostic Report

Composition Sections

5.2 Resource CapabilityStatement - Content

URL: https://hl7.org/fhir/STU3/capabilitystatement.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server that may be used as a statement of actual server functionality or a statement of required or desired server implementation.

5.2.1 Scope and Usage

The capability statement is a key part of the overall conformance framework in FHIR. It is used as a statement of the features of actual software, or of a set of rules for an application to provide. This statement connects to all the detailed statements of functionality, such as StructureDefinitions and ValueSets. This composite statement of application capability may be used for system compatibility testing, code generation, or as either the basis for a conformance assessment. For further information about Conformance testing, see Conformance Rules and Profiling FHIR.

Specifically, capability statements are used in one of three ways:

5.2.1.1 Describe an actual implementation

In this scenario, the capability statement describes the capabilities of a deployed and configured solution available at a particular access point or set of access points. The statement describes exactly how to interface with that deployed solution and thus provides for a degree of self-configuration of software solutions.

This is the type of profile that FHIR restful solutions are expected to make available on invocation of the capabilities operation. It is also the type of statement that forms a basis for the testing, certification or commissioning of specific software installations.

5.2.1.2 Describe software solution capabilities

In this scenario, the capability statement describes generic capabilities of a software application or component solution. The solution might be available for purchase or other acquisition and might be deployed and configured at any number of independent sites. Because it is not dependent on any particular implementation, the profile cannot provide specific details such as endpoint addresses. It may also need to document various configurations in which the application can be set up or describe the degree of customizability associated with the solution.

This type of statement may be used as a marketing tool by software and system developers to formally describe their capabilities. It can also be used as the basis for conformance testing of software solutions independent of a particular installation.

5.2.1.3 Describe a desired solution

In this scenario, the capability statement describes the capabilities of a desired system. It might be used as part of an architectural design process to document needed system capabilities, or might be used as part of an RFP process to formally document the requirements of a requested solution and to document the criteria by which proposals will be evaluated.

These three types of profiles can be used together. A requirements statement can be compared against the solution statements proffered by respondents to an RFP. A solution statement for a software package forms the starting point for the implementation statement associated with a particular installation of that software package.

5.2.2 Background and Context

Capability Statements provide for a degree of automatic configuration and adaptation. However, capturing absolutely every variation that could impact the interoperability of two systems, let alone keeping that detailed information up-to-date as systems evolve through maintenance and upgrades is rarely practical. Therefore, capability statements should be seen as an interim step. They provide a degree of automation. However, they also provide a great deal of human-readable content that can minimize the need for direct communication between the operators of the systems being configured to interoperate.

This resource is referenced by testscript

5.2.3 Resource Content

* Structure
* UML
* XML
* JSON
* Turtle
* R2 Diff
* All

Structure

Name | Flags | Card. | Type | Description & Constraints
CapabilityStatement | I | | DomainResource | A statement of system capabilities + There can only be one REST declaration per mode. + The set of documents must be unique by the combination of profile and mode. + Capability Statements of kind 'instance' do not have implementation elements. + Messaging end-point is required (and is only permitted) when a statement is for an implementation. + Capability Statements of kind 'requirements' do not have software or implementation elements. + A Capability Statement SHALL have at least one of description, software, or implementation element. + A Capability Statement SHALL have at least one of REST, messaging or document element. Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this capability statement (globally unique)
version | ? | 0..1 | string | Business version of the capability statement
name | ? | 0..1 | string | Name for this capability statement (computer friendly)
title | ? | 0..1 | string | Name for this capability statement (human friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 1..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
description | I | 0..1 | markdown | Natural language description of the capability statement
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for capability statement (if applicable) Jurisdiction ValueSet (Extensible)
purpose | | 0..1 | markdown | Why this capability statement is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
kind | ? | 1..1 | code | instance | capability | requirements CapabilityStatementKind (Required)
instantiates | ? | 0..* | uri | Canonical URL of another capability statement this implements
software | ?I | 0..1 | BackboneElement | Software that is covered by this capability statement
name | ? | 1..1 | string | A name the software is known by
version | ? | 0..1 | string | Version covered by this statement
releaseDate | ? | 0..1 | dateTime | Date this version released
implementation | ?I | 0..1 | BackboneElement | If this describes a specific instance
description | ? | 1..1 | string | Describes this specific instance
url | ? | 0..1 | uri | Base URL for the installation
fhirVersion | ? | 1..1 | id | FHIR Version the system uses
acceptUnknown | ? | 1..1 | code | no | extensions | elements | both UnknownContentCode (Required)
format | ? | 1..* | code | formats supported (xml | json | ttl | mime type) MimeType (Required)
patchFormat | ? | 0..* | code | Patch formats supported MimeType (Required)
implementationGuide | ? | 0..* | uri | Implementation guides supported
profile | ? | 0..* | Reference(StructureDefinition) | Profiles for use cases supported
rest | ?I | 0..* | BackboneElement | If the endpoint is a RESTful one + A given resource can only be described once per RESTful mode.
mode | ? | 1..1 | code | client | server RestfulCapabilityMode (Required)

documentation | | 0..1 | string | General description of implementation
security | ? | 0..1 | BackboneElement | Information about security of implementation
cors | ? | 0..1 | boolean | Adds CORS Headers (http://enable-cors.org/)
service | ? | 0..* | CodeableConcept | OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates RestfulSecurityService (Extensible)
description | | 0..1 | string | General description of how security works
certificate | | 0..* | BackboneElement | Certificates associated with security profiles
type | | 0..1 | code | Mime type for certificates MimeType (Required)
blob | | 0..1 | base64Binary | Actual certificate
resource | ?I | 0..* | BackboneElement | Resource served on the REST interface + Search parameter names must be unique in the context of a resource.
type | ? | 1..1 | code | A resource type that is supported ResourceType (Required)
profile | ? | 0..1 | Reference(StructureDefinition) | Base System profile for all uses of resource
documentation | | 0..1 | markdown | Additional information about the use of the resource type
interaction | | 1..* | BackboneElement | What operations are supported?
code | | 1..1 | code | read | vread | update | patch | delete | history-instance | history-type | create | search-type TypeRestfulInteraction (Required)
documentation | | 0..1 | string | Anything special about operation behavior
versioning | | 0..1 | code | no-version | versioned | versioned-update ResourceVersionPolicy (Required)
readHistory | | 0..1 | boolean | Whether vRead can return past versions
updateCreate | | 0..1 | boolean | If update can commit to a new identity
conditionalCreate | | 0..1 | boolean | If allows/uses conditional create
conditionalRead | | 0..1 | code | not-supported | modified-since | not-match | full-support ConditionalReadStatus (Required)
conditionalUpdate | | 0..1 | boolean | If allows/uses conditional update
conditionalDelete | | 0..1 | code | not-supported | single | multiple - how conditional delete is supported ConditionalDeleteStatus (Required)
referencePolicy | | 0..* | code | literal | logical | resolves | enforced | local ReferenceHandlingPolicy (Required)
searchInclude | | 0..* | string | _include values supported by the server
searchRevInclude | | 0..* | string | _revinclude values supported by the server
searchParam | | 0..* | BackboneElement | Search parameters supported by implementation
name | | 1..1 | string | Name of search parameter
definition | | 0..1 | uri | Source of definition for parameter
type | | 1..1 | code | number | date | string | token | reference | composite | quantity | uri SearchParamType (Required)
documentation | | 0..1 | string | Server-specific usage
interaction | | 0..* | BackboneElement | What operations are supported?
code | | 1..1 | code | transaction | batch | search-system | history-system SystemRestfulInteraction (Required)
documentation | | 0..1 | string | Anything special about operation behavior
searchParam | | 0..* | see searchParam | Search parameters for searching all resources
operation | ? | 0..* | BackboneElement | Definition of an operation or a custom query
name | | 1..1 | string | Name by which the operation/query is invoked
definition | ? | 1..1 | Reference(OperationDefinition) | The defined operation/query
compartment | | 0..* | uri | Compartments served/used by system
messaging | ?I | 0..* | BackboneElement | If messaging is supported + A Capability Statement messaging element SHALL have either supportedMessage or event element, but not both.
endpoint | | 0..* | BackboneElement | Where messages should be sent
protocol | | 1..1 | Coding | http | ftp | mllp + MessageTransport (Extensible)
address | | 1..1 | uri | Network address or identifier of the end-point
reliableCache | | 0..1 | unsignedInt | Reliable Message Cache Length (min)
documentation | | 0..1 | string | Messaging interface behavior details
supportedMessage | ? | 0..* | BackboneElement | Messages supported by this system
mode | ? | 1..1 | code | sender | receiver EventCapabilityMode (Required)
definition | ? | 1..1 | Reference(MessageDefinition) | Message supported by this system
event | ? | 0..* | BackboneElement | Declare support for this event
code | ? | 1..1 | Coding | Event type MessageEvent (Example)
category | | 0..1 | code | Consequence | Currency | Notification MessageSignificanceCategory (Required)
mode | | 1..1 | code | sender | receiver EventCapabilityMode (Required)
focus | | 1..1 | code | Resource that's focus of message ResourceType (Required)
request | ? | 1..1 | Reference(StructureDefinition) | Profile that describes the request
response | ? | 1..1 | Reference(StructureDefinition) | Profile that describes the response
documentation | | 0..1 | string | Endpoint-specific event documentation
document | ?I | 0..* | BackboneElement | Document definition
mode | | 1..1 | code | producer | consumer DocumentMode (Required)
documentation | | 0..1 | string | Description of document support
profile | ? | 1..1 | Reference(StructureDefinition) | Constraint on a resource used in the document
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CapabilityStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this capability statement (globally unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the capability statement -->
 <name value="[string]"/><!-- 0..1 Name for this capability statement (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this capability statement (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
```

```
<date value="[dateTime]"/><!-- 1..1 Date this was last changed -->
<publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
<contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
<description value="[markdown]"/><!--  0..1 Natural language description of the capability statement -->
<useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
<jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for capability statement (if applicable)
   --></jurisdiction>
<purpose value="[markdown]"/><!-- 0..1 Why this capability statement is defined -->
<copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
<kind value="[code]"/><!-- 1..1 instance | capability | requirements -->
<instantiates value="[uri]"/><!-- 0..* Canonical URL of another capability statement this implements -->
<software>  <!--  0..1 Software that is covered by this capability statement -->
 <name value="[string]"/><!-- 1..1 A name the software is known by -->
 <version value="[string]"/><!-- 0..1 Version covered by this statement -->
 <releaseDate value="[dateTime]"/><!-- 0..1 Date this version released -->
</software>
<implementation>  <!--  0..1 If this describes a specific instance -->
 <description value="[string]"/><!-- 1..1 Describes this specific instance -->
 <url value="[uri]"/><!-- 0..1 Base URL for the installation -->
</implementation>
<fhirVersion value="[id]"/><!-- 1..1 FHIR Version the system uses -->
<acceptUnknown value="[code]"/><!-- 1..1 no | extensions | elements | both -->
<format value="[code]"/><!-- 1..* formats supported (xml | json | ttl | mime type)  -->
<patchFormat value="[code]"/><!-- 0..* Patch formats supported  -->
<implementationGuide value="[uri]"/><!-- 0..* Implementation guides supported -->
<profile><!-- 0..* Reference(StructureDefinition) Profiles for use cases supported --></profile>
<rest>  <!--  0..* If the endpoint is a RESTful one -->
 <mode value="[code]"/><!-- 1..1 client | server -->
 <documentation value="[string]"/><!-- 0..1 General description of implementation -->
 <security>  <!-- 0..1 Information about security of implementation -->
  <cors value="[boolean]"/><!-- 0..1 Adds CORS Headers (http://enable-cors.org/) -->
  <service><!-- 0..* CodeableConcept OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates --></service>
  <description value="[string]"/><!-- 0..1 General description of how security works -->
  <certificate>  <!-- 0..* Certificates associated with security profiles -->
   <type value="[code]"/><!-- 0..1 Mime type for certificates  -->
   <blob value="[base64Binary]"/><!-- 0..1 Actual certificate -->
  </certificate>
 </security>
 <resource>  <!-- 0..* Resource served on the REST interface -->
  <type value="[code]"/><!-- 1..1 A resource type that is supported -->
  <profile><!-- 0..1 Reference(StructureDefinition) Base System profile for all uses of resource --></profile>
  <documentation value="[markdown]"/><!-- 0..1 Additional information about the use of the resource type -->
  <interaction>  <!-- 1..* What operations are supported? -->
   <code value="[code]"/><!-- 1..1 read | vread | update | patch | delete | history-instance | history-type | create |
   search-type -->
   <documentation value="[string]"/><!-- 0..1 Anything special about operation behavior -->
  </interaction>
  <versioning value="[code]"/><!-- 0..1 no-version | versioned | versioned-update -->
  <readHistory value="[boolean]"/><!-- 0..1 Whether vRead can return past versions -->
  <updateCreate value="[boolean]"/><!-- 0..1 If update can commit to a new identity -->
  <conditionalCreate value="[boolean]"/><!-- 0..1 If allows/uses conditional create -->
  <conditionalRead value="[code]"/><!-- 0..1 not-supported | modified-since | not-match | full-support -->
  <conditionalUpdate value="[boolean]"/><!-- 0..1 If allows/uses conditional update -->
  <conditionalDelete value="[code]"/><!-- 0..1 not-supported | single | multiple - how conditional delete is supported
   -->
  <referencePolicy value="[code]"/><!-- 0..* literal | logical | resolves | enforced | local -->
  <searchInclude value="[string]"/><!-- 0..* _include values supported by the server -->
  <searchRevInclude value="[string]"/><!-- 0..* _revinclude values supported by the server -->
  <searchParam>  <!-- 0..* Search parameters supported by implementation -->
   <name value="[string]"/><!-- 1..1 Name of search parameter -->
   <definition value="[uri]"/><!-- 0..1 Source of definition for parameter -->
   <type value="[code]"/><!-- 1..1 number | date | string | token | reference | composite | quantity | uri -->
   <documentation value="[string]"/><!-- 0..1 Server-specific usage -->
  </searchParam>
 </resource>
 <interaction>  <!-- 0..* What operations are supported? -->
  <code value="[code]"/><!-- 1..1 transaction | batch | search-system | history-system -->
  <documentation value="[string]"/><!-- 0..1 Anything special about operation behavior -->
 </interaction>
 <searchParam><!-- 0..* Content as for CapabilityStatement.rest.resource.searchParam Search parameters for searching
   all resources --></searchParam>
 <operation>  <!-- 0..* Definition of an operation or a custom query -->
  <name value="[string]"/><!-- 1..1 Name by which the operation/query is invoked -->
  <definition><!-- 1..1 Reference(OperationDefinition) The defined operation/query --></definition>
 </operation>
 <compartment value="[uri]"/><!-- 0..* Compartments served/used by system -->
</rest>
<messaging>  <!--  0..* If messaging is supported -->
 <endpoint>  <!-- 0..* Where messages should be sent -->
  <protocol><!-- 1..1 Coding http | ftp | mllp + --></protocol>
  <address value="[uri]"/><!-- 1..1 Network address or identifier of the end-point -->
 </endpoint>
 <reliableCache value="[unsignedInt]"/><!-- 0..1 Reliable Message Cache Length (min) -->
 <documentation value="[string]"/><!-- 0..1 Messaging interface behavior details -->
 <supportedMessage>  <!-- 0..* Messages supported by this system -->
  <mode value="[code]"/><!-- 1..1 sender | receiver -->
  <definition><!-- 1..1 Reference(MessageDefinition) Message supported by this system --></definition>
 </supportedMessage>
 <event>  <!-- 0..* Declare support for this event -->
```

```
   <code><!-- 1..1 Coding Event type --></code>
   <category value="[code]"/><!-- 0..1 Consequence | Currency | Notification -->
   <mode value="[code]"/><!-- 1..1 sender | receiver -->
   <focus value="[code]"/><!-- 1..1 Resource that's focus of message -->
   <request><!-- 1..1 Reference(StructureDefinition) Profile that describes the request --></request>
   <response><!-- 1..1 Reference(StructureDefinition) Profile that describes the response --></response>
   <documentation value="[string]"/><!-- 0..1 Endpoint-specific event documentation -->
  </event>
 </messaging>
 <document>  <!--  0..* Document definition -->
  <mode value="[code]"/><!-- 1..1 producer | consumer -->
  <documentation value="[string]"/><!-- 0..1 Description of document support -->
  <profile><!-- 1..1 Reference(StructureDefinition) Constraint on a resource used in the document --></profile>
 </document>
</CapabilityStatement>
```

JSON Template

```
{
  "resourceType" : "CapabilityStatement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this capability statement (globally unique)
  "version" : "<string>", // Business version of the capability statement
  "name" : "<string>", // Name for this capability statement (computer friendly)
  "title" : "<string>", // Name for this capability statement (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // R!  Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // C? Natural language description of the capability statement
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for capability statement (if applicable)
  "purpose" : "<markdown>", // Why this capability statement is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "kind" : "<code>", // R!  instance | capability | requirements
  "instantiates" : ["<uri>"], // Canonical URL of another capability statement this implements
  "software" : { // C? Software that is covered by this capability statement
    "name" : "<string>", // R!  A name the software is known by
    "version" : "<string>", // Version covered by this statement
    "releaseDate" : "<dateTime>" // Date this version released
  },
  "implementation" : { // C? If this describes a specific instance
    "description" : "<string>", // R!  Describes this specific instance
    "url" : "<uri>" // Base URL for the installation
  },
  "fhirVersion" : "<id>", // R!  FHIR Version the system uses
  "acceptUnknown" : "<code>", // R!  no | extensions | elements | both
  "format" : ["<code>"], // R!  formats supported (xml | json | ttl | mime type)
  "patchFormat" : ["<code>"], // Patch formats supported
  "implementationGuide" : ["<uri>"], // Implementation guides supported
  "profile" : [{ Reference(StructureDefinition) }], // Profiles for use cases supported
  "rest" : [{ // C? If the endpoint is a RESTful one
    "mode" : "<code>", // R!  client | server
    "documentation" : "<string>", // General description of implementation
    "security" : { // Information about security of implementation
      "cors" : <boolean>, // Adds CORS Headers (http://enable-cors.org/)
      "service" : [{ CodeableConcept }], // OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates
      "description" : "<string>", // General description of how security works
      "certificate" : [{ // Certificates associated with security profiles
        "type" : "<code>", // Mime type for certificates
        "blob" : "<base64Binary>" // Actual certificate
      }]
    },
    "resource" : [{ // Resource served on the REST interface
      "type" : "<code>", // R!  A resource type that is supported
      "profile" : { Reference(StructureDefinition) }, // Base System profile for all uses of resource
      "documentation" : "<markdown>", // Additional information about the use of the resource type
      "interaction" : [{ // R!  What operations are supported?
        "code" : "<code>", // R!  read | vread | update | patch | delete | history-instance | history-type | create |
    search-type
        "documentation" : "<string>" // Anything special about operation behavior
      }],
      "versioning" : "<code>", // no-version | versioned | versioned-update
      "readHistory" : <boolean>, // Whether vRead can return past versions
      "updateCreate" : <boolean>, // If update can commit to a new identity
      "conditionalCreate" : <boolean>, // If allows/uses conditional create
      "conditionalRead" : "<code>", // not-supported | modified-since | not-match | full-support
      "conditionalUpdate" : <boolean>, // If allows/uses conditional update
      "conditionalDelete" : "<code>", // not-supported | single | multiple - how conditional delete is supported
      "referencePolicy" : ["<code>"], // literal | logical | resolves | enforced | local
      "searchInclude" : ["<string>"], // _include values supported by the server
      "searchRevInclude" : ["<string>"], // _revinclude values supported by the server
      "searchParam" : [{ // Search parameters supported by implementation
        "name" : "<string>", // R!  Name of search parameter
        "definition" : "<uri>", // Source of definition for parameter
        "type" : "<code>", // R!  number | date | string | token | reference | composite | quantity | uri
        "documentation" : "<string>" // Server-specific usage
      }]
```

```
    }],
    "interaction" : [{ // What operations are supported?
      "code" : "<code>", // R!  transaction | batch | search-system | history-system
      "documentation" : "<string>" // Anything special about operation behavior
    }],
    "searchParam" : [{ Content as for CapabilityStatement.rest.resource.searchParam }], // Search parameters for
    searching all resources
    "operation" : [{ // Definition of an operation or a custom query
      "name" : "<string>", // R!  Name by which the operation/query is invoked
      "definition" : { Reference(OperationDefinition) } // R!  The defined operation/query
    }],
    "compartment" : ["<uri>"] // Compartments served/used by system
  }],
  "messaging" : [{ // C? If messaging is supported
    "endpoint" : [{ // Where messages should be sent
      "protocol" : { Coding }, // R!  http | ftp | mllp +
      "address" : "<uri>" // R!  Network address or identifier of the end-point
    }],
    "reliableCache" : "<unsignedInt>", // Reliable Message Cache Length (min)
    "documentation" : "<string>", // Messaging interface behavior details
    "supportedMessage" : [{ // Messages supported by this system
      "mode" : "<code>", // R!  sender | receiver
      "definition" : { Reference(MessageDefinition) } // R!  Message supported by this system
    }],
    "event" : [{ // Declare support for this event
      "code" : { Coding }, // R!  Event type
      "category" : "<code>", // Consequence | Currency | Notification
      "mode" : "<code>", // R!  sender | receiver
      "focus" : "<code>", // R!  Resource that's focus of message
      "request" : { Reference(StructureDefinition) }, // R!  Profile that describes the request
      "response" : { Reference(StructureDefinition) }, // R!  Profile that describes the response
      "documentation" : "<string>" // Endpoint-specific event documentation
    }]
  }],
  "document" : [{ // C? Document definition
    "mode" : "<code>", // R!  producer | consumer
    "documentation" : "<string>", // Description of document support
    "profile" : { Reference(StructureDefinition) } // R!  Constraint on a resource used in the document
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CapabilityStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CapabilityStatement.url [ uri ]; # 0..1 Logical URI to reference this capability statement (globally unique)
  fhir:CapabilityStatement.version [ string ]; # 0..1 Business version of the capability statement
  fhir:CapabilityStatement.name [ string ]; # 0..1 Name for this capability statement (computer friendly)
  fhir:CapabilityStatement.title [ string ]; # 0..1 Name for this capability statement (human friendly)
  fhir:CapabilityStatement.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:CapabilityStatement.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:CapabilityStatement.date [ dateTime ]; # 1..1 Date this was last changed
  fhir:CapabilityStatement.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:CapabilityStatement.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:CapabilityStatement.description [ markdown ]; # 0..1 Natural language description of the capability statement
  fhir:CapabilityStatement.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:CapabilityStatement.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for capability statement
     (if applicable)
  fhir:CapabilityStatement.purpose [ markdown ]; # 0..1 Why this capability statement is defined
  fhir:CapabilityStatement.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:CapabilityStatement.kind [ code ]; # 1..1 instance | capability | requirements
  fhir:CapabilityStatement.instantiates [ uri ], ... ; # 0..* Canonical URL of another capability statement this
    implements
  fhir:CapabilityStatement.software [ # 0..1 Software that is covered by this capability statement
    fhir:CapabilityStatement.software.name [ string ]; # 1..1 A name the software is known by
    fhir:CapabilityStatement.software.version [ string ]; # 0..1 Version covered by this statement
    fhir:CapabilityStatement.software.releaseDate [ dateTime ]; # 0..1 Date this version released
  ];
  fhir:CapabilityStatement.implementation [ # 0..1 If this describes a specific instance
    fhir:CapabilityStatement.implementation.description [ string ]; # 1..1 Describes this specific instance
    fhir:CapabilityStatement.implementation.url [ uri ]; # 0..1 Base URL for the installation
  ];
  fhir:CapabilityStatement.fhirVersion [ id ]; # 1..1 FHIR Version the system uses
  fhir:CapabilityStatement.acceptUnknown [ code ]; # 1..1 no | extensions | elements | both
  fhir:CapabilityStatement.format [ code ], ... ; # 1..* formats supported (xml | json | ttl | mime type)
  fhir:CapabilityStatement.patchFormat [ code ], ... ; # 0..* Patch formats supported
  fhir:CapabilityStatement.implementationGuide [ uri ], ... ; # 0..* Implementation guides supported
  fhir:CapabilityStatement.profile [ Reference(StructureDefinition) ], ... ; # 0..* Profiles for use cases supported
  fhir:CapabilityStatement.rest [ # 0..* If the endpoint is a RESTful one
    fhir:CapabilityStatement.rest.mode [ code ]; # 1..1 client | server
    fhir:CapabilityStatement.rest.documentation [ string ]; # 0..1 General description of implementation
    fhir:CapabilityStatement.rest.security [ # 0..1 Information about security of implementation
      fhir:CapabilityStatement.rest.security.cors [ boolean ]; # 0..1 Adds CORS Headers (http://enable-cors.org/)
      fhir:CapabilityStatement.rest.security.service [ CodeableConcept ], ... ; # 0..* OAuth | SMART-on-FHIR | NTLM |
```

```
    Basic | Kerberos | Certificates
      fhir:CapabilityStatement.rest.security.description [ string ]; # 0..1 General description of how security works
      fhir:CapabilityStatement.rest.security.certificate [ # 0..* Certificates associated with security profiles
        fhir:CapabilityStatement.rest.security.certificate.type [ code ]; # 0..1 Mime type for certificates
        fhir:CapabilityStatement.rest.security.certificate.blob [ base64Binary ]; # 0..1 Actual certificate
      ], ...;
    ];
    fhir:CapabilityStatement.rest.resource [ # 0..* Resource served on the REST interface
      fhir:CapabilityStatement.rest.resource.type [ code ]; # 1..1 A resource type that is supported
      fhir:CapabilityStatement.rest.resource.profile [ Reference(StructureDefinition) ]; # 0..1 Base System profile for
    all uses of resource
      fhir:CapabilityStatement.rest.resource.documentation [ markdown ]; # 0..1 Additional information about the use of
    the resource type
      fhir:CapabilityStatement.rest.resource.interaction [ # 1..* What operations are supported?
        fhir:CapabilityStatement.rest.resource.interaction.code [ code ]; # 1..1 read | vread | update | patch | delete
    | history-instance | history-type | create | search-type
        fhir:CapabilityStatement.rest.resource.interaction.documentation [ string ]; # 0..1 Anything special about
    operation behavior
      ], ...;
      fhir:CapabilityStatement.rest.resource.versioning [ code ]; # 0..1 no-version | versioned | versioned-update
      fhir:CapabilityStatement.rest.resource.readHistory [ boolean ]; # 0..1 Whether vRead can return past versions
      fhir:CapabilityStatement.rest.resource.updateCreate [ boolean ]; # 0..1 If update can commit to a new identity
      fhir:CapabilityStatement.rest.resource.conditionalCreate [ boolean ]; # 0..1 If allows/uses conditional create
      fhir:CapabilityStatement.rest.resource.conditionalRead [ code ]; # 0..1 not-supported | modified-since | not-match
     | full-support
      fhir:CapabilityStatement.rest.resource.conditionalUpdate [ boolean ]; # 0..1 If allows/uses conditional update
      fhir:CapabilityStatement.rest.resource.conditionalDelete [ code ]; # 0..1 not-supported | single | multiple - how
    conditional delete is supported
      fhir:CapabilityStatement.rest.resource.referencePolicy [ code ], ... ; # 0..* literal | logical | resolves |
    enforced | local
      fhir:CapabilityStatement.rest.resource.searchInclude [ string ], ... ; # 0..* _include values supported by the
    server
      fhir:CapabilityStatement.rest.resource.searchRevInclude [ string ], ... ; # 0..* _revinclude values supported by
    the server
      fhir:CapabilityStatement.rest.resource.searchParam [ # 0..* Search parameters supported by implementation
        fhir:CapabilityStatement.rest.resource.searchParam.name [ string ]; # 1..1 Name of search parameter
        fhir:CapabilityStatement.rest.resource.searchParam.definition [ uri ]; # 0..1 Source of definition for parameter
        fhir:CapabilityStatement.rest.resource.searchParam.type [ code ]; # 1..1 number | date | string | token |
    reference | composite | quantity | uri
        fhir:CapabilityStatement.rest.resource.searchParam.documentation [ string ]; # 0..1 Server-specific usage
      ], ...;
    ], ...;
    fhir:CapabilityStatement.rest.interaction [ # 0..* What operations are supported?
      fhir:CapabilityStatement.rest.interaction.code [ code ]; # 1..1 transaction | batch | search-system | history-
    system
      fhir:CapabilityStatement.rest.interaction.documentation [ string ]; # 0..1 Anything special about operation
    behavior
    ], ...;
    fhir:CapabilityStatement.rest.searchParam [ See CapabilityStatement.rest.resource.searchParam ], ... ; # 0..* Search
     parameters for searching all resources
    fhir:CapabilityStatement.rest.operation [ # 0..* Definition of an operation or a custom query
      fhir:CapabilityStatement.rest.operation.name [ string ]; # 1..1 Name by which the operation/query is invoked
      fhir:CapabilityStatement.rest.operation.definition [ Reference(OperationDefinition) ]; # 1..1 The defined
    operation/query
    ], ...;
    fhir:CapabilityStatement.rest.compartment [ uri ], ... ; # 0..* Compartments served/used by system
  ], ...;
  fhir:CapabilityStatement.messaging [ # 0..* If messaging is supported
    fhir:CapabilityStatement.messaging.endpoint [ # 0..* Where messages should be sent
      fhir:CapabilityStatement.messaging.endpoint.protocol [ Coding ]; # 1..1 http | ftp | mllp +
      fhir:CapabilityStatement.messaging.endpoint.address [ uri ]; # 1..1 Network address or identifier of the end-point
    ], ...;
    fhir:CapabilityStatement.messaging.reliableCache [ unsignedInt ]; # 0..1 Reliable Message Cache Length (min)
    fhir:CapabilityStatement.messaging.documentation [ string ]; # 0..1 Messaging interface behavior details
    fhir:CapabilityStatement.messaging.supportedMessage [ # 0..* Messages supported by this system
      fhir:CapabilityStatement.messaging.supportedMessage.mode [ code ]; # 1..1 sender | receiver
      fhir:CapabilityStatement.messaging.supportedMessage.definition [ Reference(MessageDefinition) ]; # 1..1 Message
    supported by this system
    ], ...;
    fhir:CapabilityStatement.messaging.event [ # 0..* Declare support for this event
      fhir:CapabilityStatement.messaging.event.code [ Coding ]; # 1..1 Event type
      fhir:CapabilityStatement.messaging.event.category [ code ]; # 0..1 Consequence | Currency | Notification

      fhir:CapabilityStatement.messaging.event.mode [ code ]; # 1..1 sender | receiver
      fhir:CapabilityStatement.messaging.event.focus [ code ]; # 1..1 Resource that's focus of message
      fhir:CapabilityStatement.messaging.event.request [ Reference(StructureDefinition) ]; # 1..1 Profile that describes
     the request
      fhir:CapabilityStatement.messaging.event.response [ Reference(StructureDefinition) ]; # 1..1 Profile that
    describes the response
      fhir:CapabilityStatement.messaging.event.documentation [ string ]; # 0..1 Endpoint-specific event documentation
    ], ...;
  ], ...;
  fhir:CapabilityStatement.document [ # 0..* Document definition
    fhir:CapabilityStatement.document.mode [ code ]; # 1..1 producer | consumer
    fhir:CapabilityStatement.document.documentation [ string ]; # 0..1 Description of document support
    fhir:CapabilityStatement.document.profile [ Reference(StructureDefinition) ]; # 1..1 Constraint on a resource used
    in the document
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

CapabilityStatement | I | | DomainResource | A statement of system capabilities + There can only be one REST declaration per mode. + The set of documents must be unique by the combination of profile and mode. + Capability Statements of kind 'instance' do not have implementation elements. + Messaging end-point is required (and is only permitted) when a statement is for an implementation. + Capability Statements of kind 'requirements' do not have software or implementation elements. + A Capability Statement SHALL have at least one of description, software, or implementation element. + A Capability Statement SHALL have at least one of REST, messaging or document element. Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | ? | 0..1 | uri | Logical URI to reference this capability statement (globally unique)

version | ? | 0..1 | string | Business version of the capability statement

name | ? | 0..1 | string | Name for this capability statement (computer friendly)

title | ? | 0..1 | string | Name for this capability statement (human friendly)

status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage

date | ? | 1..1 | dateTime | Date this was last changed

publisher | ? | 0..1 | string | Name of the publisher (organization or individual)

contact | ? | 0..* | ContactDetail | Contact details for the publisher

description | I | 0..1 | markdown | Natural language description of the capability statement

useContext | ? | 0..* | UsageContext | Context the content is intended to support

jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for capability statement (if applicable) Jurisdiction ValueSet (Extensible)

purpose | | 0..1 | markdown | Why this capability statement is defined

copyright | | 0..1 | markdown | Use and/or publishing restrictions

kind | ? | 1..1 | code | instance | capability | requirements CapabilityStatementKind (Required)

instantiates | ? | 0..* | uri | Canonical URL of another capability statement this implements

software | ?I | 0..1 | BackboneElement | Software that is covered by this capability statement

name | ? | 1..1 | string | A name the software is known by

version | ? | 0..1 | string | Version covered by this statement

releaseDate | ? | 0..1 | dateTime | Date this version released

implementation | ?I | 0..1 | BackboneElement | If this describes a specific instance

description | ? | 1..1 | string | Describes this specific instance

url | ? | 0..1 | uri | Base URL for the installation

fhirVersion | ? | 1..1 | id | FHIR Version the system uses

acceptUnknown | ? | 1..1 | code | no | extensions | elements | both UnknownContentCode (Required)

format | ? | 1..* | code | formats supported (xml | json | ttl | mime type) MimeType (Required)

patchFormat | ? | 0..* | code | Patch formats supported MimeType (Required)

implementationGuide | ? | 0..* | uri | Implementation guides supported

profile | ? | 0..* | Reference(StructureDefinition) | Profiles for use cases supported

rest | ?I | 0..* | BackboneElement | If the endpoint is a RESTful one + A given resource can only be described once per RESTful mode.

mode | ? | 1..1 | code | client | server RestfulCapabilityMode (Required)

documentation | | 0..1 | string | General description of implementation

security | ? | 0..1 | BackboneElement | Information about security of implementation

cors | ? | 0..1 | boolean | Adds CORS Headers (http://enable-cors.org/)

service | ? | 0..* | CodeableConcept | OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates RestfulSecurityService (Extensible)

description | | 0..1 | string | General description of how security works

certificate | | 0..* | BackboneElement | Certificates associated with security profiles

type | | 0..1 | code | Mime type for certificates MimeType (Required)

blob | | 0..1 | base64Binary | Actual certificate

resource | ?I | 0..* | BackboneElement | Resource served on the REST interface + Search parameter names must be unique in the context of a resource.

type | ? | 1..1 | code | A resource type that is supported ResourceType (Required)

profile | ? | 0..1 | Reference(StructureDefinition) | Base System profile for all uses of resource

documentation | | 0..1 | markdown | Additional information about the use of the resource type

interaction | | 1..* | BackboneElement | What operations are supported?

code | | 1..1 | code | read | vread | update | patch | delete | history-instance | history-type | create | search-type TypeRestfulInteraction (Required)

documentation | | 0..1 | string | Anything special about operation behavior

versioning | | 0..1 | code | no-version | versioned | versioned-update ResourceVersionPolicy (Required)

readHistory | | 0..1 | boolean | Whether vRead can return past versions

updateCreate | | 0..1 | boolean | If update can commit to a new identity

conditionalCreate | | 0..1 | boolean | If allows/uses conditional create

conditionalRead | | 0..1 | code | not-supported | modified-since | not-match | full-support ConditionalReadStatus (Required)

conditionalUpdate | | 0..1 | boolean | If allows/uses conditional update

conditionalDelete | | 0..1 | code | not-supported | single | multiple - how conditional delete is supported ConditionalDeleteStatus (Required)

referencePolicy | | 0..* | code | literal | logical | resolves | enforced | local ReferenceHandlingPolicy (Required)

searchInclude | | 0..* | string | _include values supported by the server

searchRevInclude | | 0..* | string | _revinclude values supported by the server

searchParam | | 0..* | BackboneElement | Search parameters supported by implementation

name | | 1..1 | string | Name of search parameter

definition | | 0..1 | uri | Source of definition for parameter

type | | 1..1 | code | number | date | string | token | reference | composite | quantity | uri SearchParamType (Required)
documentation | | 0..1 | string | Server-specific usage
interaction | | 0..* | BackboneElement | What operations are supported?
code | | 1..1 | code | transaction | batch | search-system | history-system SystemRestfulInteraction (Required)
documentation | | 0..1 | string | Anything special about operation behavior
searchParam | | 0..* | see searchParam | Search parameters for searching all resources
operation | ? | 0..* | BackboneElement | Definition of an operation or a custom query
name | | 1..1 | string | Name by which the operation/query is invoked
definition | ? | 1..1 | Reference(OperationDefinition) | The defined operation/query
compartment | | 0..* | uri | Compartments served/used by system
messaging | ?I | 0..* | BackboneElement | If messaging is supported + A Capability Statement messaging element SHALL have either supportedMessage or event element, but not both.
endpoint | | 0..* | BackboneElement | Where messages should be sent
protocol | | 1..1 | Coding | http | ftp | mllp + MessageTransport (Extensible)
address | | 1..1 | uri | Network address or identifier of the end-point
reliableCache | | 0..1 | unsignedInt | Reliable Message Cache Length (min)
documentation | | 0..1 | string | Messaging interface behavior details
supportedMessage | ? | 0..* | BackboneElement | Messages supported by this system
mode | ? | 1..1 | code | sender | receiver EventCapabilityMode (Required)
definition | ? | 1..1 | Reference(MessageDefinition) | Message supported by this system
event | ? | 0..* | BackboneElement | Declare support for this event
code | ? | 1..1 | Coding | Event type MessageEvent (Example)
category | | 0..1 | code | Consequence | Currency | Notification MessageSignificanceCategory (Required)
mode | | 1..1 | code | sender | receiver EventCapabilityMode (Required)
focus | | 1..1 | code | Resource that's focus of message ResourceType (Required)
request | ? | 1..1 | Reference(StructureDefinition) | Profile that describes the request
response | ? | 1..1 | Reference(StructureDefinition) | Profile that describes the response
documentation | | 0..1 | string | Endpoint-specific event documentation
document | ?I | 0..* | BackboneElement | Document definition
mode | | 1..1 | code | producer | consumer DocumentMode (Required)
documentation | | 0..1 | string | Description of document support
profile | ? | 1..1 | Reference(StructureDefinition) | Constraint on a resource used in the document
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CapabilityStatement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this capability statement (globally unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the capability statement -->
 <name value="[string]"/><!-- 0..1 Name for this capability statement (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this capability statement (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 1..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!--  0..1 Natural language description of the capability statement -->
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for capability statement (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this capability statement is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <kind value="[code]"/><!-- 1..1 instance | capability | requirements -->
 <instantiates value="[uri]"/><!-- 0..* Canonical URL of another capability statement this implements -->
 <software>  <!--  0..1 Software that is covered by this capability statement -->
  <name value="[string]"/><!-- 1..1 A name the software is known by -->
  <version value="[string]"/><!-- 0..1 Version covered by this statement -->
  <releaseDate value="[dateTime]"/><!-- 0..1 Date this version released -->
 </software>
 <implementation>  <!--  0..1 If this describes a specific instance -->
  <description value="[string]"/><!-- 1..1 Describes this specific instance -->
  <url value="[uri]"/><!-- 0..1 Base URL for the installation -->
 </implementation>
 <fhirVersion value="[id]"/><!-- 1..1 FHIR Version the system uses -->
 <acceptUnknown value="[code]"/><!-- 1..1 no | extensions | elements | both -->
 <format value="[code]"/><!-- 1..* formats supported (xml | json | ttl | mime type)  -->
 <patchFormat value="[code]"/><!-- 0..* Patch formats supported  -->
 <implementationGuide value="[uri]"/><!-- 0..* Implementation guides supported -->
 <profile><!-- 0..* Reference(StructureDefinition) Profiles for use cases supported --></profile>
 <rest>  <!--  0..* If the endpoint is a RESTful one -->
  <mode value="[code]"/><!-- 1..1 client | server -->
  <documentation value="[string]"/><!-- 0..1 General description of implementation -->
  <security>  <!-- 0..1 Information about security of implementation -->
   <cors value="[boolean]"/><!-- 0..1 Adds CORS Headers (http://enable-cors.org/) -->
   <service><!-- 0..* CodeableConcept OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates --></service>
   <description value="[string]"/><!-- 0..1 General description of how security works -->
   <certificate>  <!-- 0..* Certificates associated with security profiles -->
    <type value="[code]"/><!-- 0..1 Mime type for certificates  -->
    <blob value="[base64Binary]"/><!-- 0..1 Actual certificate -->
```

```
  </certificate>
 </security>
 <resource>  <!-- 0..* Resource served on the REST interface -->
  <type value="[code]"/><!-- 1..1 A resource type that is supported -->
  <profile><!-- 0..1 Reference(StructureDefinition) Base System profile for all uses of resource --></profile>
  <documentation value="[markdown]"/><!-- 0..1 Additional information about the use of the resource type -->
  <interaction>  <!-- 1..* What operations are supported? -->
   <code value="[code]"/><!-- 1..1 read | vread | update | patch | delete | history-instance | history-type | create |
   search-type -->
   <documentation value="[string]"/><!-- 0..1 Anything special about operation behavior -->
  </interaction>
  <versioning value="[code]"/><!-- 0..1 no-version | versioned | versioned-update -->
  <readHistory value="[boolean]"/><!-- 0..1 Whether vRead can return past versions -->
  <updateCreate value="[boolean]"/><!-- 0..1 If update can commit to a new identity -->
  <conditionalCreate value="[boolean]"/><!-- 0..1 If allows/uses conditional create -->
  <conditionalRead value="[code]"/><!-- 0..1 not-supported | modified-since | not-match | full-support -->
  <conditionalUpdate value="[boolean]"/><!-- 0..1 If allows/uses conditional update -->
  <conditionalDelete value="[code]"/><!-- 0..1 not-supported | single | multiple - how conditional delete is supported
   -->
  <referencePolicy value="[code]"/><!-- 0..* literal | logical | resolves | enforced | local -->
  <searchInclude value="[string]"/><!-- 0..* _include values supported by the server -->
  <searchRevInclude value="[string]"/><!-- 0..* _revinclude values supported by the server -->
  <searchParam>  <!-- 0..* Search parameters supported by implementation -->
   <name value="[string]"/><!-- 1..1 Name of search parameter -->
   <definition value="[uri]"/><!-- 0..1 Source of definition for parameter -->
   <type value="[code]"/><!-- 1..1 number | date | string | token | reference | composite | quantity | uri -->
   <documentation value="[string]"/><!-- 0..1 Server-specific usage -->
  </searchParam>
 </resource>
 <interaction>  <!-- 0..* What operations are supported? -->
  <code value="[code]"/><!-- 1..1 transaction | batch | search-system | history-system -->
  <documentation value="[string]"/><!-- 0..1 Anything special about operation behavior -->
 </interaction>
 <searchParam><!-- 0..* Content as for CapabilityStatement.rest.resource.searchParam Search parameters for searching
   all resources --></searchParam>
 <operation>  <!-- 0..* Definition of an operation or a custom query -->
  <name value="[string]"/><!-- 1..1 Name by which the operation/query is invoked -->
  <definition><!-- 1..1 Reference(OperationDefinition) The defined operation/query --></definition>
 </operation>
 <compartment value="[uri]"/><!-- 0..* Compartments served/used by system -->
</rest>
<messaging>  <!--  0..* If messaging is supported -->
 <endpoint>  <!-- 0..* Where messages should be sent -->
  <protocol><!-- 1..1 Coding http | ftp | mllp + --></protocol>
  <address value="[uri]"/><!-- 1..1 Network address or identifier of the end-point -->
 </endpoint>
 <reliableCache value="[unsignedInt]"/><!-- 0..1 Reliable Message Cache Length (min) -->
 <documentation value="[string]"/><!-- 0..1 Messaging interface behavior details -->
 <supportedMessage>  <!-- 0..* Messages supported by this system -->
  <mode value="[code]"/><!-- 1..1 sender | receiver -->
  <definition><!-- 1..1 Reference(MessageDefinition) Message supported by this system --></definition>
 </supportedMessage>
 <event>  <!-- 0..* Declare support for this event -->
  <code><!-- 1..1 Coding Event type --></code>
  <category value="[code]"/><!-- 0..1 Consequence | Currency | Notification -->
  <mode value="[code]"/><!-- 1..1 sender | receiver -->
  <focus value="[code]"/><!-- 1..1 Resource that's focus of message -->
  <request><!-- 1..1 Reference(StructureDefinition) Profile that describes the request --></request>
  <response><!-- 1..1 Reference(StructureDefinition) Profile that describes the response --></response>
  <documentation value="[string]"/><!-- 0..1 Endpoint-specific event documentation -->
 </event>
</messaging>
<document>  <!--  0..* Document definition -->
 <mode value="[code]"/><!-- 1..1 producer | consumer -->
 <documentation value="[string]"/><!-- 0..1 Description of document support -->
 <profile><!-- 1..1 Reference(StructureDefinition) Constraint on a resource used in the document --></profile>
</document>
</CapabilityStatement>
```

JSON Template

```
{
  "resourceType" : "CapabilityStatement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this capability statement (globally unique)
  "version" : "<string>", // Business version of the capability statement
  "name" : "<string>", // Name for this capability statement (computer friendly)
  "title" : "<string>", // Name for this capability statement (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // R!  Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // C? Natural language description of the capability statement
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for capability statement (if applicable)
  "purpose" : "<markdown>", // Why this capability statement is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "kind" : "<code>", // R!  instance | capability | requirements
```

```
"instantiates" : ["<uri>"], // Canonical URL of another capability statement this implements
"software" : { // C? Software that is covered by this capability statement
  "name" : "<string>", // R!  A name the software is known by
  "version" : "<string>", // Version covered by this statement
  "releaseDate" : "<dateTime>" // Date this version released
},
"implementation" : { // C? If this describes a specific instance
  "description" : "<string>", // R!  Describes this specific instance
  "url" : "<uri>" // Base URL for the installation
},
"fhirVersion" : "<id>", // R!  FHIR Version the system uses
"acceptUnknown" : "<code>", // R!  no | extensions | elements | both
"format" : ["<code>"], // R!  formats supported (xml | json | ttl | mime type)
"patchFormat" : ["<code>"], // Patch formats supported
"implementationGuide" : ["<uri>"], // Implementation guides supported
"profile" : [{ Reference(StructureDefinition) }], // Profiles for use cases supported
"rest" : [{ // C? If the endpoint is a RESTful one
  "mode" : "<code>", // R!  client | server
  "documentation" : "<string>", // General description of implementation
  "security" : { // Information about security of implementation
    "cors" : <boolean>, // Adds CORS Headers (http://enable-cors.org/)
    "service" : [{ CodeableConcept }], // OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates
    "description" : "<string>", // General description of how security works
    "certificate" : [{ // Certificates associated with security profiles
      "type" : "<code>", // Mime type for certificates
      "blob" : "<base64Binary>" // Actual certificate
    }]
  },
  "resource" : [{ // Resource served on the REST interface
    "type" : "<code>", // R!  A resource type that is supported
    "profile" : { Reference(StructureDefinition) }, // Base System profile for all uses of resource
    "documentation" : "<markdown>", // Additional information about the use of the resource type
    "interaction" : [{ // R!  What operations are supported?
      "code" : "<code>", // R!  read | vread | update | patch | delete | history-instance | history-type | create |
  search-type
      "documentation" : "<string>" // Anything special about operation behavior
    }],
    "versioning" : "<code>", // no-version | versioned | versioned-update
    "readHistory" : <boolean>, // Whether vRead can return past versions
    "updateCreate" : <boolean>, // If update can commit to a new identity
    "conditionalCreate" : <boolean>, // If allows/uses conditional create
    "conditionalRead" : "<code>", // not-supported | modified-since | not-match | full-support
    "conditionalUpdate" : <boolean>, // If allows/uses conditional update
    "conditionalDelete" : "<code>", // not-supported | single | multiple - how conditional delete is supported
    "referencePolicy" : ["<code>"], // literal | logical | resolves | enforced | local
    "searchInclude" : ["<string>"], // _include values supported by the server
    "searchRevInclude" : ["<string>"], // _revinclude values supported by the server
    "searchParam" : [{ // Search parameters supported by implementation
      "name" : "<string>", // R!  Name of search parameter
      "definition" : "<uri>", // Source of definition for parameter
      "type" : "<code>", // R!  number | date | string | token | reference | composite | quantity | uri
      "documentation" : "<string>" // Server-specific usage
    }]
  }],
  "interaction" : [{ // What operations are supported?
    "code" : "<code>", // R!  transaction | batch | search-system | history-system
    "documentation" : "<string>" // Anything special about operation behavior
  }],
  "searchParam" : [{ Content as for CapabilityStatement.rest.resource.searchParam }], // Search parameters for
  searching all resources
  "operation" : [{ // Definition of an operation or a custom query
    "name" : "<string>", // R!  Name by which the operation/query is invoked
    "definition" : { Reference(OperationDefinition) } // R!  The defined operation/query
  }],
  "compartment" : ["<uri>"] // Compartments served/used by system
}],
"messaging" : [{ // C? If messaging is supported
  "endpoint" : [{ // Where messages should be sent
    "protocol" : { Coding }, // R!  http | ftp | mllp +
    "address" : "<uri>" // R!  Network address or identifier of the end-point
  }],
  "reliableCache" : "<unsignedInt>", // Reliable Message Cache Length (min)
  "documentation" : "<string>", // Messaging interface behavior details
  "supportedMessage" : [{ // Messages supported by this system
    "mode" : "<code>", // R!  sender | receiver
    "definition" : { Reference(MessageDefinition) } // R!  Message supported by this system
  }],
  "event" : [{ // Declare support for this event
    "code" : { Coding }, // R!  Event type
    "category" : "<code>", // Consequence | Currency | Notification
    "mode" : "<code>", // R!  sender | receiver
    "focus" : "<code>", // R!  Resource that's focus of message
    "request" : { Reference(StructureDefinition) }, // R!  Profile that describes the request
    "response" : { Reference(StructureDefinition) }, // R!  Profile that describes the response
    "documentation" : "<string>" // Endpoint-specific event documentation
  }]
}],
"document" : [{ // C? Document definition
  "mode" : "<code>", // R!  producer | consumer
```

```
    "documentation" : "<string>", // Description of document support
    "profile" : { Reference(StructureDefinition) } // R!  Constraint on a resource used in the document
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CapabilityStatement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CapabilityStatement.url [ uri ]; # 0..1 Logical URI to reference this capability statement (globally unique)
  fhir:CapabilityStatement.version [ string ]; # 0..1 Business version of the capability statement
  fhir:CapabilityStatement.name [ string ]; # 0..1 Name for this capability statement (computer friendly)
  fhir:CapabilityStatement.title [ string ]; # 0..1 Name for this capability statement (human friendly)
  fhir:CapabilityStatement.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:CapabilityStatement.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:CapabilityStatement.date [ dateTime ]; # 1..1 Date this was last changed
  fhir:CapabilityStatement.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:CapabilityStatement.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:CapabilityStatement.description [ markdown ]; # 0..1 Natural language description of the capability statement
  fhir:CapabilityStatement.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:CapabilityStatement.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for capability statement
     (if applicable)
  fhir:CapabilityStatement.purpose [ markdown ]; # 0..1 Why this capability statement is defined
  fhir:CapabilityStatement.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:CapabilityStatement.kind [ code ]; # 1..1 instance | capability | requirements
  fhir:CapabilityStatement.instantiates [ uri ], ... ; # 0..* Canonical URL of another capability statement this
    implements
  fhir:CapabilityStatement.software [ # 0..1 Software that is covered by this capability statement
    fhir:CapabilityStatement.software.name [ string ]; # 1..1 A name the software is known by
    fhir:CapabilityStatement.software.version [ string ]; # 0..1 Version covered by this statement
    fhir:CapabilityStatement.software.releaseDate [ dateTime ]; # 0..1 Date this version released
  ];
  fhir:CapabilityStatement.implementation [ # 0..1 If this describes a specific instance
    fhir:CapabilityStatement.implementation.description [ string ]; # 1..1 Describes this specific instance
    fhir:CapabilityStatement.implementation.url [ uri ]; # 0..1 Base URL for the installation
  ];
  fhir:CapabilityStatement.fhirVersion [ id ]; # 1..1 FHIR Version the system uses
  fhir:CapabilityStatement.acceptUnknown [ code ]; # 1..1 no | extensions | elements | both
  fhir:CapabilityStatement.format [ code ], ... ; # 1..* formats supported (xml | json | ttl | mime type)
  fhir:CapabilityStatement.patchFormat [ code ], ... ; # 0..* Patch formats supported
  fhir:CapabilityStatement.implementationGuide [ uri ], ... ; # 0..* Implementation guides supported
  fhir:CapabilityStatement.profile [ Reference(StructureDefinition) ], ... ; # 0..* Profiles for use cases supported
  fhir:CapabilityStatement.rest [ # 0..* If the endpoint is a RESTful one
    fhir:CapabilityStatement.rest.mode [ code ]; # 1..1 client | server
    fhir:CapabilityStatement.rest.documentation [ string ]; # 0..1 General description of implementation
    fhir:CapabilityStatement.rest.security [ # 0..1 Information about security of implementation
      fhir:CapabilityStatement.rest.security.cors [ boolean ]; # 0..1 Adds CORS Headers (http://enable-cors.org/)
      fhir:CapabilityStatement.rest.security.service [ CodeableConcept ], ... ; # 0..* OAuth | SMART-on-FHIR | NTLM |
    Basic | Kerberos | Certificates
      fhir:CapabilityStatement.rest.security.description [ string ]; # 0..1 General description of how security works
      fhir:CapabilityStatement.rest.security.certificate [ # 0..* Certificates associated with security profiles
        fhir:CapabilityStatement.rest.security.certificate.type [ code ]; # 0..1 Mime type for certificates
        fhir:CapabilityStatement.rest.security.certificate.blob [ base64Binary ]; # 0..1 Actual certificate
      ], ...;
    ];
    fhir:CapabilityStatement.rest.resource [ # 0..* Resource served on the REST interface
      fhir:CapabilityStatement.rest.resource.type [ code ]; # 1..1 A resource type that is supported
      fhir:CapabilityStatement.rest.resource.profile [ Reference(StructureDefinition) ]; # 0..1 Base System profile for
    all uses of resource
      fhir:CapabilityStatement.rest.resource.documentation [ markdown ]; # 0..1 Additional information about the use of
    the resource type
      fhir:CapabilityStatement.rest.resource.interaction [ # 1..* What operations are supported?
        fhir:CapabilityStatement.rest.resource.interaction.code [ code ]; # 1..1 read | vread | update | patch | delete
    | history-instance | history-type | create | search-type
        fhir:CapabilityStatement.rest.resource.interaction.documentation [ string ]; # 0..1 Anything special about
    operation behavior
      ], ...;
      fhir:CapabilityStatement.rest.resource.versioning [ code ]; # 0..1 no-version | versioned | versioned-update
      fhir:CapabilityStatement.rest.resource.readHistory [ boolean ]; # 0..1 Whether vRead can return past versions
      fhir:CapabilityStatement.rest.resource.updateCreate [ boolean ]; # 0..1 If update can commit to a new identity
      fhir:CapabilityStatement.rest.resource.conditionalCreate [ boolean ]; # 0..1 If allows/uses conditional create
      fhir:CapabilityStatement.rest.resource.conditionalRead [ code ]; # 0..1 not-supported | modified-since | not-match
     | full-support
      fhir:CapabilityStatement.rest.resource.conditionalUpdate [ boolean ]; # 0..1 If allows/uses conditional update
      fhir:CapabilityStatement.rest.resource.conditionalDelete [ code ]; # 0..1 not-supported | single | multiple - how
    conditional delete is supported
      fhir:CapabilityStatement.rest.resource.referencePolicy [ code ], ... ; # 0..* literal | logical | resolves |
    enforced | local
      fhir:CapabilityStatement.rest.resource.searchInclude [ string ], ... ; # 0..* _include values supported by the
    server
      fhir:CapabilityStatement.rest.resource.searchRevInclude [ string ], ... ; # 0..* _revinclude values supported by
    the server
      fhir:CapabilityStatement.rest.resource.searchParam [ # 0..* Search parameters supported by implementation
        fhir:CapabilityStatement.rest.resource.searchParam.name [ string ]; # 1..1 Name of search parameter
```

```
      fhir:CapabilityStatement.rest.resource.searchParam.definition [ uri ]; # 0..1 Source of definition for parameter
      fhir:CapabilityStatement.rest.resource.searchParam.type [ code ]; # 1..1 number | date | string | token |
  reference | composite | quantity | uri
      fhir:CapabilityStatement.rest.resource.searchParam.documentation [ string ]; # 0..1 Server-specific usage
    ], ...;
  ], ...;
  fhir:CapabilityStatement.rest.interaction [ # 0..* What operations are supported?
    fhir:CapabilityStatement.rest.interaction.code [ code ]; # 1..1 transaction | batch | search-system | history-
  system
    fhir:CapabilityStatement.rest.interaction.documentation [ string ]; # 0..1 Anything special about operation
  behavior
  ], ...;
  fhir:CapabilityStatement.rest.searchParam [ See CapabilityStatement.rest.resource.searchParam ], ... ; # 0..* Search
   parameters for searching all resources
  fhir:CapabilityStatement.rest.operation [ # 0..* Definition of an operation or a custom query
    fhir:CapabilityStatement.rest.operation.name [ string ]; # 1..1 Name by which the operation/query is invoked
    fhir:CapabilityStatement.rest.operation.definition [ Reference(OperationDefinition) ]; # 1..1 The defined
  operation/query
  ], ...;
  fhir:CapabilityStatement.rest.compartment [ uri ], ... ; # 0..* Compartments served/used by system
], ...;
fhir:CapabilityStatement.messaging [ # 0..* If messaging is supported
  fhir:CapabilityStatement.messaging.endpoint [ # 0..* Where messages should be sent
    fhir:CapabilityStatement.messaging.endpoint.protocol [ Coding ]; # 1..1 http | ftp | mllp +
    fhir:CapabilityStatement.messaging.endpoint.address [ uri ]; # 1..1 Network address or identifier of the end-point
  ], ...;
  fhir:CapabilityStatement.messaging.reliableCache [ unsignedInt ]; # 0..1 Reliable Message Cache Length (min)
  fhir:CapabilityStatement.messaging.documentation [ string ]; # 0..1 Messaging interface behavior details
  fhir:CapabilityStatement.messaging.supportedMessage [ # 0..* Messages supported by this system
    fhir:CapabilityStatement.messaging.supportedMessage.mode [ code ]; # 1..1 sender | receiver
    fhir:CapabilityStatement.messaging.supportedMessage.definition [ Reference(MessageDefinition) ]; # 1..1 Message
  supported by this system
  ], ...;
  fhir:CapabilityStatement.messaging.event [ # 0..* Declare support for this event
    fhir:CapabilityStatement.messaging.event.code [ Coding ]; # 1..1 Event type
    fhir:CapabilityStatement.messaging.event.category [ code ]; # 0..1 Consequence | Currency | Notification
    fhir:CapabilityStatement.messaging.event.mode [ code ]; # 1..1 sender | receiver
    fhir:CapabilityStatement.messaging.event.focus [ code ]; # 1..1 Resource that's focus of message
    fhir:CapabilityStatement.messaging.event.request [ Reference(StructureDefinition) ]; # 1..1 Profile that describes
   the request
    fhir:CapabilityStatement.messaging.event.response [ Reference(StructureDefinition) ]; # 1..1 Profile that
  describes the response
    fhir:CapabilityStatement.messaging.event.documentation [ string ]; # 0..1 Endpoint-specific event documentation
  ], ...;
], ...;
fhir:CapabilityStatement.document [ # 0..* Document definition
  fhir:CapabilityStatement.document.mode [ code ]; # 1..1 producer | consumer
  fhir:CapabilityStatement.document.documentation [ string ]; # 0..1 Description of document support
  fhir:CapabilityStatement.document.profile [ Reference(StructureDefinition) ]; # 1..1 Constraint on a resource used
  in the document
], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

5.2.3.1 Terminology Bindings

Path | Definition | Type | Reference

CapabilityStatement.status | The lifecycle status of a Value Set or Concept Map. | Required | PublicationStatus

CapabilityStatement.jurisdiction | Countries and regions within which this artifact is targeted for use | Extensible | Jurisdiction ValueSet

CapabilityStatement.kind | How a capability statement is intended to be used. | Required | CapabilityStatementKind

CapabilityStatement.acceptUnknown | A code that indicates whether an application accepts unknown elements or extensions when reading resources. | Required | UnknownContentCode

CapabilityStatement.format CapabilityStatement.patchFormat CapabilityStatement.rest.security.certificate.type | The mime type of an attachment. Any valid mime type is allowed. | Required | BCP 13 (RFCs 2045, 2046, 2047, 4288, 4289 and 2049)

CapabilityStatement.rest.mode | The mode of a RESTful capability statement. | Required | RestfulCapabilityMode

CapabilityStatement.rest.security.service | Types of security services used with FHIR. | Extensible | RestfulSecurityService

CapabilityStatement.rest.resource.type CapabilityStatement.messaging.event.focus | One of the resource types defined as part of FHIR. | Required | Resource Types

CapabilityStatement.rest.resource.interaction.code | Operations supported by REST at the type or instance level. | Required | TypeRestfulInteraction

CapabilityStatement.rest.resource.versioning | How the system supports versioning for a resource. | Required | ResourceVersionPolicy

CapabilityStatement.rest.resource.conditionalRead | A code that indicates how the server supports conditional read. | Required | ConditionalReadStatus

CapabilityStatement.rest.resource.conditionalDelete | A code that indicates how the server supports conditional delete. | Required | ConditionalDeleteStatus

CapabilityStatement.rest.resource.referencePolicy | A set of flags that defines how references are supported. | Required | ReferenceHandlingPolicy

CapabilityStatement.rest.resource.searchParam.type | Data types allowed to be used for search parameters. | Required | SearchParamType

CapabilityStatement.rest.interaction.code | Operations supported by REST at the system level. | Required | SystemRestfulInteraction

CapabilityStatement.messaging.endpoint.protocol | The protocol used for message transport. | Extensible | MessageTransport

CapabilityStatement.messaging.supportedMessage.mode CapabilityStatement.messaging.event.mode | The mode of a message capability statement. | Required | EventCapabilityMode
CapabilityStatement.messaging.event.code | One of the message events defined as part of FHIR. | Example | Message Events
CapabilityStatement.messaging.event.category | The impact of the content of a message. | Required | MessageSignificanceCategory
CapabilityStatement.document.mode | Whether the application produces or consumes documents. | Required | DocumentMode

5.2.3.2 Constraints

* cpb-1: A Capability Statement SHALL have at least one of REST, messaging or document element. (expression : rest.exists() or messaging.exists() or document.exists())

* cpb-12: On CapabilityStatement.rest.resource: Search parameter names must be unique in the context of a resource. (expression on CapabilityStatement.rest.resource: searchParam.select(name).isDistinct())

* cpb-14: Capability Statements of kind 'requirements' do not have software or implementation elements. (expression : (software.empty() and implementation.empty()) or kind != 'requirements')

* cpb-15: Capability Statements of kind 'instance' do not have implementation elements. (expression : implementation.empty() or kind != 'capability')

* cpb-16: On CapabilityStatement.messaging: A Capability Statement messaging element SHALL have either supportedMessage or event element, but not both. (expression on CapabilityStatement.messaging: supportedMessage.empty() != event.empty())

* cpb-2: A Capability Statement SHALL have at least one of description, software, or implementation element. (expression : (description.count() + software.count() + implementation.count()) > 0)

* cpb-3: Messaging end-point is required (and is only permitted) when a statement is for an implementation. (expression : messaging.endpoint.empty() or kind = 'instance')

* cpb-7: The set of documents must be unique by the combination of profile and mode. (expression : document.select(profile.reference&mode).isDistinct())

* cpb-8: There can only be one REST declaration per mode. (expression : rest.select(mode).isDistinct())

* cpb-9: On CapabilityStatement.rest: A given resource can only be described once per RESTful mode. (expression on CapabilityStatement.rest: resource.select(type).isDistinct())

5.2.4 Notes:

* The CapabilityStatement resource provides for an application to describe its use of the RESTful paradigm messaging events, or FHIR documents. Usually, an application would only describe one, but more than one may be described

* RESTful CapabilityStatement rules:

* RESTful servers are required to provide this resource on demand. Servers SHALL specify what resource types and operations are supported, and SHOULD also specify profiles for each resource type.

* RESTful clients SHOULD publish a capability statement

* The search parameters that a server supports (or a client makes use of) are specified in the resource profile that the capability statement references

* Resource Types or operations that are not listed are not supported

* Messaging CapabilityStatement rules:

* The interpretation of request and response depends on the mode. If the mode is sender, then request specifies what the application sends, and response specifies what it accepts. If the mode is "receiver", then this is reversed

* If a request or response is not specified for an event, then no rules are made for it

* Events that are not listed are not supported

* The MessageDefinition resource is newly proposed and is still considered 'draft'. The supportedMessage element can be used in place of the event and the work group believes it may meet implementer needs better, however because the new mechanism has not yet been reviewed by ballot, the older 'event' mechanism has been retained. Implementers may use one or the other to define their capabilities. Feedback is welcome.

* Document CapabilityStatement rules:

* Document profiles should directly constrain the Document.information.class & type elements so that there is no ambiguity concerning which profile any given document conforms to.

* Other service based use of resources: Due to the variability of these services, the CapabilityStatement resource does not attempt to describe service based use of resources. The various service specifications will need to describe this usage in their own way.

5.2.4.1 Supporting Profiles

A CapabilityStatement declares two different kinds of profiles for the functionality it describes. For a discussion of the use of these two types of resources, see two uses for profiles.

5.2.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
date | date | The capability statement publication date | CapabilityStatement.date
description | string | The description of the capability statement | CapabilityStatement.description
event | token | Event code in a capability statement | CapabilityStatement.messaging.event.code
fhirversion | token | The version of FHIR | CapabilityStatement.version
format | token | formats supported (xml | json | ttl | mime type) | CapabilityStatement.format
guide | uri | Implementation guides supported | CapabilityStatement.implementationGuide
jurisdiction | token | Intended jurisdiction for the capability statement | CapabilityStatement.jurisdiction
mode | token | Mode - restful (server/client) or messaging (sender/receiver) | CapabilityStatement.rest.mode
name | string | Computationally friendly name of the capability statement | CapabilityStatement.name
publisher | string | Name of the publisher of the capability statement | CapabilityStatement.publisher
resource | token | Name of a resource mentioned in a capability statement | CapabilityStatement.rest.resource.type
resource-profile | reference | A profile id invoked in a capability statement | CapabilityStatement.rest.resource.profile (StructureDefinition)
security-service | token | OAuth | SMART-on-FHIR | NTLM | Basic | Kerberos | Certificates | CapabilityStatement.rest.security.service
software | string | Part of a the name of a software application | CapabilityStatement.software.name
status | token | The current status of the capability statement | CapabilityStatement.status
supported-profile | reference | Profiles for use cases supported | CapabilityStatement.profile (StructureDefinition)
title | string | The human-friendly name of the capability statement | CapabilityStatement.title
url | uri | The uri that identifies the capability statement | CapabilityStatement.url
version | token | The business version of the capability statement | CapabilityStatement.version

5.2.6 Resource CapabilityStatement - Examples

URL: https://hl7.org/fhir/STU3/capabilitystatement-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.2.6 Resource CapabilityStatement - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

General Condition Example | example | XML | JSON | Turtle
PHR Example | phr | XML | JSON | Turtle
Complete Capability Statement | capabilitystatement-base | XML | JSON | Turtle
EmptyCapabilityStatement | capabilitystatement-base2 | XML | JSON | Turtle
Terminology Server Base Capability Statement | terminology-server | XML | JSON | Turtle
Knowledge Repository Base Capability Statement | knowledge-repository | XML | JSON | Turtle
Measure Processor Base Capability Statement | measure-processor | XML | JSON | Turtle
Showing new message definition structure | messagedefinition | XML | JSON | Turtle
System which records information about EHR events | null | XML | JSON | Turtle | from EHRS Functional model - Record Lifecycle Events IG

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.2.7 Resource CapabilityStatement - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/capabilitystatement-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.2.7 Resource CapabilityStatement - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the CapabilityStatement resource.

CapabilityStatement
Definition | A Capability Statement documents a set of capabilities (behaviors) of a FHIR Server that may be used as a statement of actual server functionality or a statement of required or desired server implementation.
Control | 1..1
Invariants | Defined on this element cpb-1: A Capability Statement SHALL have at least one of REST, messaging or document element. (expression : rest.exists() or messaging.exists() or document.exists(), xpath: exists(f:rest) or exists(f:messaging) or exists(f:document)) cpb-14: Capability Statements of kind 'requirements' do not have software or implementation elements. (expression : (software.empty() and implementation.empty()) or kind != 'requirements', xpath: not(exists(f:software) or exists(f:implementation)) or (f:kind/@value != 'requirements')) cpb-15: Capability Statements of kind 'instance' do not have implementation elements. (expression : implementation.empty() or kind != 'capability', xpath: not(exists(f:implementation)) or (f:kind/@value != 'capability')) cpb-2: A Capability Statement SHALL have at least one of description, software, or implementation element. (expression : (description.count() + software.count() + implementation.count()) > 0, xpath: count(f:software | f:implementation | f:description) > 0) cpb-3: Messaging end-point is required (and is only permitted) when a statement is for an implementation. (expression : messaging.endpoint.empty() or kind = 'instance', xpath: not(exists(f:messaging/f:endpoint)) or f:kind/@value = 'instance') cpb-7: The set of documents must be unique by the combination of profile and mode. (expression : document.select(profile.reference&mode).isDistinct(), xpath: count(f:document[f:mode/@value='producer'])=count(distinct-values(f:document[f:mode/@value='producer']/f:profile/f:reference/@value)) and count(f:document[f:mode/@value='consumer'])=count(distinct-values(f:document[f:mode/@value='consumer']/f:profile/f:reference/@value))) cpb-8:

There can only be one REST declaration per mode. (expression : rest.select(mode).isDistinct(), xpath: count(f:rest)=count(distinct-values(f:rest/f:mode/@value)))

CapabilityStatement.url

Definition | An absolute URI that is used to identify this capability statement when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this capability statement is (or will be) published. The URL SHOULD include the major version of the capability statement. For more information see Technical and Business Versions.

Control | 0..1

Type | uri

Requirements | Allows the capability statement to be referenced by a single globally unique identifier.

Summary | true

Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.

CapabilityStatement.version

Definition | The identifier that is used to identify this version of the capability statement when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the capability statement author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.

Note | This is a business versionId, not a resource version id (see discussion)

Control | 0..1

Type | string

Summary | true

Comments | There may be different capability statement instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the capability statement with the format [url]|[version].

CapabilityStatement.name

Definition | A natural language name identifying the capability statement. This name should be usable as an identifier for the module by machine processing applications such as code generation.

Control | 0..1

Type | string

Requirements | Support human navigation and code generation.

Summary | true

Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.

CapabilityStatement.title

Definition | A short, descriptive, user-friendly title for the capability statement.

Control | 0..1

Type | string

Summary | true

Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.

CapabilityStatement.status

Definition | The status of this capability statement. Enables tracking the life-cycle of the content.

Control | 1..1

Terminology Binding | PublicationStatus (Required)

Type | code

Is Modifier | true

Summary | true

Comments | Allows filtering of capability statements that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired capability statement without due consideration. This is not intended for use with actual capability statements, but where capability statements are used to describe possible or desired systems.

CapabilityStatement.experimental

Definition | A boolean value to indicate that this capability statement is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.

Control | 0..1

Type | boolean

Is Modifier | true

Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level capability statement.

Summary | true

Comments | Allows filtering of capability statement that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental capability statement in production.

CapabilityStatement.date

Definition | The date (and optionally time) when the capability statement was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the capability statement changes.

Control | 1..1

Type | dateTime

Alternate Names | Revision Date

Summary | true

Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the capability statement. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.

CapabilityStatement.publisher

Definition | The name of the individual or organization that published the capability statement.

Control | 0..1

Type | string

Requirements | Helps establish the "authority/credibility" of the capability statement. May also allow for contact.
Summary | true
Comments | Usually an organization, but may be an individual. The publisher (or steward) of the capability statement is the organization or individual primarily responsible for the maintenance and upkeep of the capability statement. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the capability statement. This item SHOULD be populated unless the information is available from context.
CapabilityStatement.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
CapabilityStatement.description
Definition | A free text natural language description of the capability statement from a consumer's perspective. Typically, this is used when the capability statement describes a desired rather than an actual solution, for example as a formal expression of requirements as part of an RFP.
Control | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the capability statement was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the capability statement as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created). This does not need to be populated if the description is adequately implied by the software or implementation details.
Invariants | Affect this element cpb-2: A Capability Statement SHALL have at least one of description, software, or implementation element. (expression : (description.count() + software.count() + implementation.count()) > 0, xpath: count(f:software | f:implementation | f:description) > 0)
CapabilityStatement.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate capability statement instances.
Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
CapabilityStatement.jurisdiction
Definition | A legal or geographic region in which the capability statement is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the capability statement to be used in jurisdictions other than those for which it was originally designed or intended.
CapabilityStatement.purpose
Definition | Explaination of why this capability statement is needed and why it has been designed as it has.
Control | 0..1
Type | markdown
Comments | This element does not describe the usage of the capability statement Instead it provides traceability of "why" the resource is either needed or "why" it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this capability statement.
CapabilityStatement.copyright
Definition | A copyright statement relating to the capability statement and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the capability statement.
Control | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the capability statement and/or its content.
Alternate Names | License; Restrictions
CapabilityStatement.kind
Definition | The way that this statement is intended to be used, to describe an actual running instance of software, a particular product (kind not instance of software) or a class of implementation (e.g. a desired purchase).
Control | 1..1
Terminology Binding | CapabilityStatementKind (Required)
Type | code
Requirements | Allow searching the 3 modes.
Summary | true
CapabilityStatement.instantiates
Definition | Reference to a canonical URL of another CapabilityStatement that this software implements or uses. This capability statement is a published API description that corresponds to a business service. The rest of the capability statement does not need to repeat the details of the referenced resource, but can do so.
Control | 0..*
Type | uri
Summary | true
Comments | HL7 defines the following Services: Terminology Service. Many Implementation Guides define additional services.
CapabilityStatement.software
Definition | Software that is covered by this capability statement. It is used when the capability statement describes the capabilities of a particular software version, independent of an installation.
Control | 0..1

Summary | true
Invariants | Affect this element cpb-2: A Capability Statement SHALL have at least one of description, software, or implementation element. (expression : (description.count() + software.count() + implementation.count()) > 0, xpath: count(f:software | f:implementation | f:description) > 0)
CapabilityStatement.software.name
Definition | Name software is known by.
Control | 1..1
Type | string
Summary | true
CapabilityStatement.software.version
Definition | The version identifier for the software covered by this statement.
Note | This is a business versionId, not a resource version id (see discussion)
Control | 0..1
Type | string
Summary | true
Comments | If possible, a version should be specified, as statements are likely to be different for different versions of software.
CapabilityStatement.software.releaseDate
Definition | Date this version of the software was released.
Control | 0..1
Type | dateTime
Summary | true
CapabilityStatement.implementation
Definition | Identifies a specific implementation instance that is described by the capability statement - i.e. a particular installation, rather than the capabilities of a software program.
Control | 0..1
Summary | true
Invariants | Affect this element cpb-2: A Capability Statement SHALL have at least one of description, software, or implementation element. (expression : (description.count() + software.count() + implementation.count()) > 0, xpath: count(f:software | f:implementation | f:description) > 0)
CapabilityStatement.implementation.description
Definition | Information about the specific installation that this capability statement relates to.
Control | 1..1
Type | string
Summary | true
CapabilityStatement.implementation.url
Definition | An absolute base URL for the implementation. This forms the base for REST interfaces as well as the mailbox and document interfaces.
Control | 0..1
Type | uri
Summary | true
CapabilityStatement.fhirVersion
Definition | The version of the FHIR specification on which this capability statement is based.
Control | 1..1
Type | id
Summary | true
CapabilityStatement.acceptUnknown
Definition | A code that indicates whether the application accepts unknown elements or extensions when reading resources.
Control | 1..1
Terminology Binding | UnknownContentCode (Required)
Type | code
Summary | true
Comments | Unknown elements in a resource can only arise as later versions of the specification are published, because this is the only place where such elements can be defined. Hence accepting unknown elements is about inter-version compatibility. Applications are recommended to accept unknown extensions and elements ('both'), but this is not always possible.
CapabilityStatement.format
Definition | A list of the formats supported by this implementation using their content types.
Control | 1..*
Terminology Binding | MimeType :
Type | code
Summary | true
Comments | "xml", "json" and "ttl" are allowed, which describe the simple encodings described in the specification (and imply appropriate bundle support). Otherwise, mime types are legal here.
CapabilityStatement.patchFormat
Definition | A list of the patch formats supported by this implementation using their content types.
Control | 0..*
Terminology Binding | MimeType :
Type | code
Summary | true
Comments | At present, the patch mime types application/json-patch+json and application/xml-patch+xml are legal. Generally, if a server supports PATCH, it would be expected to support the patch formats and match the formats it supports, but this is not always possible or necessary.
CapabilityStatement.implementationGuide
Definition | A list of implementation guides that the server does (or should) support in their entirety.
Control | 0..*
Type | uri
Summary | true
CapabilityStatement.profile

Definition | A list of profiles that represent different use cases supported by the system. For a server, "supported by the system" means the system hosts/produces a set of resources that are conformant to a particular profile, and allows clients that use its services to search using this profile and to find appropriate data. For a client, it means the system will search by this profile and process data according to the guidance implicit in the profile. See further discussion in Using Profiles.
Control | 0..*
Type | Reference(StructureDefinition)
Summary | true
Comments | Supported profiles are different than the profiles that apply to a particular resource in rest.resource.profile. The resource profile is a general statement of what features of the resource are supported overall by the system - the sum total of the facilities it supports. A supported profile is a deeper statement about the functionality of the data and services provided by the server (or used by the client). A typical case is a laboratory system that produces a set of different reports - this is the list of types of data that it publishes. A key aspect of declaring profiles here is the question of how the client converts knowledge that the server publishes this data into working with the data; the client can inspect individual resources to determine whether they conform to a particular profile, but how does it find the ones that does? It does so by searching using the profile parameter, so any resources listed here must be valid values for the profile resource (using the identifier in the target profile). Typically, supported profiles cross resource types to describe a network of related resources, so they are listed here rather than by resource. However, they do not need to describe more than one resource.
CapabilityStatement.rest
Definition | A definition of the restful capabilities of the solution, if any.
Control | 0..*
Summary | true
Comments | Multiple repetitions allow definition of both client and/or server behaviors or possibly behaviors under different configuration settings (for software or requirements statements).
Invariants | Defined on this element cpb-9: A given resource can only be described once per RESTful mode. (expression : resource.select(type).isDistinct(), xpath: count(f:resource)=count(distinct-values(f:resource/f:type/@value))) Affect this element cpb-1: A Capability Statement SHALL have at least one of REST, messaging or document element. (expression : rest.exists() or messaging.exists() or document.exists(), xpath: exists(f:rest) or exists(f:messaging) or exists(f:document))
CapabilityStatement.rest.mode
Definition | Identifies whether this portion of the statement is describing the ability to initiate or receive restful operations.
Control | 1..1
Terminology Binding | RestfulCapabilityMode (Required)
Type | code
Summary | true
CapabilityStatement.rest.documentation
Definition | Information about the system's restful capabilities that apply across all applications, such as security.
Control | 0..1
Type | string
CapabilityStatement.rest.security
Definition | Information about security implementation from an interface perspective - what a client needs to know.
Control | 0..1
Summary | true
CapabilityStatement.rest.security.cors
Definition | Server adds CORS headers when responding to requests - this enables javascript applications to use the server.
Control | 0..1
Type | boolean
Summary | true
Comments | The easiest CORS headers to add are Access-Control-Allow-Origin: * & Access-Control-Request-Method: GET, POST, PUT, DELETE. All servers SHOULD support CORS.
CapabilityStatement.rest.security.service
Definition | Types of security services that are supported/required by the system.
Control | 0..*
Terminology Binding | RestfulSecurityService (Extensible)
Type | CodeableConcept
Summary | true
CapabilityStatement.rest.security.description
Definition | General description of how security works.
Control | 0..1
Type | string
CapabilityStatement.rest.security.certificate
Definition | Certificates associated with security profiles.
Control | 0..*
CapabilityStatement.rest.security.certificate.type
Definition | Mime type for a certificate.
Control | 0..1
Terminology Binding | MimeType :
Type | code
CapabilityStatement.rest.security.certificate.blob
Definition | Actual certificate.
Control | 0..1
Type | base64Binary
CapabilityStatement.rest.resource
Definition | A specification of the restful capabilities of the solution for a specific resource type.
Control | 0..*
Summary | true
Comments | Max of one repetition per resource type.

Invariants | Defined on this element cpb-12: Search parameter names must be unique in the context of a resource. (expression : searchParam.select(name).isDistinct(), xpath: count(f:searchParam)=count(distinct-values(f:searchParam/f:name/@value)))
CapabilityStatement.rest.resource.type
Definition | A type of resource exposed via the restful interface.
Control | 1..1
Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
CapabilityStatement.rest.resource.profile
Definition | A specification of the profile that describes the solution's overall support for the resource, including any constraints on cardinality, bindings, lengths or other limitations. See further discussion in Using Profiles.
Control | 0..1
Type | Reference(StructureDefinition)
Summary | true
Comments | The profile applies to all resources of this type - i.e. it is the superset of what is supported by the system.
CapabilityStatement.rest.resource.documentation
Definition | Additional information about the resource type used by the system.
Control | 0..1
Type | markdown
CapabilityStatement.rest.resource.interaction
Definition | Identifies a restful operation supported by the solution.
Control | 1..*
CapabilityStatement.rest.resource.interaction.code
Definition | Coded identifier of the operation, supported by the system resource.
Control | 1..1
Terminology Binding | TypeRestfulInteraction (Required)
Type | code
CapabilityStatement.rest.resource.interaction.documentation
Definition | Guidance specific to the implementation of this operation, such as 'delete is a logical delete' or 'updates are only allowed with version id' or 'creates permitted from pre-authorized certificates only'.
Control | 0..1
Type | string
Requirements | REST allows a degree of variability in the implementation of RESTful solutions that is useful for exchange partners to be aware of.
CapabilityStatement.rest.resource.versioning
Definition | This field is set to no-version to specify that the system does not support (server) or use (client) versioning for this resource type. If this has some other value, the server must at least correctly track and populate the versionId meta-property on resources. If the value is 'versioned-update', then the server supports all the versioning features, including using e-tags for version integrity in the API.
Control | 0..1
Terminology Binding | ResourceVersionPolicy (Required)
Type | code
Comments | If a server supports versionIds correctly, it SHOULD support vread too, but is not required to do so.
CapabilityStatement.rest.resource.readHistory
Definition | A flag for whether the server is able to return past versions as part of the vRead operation.
Control | 0..1
Type | boolean
Comments | It is useful to support the vRead operation for current operations, even if past versions aren't available.
CapabilityStatement.rest.resource.updateCreate
Definition | A flag to indicate that the server allows or needs to allow the client to create new identities on the server (e.g. that is, the client PUTs to a location where there is no existing resource). Allowing this operation means that the server allows the client to create new identities on the server.
Control | 0..1
Type | boolean
Comments | Allowing the clients to create new identities on the server means that the system administrator needs to have confidence that the clients do not create clashing identities between them. Obviously, if there is only one client, this won't happen. While creating identities on the client means that the clients need to be managed, it's much more convenient for many scenarios if such management can be put in place.
CapabilityStatement.rest.resource.conditionalCreate
Definition | A flag that indicates that the server supports conditional create.
Control | 0..1
Type | boolean
Comments | Conditional Create is mainly appropriate for interface engine scripts converting from other formats, such as v2.
CapabilityStatement.rest.resource.conditionalRead
Definition | A code that indicates how the server supports conditional read.
Control | 0..1
Terminology Binding | ConditionalReadStatus (Required)
Type | code
Comments | Conditional Read is mainly appropriate for interface engine scripts converting from other formats, such as v2.
CapabilityStatement.rest.resource.conditionalUpdate
Definition | A flag that indicates that the server supports conditional update.
Control | 0..1
Type | boolean
Comments | Conditional Update is mainly appropriate for interface engine scripts converting from other formats, such as v2.
CapabilityStatement.rest.resource.conditionalDelete
Definition | A code that indicates how the server supports conditional delete.
Control | 0..1

Terminology Binding | ConditionalDeleteStatus (Required)
Type | code
Comments | Conditional Delete is mainly appropriate for interface engine scripts converting from other formats, such as v2.
CapabilityStatement.rest.resource.referencePolicy
Definition | A set of flags that defines how references are supported.
Control | 0..*
Terminology Binding | ReferenceHandlingPolicy (Required)
Type | code
CapabilityStatement.rest.resource.searchInclude
Definition | A list of _include values supported by the server.
Control | 0..*
Type | string
Comments | If this list is empty, the server does not support includes.
CapabilityStatement.rest.resource.searchRevInclude
Definition | A list of _revinclude (reverse include) values supported by the server.
Control | 0..*
Type | string
Comments | If this list is empty, the server does not support reverse includes.
CapabilityStatement.rest.resource.searchParam
Definition | Search parameters for implementations to support and/or make use of - either references to ones defined in the specification, or additional ones defined for/by the implementation.
Control | 0..*
CapabilityStatement.rest.resource.searchParam.name
Definition | The name of the search parameter used in the interface.
Control | 1..1
Type | string
Comments | Parameter names cannot overlap with standard parameter names, and standard parameters cannot be redefined.
CapabilityStatement.rest.resource.searchParam.definition
Definition | An absolute URI that is a formal reference to where this parameter was first defined, so that a client can be confident of the meaning of the search parameter (a reference to SearchParameter.url).
Control | 0..1
Type | uri
Comments | This SHOULD be present, and matches refers to a SearchParameter by its canonical url. If systems wish to document their support for modifiers, comparators, target resource types, and chained parameters, they should do using a search parameter resource.
CapabilityStatement.rest.resource.searchParam.type
Definition | The type of value a search parameter refers to, and how the content is interpreted.
Control | 1..1
Terminology Binding | SearchParamType (Required)
Type | code
Comments | While this can be looked up from the definition, it is included here as a convenience for systems that autogenerate a query interface based on the server capability statement. It SHALL be the same as the type in the search parameter definition.
CapabilityStatement.rest.resource.searchParam.documentation
Definition | This allows documentation of any distinct behaviors about how the search parameter is used. For example, text matching algorithms.
Control | 0..1
Type | string
CapabilityStatement.rest.interaction
Definition | A specification of restful operations supported by the system.
Control | 0..*
CapabilityStatement.rest.interaction.code
Definition | A coded identifier of the operation, supported by the system.
Control | 1..1
Terminology Binding | SystemRestfulInteraction (Required)
Type | code
CapabilityStatement.rest.interaction.documentation
Definition | Guidance specific to the implementation of this operation, such as limitations on the kind of transactions allowed, or information about system wide search is implemented.
Control | 0..1
Type | string
CapabilityStatement.rest.searchParam
Definition | Search parameters that are supported for searching all resources for implementations to support and/or make use of - either references to ones defined in the specification, or additional ones defined for/by the implementation.
Control | 0..*
Type | See CapabilityStatement.rest.resource.searchParam
Comments | Typically, the only search parameters supported for all searchse are those that apply to all resources - tags, profiles, text search etc.
CapabilityStatement.rest.operation
Definition | Definition of an operation or a named query together with its parameters and their meaning and type.
Control | 0..*
Summary | true
CapabilityStatement.rest.operation.name
Definition | The name of the operation or query. For an operation, this is the name prefixed with $ and used in the URL. For a query, this is the name used in the _query parameter when the query is called.
Control | 1..1
Type | string

Comments | The name here SHOULD be the same as the name in the definition, unless there is a name clash and the name cannot be used. The name does not include the "$" portion that is always included in the URL.
CapabilityStatement.rest.operation.definition
Definition | Where the formal definition can be found.
Control | 1..1
Type | Reference(OperationDefinition)
Summary | true
Comments | This can be used to build an HTML form to invoke the operation, for instance.
CapabilityStatement.rest.compartment
Definition | An absolute URI which is a reference to the definition of a compartment that the system supports. The reference is to a CompartmentDefinition resource by its canonical URL .
Control | 0..*
Type | uri
Comments | At present, the only defined compartments are at CompartmentDefinition.
CapabilityStatement.messaging
Definition | A description of the messaging capabilities of the solution.
Control | 0..*
Summary | true
Comments | Multiple repetitions allow the documentation of multiple endpoints per solution.
Invariants | Defined on this element cpb-16: A Capability Statement messaging element SHALL have either supportedMessage or event element, but not both. (expression : supportedMessage.empty() != event.empty(), xpath: exists(f:supportedMessage) != exists(f:event)) Affect this element cpb-1: A Capability Statement SHALL have at least one of REST, messaging or document element. (expression : rest.exists() or messaging.exists() or document.exists(), xpath: exists(f:rest) or exists(f:messaging) or exists(f:document))
CapabilityStatement.messaging.endpoint
Definition | An endpoint (network accessible address) to which messages and/or replies are to be sent.
Control | 0..*
Alternate Names | 3
CapabilityStatement.messaging.endpoint.protocol
Definition | A list of the messaging transport protocol(s) identifiers, supported by this endpoint.
Control | 1..1
Terminology Binding | MessageTransport (Extensible)
Type | Coding
CapabilityStatement.messaging.endpoint.address
Definition | The network address of the end-point. For solutions that do not use network addresses for routing, it can be just an identifier.
Control | 1..1
Type | uri
CapabilityStatement.messaging.reliableCache
Definition | Length if the receiver's reliable messaging cache in minutes (if a receiver) or how long the cache length on the receiver should be (if a sender).
Control | 0..1
Type | unsignedInt
Comments | If this value is missing then the application does not implement (receiver) or depend on (sender) reliable messaging.
CapabilityStatement.messaging.documentation
Definition | Documentation about the system's messaging capabilities for this endpoint not otherwise documented by the capability statement. For example, the process for becoming an authorized messaging exchange partner.
Control | 0..1
Type | string
CapabilityStatement.messaging.supportedMessage
Definition | References to message definitions for messages this system can send or receive.
Control | 0..*
Summary | true
Comments | This is a proposed alternative to the messaging.event structure.
CapabilityStatement.messaging.supportedMessage.mode
Definition | The mode of this event declaration - whether application is sender or receiver.
Control | 1..1
Terminology Binding | EventCapabilityMode (Required)
Type | code
Summary | true
CapabilityStatement.messaging.supportedMessage.definition
Definition | Points to a message definition that identifies the messaging event, message structure, allowed responses, etc.
Control | 1..1
Type | Reference(MessageDefinition)
Summary | true
CapabilityStatement.messaging.event
Definition | A description of the solution's support for an event at this end-point.
Control | 0..*
Summary | true
Comments | The same event may be listed up to two times - once as sender and once as receiver.
CapabilityStatement.messaging.event.code
Definition | A coded identifier of a supported messaging event.
Control | 1..1
Terminology Binding | the Event List in the messaging framework
Type | Coding

Summary | true
CapabilityStatement.messaging.event.category
Definition | The impact of the content of the message.
Control | 0..1
Terminology Binding | MessageSignificanceCategory (Required)
Type | code
CapabilityStatement.messaging.event.mode
Definition | The mode of this event declaration - whether an application is a sender or receiver.
Control | 1..1
Terminology Binding | EventCapabilityMode (Required)
Type | code
CapabilityStatement.messaging.event.focus
Definition | A resource associated with the event. This is the resource that defines the event.
Control | 1..1
Terminology Binding | Any defined Resource Type name
Type | code
Comments | This SHALL be provided if the event type supports multiple different resource types.
CapabilityStatement.messaging.event.request
Definition | Information about the request for this event.
Control | 1..1
Type | Reference(StructureDefinition)
Summary | true
CapabilityStatement.messaging.event.response
Definition | Information about the response for this event.
Control | 1..1
Type | Reference(StructureDefinition)
Summary | true
CapabilityStatement.messaging.event.documentation
Definition | Guidance on how this event is handled, such as internal system trigger points, business rules, etc.
Control | 0..1
Type | string
CapabilityStatement.document
Definition | A document definition.
Control | 0..*
Summary | true
Invariants | Affect this element cpb-1: A Capability Statement SHALL have at least one of REST, messaging or document element. (expression : rest.exists() or messaging.exists() or document.exists(), xpath: exists(f:rest) or exists(f:messaging) or exists(f:document))
CapabilityStatement.document.mode
Definition | Mode of this document declaration - whether an application is a producer or consumer.
Control | 1..1
Terminology Binding | DocumentMode (Required)
Type | code
CapabilityStatement.document.documentation
Definition | A description of how the application supports or uses the specified document profile. For example, when documents are created, what action is taken with consumed documents, etc.
Control | 0..1
Type | string
CapabilityStatement.document.profile
Definition | A constraint on a resource used in the document.
Control | 1..1
Type | Reference(StructureDefinition)
Summary | true
Comments | The first resource is the document resource.

5.2.8 Resource CapabilityStatement - Mappings

URL: https://hl7.org/fhir/STU3/capabilitystatement-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.2.8 Resource CapabilityStatement - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Mappings for the capabilitystatement resource.

5.2.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

CapabilityStatement | n/a

5.2.9 Resource CapabilityStatement - Design Notes

URL: https://hl7.org/fhir/STU3/capabilitystatement-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.2.9 Resource CapabilityStatement - Design Notes

to do

5.2.10 Resource CapabilityStatement - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/capabilitystatement-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.2.10 Resource CapabilityStatement - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the CapabilityStatement resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

5.2.10.1 Profiles

No Profiles defined for this resource

5.2.10.2 Extensions

capabilitystatement-expectation | expectation | for Common CapabilityStatement extensions
capabilitystatement-prohibited | prohibited | for Common CapabilityStatement extensions
capabilitystatement-search-parameter-combination | search-parameter-combination | for Common CapabilityStatement extensions
capabilitystatement-supported-system | supported-system | for Common CapabilityStatement extensions
capabilitystatement-websocket | websocket | for Common CapabilityStatement extensions
resource-approvalDate | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | lastReviewDate | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

5.2.10.3 Search Extensions

No Search Extensions defined for this resource

5.2.11 Common CapabilityStatement extensions

URL: https://hl7.org/fhir/STU3/capabilitystatement-common.html

Common extensions for use with the CapabilityStatement resource

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.2.11 Common CapabilityStatement extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

5.2.11.1 Content

Extensions:
capabilitystatement-expectation | expectation : Defines the level of expectation associated with a given system capability.
capabilitystatement-prohibited | prohibited : If set to true, indicates that support for the specified behavior would make a system non=conformant with the specification.
capabilitystatement-supported-system | supported-system : A code system that is supported by the system that is not defined in a value set resource.
capabilitystatement-search-parameter-combination | search-parameter-combination : A container for a single allowable parameter combination.
capabilitystatement-websocket | websocket : Where the server provides its web socket end-point.

5.2.12 Resource CapabilityStatement - Operations

URL: https://hl7.org/fhir/STU3/capabilitystatement-operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.2.12 Resource CapabilityStatement - Operations

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

This resource has 3 operations associated with it:

$subset | Fetch a subset of the CapabilityStatement resource
$implements | Test if a server implements a client's required operations
$conforms | Test if a server implements a client's required operations

For more information about operations, including how they are invoked, see Operations.

5.2.12.1 Fetch a subset of the CapabilityStatement resource

This operation asks the server to return a subset of the CapabilityStatement resource - just the REST parts that relate to a set of nominated resources - the resources that the client is interested in

The official URL for this operation definition is

`http://hl7.org/fhir/OperationDefinition/CapabilityStatement-subset`

Formal Definition (as a OperationDefinition).

URL: [base]/CapabilityStatement/$subset

URL: [base]/CapabilityStatement/[id]/$subset

This is an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
server | 0..1 | uri | | | The canonical URL - use this if the subset is not invoked on an instance (or on the /metadata end-point)
resource | 1..* | code | | | A resource that the client would like to include in the return
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
return | 1..1 | CapabilityStatement | | | The subsetted CapabilityStatement resource that is returned. This should be tagged with the SUBSETTED code Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

5.2.12.1.1 Examples

Request a subset of Patient capabilities from a known CapabilityStatement (Request):

```
POST [base]/CapabilityStatement/example/$subset
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="resource"/>
    <valueCode value="Patient"/>
  </parameter>
</Parameters>
```

Response:

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<CapabilityStatement xmlns="http://hl7.org/fhir">
  <id value="example"/>
  <meta>
    <tag>
      <system value="http://hl7.org/fhir/v3/ObservationValue"/>
      <code value="SUBSETTED"/> <display value="subsetted"/>
    </tag>
  </meta>
  <rest>
    <resource>
      <type value="Patient"/>
      <interaction>
        <code value="read"/>
      </interaction>
      <interaction>
        <code value="vread"/>
        <documentation value="Only supported for patient records since 12-Dec 2012"/>
      </interaction>
      <interaction>
        <code value="update"/>
      </interaction>
      <interaction>
        <code value="history-instance"/>
      </interaction>
      <interaction>
        <code value="create"/>
      </interaction>
      <interaction>
        <code value="history-type"/>
      </interaction>
      <versioning value="versioned-update"/>
      <readHistory value="true"/>
      <updateCreate value="false"/>
      <conditionalCreate value="true"/>
      <conditionalUpdate value="false"/>
      <conditionalDelete value="not-supported"/>
      <searchInclude value="Organization"/>
```

```
      <searchRevInclude value="Person"/>
      <searchParam>
        <name value="identifier"/>
        <definition value="http://hl7.org/fhir/SearchParameter/Patient-identifier"/>
        <type value="token"/>
        <documentation value="Only supports search by institution MRN"/>
      </searchParam>
      <searchParam>
        <name value="careprovider"/>
        <definition value="http://hl7.org/fhir/SearchParameter/Patient-careprovider"/>
        <type value="reference"/>
      </searchParam>
    </resource>
  </rest>
</CapabilityStatement>
```

5.2.12.2 Test if a server implements a client's required operations

This operation asks the server to check that it implements all the resources, interactions, search parameters, and operations that the client provides in its capability statement. The client provides its capability statement inline, or by referring the server to the canonical URL of its capability statement

The official URL for this operation definition is

`http://hl7.org/fhir/OperationDefinition/CapabilityStatement-implements`

Formal Definition (as a OperationDefinition).

URL: [base]/CapabilityStatement/$implements

URL: [base]/CapabilityStatement/[id]/$implements

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

server | 0..1 | uri | | | The canonical URL for the server capability statement - use this if the implements is not invoked on an instance (or on the /metadata end-point)

client | 0..1 | uri | | | The canonical URL for the client capability statement - use this if the implements is not invoked on an instance (or on the /metadata end-point)

resource | 0..1 | CapabilityStatement | | | The client capability statement, provided inline

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | OperationOutcome | | | Outcome of the CapabilityStatement test Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The operation does not perform a full conformance check; in particular it does not check that the profiles align. It merely checks that the behaviors the client wishes to use are provided

Technically, this operation is implemented as follows: * The server's capability statement must have an entry for each resource in the client's capability statement * The servers' resource support must have matching flags for updateCreate, conditionalCreate, conditionalRead, conditionalUpdate, conditionalDelete, searchInclude, searchRevInclude * The server capability statement must have a matching interaction for each interaction in the client capability statement (whether or not it is on a resource) * The server must have a search parameter with matching name and definition for any search parameters in the client capability statement * The server must have an operation definitions with a matching reference for any operations in the client capability statement

If the capability statements match by these rules, then the return value is a 200 OK with an operation outcome that contains no issues with severity >= error. If the capability statement doesn't match, the return value is a 4xx error, with an OperationOutcome with at least one issue with severity >= error

5.2.12.2.1 Examples

Request a CapabilityStatement implements comparison between a client and server systems (Request):

```
POST [base]/CapabilityStatement/$implements
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="server"/>
    <valueUri value="http://server.example.org/fhir/metadata"/>
  </parameter>
  <parameter>
    <name value="client"/>
    <valueUri value="http://client.example.org/fhir/metadata"/>
  </parameter>
</Parameters>
```

Response:

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<OperationOutcome xmlns="http://hl7.org/fhir">
  <id value="implementsok"/>
  <text>
```

```
    <status value="additional"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Server http://server.example.org/fhir/metadata implements client http://client.example.org/fhir/metadata
    capabilities.</p>
    </div>
  </text>
  <issue>
    <severity value="information"/>
    <code value="informational"/>
    <details>
      <text value="Server http://server.example.org/fhir/metadata implements client
    http://client.example.org/fhir/metadata capabilities."/>
    </details>
 </issue>
</OperationOutcome>
```

5.2.12.3 Test if a server implements a client's required operations

This operation asks the server to check that it implements all the resources, interactions, search parameters, and operations that the client provides in its capability statement. The client provides both capability statements by reference, and must ensure that all the referenced resources are available to the conformance server

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/CapabilityStatement-conforms
```

Formal Definition (as a OperationDefinition).

URL: [base]/CapabilityStatement/$conforms

This is an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
left	0..1	uri			The canonical URL for the left-hand system's capability statement
right	0..1	uri			The canonical URL for the right-hand system's capability statement
mode	0..1	code			What kind of comparison to perform - server to server, or client to server (use the codes 'server/server' or 'client/server')

Out Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
issues	1..1	OperationOutcome			Outcome of the CapabilityStatement test
union	0..1	CapabilityStatement			The intersection of the functionality described by the CapabilityStatement resources
intersection	0..1	CapabilityStatement			The union of the functionality described by the CapabilityStatement resources

The operation performs a full comparison of the functionality described by the two capability statements, including the profiles and value sets they reference, and also including concept maps and structure maps.

The full execution of this operation is still a matter of research, but it is intended to support comparison of systems to see if they will interoperate

If the capability statements can be successfully compared, then the return value is a 200 OK with an OperationOutcome along with intersection and union capability statements. The operation outcome can contain errors relating to differences between the capability statements. If the capability statements cannot be compared, because dependencies cannot be located, the return value is a 4xx error, with an OperationOutcome with at least one issue with severity >= error

5.2.12.3.1 Examples

Request a CapabilityStatement conforms comparison between two server systems (Request):

```
POST [base]/CapabilityStatement/$conforms
[other headers]

<Parameters xmlns="http://hl7.org/fhir">
  <parameter>
    <name value="left"/>
    <valueUri value="http://server1.example.org/fhir/metadata"/>
  </parameter>
  <parameter>
    <name value="right"/>
    <valueUri value="http://server2.example.org/fhir/metadata"/>
  </parameter>
  <parameter>
    <name value="mode"/>
    <valueCode value="server/server"/>
  </parameter>
</Parameters>
```

Response:

```
HTTP/1.1 200 OK
[other headers]

<?xml version="1.0" encoding="UTF-8"?>
<OperationOutcome xmlns="http://hl7.org/fhir">
  <id value="conformsnotok"/>
  <text>
    <status value="additional"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
      <p>Request timeout waiting for server http://server1.example.org/fhir/metadata CapabilitieStatement.</p>
    </div>
```

```
  </text>
  <issue>
    <severity value="error"/>
    <code value="timeout"/>
    <diagnostics value="Request timeout waiting for server http://server1.example.org/fhir/metadata
    CapabilitieStatement."/>
 </issue>
</OperationOutcome>
```

5.2.13 Resource CapabilityStatement - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/capabilitystatement-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.2.13 Resource CapabilityStatement - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for CapabilityStatement.

Functional status for this map: 19 tests that all execute ok. 19 fail round-trip testing and 12 r3 resources are invalid (453 errors). (see documentation)

5.2.13.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/Conformance2to3" = "R2 to R3 Conversions for Conformance"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Conformance" alias ConformanceR2 as source
uses "http://hl7.org/fhir/StructureDefinition/CapabilityStatement" alias CapabilityStatement as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  Conformance extends DomainResource
  input src : ConformanceR2 as source
  input tgt : CapabilityStatement as target

  "conf-url" : for src.url make tgt.url
  "conf-version" : for src.version make tgt.version
  "conf-name" : for src.name make tgt.name
  "conf-status" : for src.status make tgt.status
  "conf-experimental" : for src.experimental make tgt.experimental
  "conf-publisher" : for src.publisher make tgt.publisher
  "conf-contact" : for src.contact as vs make tgt.contact as vt then contact(vs, vt)
  "conf-date" : for src.date make tgt.date
  "conf-description" : for src.description make tgt.description
  "conf-requirements" : for src.requirements make tgt.purpose
  "conf-copyright" : for src.copyright make tgt.copyright
  "conf-kind" : for src.kind make tgt.kind
  "conf-software" : for src.software as vs make tgt.software as vt then software(vs, vt)
  "conf-implementation" : for src.implementation as vs make tgt.implementation as vt then implementation(vs, vt)
  "conf-fhirVersion" : for src.fhirVersion make tgt.fhirVersion
  "conf-acceptUnknown" : for src.acceptUnknown make tgt.acceptUnknown
  "conf-format" : for src.format make tgt.format
  "conf-profile " : for src.profile make tgt.profile
  "conf-rest" : for src.rest as vs make tgt.rest as vt then rest(vs, vt)
  "conf-messaging" : for src.messaging as vs make tgt.messaging as vt then messaging(vs, vt)
  "conf-document" : for src.document as vs make tgt.document as vt then document(vs, vt)
endgroup

group contact extends BackboneElement
  input src as source
  input tgt as target

 "conf-name" : for src.name make tgt.name
 "conf-telecom" : for src.telecom make tgt.telecom
endgroup

group software extends BackboneElement
  input src as source
  input tgt as target

  "conf-name" : for src.name make tgt.name
  "conf-version" : for src.version make tgt.version
  "conf-releaseDate" : for src.releaseDate make tgt.releaseDate
endgroup

group implementation extends BackboneElement
  input src as source
  input tgt as target

  "conf-description" : for src.description make tgt.description
  "conf-url" : for src.url make tgt.url
endgroup

group rest extends BackboneElement
  input src as source
  input tgt as target
```

```
  "conf-mode" : for src.mode make tgt.mode
  "conf-documentation" : for src.documentation make tgt.documentation
  "conf-security" : for src.security as vs make tgt.security as vt then security(vs, vt)
  "conf-resource" : for src.resource as vs make tgt.resource as vt then resource(vs, vt)
  "conf-interaction" : for src.interaction where code != 'transaction' as vs make tgt.interaction as vt then
    interaction(vs, vt)
  "conf-transactionMode" : for src.transactionMode where $this = 'batch'  or $this = 'both' make tgt.interaction as i,
    i.code = 'batch'
  "conf-transactionMode" : for src.transactionMode where $this = 'transaction'  or $this = 'both' make tgt.interaction
    as i, i.code = 'transaction'
  "conf-searchParam" : for src.searchParam as vs make tgt.searchParam as vt then searchParam(vs, vt)
  "conf-operation" : for src.operation as vs make tgt.operation as vt then operation(vs, vt)
  "conf-compartment" : for src.compartment make tgt.compartment
endgroup

group security extends BackboneElement
  input src as source
  input tgt as target

  "conf-cors" : for src.cors make tgt.cors
  "conf-service" : for src.service make tgt.service
  "conf-description" : for src.description make tgt.description
  "conf-certificate" : for src.certificate as vs make tgt.certificate as vt then certificate(vs, vt)
endgroup

group certificate extends BackboneElement
  input src as source
  input tgt as target

  "conf-type" : for src.type make tgt.type
  "conf-blob" : for src.blob make tgt.blob
endgroup

group resource extends BackboneElement
  input src as source
  input tgt as target

  "conf-type2" : for src.type where $this != 'Conformance' make tgt.type
  "conf-type3b" : for src.type where $this = 'Conformance' make tgt.type = 'CapabilityStatement'
  "conf-profile" : for src.profile make tgt.profile
  "conf-documentation" : for src.documentation make tgt.documentation
  "conf-interaction" : for src.interaction as vs where code != 'validate' make tgt.interaction as vt then
    interaction(vs, vt)
  "conf-versioning" : for src.versioning make tgt.versioning
  "conf-readHistory" : for src.readHistory make tgt.readHistory
  "conf-updateCreate" : for src.updateCreate make tgt.updateCreate
  "conf-conditionalCreate" : for src.conditionalCreate make tgt.conditionalCreate
  "conf-conditionalRead" : for src.conditionalRead make tgt.conditionalRead
  "conf-conditionalUpdate" : for src.conditionalUpdate make tgt.conditionalUpdate
  "conf-conditionalDelete" : for src.conditionalDelete make tgt.conditionalDelete
  "conf-searchInclude" : for src.searchInclude make tgt.searchInclude
  "conf-searchRevInclude" : for src.searchRevInclude make tgt.searchRevInclude
  "conf-searchParam" : for src.searchParam as vs make tgt.searchParam as vt then searchParam(vs, vt)
endgroup

group interaction extends BackboneElement
  input src as source
  input tgt as target

  "conf-code" : for src.code make tgt.code
  "conf-documentation" : for src.documentation make tgt.documentation
endgroup

group searchParam extends BackboneElement
  input src as source
  input tgt as target

  "conf-name" : for src.name make tgt.name
  "conf-definition" : for src.definition make tgt.definition
  "conf-type4" : for src.type make tgt.type
  "conf-documentation" : for src.documentation make tgt.documentation
//?  target
//?  modifier
//?  chain
endgroup

group operation extends BackboneElement
  input src as source
  input tgt as target

  "conf-name" : for src.name make tgt.name
  "conf-definition" : for src.definition make tgt.definition
endgroup

group messaging extends BackboneElement
  input src as source
  input tgt as target

  "conf-endpoint" : for src.endpoint as vs make tgt.endpoint as vt then endpoint(vs, vt)
```

```
  "conf-reliableCache" : for src.reliableCache make tgt.reliableCache
  "conf-documentation" : for src.documentation make tgt.documentation
  "conf-event" : for src.event as vs make tgt.event as vt then event(vs, vt)
endgroup

group endpoint extends BackboneElement
  input src as source
  input tgt as target

  "conf-protocol" : for src.protocol make tgt.protocol
  "conf-address" : for src.address make tgt.address
endgroup

group event extends BackboneElement
  input src as source
  input tgt as target

  "conf-code" : for src.code make tgt.code
  "conf-category" : for src.category make tgt.category
  "conf-mode" : for src.mode make tgt.mode
  "conf-focus" : for src.focus make tgt.focus
  "conf-request" : for src.request make tgt.request
  "conf-response" : for src.response make tgt.response
  "conf-documentation" : for src.documentation make tgt.documentation
endgroup

group document extends BackboneElement
  input src as source
  input tgt as target

  "conf-mode" : for src.mode make tgt.mode
  "conf-documentation" : for src.documentation make tgt.documentation
  "conf-profile" : for src.profile make tgt.profile
endgroup
```

5.2.13.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/CapabilityStatement3to2" = "R2 to R3 Conversions for CapabilityStatement"


uses "http://hl7.org/fhir/StructureDefinition/CapabilityStatement" alias CapabilityStatement as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/Conformance" alias ConformanceR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  CapabilityStatement extends DomainResource
  input src : CapabilityStatement as source
  input tgt : ConformanceR2 as target

  "conf-url" : for src.url make tgt.url
  "conf-version" : for src.version make tgt.version
  "conf-name" : for src.name make tgt.name
  "conf-status" : for src.status make tgt.status
  "conf-experimental" : for src.experimental make tgt.experimental
  "conf-publisher" : for src.publisher make tgt.publisher
  "conf-contact" : for src.contact as vs make tgt.contact as vt then contact(vs, vt)
  "conf-date" : for src.date make tgt.date
  "conf-description" : for src.description make tgt.description
  "conf-requirements" : for src.purpose make tgt.requirements
  "conf-copyright" : for src.copyright make tgt.copyright
  "conf-kind" : for src.kind make tgt.kind
  "conf-software" : for src.software as vs make tgt.software as vt then software(vs, vt)
  "conf-implementation" : for src.implementation as vs make tgt.implementation as vt then implementation(vs, vt)
  "conf-fhirVersion" : for src.fhirVersion make tgt.fhirVersion
  "conf-acceptUnknown" : for src.acceptUnknown make tgt.acceptUnknown
  "conf-format" : for src.format make tgt.format
  "conf-profile " : for src.profile make tgt.profile
  "conf-rest" : for src.rest as vs make tgt.rest as vt then rest(vs, vt)
  "conf-messaging" : for src.messaging as vs make tgt.messaging as vt then messaging(vs, vt)
  "conf-document" : for src.document as vs make tgt.document as vt then document(vs, vt)
endgroup

group contact extends Element
  input src as source
  input tgt as target

 "conf-name" : for src.name make tgt.name
 "conf-telecom" : for src.telecom make tgt.telecom
endgroup

group software extends BackboneElement
  input src as source
  input tgt as target

  "conf-name" : for src.name make tgt.name
  "conf-version" : for src.version make tgt.version
  "conf-releaseDate" : for src.releaseDate make tgt.releaseDate
endgroup

group implementation extends BackboneElement
```

```
  input src as source
  input tgt as target

  "conf-description" : for src.description make tgt.description
  "conf-url" : for src.url make tgt.url
endgroup

group rest extends BackboneElement
  input src as source
  input tgt as target

  "conf-mode" : for src.mode make tgt.mode
  "conf-documentation" : for src.documentation make tgt.documentation
  "conf-security" : for src.security as vs make tgt.security as vt then security(vs, vt)
  "conf-resource" : for src.resource as vs make tgt.resource as vt then resource(vs, vt)
  "conf-interaction" : for src.interaction as vs where ($this = 'batch' or $this = 'transaction').not() make
    tgt.interaction as vt then interaction(vs, vt)
  "conf-transactionMode-1" : for src where src.interaction.where($this = 'transaction').exists() and
    src.interaction.where($this = 'batch').exists() make tgt.transactionMode = 'both'
  "conf-transactionMode-2" : for src where src.interaction.where($this = 'transaction').exists() and
    src.interaction.where($this = 'batch').empty() make tgt.transactionMode = 'transaction'
  "conf-transactionMode-3" : for src where src.interaction.where($this = 'transaction').empty() and
    src.interaction.where($this = 'batch').exists() make tgt.transactionMode = 'batch'
  "conf-transactionMode-4" : for src where src.interaction.where($this = 'transaction').empty() and
    src.interaction.where($this = 'batch').empty() make tgt.transactionMode = 'none'
  "conf-searchParam" : for src.searchParam as vs make tgt.searchParam as vt then searchParam(vs, vt)
  "conf-operation" : for src.operation as vs make tgt.operation as vt then operation(vs, vt)
  "conf-compartment" : for src.compartment make tgt.compartment
endgroup

group security extends BackboneElement
  input src as source
  input tgt as target

  "conf-cors" : for src.cors make tgt.cors
  "conf-service" : for src.service make tgt.service
  "conf-description" : for src.description make tgt.description
  "conf-certificate" : for src.certificate as vs make tgt.certificate as vt then certificate(vs, vt)
endgroup

group certificate extends BackboneElement
  input src as source
  input tgt as target

  "conf-type" : for src.type make tgt.type
  "conf-blob" : for src.blob make tgt.blob
endgroup

group resource extends BackboneElement
  input src as source
  input tgt as target

  "conf-type2" : for src.type where $this != 'CapabilityStatement' make tgt.type
  "conf-type3" : for src.type where $this = 'CapabilityStatement' make tgt.type = 'Conformance'
  "conf-profile" : for src.profile make tgt.profile
  "conf-documentation" : for src.documentation make tgt.documentation
  "conf-interaction" : for src.interaction as vs make tgt.interaction as vt then interaction(vs, vt)
  "conf-versioning" : for src.versioning make tgt.versioning
  "conf-readHistory" : for src.readHistory make tgt.readHistory
  "conf-updateCreate" : for src.updateCreate make tgt.updateCreate
  "conf-conditionalCreate" : for src.conditionalCreate make tgt.conditionalCreate
  "conf-conditionalRead" : for src.conditionalRead make tgt.conditionalRead
  "conf-conditionalUpdate" : for src.conditionalUpdate make tgt.conditionalUpdate
  "conf-conditionalDelete" : for src.conditionalDelete make tgt.conditionalDelete
  "conf-searchInclude" : for src.searchInclude make tgt.searchInclude
  "conf-searchRevInclude" : for src.searchRevInclude make tgt.searchRevInclude
  "conf-searchParam" : for src.searchParam as vs make tgt.searchParam as vt then searchParam(vs, vt)
endgroup

group interaction extends BackboneElement
  input src as source
  input tgt as target

  "conf-code" : for src.code make tgt.code
  "conf-documentation" : for src.documentation make tgt.documentation
endgroup

group searchParam extends BackboneElement
  input src as source
  input tgt as target

  "conf-name" : for src.name make tgt.name
  "conf-definition" : for src.definition make tgt.definition
  "conf-type4" : for src.type make tgt.type
  "conf-documentation" : for src.documentation make tgt.documentation
//?  target
//?  modifier
//?  chain
endgroup
```

```
group operation extends BackboneElement
  input src as source
  input tgt as target

  "conf-name" : for src.name make tgt.name
  "conf-definition" : for src.definition make tgt.definition
endgroup

group messaging extends BackboneElement
  input src as source
  input tgt as target

  "conf-endpoint" : for src.endpoint as vs make tgt.endpoint as vt then endpoint(vs, vt)
  "conf-reliableCache" : for src.reliableCache make tgt.reliableCache
  "conf-documentation" : for src.documentation make tgt.documentation
  "conf-event" : for src.event as vs make tgt.event as vt then event(vs, vt)
endgroup

group endpoint extends BackboneElement
  input src as source
  input tgt as target

  "conf-protocol" : for src.protocol make tgt.protocol
  "conf-address" : for src.address make tgt.address
endgroup

group event extends BackboneElement
  input src as source
  input tgt as target

  "conf-code" : for src.code make tgt.code
  "conf-category" : for src.category make tgt.category
  "conf-mode" : for src.mode make tgt.mode
  "conf-focus" : for src.focus make tgt.focus
  "conf-request" : for src.request make tgt.request
  "conf-response" : for src.response make tgt.response
  "conf-documentation" : for src.documentation make tgt.documentation
endgroup

group document extends BackboneElement
  input src as source
  input tgt as target

  "conf-mode" : for src.mode make tgt.mode
  "conf-documentation" : for src.documentation make tgt.documentation
  "conf-profile" : for src.profile make tgt.profile
endgroup
```

5.2.13.3 R3 Validation Errors

conformance-daf-query-requestor | ERROR @ CapabilityStatement.rest.resource[1].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[7].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[8].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[9].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[10].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[11].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[12].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[13].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[14].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[15].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @

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CapabilityStatement.rest.resource[12].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[12].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[12].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[12].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[12].searchParam[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[12].searchParam[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[12].searchParam[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[12].searchParam[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[13].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[13].interaction[2].extension: The extension 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CapabilityStatement.rest.resource[13].searchParam[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[13].searchParam[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[13].searchParam[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[14].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[14].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[14].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[14].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[14].searchParam[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[14].searchParam[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[14].searchParam[3].extension: The extension 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extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[14].searchParam[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[15].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[15].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[15].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[15].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[15].searchParam[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[15].searchParam[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[15].searchParam[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[15].searchParam[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[15].searchParam[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[16].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[16].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[16].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[16].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[16].searchParam[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[16].searchParam[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[16].searchParam[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[16].searchParam[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator)

conformance-ehrs-rle | ERROR @ CapabilityStatement.rest.resource[1].interaction.extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction.extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator)

phr | ERROR @ CapabilityStatement: Profile http://hl7.org/fhir/StructureDefinition/CapabilityStatement, Element 'CapabilityStatement.status': minimum required = 1, but only found 0 (src = InstanceValidator)

conformance-sdc-de-curator | ERROR @ CapabilityStatement.rest.resource[1].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[7].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[7].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator)

conformance-sdc-de-registry | ERROR @ CapabilityStatement.rest.resource[1].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[7].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[7].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[8].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @

CapabilityStatement.rest.resource[1].searchParam[9].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[10].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[11].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].searchParam[12].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[7].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].searchParam[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].searchParam[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].searchParam[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].searchParam[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].searchParam[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].searchParam[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].searchParam[7].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].searchParam[8].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].searchParam[9].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].searchParam[10].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator)

conformance-sdc-form-archiver | ERROR @ CapabilityStatement.rest.resource[1].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[4].interaction.extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator)

conformance-sdc-form-designer | ERROR @ CapabilityStatement.rest.resource[1].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[7].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[7].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[7].extension: The extension

http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator)
conformance-sdc-form-filler | ERROR @ CapabilityStatement.rest.resource[1].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[2].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[3].interaction[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[4].interaction.extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[5].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[5].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.operation.extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator)
conformance-sdc-form-manager | ERROR @ CapabilityStatement.rest.resource[1].interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource[1].interaction[5].extension: The extension 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unknown, and not allowed here (src = InstanceValidator)
conformance-sdc-form-receiver | ERROR @ CapabilityStatement.rest.resource.interaction[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource.interaction[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown,

and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource.interaction[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource.interaction[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource.interaction[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource.searchParam[1].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource.searchParam[2].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource.searchParam[3].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource.searchParam[4].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource.searchParam[5].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator) ERROR @ CapabilityStatement.rest.resource.searchParam[6].extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-expectation is unknown, and not allowed here (src = InstanceValidator)

conformance-terminology-server | ERROR @ CapabilityStatement.extension: The extension http://hl7.org/fhir/StructureDefinition/conformance-supported-system is unknown, and not allowed here (src = InstanceValidator)

5.3 Resource StructureDefinition - Content

URL: https://hl7.org/fhir/STU3/structuredefinition.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 5 | Trial Use | Compartments: Not linked to any defined compartments

A definition of a FHIR structure. This resource is used to describe the underlying resources, data types defined in FHIR, and also for describing extensions and constraints on resources and data types.

5.3.1 Scope and Usage

The StructureDefinition resource describes a structure - a set of data element definitions, and their associated rules of usage. These structure definitions are used to describe both the content defined in the FHIR specification itself - Resources, data types, the underlying infrastructural types, and also are used to describe how these structures are used in implementations. This allows the definitions of the structures to be shared and published through repositories of structure definitions, compared with each other, and used as the basis for code, report and UI generation.

Note that as part of the specification itself, a full set of structure definitions for all resources and data types is published.

5.3.2 Boundaries and Relationships

* StructureDefinitions are used by CapabilityStatement instances for specifying how resources are used

* StructureDefinitions use Value Sets to specify the content of coded elements

* StructureDefinitions define concrete elements and structures for use with FHIR alone and that have defined wire representations (XML, JSON, etc.). This is distinct from DataElement which defines abstract elements that might appear anywhere - FHIR, questionnaire questions, CDA , HL7 v2 , X12, OpenEHR, a proprietary database, etc. Data elements may map to FHIR resources, data types and/or extensions but do not have any defined serialization format of their own.

5.3.3 Background and Context

Implementers should be familiar with the background and concepts described in Profiling FHIR before working with this resource.

5.3.3.1 Metadata

The StructureDefinition resource has a set of metadata that is mostly shared with the Value Set, CapabilityStatement and other infrastructure resources. The metadata describes the structure, and helps find the structure when registered in profile repositories.

url | The identifier that is used to identify this structure when it is referenced in a specification, model, design or an instance. This URL is where the structure can be accessed

identifier | Other identifiers that are used to identify this structure

version | The identifier that is used to identify this version of the structure when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the structure author manually and the value should be a timestamp. Note that there may be multiple resource versions of the structure that have this same identifier. The resource will have updates that create new versions for technical reasons, whereas the stated version number needs to be under the author's control

name | A free text natural language name identifying the structure

publisher | Details of the individual or organization who accepts responsibility for publishing the structure. This helps establish the "authority/credibility" of the structure.

telecom | Contact details to assist a user in finding and communicating with the publisher

description | A free text natural language description of the structure and its use

requirements | The Scope and Usage that this structure was created to meet

code | A set of terms from external terminologies that may be used to assist with indexing and searching of profiles

status | The status of the structure allows filtering of profiles that are appropriate for use vs. not. See the Status Codes

experimental | This structure was authored for testing purposes (or education/evaluation/marketing), and is not intended for genuine usage

date | The date this version of the structure was published

fhirVersion | The version of the FHIR specification on which this structure is based. It is not necessary to specify the version, as most profiles are valid across multiple versions, and the validity of a structure against a particular version of FHIR can easily be checked by tooling.
type | The type the structure describes.
contextType | For extensions, the types of contexts in which the extension can be used (resources, datatypes, or extensions).
context | For extensions, the specific types of resources or data types on which the extension can be used.

Notes:

* The name of the structure is not required to be globally unique, but the name should have some scoping information (e.g. Acme Inc. (USA), Allergy List)

* Multiple codes may be assigned to the structure. These may either describe the structure, the focus of the structure or both. They are solely to help find the structure by searching for structured concepts

* The 3 status codes (draft, active, and retired) are the codes that are relevant to structure consumers. Authors may wish to use the authoring-status extension to track the life cycle of a structure as it is prepared

This resource is referenced by CapabilityStatement, ImplementationGuide, MessageDefinition and OperationDefinition

5.3.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
StructureDefinition | I | | DomainResource | Structural Definition + In any snapshot or differential, no label, code or requirements on the an element without a "." in the path (e.g. the first element) + Custom types can only be used in logical models + All element definitions must have unique ids (snapshot) + Context Invariants can only be used for extensions + All element definitions must have unique ids (diff) + If there's a type, its content must match the path name in the first element of a snapshot + All element definitions must have an id + Element paths must be unique unless the structure is a constraint + If the structure describes a base Resource or Type, the URL has to start with "http://hl7.org/fhir/StructureDefinition/" and the tail must match the id + A structure must have either a differential, or a snapshot (or both) + If the structure defines an extension then the structure must have context information + If the structure is not abstract, then there SHALL be a baseDefinition Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 1..1 | uri | Logical URI to reference this structure definition (globally unique)
identifier | ? | 0..* | Identifier | Additional identifier for the structure definition
version | ? | 0..1 | string | Business version of the structure definition
name | ? | 1..1 | string | Name for this structure definition (computer friendly)
title | ? | 0..1 | string | Name for this structure definition (human friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the structure definition
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for structure definition (if applicable) Jurisdiction ValueSet (Extensible)
purpose | | 0..1 | markdown | Why this structure definition is defined
copyright | | 0..1 | markdown | Use and/or publishing restrictions
keyword | ? | 0..* | Coding | Assist with indexing and finding Structure Definition Codes (Example)
fhirVersion | ? | 0..1 | id | FHIR Version this StructureDefinition targets
mapping | I | 0..* | BackboneElement | External specification that the content is mapped to + Must have at least a name or a uri (or both)
identity | | 1..1 | id | Internal id when this mapping is used
uri | I | 0..1 | uri | Identifies what this mapping refers to
name | I | 0..1 | string | Names what this mapping refers to
comment | | 0..1 | string | Versions, Issues, Scope limitations etc.
kind | ? | 1..1 | code | primitive-type | complex-type | resource | logical StructureDefinitionKind (Required)
abstract | ? | 1..1 | boolean | Whether the structure is abstract
contextType | ?I | 0..1 | code | resource | datatype | extension ExtensionContext (Required)
context | ?I | 0..* | string | Where the extension can be used in instances
contextInvariant | ?I | 0..* | string | FHIRPath invariants - when the extension can be used
type | ?I | 1..1 | code | Type defined or constrained by this structure FHIRDefinedType (Extensible)
baseDefinition | ?I | 0..1 | uri | Definition that this type is constrained/specialized from
derivation | ?I | 0..1 | code | specialization | constraint - How relates to base definition TypeDerivationRule (Required)
snapshot | I | 0..1 | BackboneElement | Snapshot view of the structure + The first element in a snapshot has no type + In any snapshot, all the elements must be in the specified type + Each element definition in a snapshot must have a formal definition and cardinalities
element | I | 1..* | ElementDefinition | Definition of elements in the resource (if no StructureDefinition)
differential | I | 0..1 | BackboneElement | Differential view of the structure + If the first element in a differential has no "." in the path, it has no type + No slicing on the root element + In any differential, all the elements must be in the specified type

element | | 1..* | ElementDefinition | Definition of elements in the resource (if no StructureDefinition)
Documentation for this format

UML Diagram (Legend)

XML Template

```
<StructureDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Logical URI to reference this structure definition (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the structure definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the structure definition -->
 <name value="[string]"/><!-- 1..1 Name for this structure definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this structure definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the structure definition -->
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for structure definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this structure definition is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <keyword><!-- 0..* Coding Assist with indexing and finding --></keyword>
 <fhirVersion value="[id]"/><!-- 0..1 FHIR Version this StructureDefinition targets -->
 <mapping>  <!-- 0..* External specification that the content is mapped to -->
  <identity value="[id]"/><!-- 1..1 Internal id when this mapping is used -->
  <uri value="[uri]"/><!--  0..1 Identifies what this mapping refers to -->
  <name value="[string]"/><!--  0..1 Names what this mapping refers to -->
  <comment value="[string]"/><!-- 0..1 Versions, Issues, Scope limitations etc. -->
 </mapping>
 <kind value="[code]"/><!-- 1..1 primitive-type | complex-type | resource | logical -->
 <abstract value="[boolean]"/><!-- 1..1 Whether the structure is abstract -->
 <contextType value="[code]"/><!--  0..1 resource | datatype | extension -->
 <context value="[string]"/><!--  0..* Where the extension can be used in instances -->
 <contextInvariant value="[string]"/><!--  0..* FHIRPath invariants - when the extension can be used -->
 <type value="[code]"/><!--  1..1 Type defined or constrained by this structure -->
 <baseDefinition value="[uri]"/><!--  0..1 Definition that this type is constrained/specialized from -->
 <derivation value="[code]"/><!--  0..1 specialization | constraint - How relates to base definition -->
 <snapshot>  <!--  0..1 Snapshot view of the structure -->
  <element><!--  1..* ElementDefinition Definition of elements in the resource (if no StructureDefinition) --></element>
 </snapshot>
 <differential>  <!--  0..1 Differential view of the structure -->
  <element><!-- 1..* ElementDefinition Definition of elements in the resource (if no StructureDefinition) --></element>
 </differential>
</StructureDefinition>
```

JSON Template

```
{
  "resourceType" : "StructureDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Logical URI to reference this structure definition (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the structure definition
  "version" : "<string>", // Business version of the structure definition
  "name" : "<string>", // R!  Name for this structure definition (computer friendly)
  "title" : "<string>", // Name for this structure definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the structure definition
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for structure definition (if applicable)
  "purpose" : "<markdown>", // Why this structure definition is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "keyword" : [{ Coding }], // Assist with indexing and finding
  "fhirVersion" : "<id>", // FHIR Version this StructureDefinition targets
  "mapping" : [{ // External specification that the content is mapped to
    "identity" : "<id>", // R!  Internal id when this mapping is used
    "uri" : "<uri>", // C? Identifies what this mapping refers to
    "name" : "<string>", // C? Names what this mapping refers to
    "comment" : "<string>" // Versions, Issues, Scope limitations etc.
  }],
  "kind" : "<code>", // R!  primitive-type | complex-type | resource | logical
  "abstract" : <boolean>, // R!  Whether the structure is abstract
  "contextType" : "<code>", // C? resource | datatype | extension
  "context" : ["<string>"], // C? Where the extension can be used in instances
  "contextInvariant" : ["<string>"], // C? FHIRPath invariants - when the extension can be used
  "type" : "<code>", // C? R!  Type defined or constrained by this structure
  "baseDefinition" : "<uri>", // C? Definition that this type is constrained/specialized from
  "derivation" : "<code>", //  specialization | constraint - How relates to base definition
  "snapshot" : { // C? Snapshot view of the structure
    "element" : [{ ElementDefinition }] // C? R!  Definition of elements in the resource (if no StructureDefinition)
  },
  "differential" : { // C? Differential view of the structure
```

```
    "element" : [{ ElementDefinition }] // R!  Definition of elements in the resource (if no StructureDefinition)
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:StructureDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:StructureDefinition.url [ uri ]; # 1..1 Logical URI to reference this structure definition (globally unique)
  fhir:StructureDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the structure definition
  fhir:StructureDefinition.version [ string ]; # 0..1 Business version of the structure definition
  fhir:StructureDefinition.name [ string ]; # 1..1 Name for this structure definition (computer friendly)
  fhir:StructureDefinition.title [ string ]; # 0..1 Name for this structure definition (human friendly)
  fhir:StructureDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:StructureDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:StructureDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:StructureDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:StructureDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:StructureDefinition.description [ markdown ]; # 0..1 Natural language description of the structure definition
  fhir:StructureDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:StructureDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for structure definition
     (if applicable)
  fhir:StructureDefinition.purpose [ markdown ]; # 0..1 Why this structure definition is defined
  fhir:StructureDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:StructureDefinition.keyword [ Coding ], ... ; # 0..* Assist with indexing and finding
  fhir:StructureDefinition.fhirVersion [ id ]; # 0..1 FHIR Version this StructureDefinition targets
  fhir:StructureDefinition.mapping [ # 0..* External specification that the content is mapped to
    fhir:StructureDefinition.mapping.identity [ id ]; # 1..1 Internal id when this mapping is used
    fhir:StructureDefinition.mapping.uri [ uri ]; # 0..1 Identifies what this mapping refers to
    fhir:StructureDefinition.mapping.name [ string ]; # 0..1 Names what this mapping refers to
    fhir:StructureDefinition.mapping.comment [ string ]; # 0..1 Versions, Issues, Scope limitations etc.
  ], ...;
  fhir:StructureDefinition.kind [ code ]; # 1..1 primitive-type | complex-type | resource | logical
  fhir:StructureDefinition.abstract [ boolean ]; # 1..1 Whether the structure is abstract
  fhir:StructureDefinition.contextType [ code ]; # 0..1 resource | datatype | extension
  fhir:StructureDefinition.context [ string ], ... ; # 0..* Where the extension can be used in instances
  fhir:StructureDefinition.contextInvariant [ string ], ... ; # 0..* FHIRPath invariants - when the extension can be
    used
  fhir:StructureDefinition.type [ code ]; # 1..1 Type defined or constrained by this structure
  fhir:StructureDefinition.baseDefinition [ uri ]; # 0..1 Definition that this type is constrained/specialized from
  fhir:StructureDefinition.derivation [ code ]; # 0..1 specialization | constraint - How relates to base definition
  fhir:StructureDefinition.snapshot [ # 0..1 Snapshot view of the structure
    fhir:StructureDefinition.snapshot.element [ ElementDefinition ], ... ; # 1..* Definition of elements in the resource
     (if no StructureDefinition)
  ];
  fhir:StructureDefinition.differential [ # 0..1 Differential view of the structure
    fhir:StructureDefinition.differential.element [ ElementDefinition ], ... ; # 1..* Definition of elements in the
    resource (if no StructureDefinition)
  ];
]
```

Changes since DSTU2

StructureDefinition

StructureDefinition | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

StructureDefinition.title | Renamed from display to title

StructureDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status

StructureDefinition.experimental | Now marked as Modifier

StructureDefinition.contact | Type changed from BackboneElement to ContactDetail

StructureDefinition.description | Type changed from string to markdown

StructureDefinition.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)

StructureDefinition.jurisdiction | Added Element

StructureDefinition.purpose | Renamed from requirements to purpose Type changed from string to markdown

StructureDefinition.copyright | Type changed from string to markdown

StructureDefinition.keyword | Renamed from code to keyword

StructureDefinition.mapping.comment | Renamed from comments to comment

StructureDefinition.contextInvariant | Added Element

StructureDefinition.type | Renamed from constrainedType to type Min Cardinality changed from 0 to 1 Change binding strength from required to extensible

StructureDefinition.baseDefinition | Renamed from base to baseDefinition

StructureDefinition.derivation | Added Element

StructureDefinition.contact.name | deleted

StructureDefinition.contact.telecom | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 248 tests that all execute ok. 60 fail round-trip testing and 145 r3 resources are invalid (441 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints

StructureDefinition | I | | DomainResource | Structural Definition + In any snapshot or differential, no label, code or requirements on the an element without a "." in the path (e.g. the first element) + Custom types can only be used in logical models + All element definitions must have unique ids (snapshot) + Context Invariants can only be used for extensions + All element definitions must have unique ids (diff) + If there's a type, its content must match the path name in the first element of a snapshot + All element definitions must have an id + Element paths must be unique unless the structure is a constraint + If the structure describes a base Resource or Type, the URL has to start with "http://hl7.org/fhir/StructureDefinition/" and the tail must match the id + A structure must have either a differential, or a snapshot (or both) + If the structure defines an extension then the structure must have context information + If the structure is not abstract, then there SHALL be a baseDefinition Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | ? | 1..1 | uri | Logical URI to reference this structure definition (globally unique)

identifier | ? | 0..* | Identifier | Additional identifier for the structure definition

version | ? | 0..1 | string | Business version of the structure definition

name | ? | 1..1 | string | Name for this structure definition (computer friendly)

title | ? | 0..1 | string | Name for this structure definition (human friendly)

status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage

date | ? | 0..1 | dateTime | Date this was last changed

publisher | ? | 0..1 | string | Name of the publisher (organization or individual)

contact | ? | 0..* | ContactDetail | Contact details for the publisher

description | | 0..1 | markdown | Natural language description of the structure definition

useContext | ? | 0..* | UsageContext | Context the content is intended to support

jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for structure definition (if applicable) Jurisdiction ValueSet (Extensible)

purpose | | 0..1 | markdown | Why this structure definition is defined

copyright | | 0..1 | markdown | Use and/or publishing restrictions

keyword | ? | 0..* | Coding | Assist with indexing and finding Structure Definition Codes (Example)

fhirVersion | ? | 0..1 | id | FHIR Version this StructureDefinition targets

mapping | I | 0..* | BackboneElement | External specification that the content is mapped to + Must have at least a name or a uri (or both)

identity | | 1..1 | id | Internal id when this mapping is used

uri | I | 0..1 | uri | Identifies what this mapping refers to

name | I | 0..1 | string | Names what this mapping refers to

comment | | 0..1 | string | Versions, Issues, Scope limitations etc.

kind | ? | 1..1 | code | primitive-type | complex-type | resource | logical StructureDefinitionKind (Required)

abstract | ? | 1..1 | boolean | Whether the structure is abstract

contextType | ?I | 0..1 | code | resource | datatype | extension ExtensionContext (Required)

context | ?I | 0..* | string | Where the extension can be used in instances

contextInvariant | ?I | 0..* | string | FHIRPath invariants - when the extension can be used

type | ?I | 1..1 | code | Type defined or constrained by this structure FHIRDefinedType (Extensible)

baseDefinition | ?I | 0..1 | uri | Definition that this type is constrained/specialized from

derivation | ?I | 0..1 | code | specialization | constraint - How relates to base definition TypeDerivationRule (Required)

snapshot | I | 0..1 | BackboneElement | Snapshot view of the structure + The first element in a snapshot has no type + In any snapshot, all the elements must be in the specified type + Each element definition in a snapshot must have a formal definition and cardinalities

element | I | 1..* | ElementDefinition | Definition of elements in the resource (if no StructureDefinition)

differential | I | 0..1 | BackboneElement | Differential view of the structure + If the first element in a differential has no "." in the path, it has no type + No slicing on the root element + In any differential, all the elements must be in the specified type

element | | 1..* | ElementDefinition | Definition of elements in the resource (if no StructureDefinition)

Documentation for this format

UML Diagram (Legend)

XML Template

```
<StructureDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Logical URI to reference this structure definition (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the structure definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the structure definition -->
 <name value="[string]"/><!-- 1..1 Name for this structure definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this structure definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the structure definition -->
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for structure definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this structure definition is defined -->
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <keyword><!-- 0..* Coding Assist with indexing and finding --></keyword>
 <fhirVersion value="[id]"/><!-- 0..1 FHIR Version this StructureDefinition targets -->
 <mapping>  <!-- 0..* External specification that the content is mapped to -->
  <identity value="[id]"/><!-- 1..1 Internal id when this mapping is used -->
  <uri value="[uri]"/><!--  0..1 Identifies what this mapping refers to -->
  <name value="[string]"/><!--  0..1 Names what this mapping refers to -->
```

```
  <comment value="[string]"/><!-- 0..1 Versions, Issues, Scope limitations etc. -->
 </mapping>
 <kind value="[code]"/><!-- 1..1 primitive-type | complex-type | resource | logical -->
 <abstract value="[boolean]"/><!-- 1..1 Whether the structure is abstract -->
 <contextType value="[code]"/><!--  0..1 resource | datatype | extension -->
 <context value="[string]"/><!--  0..* Where the extension can be used in instances -->
 <contextInvariant value="[string]"/><!--  0..* FHIRPath invariants - when the extension can be used -->
 <type value="[code]"/><!--  1..1 Type defined or constrained by this structure -->
 <baseDefinition value="[uri]"/><!--  0..1 Definition that this type is constrained/specialized from -->
 <derivation value="[code]"/><!--  0..1 specialization | constraint - How relates to base definition -->
 <snapshot>  <!--  0..1 Snapshot view of the structure -->
  <element><!--  1..* ElementDefinition Definition of elements in the resource (if no StructureDefinition) --></element>
 </snapshot>
 <differential>  <!--  0..1 Differential view of the structure -->
  <element><!-- 1..* ElementDefinition Definition of elements in the resource (if no StructureDefinition) --></element>
 </differential>
</StructureDefinition>
```

JSON Template

```
{
  "resourceType" : "StructureDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Logical URI to reference this structure definition (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the structure definition
  "version" : "<string>", // Business version of the structure definition
  "name" : "<string>", // R!  Name for this structure definition (computer friendly)
  "title" : "<string>", // Name for this structure definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the structure definition
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for structure definition (if applicable)
  "purpose" : "<markdown>", // Why this structure definition is defined
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "keyword" : [{ Coding }], // Assist with indexing and finding
  "fhirVersion" : "<id>", // FHIR Version this StructureDefinition targets
  "mapping" : [{ // External specification that the content is mapped to
    "identity" : "<id>", // R!  Internal id when this mapping is used
    "uri" : "<uri>", // C? Identifies what this mapping refers to
    "name" : "<string>", // C? Names what this mapping refers to
    "comment" : "<string>" // Versions, Issues, Scope limitations etc.
  }],
  "kind" : "<code>", // R!  primitive-type | complex-type | resource | logical
  "abstract" : <boolean>, // R!  Whether the structure is abstract
  "contextType" : "<code>", // C? resource | datatype | extension
  "context" : ["<string>"], // C? Where the extension can be used in instances
  "contextInvariant" : ["<string>"], // C? FHIRPath invariants - when the extension can be used
  "type" : "<code>", // C? R!  Type defined or constrained by this structure
  "baseDefinition" : "<uri>", // C? Definition that this type is constrained/specialized from
  "derivation" : "<code>", //  specialization | constraint - How relates to base definition
  "snapshot" : { // C? Snapshot view of the structure
    "element" : [{ ElementDefinition }] // C? R!  Definition of elements in the resource (if no StructureDefinition)
  },
  "differential" : { // C? Differential view of the structure
    "element" : [{ ElementDefinition }] // R!  Definition of elements in the resource (if no StructureDefinition)
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:StructureDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:StructureDefinition.url [ uri ]; # 1..1 Logical URI to reference this structure definition (globally unique)
  fhir:StructureDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the structure definition
  fhir:StructureDefinition.version [ string ]; # 0..1 Business version of the structure definition
  fhir:StructureDefinition.name [ string ]; # 1..1 Name for this structure definition (computer friendly)
  fhir:StructureDefinition.title [ string ]; # 0..1 Name for this structure definition (human friendly)
  fhir:StructureDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:StructureDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:StructureDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:StructureDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:StructureDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:StructureDefinition.description [ markdown ]; # 0..1 Natural language description of the structure definition
  fhir:StructureDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:StructureDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for structure definition
     (if applicable)
  fhir:StructureDefinition.purpose [ markdown ]; # 0..1 Why this structure definition is defined
  fhir:StructureDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:StructureDefinition.keyword [ Coding ], ... ; # 0..* Assist with indexing and finding
  fhir:StructureDefinition.fhirVersion [ id ]; # 0..1 FHIR Version this StructureDefinition targets
```

```
  fhir:StructureDefinition.mapping [ # 0..* External specification that the content is mapped to
    fhir:StructureDefinition.mapping.identity [ id ]; # 1..1 Internal id when this mapping is used
    fhir:StructureDefinition.mapping.uri [ uri ]; # 0..1 Identifies what this mapping refers to
    fhir:StructureDefinition.mapping.name [ string ]; # 0..1 Names what this mapping refers to
    fhir:StructureDefinition.mapping.comment [ string ]; # 0..1 Versions, Issues, Scope limitations etc.
  ], ...;
  fhir:StructureDefinition.kind [ code ]; # 1..1 primitive-type | complex-type | resource | logical
  fhir:StructureDefinition.abstract [ boolean ]; # 1..1 Whether the structure is abstract
  fhir:StructureDefinition.contextType [ code ]; # 0..1 resource | datatype | extension
  fhir:StructureDefinition.context [ string ], ... ; # 0..* Where the extension can be used in instances
  fhir:StructureDefinition.contextInvariant [ string ], ... ; # 0..* FHIRPath invariants - when the extension can be
    used
  fhir:StructureDefinition.type [ code ]; # 1..1 Type defined or constrained by this structure
  fhir:StructureDefinition.baseDefinition [ uri ]; # 0..1 Definition that this type is constrained/specialized from
  fhir:StructureDefinition.derivation [ code ]; # 0..1 specialization | constraint - How relates to base definition
  fhir:StructureDefinition.snapshot [ # 0..1 Snapshot view of the structure
    fhir:StructureDefinition.snapshot.element [ ElementDefinition ], ... ; # 1..* Definition of elements in the resource
     (if no StructureDefinition)
  ];
  fhir:StructureDefinition.differential [ # 0..1 Differential view of the structure
    fhir:StructureDefinition.differential.element [ ElementDefinition ], ... ; # 1..* Definition of elements in the
    resource (if no StructureDefinition)
  ];
]
```

Changes since DSTU2

StructureDefinition

StructureDefinition | Min Cardinality changed from 1 to 0 Max Cardinality changed from 1 to *

StructureDefinition.title | Renamed from display to title

StructureDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status

StructureDefinition.experimental | Now marked as Modifier

StructureDefinition.contact | Type changed from BackboneElement to ContactDetail

StructureDefinition.description | Type changed from string to markdown

StructureDefinition.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)

StructureDefinition.jurisdiction | Added Element

StructureDefinition.purpose | Renamed from requirements to purpose Type changed from string to markdown

StructureDefinition.copyright | Type changed from string to markdown

StructureDefinition.keyword | Renamed from code to keyword

StructureDefinition.mapping.comment | Renamed from comments to comment

StructureDefinition.contextInvariant | Added Element

StructureDefinition.type | Renamed from constrainedType to type Min Cardinality changed from 0 to 1 Change binding strength from required to extensible

StructureDefinition.baseDefinition | Renamed from base to baseDefinition

StructureDefinition.derivation | Added Element

StructureDefinition.contact.name | deleted

StructureDefinition.contact.telecom | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 248 tests that all execute ok. 60 fail round-trip testing and 145 r3 resources are invalid (441 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

5.3.4.1 Terminology Bindings

Path | Definition | Type | Reference

StructureDefinition.status | The lifecycle status of a Value Set or Concept Map. | Required | PublicationStatus

StructureDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use | Extensible | Jurisdiction ValueSet

StructureDefinition.keyword | Codes for the meaning of the defined structure (SNOMED CT and LOINC codes, as an example). | Example | Structure Definition Codes

StructureDefinition.kind | Defines the type of structure that a definition is describing. | Required | StructureDefinitionKind

StructureDefinition.contextType | How an extension context is interpreted. | Required | ExtensionContext

StructureDefinition.type | Either a resource or a data type, including logical model types | Extensible | FHIRDefinedType

StructureDefinition.derivation | How a type relates to its baseDefinition. | Required | TypeDerivationRule

5.3.4.2 Constraints

* sdf-1: Element paths must be unique unless the structure is a constraint (expression : derivation = 'constraint' or snapshot.element.select(path).isDistinct())

* sdf-11: If there's a type, its content must match the path name in the first element of a snapshot (expression : snapshot.empty() or snapshot.element.first().path = type)

* sdf-14: All element definitions must have an id (expression : snapshot.element.all(id.exists()) and differential.element.all(id.exists()))

* sdf-15: On StructureDefinition.snapshot: The first element in a snapshot has no type (expression on StructureDefinition.snapshot: element.first().type.empty())

* sdf-15a: On StructureDefinition.differential: If the first element in a differential has no "." in the path, it has no type (expression on StructureDefinition.differential: element.first().path.contains('.').not() implies element.first().type.empty())

* sdf-16: All element definitions must have unique ids (snapshot) (expression : snapshot.element.id.trace('ids').isDistinct())

* sdf-17: All element definitions must have unique ids (diff) (expression : differential.element.id.trace('ids').isDistinct())

* sdf-18: Context Invariants can only be used for extensions (expression : contextInvariant.exists() implies type = 'Extension')

* sdf-19: Custom types can only be used in logical models (expression : kind = 'logical' or (differential.element.type.code.all(hasValue() implies matches('^[a-zA-Z0-9]+$')) and snapshot.element.type.code.all(hasValue() implies matches('^[a-zA-Z0-9]+$'))))

* sdf-2: On StructureDefinition.mapping: Must have at least a name or a uri (or both) (expression on StructureDefinition.mapping: name.exists() or uri.exists())

* sdf-20: On StructureDefinition.differential: No slicing on the root element (expression on StructureDefinition.differential: element.where(path.contains('.').not()).slicing.empty())

* sdf-3: On StructureDefinition.snapshot: Each element definition in a snapshot must have a formal definition and cardinalities (expression on StructureDefinition.snapshot: element.all(definition and min and max))

* sdf-4: If the structure is not abstract, then there SHALL be a baseDefinition (expression : abstract = true or baseDefinition.exists())

* sdf-5: If the structure defines an extension then the structure must have context information (expression : type != 'Extension' or derivation = 'specialization' or (context.exists() and contextType.exists()))

* sdf-6: A structure must have either a differential, or a snapshot (or both) (expression : snapshot.exists() or differential.exists())

* sdf-7: If the structure describes a base Resource or Type, the URL has to start with "http://hl7.org/fhir/StructureDefinition/" and the tail must match the id (expression : (derivation = 'constraint') or (kind = 'logical') or (url = 'http://hl7.org/fhir/StructureDefinition/'+id))

* sdf-8: On StructureDefinition.snapshot: In any snapshot, all the elements must be in the specified type (expression on StructureDefinition.snapshot: (%resource.kind = 'logical' or element.first().path = %resource.type) and element.tail().all(path.startsWith(%resource.snapshot.element.first().path&'.')))

* sdf-8a: On StructureDefinition.differential: In any differential, all the elements must be in the specified type (expression on StructureDefinition.differential: element.first().path.startsWith(%resource.type) and element.tail().all(path.startsWith(%resource.type&'.')))

* sdf-9: In any snapshot or differential, no label, code or requirements on the an element without a "." in the path (e.g. the first element) (expression : children().element.where(path.contains('.').not()).label.empty() and children().element.where(path.contains('.').not()).code.empty() and children().element.where(path.contains('.').not()).requirements.empty())

5.3.5 Interpretation Notes:

* A structure is represented as a flat list of elements. The element.path provides the overall structure.

* Differentials in constraints need only specify elements that they are making rules about. Other elements can be inferred as defined in the base resource

* Elements specified in the differential (and all elements in the snapshot) must be ordered as such:

* Elements from the baseDefinition appear before new elements in a specialization profile

* Elements must be in the same order as the baseDefinition, and child elements appear in depth-first order.

* Unsliced descendants of sliced elements appear before slices

* element.short, element.formal, element.comments and element.mapping provide the specific definition for the field in a specific context.

* element.type is used to specify which types may be used for this element. If there is more than one type, the element offers a choice of types and must have a name that terminates in "[x]". When profiling a resource and only one type is specified, the name of the element is changed to include the type instead of '[x]'.

* When using XPath to define constraints about the relationship between the contents of the narrative and the contents of the data elements, the element against which the rule is expressed is the one against which the error or warning will be reported by a validator.

* When expression constraints are placed on elements that offer a choice of types, the expression has to select the type to which it applies using the fluentpath is operator .

* For XPath constraints, the prefix "f" should be used for "http://hl7.org/fhir", and "h" for "http://www.w3.org/1999/xhtml". XPath constraints are written against the XML representation

* The condition element is used to assert that a constraint defined on another element affects the allowed cardinality of this element.

5.3.5.1 Different Uses for StructureDefinition

The base structure definition is used in a number of different ways to support the FHIR specification. The various uses of the StructureDefinition are controlled by the elements kind, constrainedType, base, and url, using this basic pattern:

```
{
  "url": the identity of this structure definition,
  "kind": (datatype | resource),
  "type": the type being constrained (if it's a constraint)
  "baseDefinition": the structure definition from which this is derived
}
```

This list shows a number of examples, with links to real examples for each:

* Base definition of a data type (example: - ,):

Quantity

XML

JSON

```
{
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/Quantity",
  "name": "Quantity",
  "kind": "datatype",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/Element"
}
```

* A constrained data type (example: - ,):

Money

XML

JSON

```
{
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/Money",
  "name": "Money",
  "kind": "datatype",
  "type": "Quantity",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/Quantity"
}
```

* Base definition of a resource (example: - ,):

Patient

XML

JSON

```
{
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/Patient",
  "name": "Patient",
  "kind": "resource",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/DomainResource"
}
```

* Constraint on a resource (example: - ,):

Clinical Document Profile for Composition

XML

JSON

```
{
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/clinicaldocument",
  "name": "Clinical Document Profile for Composition",
  "kind": "resource",
  "type": "Composition",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/Composition"
}
```

* Base Extension (a standard data type) (example: - ,):

Extension

XML

JSON

```
{
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/Extension",
  "name": "Extension",
  "kind": "datatype",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/Element"
}
```

* A defined Extension (example: - ,):

Extension Data Absent Reason

XML

JSON

```
{
  "resourceType": "StructureDefinition",
  "url": "http://hl7.org/fhir/StructureDefinition/data-absent-reason",
  "name": "Data Absent Reason",
  "kind": "complex-type",
  "type": "Extension",
  "baseDefinition": "http://hl7.org/fhir/StructureDefinition/Extension"
}
```

* A constraint on a defined extension (no examples currently defined):

```
{
  "resourceType": "StructureDefinition",
  "url": "http://example.org/fhir/StructureDefinition/race",
  "name": "Race codes used by institution (a subset of meaningful use codes)",
  "kind": "datatype",
  "type": "Extension",
  "baseDefinition": "http://hl7.org/fhir/us/core/StructureDefinition/us-core-race"
}
```

On this list, structure definitions of type 1, 3, and 5 can only be defined by the FHIR specification itself. The other kinds of structure definitions are (or may be) created by the specification, but can also be defined by other implementers.

5.3.5.2 Rules for Constrained Types

When the structure is a constraint (constrainedType is not null), see Extending and Restricting Resources for the rules that apply.

5.3.5.3 Common Mapping Targets

Structures are able to map elements to concepts in other definition systems. Generally these are used to map the elements to local implementation models, data paths, and concepts. However they are also used to map to other standards and published terminologies. These are the standard URIs used for common targets of the mapping:

http://hl7.org/fhir/logical | Logical Mapping
http://clinicaltrials.gov | ClinicalTrials.gov Mapping
http://www.cdisc.org/define-xml | CDISC Define XML 2.0 metadata standard
http://hl7.org/fhir/workflow | Workflow Mapping
http://wiki.siframework.org/IHE+SDC+Profile | IHE Structured Data Capture
http://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/UniqueDeviceIdentification/default.htm | UDI Mapping
http://ncpdp.org/SCRIPT10_6 | Mapping to NCPDP SCRIPT 10.6
http://www.cda-adc.ca/en/services/cdanet/ | Canadian Dental Association eclaims standard
http://snomed.info/conceptdomain | SNOMED CT Concept Domain Binding
http://wiki.ihe.net/index.php?title=Data_Element_Exchange | IHE Data Element Exchange (DEX)
http://hl7.org/v2 | HL7 v2 Mapping
http://hl7.org/v3 | RIM Mapping
http://metadata-standards.org/11179/ | ISO 11179
http://snomed.info/sct | SNOMED CT Attribute Binding
http://nema.org/dicom | DICOM Tag Mapping
http://github.com/MDMI/ReferentIndexContent | MDMI Referent Index
http://ihe.net/xds | XDS metadata equivalent
http://cap.org/ecc | CAP Code
http://hl7.org/v3/cda | CDA (R2)
http://www.hl7.org/implement/standards/product_brief.cfm?product_id=378 | Quality Improvement Domain Analysis Model
http://www.pharmacists.ca/ | Canadian Pharmacy Associaiton eclaims standard
http://www.healthit.gov/quality-data-model | Quality Data Model
http://hl7.org/fhir/auditevent | FHIR AuditEvent
http://www.w3.org/ns/prov | W3C PROV
http://hl7.org/orim | Ontological RIM Mapping
http://loinc.org | LOINC code for the element
http://w3.org/vcard | vCard Mapping
http://openehr.org | Open EHR Archetype Mapping
http://www.omg.org/spec/ServD/1.0/ | ServD
http://wiki.siframework.org/CQF | Quality Improvement and Clinical Knowledge (QUICK)
http://hl7.org/fhir/provenance | FHIR Provenance
http://www.ietf.org/rfc/rfc2445.txt | iCalendar

5.3.5.4 Logical Models

StructureDefinitions are used to define the basic structures of FHIR: data types, resources, extensions, and profiles. The same definition structure can also be used to define any arbitrary structures that are a directed acyclic graph with typed nodes, where the primitive types are those defined by the FHIR specification.

This technique has many uses:

* Describing any arbitrary content model

* Describing existing HL7 content models (e.g. v2, CDA) using FHIR

* Describing common design patterns used in FHIR

* Defining a content model to support the mapping language

5.3.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
abstract | token | Whether the structure is abstract | StructureDefinition.abstract
base | uri | Definition that this type is constrained/specialized from | StructureDefinition.baseDefinition
base-path | token | Path that identifies the base element | StructureDefinition.snapshot.element.base.path | StructureDefinition.differential.element.base.path
context-type | token | resource | datatype | extension | StructureDefinition.contextType
date | date | The structure definition publication date | StructureDefinition.date
derivation | token | specialization | constraint - How relates to base definition | StructureDefinition.derivation
description | string | The description of the structure definition | StructureDefinition.description
experimental | token | For testing purposes, not real usage | StructureDefinition.experimental
ext-context | string | Where the extension can be used in instances | StructureDefinition.context
identifier | token | External identifier for the structure definition | StructureDefinition.identifier
jurisdiction | token | Intended jurisdiction for the structure definition | StructureDefinition.jurisdiction
keyword | token | A code for the profile | StructureDefinition.keyword
kind | token | primitive-type | complex-type | resource | logical | StructureDefinition.kind
name | string | Computationally friendly name of the structure definition | StructureDefinition.name
path | token | A path that is constrained in the profile | StructureDefinition.snapshot.element.path | StructureDefinition.differential.element.path
publisher | string | Name of the publisher of the structure definition | StructureDefinition.publisher
status | token | The current status of the structure definition | StructureDefinition.status
title | string | The human-friendly name of the structure definition | StructureDefinition.title
type | token | Type defined or constrained by this structure | StructureDefinition.type
url | uri | The uri that identifies the structure definition | StructureDefinition.url
valueset | reference | A vocabulary binding reference | StructureDefinition.snapshot.element.binding.valueSet (ValueSet)
version | token | The business version of the structure definition | StructureDefinition.version

5.3.7 Resource StructureDefinition - Examples

URL: https://hl7.org/fhir/STU3/structuredefinition-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.3.7 Resource StructureDefinition - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

* Base Types
* Resources
* Constraints
* Extensions

Abstract Types
Element | XML | JSON
BackBoneElement | XML | JSON
Resource | XML | JSON
DomainResource | XML | JSON
Primitive Types
base64Binary | XML | JSON
boolean | XML | JSON
code | XML | JSON
date | XML | JSON
dateTime | XML | JSON
decimal | XML | JSON
id | XML | JSON
instant | XML | JSON
integer | XML | JSON
markdown | XML | JSON
oid | XML | JSON
positiveInt | XML | JSON
string | XML | JSON
time | XML | JSON
unsignedInt | XML | JSON
uri | XML | JSON
uuid | XML | JSON
Data Types
Address | XML | JSON
Age | XML | JSON

Annotation | XML | JSON
Attachment | XML | JSON
BackboneElement | XML | JSON
CodeableConcept | XML | JSON
Coding | XML | JSON
ContactDetail | XML | JSON
ContactPoint | XML | JSON
Contributor | XML | JSON
Count | XML | JSON
DataRequirement | XML | JSON
Distance | XML | JSON
Dosage | XML | JSON
Duration | XML | JSON
Element | XML | JSON
ElementDefinition | XML | JSON
Extension | XML | JSON
HumanName | XML | JSON
Identifier | XML | JSON
Meta | XML | JSON
Money | XML | JSON
Narrative | XML | JSON
ParameterDefinition | XML | JSON
Period | XML | JSON
Quantity | XML | JSON
Range | XML | JSON
Ratio | XML | JSON
Reference | XML | JSON
RelatedArtifact | XML | JSON
SampledData | XML | JSON
Signature | XML | JSON
Timing | XML | JSON
TriggerDefinition | XML | JSON
UsageContext | XML | JSON

Resources
Account | XML | JSON
ActivityDefinition | XML | JSON
AdverseEvent | XML | JSON
AllergyIntolerance | XML | JSON
Appointment | XML | JSON
AppointmentResponse | XML | JSON
AuditEvent | XML | JSON
Basic | XML | JSON
Binary | XML | JSON
BodySite | XML | JSON
Bundle | XML | JSON
CapabilityStatement | XML | JSON
CarePlan | XML | JSON
CareTeam | XML | JSON
ChargeItem | XML | JSON
Claim | XML | JSON
ClaimResponse | XML | JSON
ClinicalImpression | XML | JSON
CodeSystem | XML | JSON
Communication | XML | JSON
CommunicationRequest | XML | JSON
CompartmentDefinition | XML | JSON
Composition | XML | JSON
ConceptMap | XML | JSON
Condition | XML | JSON
Consent | XML | JSON
Contract | XML | JSON
Coverage | XML | JSON
DataElement | XML | JSON
DetectedIssue | XML | JSON
Device | XML | JSON
DeviceComponent | XML | JSON
DeviceMetric | XML | JSON
DeviceRequest | XML | JSON
DeviceUseStatement | XML | JSON
DiagnosticReport | XML | JSON
DocumentManifest | XML | JSON
DocumentReference | XML | JSON

EligibilityRequest | XML | JSON
EligibilityResponse | XML | JSON
Encounter | XML | JSON
Endpoint | XML | JSON
EnrollmentRequest | XML | JSON
EnrollmentResponse | XML | JSON
EpisodeOfCare | XML | JSON
ExpansionProfile | XML | JSON
ExplanationOfBenefit | XML | JSON
FamilyMemberHistory | XML | JSON
Flag | XML | JSON
Goal | XML | JSON
GraphDefinition | XML | JSON
Group | XML | JSON
GuidanceResponse | XML | JSON
HealthcareService | XML | JSON
ImagingManifest | XML | JSON
ImagingStudy | XML | JSON
Immunization | XML | JSON
ImmunizationRecommendation | XML | JSON
ImplementationGuide | XML | JSON
Library | XML | JSON
Linkage | XML | JSON
List | XML | JSON
Location | XML | JSON
Measure | XML | JSON
MeasureReport | XML | JSON
Media | XML | JSON
Medication | XML | JSON
MedicationAdministration | XML | JSON
MedicationDispense | XML | JSON
MedicationRequest | XML | JSON
MedicationStatement | XML | JSON
MessageDefinition | XML | JSON
MessageHeader | XML | JSON
NamingSystem | XML | JSON
NutritionOrder | XML | JSON
Observation | XML | JSON
OperationDefinition | XML | JSON
OperationOutcome | XML | JSON
Organization | XML | JSON
Patient | XML | JSON
PaymentNotice | XML | JSON
PaymentReconciliation | XML | JSON
Person | XML | JSON
PlanDefinition | XML | JSON
Practitioner | XML | JSON
PractitionerRole | XML | JSON
Procedure | XML | JSON
ProcedureRequest | XML | JSON
ProcessRequest | XML | JSON
ProcessResponse | XML | JSON
Provenance | XML | JSON
Questionnaire | XML | JSON
QuestionnaireResponse | XML | JSON
ReferralRequest | XML | JSON
RelatedPerson | XML | JSON
RequestGroup | XML | JSON
ResearchStudy | XML | JSON
ResearchSubject | XML | JSON
RiskAssessment | XML | JSON
Schedule | XML | JSON
SearchParameter | XML | JSON
Sequence | XML | JSON
ServiceDefinition | XML | JSON
Slot | XML | JSON
Specimen | XML | JSON
StructureDefinition | XML | JSON
StructureMap | XML | JSON
Subscription | XML | JSON
Substance | XML | JSON
SupplyDelivery | XML | JSON

SupplyRequest | XML | JSON
Task | XML | JSON
TestReport | XML | JSON
TestScript | XML | JSON
ValueSet | XML | JSON
VisionPrescription | XML | JSON

FHIR Specification Core
BMI | XML | JSON
BodyHeight | XML | JSON
BodyLength | XML | JSON
BodyTemp | XML | JSON
BodyWeight | XML | JSON
BP | XML | JSON
Cholesterol | XML | JSON
ClinicalDocument | XML | JSON
Consent Directive | XML | JSON
DeviceMetricObservation | XML | JSON
DiagnosticReport-Genetics | XML | JSON
elementdefinition-de | XML | JSON
FamilyMemberHistory-Genetic | XML | JSON
HDLCholesterol | XML | JSON
HeadCircum | XML | JSON
HeartRate | XML | JSON
HLAResult | XML | JSON
LDLCholesterol | XML | JSON
LipidProfile | XML | JSON
Observation-Genetics | XML | JSON
OxygenSat | XML | JSON
ProcedureRequest-Genetics | XML | JSON
RespRate | XML | JSON
ShareableCodeSystem | XML | JSON
ShareableValueSet | XML | JSON
Triglyceride | XML | JSON
VitalSigns | XML | JSON
VitalsPanel | XML | JSON
EHRS Functional model - Record Lifecycle Events
ehrsrle-auditevent | XML | JSON
ehrsrle-provenance | XML | JSON
Clinical Quality Improvement Framework
CQIF-Questionnaire | XML | JSON

FHIR Specification Core
11179-de-administrative-status administrative-status | XML | JSON
11179-de-change-description change-description | XML | JSON
11179-de-classification-or-context classification-or-context | XML | JSON
11179-de-contact-address contact-address | XML | JSON
11179-de-document-reference document-reference | XML | JSON
11179-de-effective-period effective-period | XML | JSON
11179-de-is-data-element-concept is-data-element-concept | XML | JSON
11179-de-registry-org registry-org | XML | JSON
11179-de-submitter-org submitter-org | XML | JSON
11179-objectClass objectClass | XML | JSON
11179-objectClassProperty objectClassProperty | XML | JSON
11179-permitted-value-conceptmap permitted-value-conceptmap | XML | JSON
11179-permitted-value-valueset permitted-value-valueset | XML | JSON
DiagnosticReport-geneticsAnalysis Analysis | XML | JSON
DiagnosticReport-geneticsAssessedCondition AssessedCondition | XML | JSON
DiagnosticReport-geneticsFamilyMemberHistory FamilyMemberHistory | XML | JSON
allergyintolerance-certainty certainty | XML | JSON
allergyintolerance-duration duration | XML | JSON
allergyintolerance-reasonRefuted reasonRefuted | XML | JSON
allergyintolerance-resolutionAge resolutionAge | XML | JSON
allergyintolerance-substanceExposureRisk substanceExposureRisk | XML | JSON
auditevent-Accession Accession | XML | JSON
auditevent-Anonymized Anonymized | XML | JSON
auditevent-Encrypted Encrypted | XML | JSON
auditevent-Instance Instance | XML | JSON
auditevent-MPPS MPPS | XML | JSON
auditevent-NumberOfInstances NumberOfInstances | XML | JSON
auditevent-ParticipantObjectContainsStudy ParticipantObjectContainsStudy | XML | JSON
auditevent-SOPClass SOPClass | XML | JSON
birthPlace Birth Place | XML | JSON
body-site-instance Body Site Instance | XML | JSON

capabilitystatement-expectation expectation | XML | JSON
capabilitystatement-prohibited prohibited | XML | JSON
capabilitystatement-search-parameter-combination search-parameter-combination | XML | JSON
capabilitystatement-supported-system supported-system | XML | JSON
capabilitystatement-websocket websocket | XML | JSON
careplan-activity-title activity-title | XML | JSON
codesystem-author author | XML | JSON
codesystem-comment comment | XML | JSON
codesystem-comments comments | XML | JSON
codesystem-conceptOrder conceptOrder | XML | JSON
codesystem-deprecated deprecated | XML | JSON
codesystem-effectiveDate effectiveDate | XML | JSON
codesystem-expirationDate expirationDate | XML | JSON
codesystem-history history | XML | JSON
codesystem-keyWord keyWord | XML | JSON
codesystem-label label | XML | JSON
codesystem-map map | XML | JSON
codesystem-ordinalValue ordinalValue | XML | JSON
codesystem-otherName otherName | XML | JSON
codesystem-reference reference | XML | JSON
codesystem-replacedby replacedby | XML | JSON
codesystem-sourceReference sourceReference | XML | JSON
codesystem-subsumes subsumes | XML | JSON
codesystem-trusted-expansion trusted-expansion | XML | JSON
codesystem-usage usage | XML | JSON
codesystem-warning warning | XML | JSON
codesystem-workflowStatus workflowStatus | XML | JSON
coding-sctdescid sctdescid | XML | JSON
communication-media media | XML | JSON
communication-reasonNotPerformed reasonNotPerformed | XML | JSON
communicationrequest-definition definition | XML | JSON
communicationrequest-orderedBy orderedBy | XML | JSON
communicationrequest-reasonRejected reasonRejected | XML | JSON
communicationrequest-relevantHistory relevantHistory | XML | JSON
communicationrequest-supportingInfo supportingInfo | XML | JSON
composition-clindoc-otherConfidentiality otherConfidentiality | XML | JSON
concept-bidirectional bidirectional | XML | JSON
condition-basedOn basedOn | XML | JSON
condition-criticality criticality | XML | JSON
condition-definition definition | XML | JSON
condition-dueTo dueTo | XML | JSON
condition-occurredFollowing occurredFollowing | XML | JSON
condition-outcome outcome | XML | JSON
condition-partOf partOf | XML | JSON
condition-ruledOut ruledOut | XML | JSON
condition-targetBodySite targetBodySite | XML | JSON
consent-NotificationEndpoint NotificationEndpoint | XML | JSON
consent-Witness Witness | XML | JSON
consent-location location | XML | JSON
data-absent-reason Data Absent Reason | XML | JSON
datadictionary datadictionary | XML | JSON
device-din din | XML | JSON
device-implant-status implant-status | XML | JSON
devicerequest-patientInstruction patientInstruction | XML | JSON
devicerequest-reasonRejected reasonRejected | XML | JSON
diagnosticReport-addendumOf addendumOf | XML | JSON
diagnosticReport-extends extends | XML | JSON
diagnosticReport-locationPerformed locationPerformed | XML | JSON
diagnosticReport-replaces replaces | XML | JSON
diagnosticReport-summaryOf summaryOf | XML | JSON
elementdefinition-allowedUnits allowedUnits | XML | JSON
elementdefinition-bestpractice bestpractice | XML | JSON
elementdefinition-bindingName bindingName | XML | JSON
elementdefinition-equivalence equivalence | XML | JSON
elementdefinition-identifier identifier | XML | JSON
elementdefinition-inheritedExtensibleValueSet inheritedExtensibleValueSet | XML | JSON
elementdefinition-isCommonBinding isCommonBinding | XML | JSON
elementdefinition-maxValueSet maxValueSet | XML | JSON
elementdefinition-minValueSet minValueSet | XML | JSON
elementdefinition-namespace namespace | XML | JSON
elementdefinition-question question | XML | JSON
elementdefinition-selector selector | XML | JSON

elementdefinition-translatable translatable | XML | JSON
encounter-associatedEncounter associatedEncounter | XML | JSON
encounter-modeOfArrival modeOfArrival | XML | JSON
encounter-primaryDiagnosis primaryDiagnosis | XML | JSON
encounter-reasonCancelled reasonCancelled | XML | JSON
entryFormat entryFormat | XML | JSON
event-OnBehalfOf OnBehalfOf | XML | JSON
event-definition definition | XML | JSON
event-notDone notDone | XML | JSON
event-partOf partOf | XML | JSON
event-performerRole performerRole | XML | JSON
event-reasonCode reasonCode | XML | JSON
event-reasonReference reasonReference | XML | JSON
family-member-history-genetics-observation observation | XML | JSON
family-member-history-genetics-parent parent | XML | JSON
family-member-history-genetics-sibling sibling | XML | JSON
familymemberhistory-abatement abatement | XML | JSON
familymemberhistory-patient-record patient-record | XML | JSON
familymemberhistory-severity severity | XML | JSON
familymemberhistory-type type | XML | JSON
flag-detail detail | XML | JSON
flag-priority priority | XML | JSON
geolocation Geolocation | XML | JSON
goal-acceptance acceptance | XML | JSON
goal-pertainsToGoal pertainsToGoal | XML | JSON
goal-reasonRejected reasonRejected | XML | JSON
goal-relationship relationship | XML | JSON
hla-genotyping-results-allele-database allele-database | XML | JSON
hla-genotyping-results-glstring glstring | XML | JSON
hla-genotyping-results-haploid haploid | XML | JSON
hla-genotyping-results-method method | XML | JSON
http-response-header http-response-header | XML | JSON
11179-de-administrative-status administrative-status | XML | JSON
11179-de-change-description change-description | XML | JSON
11179-de-classification-or-context classification-or-context | XML | JSON
11179-de-contact-address contact-address | XML | JSON
11179-de-document-reference document-reference | XML | JSON
11179-de-effective-period effective-period | XML | JSON
11179-de-is-data-element-concept is-data-element-concept | XML | JSON
11179-de-registry-org registry-org | XML | JSON
11179-de-submitter-org submitter-org | XML | JSON
11179-objectClass objectClass | XML | JSON
11179-objectClassProperty objectClassProperty | XML | JSON
11179-permitted-value-conceptmap permitted-value-conceptmap | XML | JSON
11179-permitted-value-valueset permitted-value-valueset | XML | JSON
DiagnosticReport-geneticsAnalysis Analysis | XML | JSON
DiagnosticReport-geneticsAssessedCondition AssessedCondition | XML | JSON
DiagnosticReport-geneticsFamilyMemberHistory FamilyMemberHistory | XML | JSON
allergyintolerance-certainty certainty | XML | JSON
allergyintolerance-duration duration | XML | JSON
allergyintolerance-reasonRefuted reasonRefuted | XML | JSON
allergyintolerance-resolutionAge resolutionAge | XML | JSON
allergyintolerance-substanceExposureRisk substanceExposureRisk | XML | JSON
auditevent-Accession Accession | XML | JSON
auditevent-Anonymized Anonymized | XML | JSON
auditevent-Encrypted Encrypted | XML | JSON
auditevent-Instance Instance | XML | JSON
auditevent-MPPS MPPS | XML | JSON
auditevent-NumberOfInstances NumberOfInstances | XML | JSON
auditevent-ParticipantObjectContainsStudy ParticipantObjectContainsStudy | XML | JSON
auditevent-SOPClass SOPClass | XML | JSON
birthPlace Birth Place | XML | JSON
body-site-instance Body Site Instance | XML | JSON
capabilitystatement-expectation expectation | XML | JSON
capabilitystatement-prohibited prohibited | XML | JSON
capabilitystatement-search-parameter-combination search-parameter-combination | XML | JSON
capabilitystatement-supported-system supported-system | XML | JSON
capabilitystatement-websocket websocket | XML | JSON
careplan-activity-title activity-title | XML | JSON
codesystem-author author | XML | JSON
codesystem-comment comment | XML | JSON
codesystem-comments comments | XML | JSON

codesystem-conceptOrder conceptOrder | XML | JSON
codesystem-deprecated deprecated | XML | JSON
codesystem-effectiveDate effectiveDate | XML | JSON
codesystem-expirationDate expirationDate | XML | JSON
codesystem-history history | XML | JSON
codesystem-keyWord keyWord | XML | JSON
codesystem-label label | XML | JSON
codesystem-map map | XML | JSON
codesystem-ordinalValue ordinalValue | XML | JSON
codesystem-otherName otherName | XML | JSON
codesystem-reference reference | XML | JSON
codesystem-replacedby replacedby | XML | JSON
codesystem-sourceReference sourceReference | XML | JSON
codesystem-subsumes subsumes | XML | JSON
codesystem-trusted-expansion trusted-expansion | XML | JSON
codesystem-usage usage | XML | JSON
codesystem-warning warning | XML | JSON
codesystem-workflowStatus workflowStatus | XML | JSON
coding-sctdescid sctdescid | XML | JSON
communication-media media | XML | JSON
communication-reasonNotPerformed reasonNotPerformed | XML | JSON
communicationrequest-definition definition | XML | JSON
communicationrequest-orderedBy orderedBy | XML | JSON
communicationrequest-reasonRejected reasonRejected | XML | JSON
communicationrequest-relevantHistory relevantHistory | XML | JSON
communicationrequest-supportingInfo supportingInfo | XML | JSON
composition-clindoc-otherConfidentiality otherConfidentiality | XML | JSON
concept-bidirectional bidirectional | XML | JSON
condition-basedOn basedOn | XML | JSON
condition-criticality criticality | XML | JSON
condition-definition definition | XML | JSON
condition-dueTo dueTo | XML | JSON
condition-occurredFollowing occurredFollowing | XML | JSON
condition-outcome outcome | XML | JSON
condition-partOf partOf | XML | JSON
condition-ruledOut ruledOut | XML | JSON
condition-targetBodySite targetBodySite | XML | JSON
consent-NotificationEndpoint NotificationEndpoint | XML | JSON
consent-Witness Witness | XML | JSON
consent-location location | XML | JSON
data-absent-reason Data Absent Reason | XML | JSON
datadictionary datadictionary | XML | JSON
device-din din | XML | JSON
device-implant-status implant-status | XML | JSON
devicerequest-patientInstruction patientInstruction | XML | JSON
devicerequest-reasonRejected reasonRejected | XML | JSON
diagnosticReport-addendumOf addendumOf | XML | JSON
diagnosticReport-extends extends | XML | JSON
diagnosticReport-locationPerformed locationPerformed | XML | JSON
diagnosticReport-replaces replaces | XML | JSON
diagnosticReport-summaryOf summaryOf | XML | JSON
elementdefinition-allowedUnits allowedUnits | XML | JSON
elementdefinition-bestpractice bestpractice | XML | JSON
elementdefinition-bindingName bindingName | XML | JSON
elementdefinition-equivalence equivalence | XML | JSON
elementdefinition-identifier identifier | XML | JSON
elementdefinition-inheritedExtensibleValueSet inheritedExtensibleValueSet | XML | JSON
elementdefinition-isCommonBinding isCommonBinding | XML | JSON
elementdefinition-maxValueSet maxValueSet | XML | JSON
elementdefinition-minValueSet minValueSet | XML | JSON
elementdefinition-namespace namespace | XML | JSON
elementdefinition-question question | XML | JSON
elementdefinition-selector selector | XML | JSON
elementdefinition-translatable translatable | XML | JSON
encounter-associatedEncounter associatedEncounter | XML | JSON
encounter-modeOfArrival modeOfArrival | XML | JSON
encounter-primaryDiagnosis primaryDiagnosis | XML | JSON
encounter-reasonCancelled reasonCancelled | XML | JSON
entryFormat entryFormat | XML | JSON
event-OnBehalfOf OnBehalfOf | XML | JSON
event-definition definition | XML | JSON
event-notDone notDone | XML | JSON

event-partOf partOf | XML | JSON
event-performerRole performerRole | XML | JSON
event-reasonCode reasonCode | XML | JSON
event-reasonReference reasonReference | XML | JSON
family-member-history-genetics-observation observation | XML | JSON
family-member-history-genetics-parent parent | XML | JSON
family-member-history-genetics-sibling sibling | XML | JSON
familymemberhistory-abatement abatement | XML | JSON
familymemberhistory-patient-record patient-record | XML | JSON
familymemberhistory-severity severity | XML | JSON
familymemberhistory-type type | XML | JSON
flag-detail detail | XML | JSON
flag-priority priority | XML | JSON
geolocation Geolocation | XML | JSON
goal-acceptance acceptance | XML | JSON
goal-pertainsToGoal pertainsToGoal | XML | JSON
goal-reasonRejected reasonRejected | XML | JSON
goal-relationship relationship | XML | JSON
hla-genotyping-results-allele-database allele-database | XML | JSON
hla-genotyping-results-glstring glstring | XML | JSON
hla-genotyping-results-haploid haploid | XML | JSON
hla-genotyping-results-method method | XML | JSON
http-response-header http-response-header | XML | JSON
humanname-assembly-order assembly-order | XML | JSON
humanname-fathers-family fathers-family | XML | JSON
humanname-mothers-family mothers-family | XML | JSON
humanname-own-name own-name | XML | JSON
humanname-own-prefix own-prefix | XML | JSON
humanname-partner-name partner-name | XML | JSON
humanname-partner-prefix partner-prefix | XML | JSON
identifier-validDate validDate | XML | JSON
implementationguide-page page | XML | JSON
iso21090-AD-use AD-use | XML | JSON
iso21090-ADXP-additionalLocator ADXP-additionalLocator | XML | JSON
iso21090-ADXP-buildingNumberSuffix ADXP-buildingNumberSuffix | XML | JSON
iso21090-ADXP-careOf ADXP-careOf | XML | JSON
iso21090-ADXP-censusTract ADXP-censusTract | XML | JSON
iso21090-ADXP-delimiter ADXP-delimiter | XML | JSON
iso21090-ADXP-deliveryAddressLine ADXP-deliveryAddressLine | XML | JSON
iso21090-ADXP-deliveryInstallationArea ADXP-deliveryInstallationArea | XML | JSON
iso21090-ADXP-deliveryInstallationQualifier ADXP-deliveryInstallationQualifier | XML | JSON
iso21090-ADXP-deliveryInstallationType ADXP-deliveryInstallationType | XML | JSON
iso21090-ADXP-deliveryMode ADXP-deliveryMode | XML | JSON
iso21090-ADXP-deliveryModeIdentifier ADXP-deliveryModeIdentifier | XML | JSON
iso21090-ADXP-direction ADXP-direction | XML | JSON
iso21090-ADXP-houseNumber ADXP-houseNumber | XML | JSON
iso21090-ADXP-houseNumberNumeric ADXP-houseNumberNumeric | XML | JSON
iso21090-ADXP-postBox ADXP-postBox | XML | JSON
iso21090-ADXP-precinct ADXP-precinct | XML | JSON
iso21090-ADXP-streetAddressLine ADXP-streetAddressLine | XML | JSON
iso21090-ADXP-streetName ADXP-streetName | XML | JSON
iso21090-ADXP-streetNameBase ADXP-streetNameBase | XML | JSON
iso21090-ADXP-streetNameType ADXP-streetNameType | XML | JSON
iso21090-ADXP-unitID ADXP-unitID | XML | JSON
iso21090-ADXP-unitType ADXP-unitType | XML | JSON
iso21090-CO-value CO-value | XML | JSON
iso21090-EN-qualifier EN-qualifier | XML | JSON
iso21090-EN-representation EN-representation | XML | JSON
iso21090-SC-coding SC-coding | XML | JSON
iso21090-TEL-address TEL-address | XML | JSON
iso21090-nullFlavor nullFlavor | XML | JSON
iso21090-preferred preferred | XML | JSON
iso21090-uncertainty uncertainty | XML | JSON
iso21090-uncertaintyType uncertaintyType | XML | JSON
iso21090-verification verification | XML | JSON
location-alias alias | XML | JSON
location-distance location-distance | XML | JSON
mapSourcePublisher mapSourcePublisher | XML | JSON
match-grade match-grade | XML | JSON
maxDecimalPlaces maxDecimalPlaces | XML | JSON
maxSize maxSize | XML | JSON
maxValue maxValue | XML | JSON

medication-usualRoute usualRoute | XML | JSON
medicationdispense-validityPeriod validityPeriod | XML | JSON
medicationstatement-Prescriber Prescriber | XML | JSON
messageheader-response-request messageheader-response-request | XML | JSON
mimeType mimeType | XML | JSON
minLength minLength | XML | JSON
minValue minValue | XML | JSON
observation-bodyPosition bodyPosition | XML | JSON
observation-delta delta | XML | JSON
observation-eventTiming eventTiming | XML | JSON
observation-focal-subject focal-subject | XML | JSON
observation-geneticsAlleleName AlleleName | XML | JSON
observation-geneticsAllelicFrequency AllelicFrequency | XML | JSON
observation-geneticsAllelicState AllelicState | XML | JSON
observation-geneticsAminoAcidChangeName AminoAcidChangeName | XML | JSON
observation-geneticsAminoAcidChangeType AminoAcidChangeType | XML | JSON
observation-geneticsCopyNumberEvent CopyNumberEvent | XML | JSON
observation-geneticsDNARegionName DNARegionName | XML | JSON
observation-geneticsDNASequenceVariantName DNASequenceVariantName | XML | JSON
observation-geneticsDNASequenceVariantType DNASequenceVariantType | XML | JSON
observation-geneticsDNAVariantId DNAVariantId | XML | JSON
observation-geneticsGene Gene | XML | JSON
observation-geneticsGenomicSourceClass GenomicSourceClass | XML | JSON
observation-geneticsInterpretation Interpretation | XML | JSON
observation-geneticsPhaseSet PhaseSet | XML | JSON
observation-geneticsSequence Sequence | XML | JSON
observation-time-offset time-offset | XML | JSON
operationoutcome-authority authority | XML | JSON
operationoutcome-detectedIssue detectedIssue | XML | JSON
operationoutcome-issue-source issue-source | XML | JSON
organization-alias alias | XML | JSON
organization-period period | XML | JSON
organization-preferredContact preferredContact | XML | JSON
patient-adoptionInfo adoptionInfo | XML | JSON
patient-birthTime birthTime | XML | JSON
patient-cadavericDonor cadavericDonor | XML | JSON
patient-citizenship citizenship | XML | JSON
patient-clinicalTrial clinicalTrial | XML | JSON
patient-congregation congregation | XML | JSON
patient-disability disability | XML | JSON
patient-importance importance | XML | JSON
patient-interpreterRequired interpreterRequired | XML | JSON
patient-mothersMaidenName mothersMaidenName | XML | JSON
patient-nationality nationality | XML | JSON
patient-religion religion | XML | JSON
pharmacy-core-doseType doseType | XML | JSON
pharmacy-core-infuseOver infuseOver | XML | JSON
pharmacy-core-maxDeliveryRate maxDeliveryRate | XML | JSON
pharmacy-core-maxDeliveryVolume maxDeliveryVolume | XML | JSON
pharmacy-core-minDosePerPeriod minDosePerPeriod | XML | JSON
pharmacy-core-rateGoal rateGoal | XML | JSON
pharmacy-core-rateIncrement rateIncrement | XML | JSON
pharmacy-core-rateIncrementInterval rateIncrementInterval | XML | JSON
pharmacy-core-refillsRemaining refillsRemaining | XML | JSON
practitioner-animalSpecies animalSpecies | XML | JSON
practitioner-classification classification | XML | JSON
practitionerrole-primaryInd primaryInd | XML | JSON
procedure-approachBodySite approachBodySite | XML | JSON
procedure-causedBy causedBy | XML | JSON
procedure-incisionDateTime incisionDateTime | XML | JSON
procedure-method method | XML | JSON
procedure-progressStatus progressStatus | XML | JSON
procedure-schedule schedule | XML | JSON
procedure-targetBodySite targetBodySite | XML | JSON
procedurerequest-approachBodySite approachBodySite | XML | JSON
procedurerequest-authorizedBy authorizedBy | XML | JSON
procedurerequest-geneticsItem Item | XML | JSON
procedurerequest-precondition precondition | XML | JSON
procedurerequest-questionnaireRequest questionnaireRequest | XML | JSON
procedurerequest-reasonRefused reasonRefused | XML | JSON
procedurerequest-reasonRejected reasonRejected | XML | JSON
procedurerequest-targetBodySite targetBodySite | XML | JSON

questionnaire-allowedProfile allowedProfile | XML | JSON
questionnaire-allowedResource allowedResource | XML | JSON
questionnaire-baseType baseType | XML | JSON
questionnaire-choiceOrientation choiceOrientation | XML | JSON
questionnaire-deMap deMap | XML | JSON
questionnaire-displayCategory displayCategory | XML | JSON
questionnaire-fhirType fhirType | XML | JSON
questionnaire-hidden hidden | XML | JSON
questionnaire-itemControl itemControl | XML | JSON
questionnaire-lookupQuestionnaire lookupQuestionnaire | XML | JSON
questionnaire-maxOccurs maxOccurs | XML | JSON
questionnaire-minOccurs minOccurs | XML | JSON
questionnaire-optionExclusive optionExclusive | XML | JSON
questionnaire-optionPrefix optionPrefix | XML | JSON
questionnaire-ordinalValue ordinalValue | XML | JSON
questionnaire-referenceFilter referenceFilter | XML | JSON
questionnaire-sourceStructureMap sourceStructureMap | XML | JSON
questionnaire-studyprotocolIdentifier studyprotocolIdentifier | XML | JSON
questionnaire-supportLink supportLink | XML | JSON
questionnaire-targetStructureMap targetStructureMap | XML | JSON
questionnaire-unit unit | XML | JSON
questionnaire-usageMode usageMode | XML | JSON
questionnaireresponse-author author | XML | JSON
questionnaireresponse-note note | XML | JSON
questionnaireresponse-reason reason | XML | JSON
questionnaireresponse-reviewer reviewer | XML | JSON
referralrequest-reasonRefused reasonRefused | XML | JSON
regex regex | XML | JSON
rendered-value Rendered Value | XML | JSON
rendering-markdown markdown | XML | JSON
rendering-style style | XML | JSON
rendering-styleSensitive styleSensitive | XML | JSON
rendering-xhtml xhtml | XML | JSON
resource-approvalDate approvalDate | XML | JSON
resource-effectivePeriod effectivePeriod | XML | JSON
resource-lastReviewDate lastReviewDate | XML | JSON
specimen-collectionPriority collectionPriority | XML | JSON
specimen-isDryWeight isDryWeight | XML | JSON
specimen-processingTime processingTime | XML | JSON
specimen-sequenceNumber sequenceNumber | XML | JSON
specimen-specialHandling specialHandling | XML | JSON
structuredefinition-ancestor ancestor | XML | JSON
structuredefinition-annotation annotation | XML | JSON
structuredefinition-ballot-status ballot-status | XML | JSON
structuredefinition-category category | XML | JSON
structuredefinition-display-hint display-hint | XML | JSON
structuredefinition-explicit-type-name explicit-type-name | XML | JSON
structuredefinition-fmm fmm | XML | JSON
structuredefinition-fmm-no-warnings fmm-no-warnings | XML | JSON
structuredefinition-json-type json-type | XML | JSON
structuredefinition-rdf-type rdf-type | XML | JSON
structuredefinition-regex regex | XML | JSON
structuredefinition-summary summary | XML | JSON
structuredefinition-table-name table-name | XML | JSON
structuredefinition-template-status template-status | XML | JSON
structuredefinition-wg wg | XML | JSON
structuredefinition-xml-type xml-type | XML | JSON
task-candidateList candidateList | XML | JSON
task-replaces replaces | XML | JSON
timing-daysOfCycle daysOfCycle | XML | JSON
timing-exact exact | XML | JSON
translation Translation | XML | JSON
usagecontext-group group | XML | JSON
valueset-author author | XML | JSON
valueset-caseSensitive caseSensitive | XML | JSON
valueset-comment comment | XML | JSON
valueset-comments comments | XML | JSON
valueset-conceptOrder conceptOrder | XML | JSON
valueset-definition definition | XML | JSON
valueset-effectiveDate effectiveDate | XML | JSON
valueset-expansionSource expansionSource | XML | JSON
valueset-expirationDate expirationDate | XML | JSON

valueset-history history | XML | JSON
valueset-keyWord keyWord | XML | JSON
valueset-label label | XML | JSON
valueset-map map | XML | JSON
valueset-ordinalValue ordinalValue | XML | JSON
valueset-otherName otherName | XML | JSON
valueset-reference reference | XML | JSON
valueset-sourceReference sourceReference | XML | JSON
valueset-system system | XML | JSON
valueset-systemName systemName | XML | JSON
valueset-systemRef systemRef | XML | JSON
valueset-toocostly toocostly | XML | JSON
valueset-trusted-expansion trusted-expansion | XML | JSON
valueset-unclosed unclosed | XML | JSON
valueset-usage usage | XML | JSON
valueset-warning warning | XML | JSON
valueset-workflowStatus workflowStatus | XML | JSON
humanname-assembly-order assembly-order | XML | JSON
humanname-fathers-family fathers-family | XML | JSON
humanname-mothers-family mothers-family | XML | JSON
humanname-own-name own-name | XML | JSON
humanname-own-prefix own-prefix | XML | JSON
humanname-partner-name partner-name | XML | JSON
humanname-partner-prefix partner-prefix | XML | JSON
identifier-validDate validDate | XML | JSON
implementationguide-page page | XML | JSON
iso21090-AD-use AD-use | XML | JSON
iso21090-ADXP-additionalLocator ADXP-additionalLocator | XML | JSON
iso21090-ADXP-buildingNumberSuffix ADXP-buildingNumberSuffix | XML | JSON
iso21090-ADXP-careOf ADXP-careOf | XML | JSON
iso21090-ADXP-censusTract ADXP-censusTract | XML | JSON
iso21090-ADXP-delimiter ADXP-delimiter | XML | JSON
iso21090-ADXP-deliveryAddressLine ADXP-deliveryAddressLine | XML | JSON
iso21090-ADXP-deliveryInstallationArea ADXP-deliveryInstallationArea | XML | JSON
iso21090-ADXP-deliveryInstallationQualifier ADXP-deliveryInstallationQualifier | XML | JSON
iso21090-ADXP-deliveryInstallationType ADXP-deliveryInstallationType | XML | JSON
iso21090-ADXP-deliveryMode ADXP-deliveryMode | XML | JSON
iso21090-ADXP-deliveryModeIdentifier ADXP-deliveryModeIdentifier | XML | JSON
iso21090-ADXP-direction ADXP-direction | XML | JSON
iso21090-ADXP-houseNumber ADXP-houseNumber | XML | JSON
iso21090-ADXP-houseNumberNumeric ADXP-houseNumberNumeric | XML | JSON
iso21090-ADXP-postBox ADXP-postBox | XML | JSON
iso21090-ADXP-precinct ADXP-precinct | XML | JSON
iso21090-ADXP-streetAddressLine ADXP-streetAddressLine | XML | JSON
iso21090-ADXP-streetName ADXP-streetName | XML | JSON
iso21090-ADXP-streetNameBase ADXP-streetNameBase | XML | JSON
iso21090-ADXP-streetNameType ADXP-streetNameType | XML | JSON
iso21090-ADXP-unitID ADXP-unitID | XML | JSON
iso21090-ADXP-unitType ADXP-unitType | XML | JSON
iso21090-CO-value CO-value | XML | JSON
iso21090-EN-qualifier EN-qualifier | XML | JSON
iso21090-EN-representation EN-representation | XML | JSON
iso21090-SC-coding SC-coding | XML | JSON
iso21090-TEL-address TEL-address | XML | JSON
iso21090-nullFlavor nullFlavor | XML | JSON
iso21090-preferred preferred | XML | JSON
iso21090-uncertainty uncertainty | XML | JSON
iso21090-uncertaintyType uncertaintyType | XML | JSON
iso21090-verification verification | XML | JSON
location-alias alias | XML | JSON
location-distance location-distance | XML | JSON
mapSourcePublisher mapSourcePublisher | XML | JSON
match-grade match-grade | XML | JSON
maxDecimalPlaces maxDecimalPlaces | XML | JSON
maxSize maxSize | XML | JSON
maxValue maxValue | XML | JSON
medication-usualRoute usualRoute | XML | JSON
medicationdispense-validityPeriod validityPeriod | XML | JSON
medicationstatement-Prescriber Prescriber | XML | JSON
messageheader-response-request messageheader-response-request | XML | JSON
mimeType mimeType | XML | JSON
minLength minLength | XML | JSON

minValue minValue | XML | JSON
observation-bodyPosition bodyPosition | XML | JSON
observation-delta delta | XML | JSON
observation-eventTiming eventTiming | XML | JSON
observation-focal-subject focal-subject | XML | JSON
observation-geneticsAlleleName AlleleName | XML | JSON
observation-geneticsAllelicFrequency AllelicFrequency | XML | JSON
observation-geneticsAllelicState AllelicState | XML | JSON
observation-geneticsAminoAcidChangeName AminoAcidChangeName | XML | JSON
observation-geneticsAminoAcidChangeType AminoAcidChangeType | XML | JSON
observation-geneticsCopyNumberEvent CopyNumberEvent | XML | JSON
observation-geneticsDNARegionName DNARegionName | XML | JSON
observation-geneticsDNASequenceVariantName DNASequenceVariantName | XML | JSON
observation-geneticsDNASequenceVariantType DNASequenceVariantType | XML | JSON
observation-geneticsDNAVariantId DNAVariantId | XML | JSON
observation-geneticsGene Gene | XML | JSON
observation-geneticsGenomicSourceClass GenomicSourceClass | XML | JSON
observation-geneticsInterpretation Interpretation | XML | JSON
observation-geneticsPhaseSet PhaseSet | XML | JSON
observation-geneticsSequence Sequence | XML | JSON
observation-time-offset time-offset | XML | JSON
operationoutcome-authority authority | XML | JSON
operationoutcome-detectedIssue detectedIssue | XML | JSON
operationoutcome-issue-source issue-source | XML | JSON
organization-alias alias | XML | JSON
organization-period period | XML | JSON
organization-preferredContact preferredContact | XML | JSON
patient-adoptionInfo adoptionInfo | XML | JSON
patient-birthTime birthTime | XML | JSON
patient-cadavericDonor cadavericDonor | XML | JSON
patient-citizenship citizenship | XML | JSON
patient-clinicalTrial clinicalTrial | XML | JSON
patient-congregation congregation | XML | JSON
patient-disability disability | XML | JSON
patient-importance importance | XML | JSON
patient-interpreterRequired interpreterRequired | XML | JSON
patient-mothersMaidenName mothersMaidenName | XML | JSON
patient-nationality nationality | XML | JSON
patient-religion religion | XML | JSON
pharmacy-core-doseType doseType | XML | JSON
pharmacy-core-infuseOver infuseOver | XML | JSON
pharmacy-core-maxDeliveryRate maxDeliveryRate | XML | JSON
pharmacy-core-maxDeliveryVolume maxDeliveryVolume | XML | JSON
pharmacy-core-minDosePerPeriod minDosePerPeriod | XML | JSON
pharmacy-core-rateGoal rateGoal | XML | JSON
pharmacy-core-rateIncrement rateIncrement | XML | JSON
pharmacy-core-rateIncrementInterval rateIncrementInterval | XML | JSON
pharmacy-core-refillsRemaining refillsRemaining | XML | JSON
practitioner-animalSpecies animalSpecies | XML | JSON
practitioner-classification classification | XML | JSON
practitionerrole-primaryInd primaryInd | XML | JSON
procedure-approachBodySite approachBodySite | XML | JSON
procedure-causedBy causedBy | XML | JSON
procedure-incisionDateTime incisionDateTime | XML | JSON
procedure-method method | XML | JSON
procedure-progressStatus progressStatus | XML | JSON
procedure-schedule schedule | XML | JSON
procedure-targetBodySite targetBodySite | XML | JSON
procedurerequest-approachBodySite approachBodySite | XML | JSON
procedurerequest-authorizedBy authorizedBy | XML | JSON
procedurerequest-geneticsItem Item | XML | JSON
procedurerequest-precondition precondition | XML | JSON
procedurerequest-questionnaireRequest questionnaireRequest | XML | JSON
procedurerequest-reasonRefused reasonRefused | XML | JSON
procedurerequest-reasonRejected reasonRejected | XML | JSON
procedurerequest-targetBodySite targetBodySite | XML | JSON
questionnaire-allowedProfile allowedProfile | XML | JSON
questionnaire-allowedResource allowedResource | XML | JSON
questionnaire-baseType baseType | XML | JSON
questionnaire-choiceOrientation choiceOrientation | XML | JSON
questionnaire-deMap deMap | XML | JSON
questionnaire-displayCategory displayCategory | XML | JSON

questionnaire-fhirType fhirType | XML | JSON
questionnaire-hidden hidden | XML | JSON
questionnaire-itemControl itemControl | XML | JSON
questionnaire-lookupQuestionnaire lookupQuestionnaire | XML | JSON
questionnaire-maxOccurs maxOccurs | XML | JSON
questionnaire-minOccurs minOccurs | XML | JSON
questionnaire-optionExclusive optionExclusive | XML | JSON
questionnaire-optionPrefix optionPrefix | XML | JSON
questionnaire-ordinalValue ordinalValue | XML | JSON
questionnaire-referenceFilter referenceFilter | XML | JSON
questionnaire-sourceStructureMap sourceStructureMap | XML | JSON
questionnaire-studyprotocolIdentifier studyprotocolIdentifier | XML | JSON
questionnaire-supportLink supportLink | XML | JSON
questionnaire-targetStructureMap targetStructureMap | XML | JSON
questionnaire-unit unit | XML | JSON
questionnaire-usageMode usageMode | XML | JSON
questionnaireresponse-author author | XML | JSON
questionnaireresponse-note note | XML | JSON
questionnaireresponse-reason reason | XML | JSON
questionnaireresponse-reviewer reviewer | XML | JSON
referralrequest-reasonRefused reasonRefused | XML | JSON
regex regex | XML | JSON
rendered-value Rendered Value | XML | JSON
rendering-markdown markdown | XML | JSON
rendering-style style | XML | JSON
rendering-styleSensitive styleSensitive | XML | JSON
rendering-xhtml xhtml | XML | JSON
resource-approvalDate approvalDate | XML | JSON
resource-effectivePeriod effectivePeriod | XML | JSON
resource-lastReviewDate lastReviewDate | XML | JSON
specimen-collectionPriority collectionPriority | XML | JSON
specimen-isDryWeight isDryWeight | XML | JSON
specimen-processingTime processingTime | XML | JSON
specimen-sequenceNumber sequenceNumber | XML | JSON
specimen-specialHandling specialHandling | XML | JSON
structuredefinition-ancestor ancestor | XML | JSON
structuredefinition-annotation annotation | XML | JSON
structuredefinition-ballot-status ballot-status | XML | JSON
structuredefinition-category category | XML | JSON
structuredefinition-display-hint display-hint | XML | JSON
structuredefinition-explicit-type-name explicit-type-name | XML | JSON
structuredefinition-fmm fmm | XML | JSON
structuredefinition-fmm-no-warnings fmm-no-warnings | XML | JSON
structuredefinition-json-type json-type | XML | JSON
structuredefinition-rdf-type rdf-type | XML | JSON
structuredefinition-regex regex | XML | JSON
structuredefinition-summary summary | XML | JSON
structuredefinition-table-name table-name | XML | JSON
structuredefinition-template-status template-status | XML | JSON
structuredefinition-wg wg | XML | JSON
structuredefinition-xml-type xml-type | XML | JSON
task-candidateList candidateList | XML | JSON
task-replaces replaces | XML | JSON
timing-daysOfCycle daysOfCycle | XML | JSON
timing-exact exact | XML | JSON
translation Translation | XML | JSON
usagecontext-group group | XML | JSON
valueset-author author | XML | JSON
valueset-caseSensitive caseSensitive | XML | JSON
valueset-comment comment | XML | JSON
valueset-comments comments | XML | JSON
valueset-conceptOrder conceptOrder | XML | JSON
valueset-definition definition | XML | JSON
valueset-effectiveDate effectiveDate | XML | JSON
valueset-expansionSource expansionSource | XML | JSON
valueset-expirationDate expirationDate | XML | JSON
valueset-history history | XML | JSON
valueset-keyWord keyWord | XML | JSON
valueset-label label | XML | JSON
valueset-map map | XML | JSON
valueset-ordinalValue ordinalValue | XML | JSON
valueset-otherName otherName | XML | JSON

valueset-reference reference | XML | JSON
valueset-sourceReference sourceReference | XML | JSON
valueset-system system | XML | JSON
valueset-systemName systemName | XML | JSON
valueset-systemRef systemRef | XML | JSON
valueset-toocostly toocostly | XML | JSON
valueset-trusted-expansion trusted-expansion | XML | JSON
valueset-unclosed unclosed | XML | JSON
valueset-usage usage | XML | JSON
valueset-warning warning | XML | JSON
valueset-workflowStatus workflowStatus | XML | JSON
openEHR Profiles
openEHR-administration administration | XML | JSON
openEHR-careplan careplan | XML | JSON
openEHR-exposureDate exposureDate | XML | JSON
openEHR-exposureDescription exposureDescription | XML | JSON
openEHR-exposureDuration exposureDuration | XML | JSON
openEHR-location location | XML | JSON
openEHR-management management | XML | JSON
openEHR-test test | XML | JSON
openEHR-administration administration | XML | JSON
openEHR-careplan careplan | XML | JSON
openEHR-exposureDate exposureDate | XML | JSON
openEHR-exposureDescription exposureDescription | XML | JSON
openEHR-exposureDuration exposureDuration | XML | JSON
openEHR-location location | XML | JSON
openEHR-management management | XML | JSON
openEHR-test test | XML | JSON
Clinical Quality Improvement Framework
cqif-calculatedValue calculatedValue | XML | JSON
cqif-citation citation | XML | JSON
cqif-condition condition | XML | JSON
cqif-cqlExpression cqlExpression | XML | JSON
cqif-fhirPathExpression fhirPathExpression | XML | JSON
cqif-guidanceencounterClass encounterClass | XML | JSON
cqif-guidanceencounterType encounterType | XML | JSON
cqif-guidanceinitiatingOrganization initiatingOrganization | XML | JSON
cqif-guidanceinitiatingPerson initiatingPerson | XML | JSON
cqif-guidancereceivingOrganization receivingOrganization | XML | JSON
cqif-guidancereceivingPerson receivingPerson | XML | JSON
cqif-guidancerecipientLanguage recipientLanguage | XML | JSON
cqif-guidancerecipientType recipientType | XML | JSON
cqif-guidancesystemUserLanguage systemUserLanguage | XML | JSON
cqif-guidancesystemUserTaskContext systemUserTaskContext | XML | JSON
cqif-guidancesystemUserType systemUserType | XML | JSON
cqif-initialValue initialValue | XML | JSON
cqif-library library | XML | JSON
cqif-measureInfo measureInfo | XML | JSON
cqif-optionCode optionCode | XML | JSON
cqif-qualityOfEvidence qualityOfEvidence | XML | JSON
cqif-sourceValueSet sourceValueSet | XML | JSON
cqif-strengthOfRecommendation strengthOfRecommendation | XML | JSON
cqif-guidanceencounterType encounterType | XML | JSON
cqif-calculatedValue calculatedValue | XML | JSON
cqif-citation citation | XML | JSON
cqif-condition condition | XML | JSON
cqif-cqlExpression cqlExpression | XML | JSON
cqif-fhirPathExpression fhirPathExpression | XML | JSON
cqif-guidanceencounterClass encounterClass | XML | JSON
cqif-guidanceencounterType encounterType | XML | JSON
cqif-guidanceinitiatingOrganization initiatingOrganization | XML | JSON
cqif-guidanceinitiatingPerson initiatingPerson | XML | JSON
cqif-guidancereceivingOrganization receivingOrganization | XML | JSON
cqif-guidancereceivingPerson receivingPerson | XML | JSON
cqif-guidancerecipientLanguage recipientLanguage | XML | JSON
cqif-guidancerecipientType recipientType | XML | JSON
cqif-guidancesystemUserLanguage systemUserLanguage | XML | JSON
cqif-guidancesystemUserTaskContext systemUserTaskContext | XML | JSON
cqif-guidancesystemUserType systemUserType | XML | JSON
cqif-initialValue initialValue | XML | JSON
cqif-library library | XML | JSON
cqif-measureInfo measureInfo | XML | JSON

cqif-optionCode optionCode | XML | JSON
cqif-qualityOfEvidence qualityOfEvidence | XML | JSON
cqif-sourceValueSet sourceValueSet | XML | JSON
cqif-strengthOfRecommendation strengthOfRecommendation | XML | JSON

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.3.8 Resource StructureDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/structuredefinition-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.3.8 Resource StructureDefinition - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 5 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the StructureDefinition resource.

StructureDefinition

Definition | A definition of a FHIR structure. This resource is used to describe the underlying resources, data types defined in FHIR, and also for describing extensions and constraints on resources and data types.

Control | 1..1

Alternate Names | Template; Profile

Invariants | Defined on this element sdf-1: Element paths must be unique unless the structure is a constraint (expression : derivation = 'constraint' or snapshot.element.select(path).isDistinct(), xpath: (f:derivation/@value = 'constraint') or (count(f:snapshot/f:element) = count(distinct-values(f:snapshot/f:element/f:path/@value)))) sdf-11: If there's a type, its content must match the path name in the first element of a snapshot (expression : snapshot.empty() or snapshot.element.first().path = type, xpath: not(exists(f:snapshot)) or (f:type/@value = f:snapshot/f:element[1]/f:path/@value)) sdf-14: All element definitions must have an id (expression : snapshot.element.all(id.exists()) and differential.element.all(id.exists()), xpath: count(*/f:element)=count(*/f:element/@id)) sdf-16: All element definitions must have unique ids (snapshot) (expression : snapshot.element.id.trace('ids').isDistinct(), xpath: count(*/f:element)=count(*/f:element/@id)) sdf-17: All element definitions must have unique ids (diff) (expression : differential.element.id.trace('ids').isDistinct(), xpath: count(*/f:element)=count(*/f:element/@id)) sdf-18: Context Invariants can only be used for extensions (expression : contextInvariant.exists() implies type = 'Extension', xpath: not(exists(f:contextInvariant)) or (f:type/@value = 'Extension')) sdf-19: Custom types can only be used in logical models (expression : kind = 'logical' or (differential.element.type.code.all(hasValue() implies matches('^[a-zA-Z0-9]+$')) and snapshot.element.type.code.all(hasValue() implies matches('^[a-zA-Z0-9]+$'))), xpath: f:kind/@value = 'logical' or count(f:differential/f:element/f:type/f:code[@value and not(matches(string(@value), '^[a-zA-Z0-9]+$'))]|f:snapshot/f:element/f:type/f:code[@value and not(matches(string(@value), '^[a-zA-Z0-9]+$'))]) =0) sdf-4: If the structure is not abstract, then there SHALL be a baseDefinition (expression : abstract = true or baseDefinition.exists(), xpath: (f:abstract/@value=true()) or exists(f:baseDefinition)) sdf-5: If the structure defines an extension then the structure must have context information (expression : type != 'Extension' or derivation = 'specialization' or (context.exists() and contextType.exists()), xpath: not(f:type/@value = 'extension') or (f:derivation/@value = 'specialization') or (exists(f:context) and exists(f:contextType))) sdf-6: A structure must have either a differential, or a snapshot (or both) (expression : snapshot.exists() or differential.exists(), xpath: exists(f:snapshot) or exists(f:differential)) sdf-7: If the structure describes a base Resource or Type, the URL has to start with "http://hl7.org/fhir/StructureDefinition/" and the tail must match the id (expression : (derivation = 'constraint') or (kind = 'logical') or (url = 'http://hl7.org/fhir/StructureDefinition/'+id), xpath: (f:derivation/@value = 'constraint') or (f:kind/@value = 'logical') or (f:url/@value=concat('http://hl7.org/fhir/StructureDefinition/', f:id/@value))) sdf-9: In any snapshot or differential, no label, code or requirements on the an element without a "." in the path (e.g. the first element) (expression : children().element.where(path.contains('.').not()).label.empty() and children().element.where(path.contains('.').not()).code.empty() and children().element.where(path.contains('.').not()).requirements.empty(), xpath: not(exists(f:snapshot/f:element[not(contains(f:path/@value, '.')) and (f:label or f:code or f:requirements)])) and not(exists(f:differential/f:element[not(contains(f:path/@value, '.')) and (f:label or f:code or f:requirements)]))

StructureDefinition.url

Definition | An absolute URI that is used to identify this structure definition when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this structure definition is (or will be) published. The URL SHOULD include the major version of the structure definition. For more information see Technical and Business Versions.

Control | 1..1

Type | uri

Requirements | Allows the structure definition to be referenced by a single globally unique identifier.

Alternate Names | url; authoritative-url; destination; identity

Summary | true

Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.

StructureDefinition.identifier

Definition | A formal identifier that is used to identify this structure definition when it is represented in other formats, or referenced in a specification, model, design or an instance.

Note | This is a business identifer, not a resource identifier (see discussion)

Control | 0..*

Type | Identifier

Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.

Summary | true

Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, e.g., to identify this structure definition outside of FHIR, where it is not possible to use the logical URI.

StructureDefinition.version

Definition | The identifier that is used to identify this version of the structure definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the structure definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.

Note | This is a business versionId, not a resource version id (see discussion)

Control | 0..1
Type | string
Summary | true
Comments | There may be different structure definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the structure definition with the format [url]|[version].
StructureDefinition.name
Definition | A natural language name identifying the structure definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Control | 1..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.
StructureDefinition.title
Definition | A short, descriptive, user-friendly title for the structure definition.
Control | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc. Applications don't have to use this name, but can always fall back to it. The title also corresponds to the label for the root element.
StructureDefinition.status
Definition | The status of this structure definition. Enables tracking the life-cycle of the content.
Control | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Allows filtering of structure definitions that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired structure definition without due consideration.
StructureDefinition.experimental
Definition | A boolean value to indicate that this structure definition is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.
Control | 0..1
Type | boolean
Is Modifier | true
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level structure definition.
Summary | true
Comments | Allows filtering of structure definition that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental structure definition in production.
StructureDefinition.date
Definition | The date (and optionally time) when the structure definition was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the structure definition changes.
Control | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the structure definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
StructureDefinition.publisher
Definition | The name of the individual or organization that published the structure definition.
Control | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the structure definition. May also allow for contact.
Summary | true
Comments | Usually an organization, but may be an individual. The publisher (or steward) of the structure definition is the organization or individual primarily responsible for the maintenance and upkeep of the structure definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the structure definition. This item SHOULD be populated unless the information is available from context.
StructureDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
StructureDefinition.description
Definition | A free text natural language description of the structure definition from a consumer's perspective.
Control | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the structure definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the structure definition as conveyed in

the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created).
StructureDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate structure definition instances.
Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
StructureDefinition.jurisdiction
Definition | A legal or geographic region in which the structure definition is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the structure definition to be used in jurisdictions other than those for which it was originally designed or intended.
StructureDefinition.purpose
Definition | Explaination of why this structure definition is needed and why it has been designed as it has.
Control | 0..1
Type | markdown
Comments | This element does not describe the usage of the structure definition Instead it provides traceability of "why" the resource is either needed or "why" it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this structure definition.
StructureDefinition.copyright
Definition | A copyright statement relating to the structure definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the structure definition.
Control | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the structure definition and/or its content.
Alternate Names | License; Restrictions
StructureDefinition.keyword
Definition | A set of key words or terms from external terminologies that may be used to assist with indexing and searching of templates.
Control | 0..*
Terminology Binding | Structure Definition Codes (Example)
Type | Coding
Requirements | Assist in searching for appropriate StructureDefinitions.
Summary | true
StructureDefinition.fhirVersion
Definition | The version of the FHIR specification on which this StructureDefinition is based - this is the formal version of the specification, without the revision number, e.g. [publication].[major].[minor], which is 3.0.2 for this version.
Control | 0..1
Type | id
Summary | true
Comments | A StructureDefinition does not need to specify the target it applies to,as StructureDefinitions will often be valid across multiple versions of FHIR. FHIR tooling can determine whether a StructureDefinition is consistent with a particular StructureDefinition if desired.
StructureDefinition.mapping
Definition | An external specification that the content is mapped to.
Control | 0..*
Invariants | Defined on this element sdf-2: Must have at least a name or a uri (or both) (expression : name.exists() or uri.exists(), xpath: exists(f:uri) or exists(f:name))
StructureDefinition.mapping.identity
Definition | An Internal id that is used to identify this mapping set when specific mappings are made.
Control | 1..1
Type | id
Comments | The specification is described once, with general comments, and then specific mappings are made that reference this declaration.
StructureDefinition.mapping.uri
Definition | An absolute URI that identifies the specification that this mapping is expressed to.
Control | 0..1
Type | uri
Comments | A formal identity for the specification being mapped to helps with identifying maps consistently.
Invariants | Affect this element sdf-2: Must have at least a name or a uri (or both) (expression : name.exists() or uri.exists(), xpath: exists(f:uri) or exists(f:name))
StructureDefinition.mapping.name
Definition | A name for the specification that is being mapped to.
Control | 0..1
Type | string
Invariants | Affect this element sdf-2: Must have at least a name or a uri (or both) (expression : name.exists() or uri.exists(), xpath: exists(f:uri) or exists(f:name))
StructureDefinition.mapping.comment
Definition | Comments about this mapping, including version notes, issues, scope limitations, and other important notes for usage.
Control | 0..1
Type | string

StructureDefinition.kind
Definition | Defines the kind of structure that this definition is describing.
Control | 1..1
Terminology Binding | StructureDefinitionKind (Required)
Type | code
Summary | true
StructureDefinition.abstract
Definition | Whether structure this definition describes is abstract or not - that is, whether the structure is not intended to be instantiated. For Resources and Data types, abstract types will never be exchanged between systems.
Control | 1..1
Type | boolean
Summary | true
Comments | Abstract Resources cannot be instantiated - a concrete sub-type must be used. Abstract datatypes and extensions cannot be used in an instance. For logical models, the exact implication of "abstract" will rest with the author, depending how the logical model is used. Flagging a constraint structure as abstract conveys design intent but makes no difference to how the structure definition is handled. Note that inline declared elements that are given the type "Element" in the profile, but have children described, are anonymous concrete types that specialise Element.
StructureDefinition.contextType
Definition | If this is an extension, Identifies the context within FHIR resources where the extension can be used.
Control | 0..1
Terminology Binding | ExtensionContext (Required)
Type | code
Summary | true
Invariants | Affect this element sdf-5: If the structure defines an extension then the structure must have context information (expression : type != 'Extension' or derivation = 'specialization' or (context.exists() and contextType.exists()), xpath: not(f:type/@value = 'extension') or (f:derivation/@value = 'specialization') or (exists(f:context) and exists(f:contextType)))
StructureDefinition.context
Definition | Identifies the types of resource or data type elements to which the extension can be applied.
Control | 0..*
Type | string
Summary | true
Comments | If the context is an element that can have multiple types, then use (e.g.) value[x] if the extension works on all choice types, or otherwise an enumeration of explicitly named elements if not. Note that a context of "string" doesn't mean that the extension can be used with one of the string patterns such as "id" etc.
Invariants | Affect this element sdf-5: If the structure defines an extension then the structure must have context information (expression : type != 'Extension' or derivation = 'specialization' or (context.exists() and contextType.exists()), xpath: not(f:type/@value = 'extension') or (f:derivation/@value = 'specialization') or (exists(f:context) and exists(f:contextType)))
StructureDefinition.contextInvariant
Definition | A set of rules as Fluent Invariants about when the extension can be used (e.g. co-occurrence variants for the extension).
Control | 0..*
Type | string
Summary | true
Comments | All the rules must be true. When evaluating the invariant, the fluentpath focus is the element that holds the extension, and %extension refers to the extension itself.
Invariants | Affect this element sdf-18: Context Invariants can only be used for extensions (expression : contextInvariant.exists() implies type = 'Extension', xpath: not(exists(f:contextInvariant)) or (f:type/@value = 'Extension'))
StructureDefinition.type
Definition | The type this structure describes. If the derivation kind is 'specialization' then this is the master definition for a type, and there is always one of these (a data type, an extension, a resource, including abstract ones). Otherwise the structure definition is a constraint on the stated type (and in this case, the type cannot be an abstract type).
Control | 1..1
Terminology Binding | FHIRDefinedType (Extensible)
Type | code
Summary | true
Comments | Note that in the case of constraints, the type could be determined by chasing through the baseDefinition references until a concrete structure (derivation = specialization) is reached, or by looking at the path of the first element in the snapshot - if present - but providing the type directly makes for simpler tooling and indexing.
Invariants | Affect this element sdf-4: If the structure is not abstract, then there SHALL be a baseDefinition (expression : abstract = true or baseDefinition.exists(), xpath: (f:abstract/@value=true()) or exists(f:baseDefinition)) sdf-11: If there's a type, its content must match the path name in the first element of a snapshot (expression : snapshot.empty() or snapshot.element.first().path = type, xpath: not(exists(f:snapshot)) or (f:type/@value = f:snapshot/f:element[1]/f:path/@value))
StructureDefinition.baseDefinition
Definition | An absolute URI that is the base structure from which this type is derived, either by specialization or constraint.
Control | 0..1
Type | uri
Summary | true
Comments | If differential constraints are specified in this structure, they are applied to the base in a "differential" fashion. If there is no base, then the differential constraints cannot be provided (snapshot only). Differential structures are useful for the editing perspective, and snapshot structures are suitable for operational use. The FHIR Project provides a number of tools/services to populate snapshots from differential constraints. Logical Models have a base of "Element" or another logical model.
Invariants | Affect this element sdf-4: If the structure is not abstract, then there SHALL be a baseDefinition (expression : abstract = true or baseDefinition.exists(), xpath: (f:abstract/@value=true()) or exists(f:baseDefinition))
StructureDefinition.derivation
Definition | How the type relates to the baseDefinition.
Control | 0..1

Terminology Binding | TypeDerivationRule (Required)
Type | code
Summary | true
Comments | If the definition is a specialization, then it adds new elements in the differential, and the snapshot includes the inherited elements. If the definition is a constraint, then it can not define new elements, it can only make new rules about existing content (see Profiling Resources).
StructureDefinition.snapshot
Definition | A snapshot view is expressed in a stand alone form that can be used and interpreted without considering the base StructureDefinition.
Control | 0..1
Invariants | Defined on this element sdf-15: The first element in a snapshot has no type (expression : kind!='logical' implies snapshot.element.first().type.empty(), xpath: not(f:element[1]/f:type)) sdf-3: Each element definition in a snapshot must have a formal definition and cardinalities (expression : element.all(definition and min and max), xpath: count(f:element) = count(f:element[exists(f:definition) and exists(f:min) and exists(f:max)])) sdf-8: In any snapshot, all the elements must be in the specified type (expression : (%resource.kind = 'logical' or element.first().path = %resource.type) and element.tail().all(path.startsWith(%resource.snapshot.element.first().path&.')), xpath: f:element[1]/f:path/@value=parent::f:StructureDefinition/f:type/@value and count(f:element[position()!=1])=count(f:element[position()!=1][starts-with(f:path/@value, concat(ancestor::f:StructureDefinition/f:type/@value, '.'))])) Affect this element sdf-6: A structure must have either a differential, or a snapshot (or both) (expression : snapshot.exists() or differential.exists(), xpath: exists(f:snapshot) or exists(f:differential))
StructureDefinition.snapshot.element
Definition | Captures constraints on each element within the resource.
Control | 1..*
Type | ElementDefinition
Invariants | Affect this element sdf-3: Each element definition in a snapshot must have a formal definition and cardinalities (expression : element.all(definition and min and max), xpath: count(f:element) = count(f:element[exists(f:definition) and exists(f:min) and exists(f:max)]))
StructureDefinition.differential
Definition | A differential view is expressed relative to the base StructureDefinition - a statement of differences that it applies.
Control | 0..1
Invariants | Defined on this element sdf-15a: If the first element in a differential has no "." in the path, it has no type (expression : element.first().path.contains('.').not() implies element.first().type.empty(), xpath: not(f:element[1][not(contains(f:path/@value, '.'))]/f:type)) sdf-20: No slicing on the root element (expression : element.where(path.contains('.').not()).slicing.empty(), xpath: not(f:element[1]/f:slicing)) sdf-8a: In any differential, all the elements must be in the specified type (expression : element.first().path.startsWith(%resource.type) and element.tail().all(path.startsWith(%resource.type&.')), xpath: count(f:element)=count(f:element[f:path/@value=ancestor::f:StructureDefinition/f:type/@value or starts-with(f:path/@value, concat(ancestor::f:StructureDefinition/f:type/@value, '.'))])) Affect this element sdf-6: A structure must have either a differential, or a snapshot (or both) (expression : snapshot.exists() or differential.exists(), xpath: exists(f:snapshot) or exists(f:differential))
StructureDefinition.differential.element
Definition | Captures constraints on each element within the resource.
Control | 1..*
Type | ElementDefinition

5.3.9 Resource StructureDefinition - Mappings

URL: https://hl7.org/fhir/STU3/structuredefinition-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.3.9 Resource StructureDefinition - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

ISO 11179

Mappings for the structuredefinition resource.

5.3.9.1 Mappings for RIM Mapping (http://hl7.org/v3)

StructureDefinition | n/a

5.3.9.2 Mappings for ISO 11179 (http://metadata-standards.org/11179/)

These mappings are included to indicate where properties of the data elements defined by the FHIR specification relate to similar fields in the ISO 11179 specification.

StructureDefinition
url
identifier | Identifier part
version | Version
name
title
status | Registration status
experimental | Registration status
date
publisher | Registration Authority
contact
description

useContext
jurisdiction
purpose
copyright
keyword | keywords
fhirVersion
mapping
identity
uri
name
comment
kind
abstract
contextType
context
contextInvariant
type
baseDefinition
derivation
snapshot
element | --
differential
element | --

5.3.10 Resource StructureDefinition - Design Notes

URL: https://hl7.org/fhir/STU3/structuredefinition-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.3.10 Resource StructureDefinition - Design Notes

to do

5.3.11 Resource StructureDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/structuredefinition-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.3.11 Resource StructureDefinition - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the StructureDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

5.3.11.1 Profiles

No Profiles defined for this resource

5.3.11.2 Extensions

11179-permitted-value-conceptmap | permitted-value-conceptmap | for ISO 11179 Element Definition Profile
11179-permitted-value-valueset | permitted-value-valueset | for ISO 11179 Element Definition Profile
datadictionary | datadictionary | for Standard DataElement Extensions
resource-approvalDate | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | lastReviewDate | for Resource HL7 Extensions
structuredefinition-ancestor | ancestor | for Standard Structure Definition Extensions
structuredefinition-annotation | annotation | for Standard Structure Definition Extensions
structuredefinition-category | category | for Standard Structure Definition Extensions
structuredefinition-fmm-no-warnings | fmm-no-warnings | for Standard Structure Definition Extensions
structuredefinition-summary | summary | for Standard Structure Definition Extensions
structuredefinition-table-name | table-name | for Standard Structure Definition Extensions
structuredefinition-template-status | template-status | for Standard Structure Definition Extensions

Extensions that reference this resource:

questionnaire-allowedProfile | allowedProfile | for Core extensions for Questionnaire

5.3.11.3 Search Extensions

No Search Extensions defined for this resource

5.3.12 Standard Structure Definition Extensions

URL: https://hl7.org/fhir/STU3/structuredefinition-extensions.html

This profile describes common extensions that are used with Structure Definitions

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.3.12 Standard Structure Definition Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

5.3.12.1 Content

Extensions:

structuredefinition-display-hint | display-hint : Hinting information for the narrative generator - a series of name: value; pairs.

structuredefinition-template-status | template-status : Status code taken from HL7 template specification - allows for alignment with the template DSTU, and has more authoring status codes.

structuredefinition-fmm | fmm : The FMM level assigned to the artifact.

structuredefinition-fmm-no-warnings | fmm-no-warnings : The FMM level that would be assigned to the artifact if it had no warnings.

structuredefinition-wg | wg : The work group that owns and maintains this resource.

structuredefinition-explicit-type-name | explicit-type-name : A name to use for the type, in implementations. This is a suggestion; it's not a normative part of the FHIR specification, but it does appear in the UML diagrams, and is used in generated code, schemas, etc.to identify the type.

structuredefinition-json-type | json-type : The JSON type of a property - used for the value property of a primitive type (for which there is no type in the FHIR typing system).

structuredefinition-xml-type | xml-type : The XML (schema) type of a property - used for the value attribute of a primitive type (for which there is no type in the FHIR typing system).

structuredefinition-rdf-type | rdf-type : The XML (schema) type of a property as used in RDF - used for the value attribute of a primitive type (for which there is no type in the FHIR typing system).

structuredefinition-regex | regex : The regular expression that applies to the value, if one exists.

structuredefinition-summary | summary : Additional text for the summary presentation.

structuredefinition-annotation | annotation : Additional notes about any item in a structure definition.

structuredefinition-ancestor | ancestor : A canonical URL for a StructureDefinition that this is derived from. This is a de-normalization of a chain of StructureDefinition.baseDefinition.

structuredefinition-category | category : The category under which the resource type is presented on the official resource list.

structuredefinition-table-name | table-name : A name to use to show mappings of this type in the generated summary tables.

structuredefinition-ballot-status | ballot-status : The Current HL7 ballot status of this artifact.

5.3.13 Resource StructureDefinition - Operations

URL: https://hl7.org/fhir/STU3/structuredefinition-operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.3.13 Resource StructureDefinition - Operations

FHIR Infrastructure Work Group | Maturity Level: 5 | Trial Use | Compartments: Not linked to any defined compartments

This resource has 1 operation associated with it:

$questionnaire | Build Questionnaire

For more information about operations, including how they are invoked, see Operations.

5.3.13.1 Build Questionnaire

Generates a Questionnaire instance based on a specified StructureDefinition, creating questions for each core element or extension element found in the StructureDefinition. If the operation is not called at the instance level, one of the identifier, profile or url 'in' parameters must be provided. If more than one is specified, servers may raise an error or may resolve with the parameter of their choice. If called at the instance level, these parameters will be ignored. The response will contain a Questionnaire instance based on the specified StructureDefinition and/or an OperationOutcome resource with errors or warnings. Nested groups are used to handle complex structures and data types. If the 'supportedOnly' parameter is set to true, only those elements marked as "must support" will be included. This operation is intended to enable auto-generation of simple interfaces for arbitrary profiles. The 'questionnaire' approach to data entry has limitations that will make it less optimal than custom-defined interfaces. However, this function may be useful for simple applications or for systems that wish to support "non-core" resources with minimal development effort.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/StructureDefinition-questionnaire
```

Formal Definition (as a OperationDefinition).

URL: [base]/StructureDefinition/$questionnaire

URL: [base]/StructureDefinition/[id]/$questionnaire

This is an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

identifier | 0..1 | uri | | | A logical profile identifier (i.e. 'StructureDefinition.identifier"). The server must know the profile or be able to retrieve it from other known repositories.
profile | 0..1 | string(token) | | | The StructureDefinition is provided directly as part of the request. Servers may choose not to accept profiles in this fashion
url | 0..1 | uri | | | The profile's official URL (i.e. 'StructureDefinition.url'). The server must know the profile or be able to retrieve it from other known repositories.
supportedOnly | 0..1 | boolean | | | If true, the questionnaire will only include those elements marked as "mustSupport='true'" in the StructureDefinition.
Out Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation
return | 1..1 | Questionnaire | | | The questionnaire form generated based on the StructureDefinition. Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

Open Issue: Ideally, extensions should be populated in the generated Questionnaire that will support taking QuestionnaireResponse resources generated from the Questionnaire and turning them back into the appropriate resources.

5.3.13.1.1 Examples

Request:

```
GET /fhir/StructureDefinition/daf-patient/$questionnaire
```

Response:

```
HTTP/1.1 200 OK
[other headers]

See DAF Patient Questionnaire
```

5.3.14 Resource StructureDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/structuredefinition-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.3.14 Resource StructureDefinition - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for StructureDefinition.

Functional status for this map: 248 tests that all execute ok. 60 fail round-trip testing and 145 r3 resources are invalid (441 errors). (see documentation)

5.3.14.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/StructureDefinition2to3" = "R2 to R3 Conversions for StructureDefinition"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/StructureDefinition" alias StructureDefinitionR2 as source
uses "http://hl7.org/fhir/StructureDefinition/StructureDefinition" alias StructureDefinition as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  StructureDefinition extends DomainResource
  input src : StructureDefinitionR2 as source
  input tgt : StructureDefinition as target

  "StructureDefinition-url" : for src.url make tgt.url
  "StructureDefinition-identifier" : for src.identifier make tgt.identifier
  "StructureDefinition-version" : for src.version make tgt.version
  "StructureDefinition-name" : for src.name make tgt.name
  "StructureDefinition-title" : for src.display make tgt.title
  "StructureDefinition-status" : for src.status make tgt.status
  "StructureDefinition-experimental" : for src.experimental make tgt.experimental
  "StructureDefinition-publisher" : for src.publisher make tgt.publisher
  "StructureDefinition-contact" : for src.contact as vs0 make tgt.contact as vt0 then {
    "ContactDetail-name" : for vs0.name make vt0.name
    "ContactDetail-telecom" : for vs0.telecom make vt0.telecom
  }

  "StructureDefinition-date" : for src.date make tgt.date
  "StructureDefinition-description" : for src.description make tgt.description
  "StructureDefinition-purpose" : for src.requirements make tgt.purpose
  "StructureDefinition-copyright" : for src.copyright make tgt.copyright
  "StructureDefinition-jurisdiction" : for src.useContext where coding.system = 'urn:iso:std:iso:3166' make
    tgt.jurisdiction
  "StructureDefinition-useContext" : for src.useContext as vs where (coding.system = 'urn:iso:std:iso:3166').not() make
      tgt.useContext as uc,  uc.code as ucc,  ucc.system = "http://hl7.org/fhir/usage-context-type",  ucc.code = "task",
      uc.value as vt then CodeableConcept(vs, vt)
  "StructureDefinition-keyword" : for src.code make tgt.keyword
  "StructureDefinition-fhirVersion" : for src.fhirVersion make tgt.fhirVersion
  "StructureDefinition-mapping" : for src.mapping as vs0 make tgt.mapping as vt0 then {
    "StructureDefinition.mapping-identity" : for vs0.identity make vt0.identity
    "StructureDefinition.mapping-uri" : for vs0.uri make vt0.uri
    "StructureDefinition.mapping-name" : for vs0.name make vt0.name
    "StructureDefinition.mapping-comments" : for vs0.comments make vt0.comment
  }
```

```
  "StructureDefinition-derivation-1" : for src where constrainedType.empty() make tgt.derivation = "specialization"
  "StructureDefinition-type-1" : for src.id as vs where src.constrainedType.empty() make tgt.type = vs
  "StructureDefinition-derivation-2" : for src where constrainedType.exists() make tgt.derivation = "constraint"
  "StructureDefinition-type-2" : for src.constrainedType as vs make tgt.type = vs
  "StructureDefinition-kind-1" : for src.kind where $this = 'logical' make tgt.kind
  "StructureDefinition-kind-2" : for src.kind where $this = 'resource' make tgt.kind
  "StructureDefinition-kind-3" : for src.kind as vt where ($this = 'datatype') and (src.id in ('boolean' | 'integer' |
    'string' | 'decimal' | 'uri' | 'base64Binary' | 'instant' | 'date' | 'dateTime' | 'time' | 'code' | 'oid' | 'id' |
    'markdown' | 'unsignedInt' | 'positiveInt')) make tgt.kind = "primitive-type" as vt
  "StructureDefinition-kind-4" : for src.kind as vt where ($this = 'datatype') and (src.id in ('boolean' | 'integer' |
    'string' | 'decimal' | 'uri' | 'base64Binary' | 'instant' | 'date' | 'dateTime' | 'time' | 'code' | 'oid' | 'id' |
    'markdown' | 'unsignedInt' | 'positiveInt')).not() make tgt.kind = "complex-type" as vt
  "StructureDefinition-abstract" : for src.abstract make tgt.abstract
  "StructureDefinition-contextType" : for src.contextType make tgt.contextType
  "StructureDefinition-context" : for src.context make tgt.context
  "StructureDefinition-type" : for src.constrainedType make tgt.type
  "StructureDefinition-baseDefinition" : for src.base make tgt.baseDefinition
  "StructureDefinition-snapshot" : for src.snapshot as vs0 make tgt.snapshot as vt0 then {
    "StructureDefinition.snapshot-element-1" : for vs0.element first as vs make  vt0.element as vt,  create("boolean")
    as first,  first.value = "true" then ElementDefinition(vs, vt, first)
    "StructureDefinition.snapshot-element-2" : for vs0.element not_first as vs make  vt0.element as vt,
    create("boolean") as first,  first.value = "false" then ElementDefinition(vs, vt, first)
  }

  "StructureDefinition-differential" : for src.differential first as vs0 make tgt.differential as vt0 then {
    "StructureDefinition.differential-element-1" : for vs0.element first as vs make  vt0.element as vt,
    create("boolean") as first,  first.value = "true" then ElementDefinition(vs, vt, first)
    "StructureDefinition.differential-element-2" : for vs0.element not_first as vs make  vt0.element as vt,
    create("boolean") as first,  first.value = "false" then ElementDefinition(vs, vt, first)
  }


endgroup
```

5.3.14.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/StructureDefinition3to2" = "R3 to R2 Conversion for StructureDefinition"

conceptmap "StructureDefinitionKind" {
  prefix s = "http://hl7.org/fhir/structure-definition-kind"
  prefix t = "http://hl7.org/fhir/structure-definition-kind"

  s:"primitive-type" - t:datatype
  s:"complex-type" - t:datatype
  s:resource - t:resource
  s:logical - t:logical
}

uses "http://hl7.org/fhir/StructureDefinition/StructureDefinition" alias StructureDefinition as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/StructureDefinition" alias StructureDefinitionR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  StructureDefinition extends DomainResource
  input src : StructureDefinition as source
  input tgt : StructureDefinitionR2 as target

  "StructureDefinition-url" : for src.url make tgt.url
  "StructureDefinition-identifier" : for src.identifier make tgt.identifier
  "StructureDefinition-version" : for src.version make tgt.version
  "StructureDefinition-name" : for src.name make tgt.name
  "StructureDefinition-display" : for src.title make tgt.display
  "StructureDefinition-status" : for src.status make tgt.status
  "StructureDefinition-experimental" : for src.experimental make tgt.experimental
  "StructureDefinition-publisher" : for src.publisher make tgt.publisher
  "StructureDefinition-contact" : for src.contact as vs0 make tgt.contact as vt0 then {
    "StructureDefinition.contact-name" : for vs0.name make vt0.name
    "StructureDefinition.contact-telecom" : for vs0.telecom make vt0.telecom
  }

  "StructureDefinition-date" : for src.date make tgt.date
  "StructureDefinition-description" : for src.description make tgt.description
  "StructureDefinition-useContext" : for src.jurisdiction make tgt.useContext
  "StructureDefinition-requirements" : for src.purpose make tgt.requirements
  "StructureDefinition-copyright" : for src.copyright make tgt.copyright
  "StructureDefinition-code" : for src.keyword make tgt.code
  "StructureDefinition-fhirVersion" : for src.fhirVersion make tgt.fhirVersion
  "StructureDefinition-mapping" : for src.mapping as vs0 make tgt.mapping as vt0 then {
    "StructureDefinition.mapping-identity" : for vs0.identity make vt0.identity
    "StructureDefinition.mapping-uri" : for vs0.uri make vt0.uri
    "StructureDefinition.mapping-name" : for vs0.name make vt0.name
    "StructureDefinition.mapping-comments" : for vs0.comment make vt0.comments
  }

  "StructureDefinition-kind" : for src.kind as v make tgt.kind = translate(v, "#StructureDefinitionKind", "code")
  "StructureDefinition-contextType" : for src.contextType make tgt.contextType
  "StructureDefinition-context" : for src.context make tgt.context
  "StructureDefinition-constrainedType" : for src.type as vs where src.derivation = 'constraint' make
    tgt.constrainedType as vt then code(vs, vt)
  "StructureDefinition-abstract" : for src.abstract make tgt.abstract
```

```
 "StructureDefinition-base" : for src.baseDefinition make tgt.base
 "StructureDefinition-snapshot" : for src.snapshot as vs0 make tgt.snapshot as vt0 then {
   "StructureDefinition.snapshot-element" : for vs0.element as vs make vt0.element as vt then ElementDefinition(src,
   vs, vt)
 }

 "StructureDefinition-differential" : for src.differential as vs0 make tgt.differential as vt0 then {
   "StructureDefinition.differential-element" : for vs0.element as vs make vt0.element as vt then
   ElementDefinition(src, vs, vt)
 }


endgroup
```

5.3.14.3 R3 Validation Errors

Account | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Address | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
AllergyIntolerance | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Annotation | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Appointment | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
AppointmentResponse | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Attachment | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
AuditEvent | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
BackboneElement | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Basic | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Binary | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
BodySite | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Bundle | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
CarePlan | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Claim | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ClaimResponse | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ClinicalImpression | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
CodeableConcept | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Coding | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Communication | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
CommunicationRequest | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Composition | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ConceptMap | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Condition | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Conformance | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
consentdirective | ERROR @ StructureDefinition.snapshot.element[10]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[11]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[12]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[13]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[14]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[15]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[16]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[17]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[18]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[19]: sliceName must be a proper token [sliceName.empty() or

sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[20]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[24]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[25]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[26]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[30]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[31]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[32]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[33]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[34]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[35]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[36]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[37]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[38]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[42]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[43]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[44]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[45]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[49]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[50]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[51]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[52]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[53]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[54]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[55]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[56]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[57]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[61]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[62]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[63]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[64]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[68]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[69]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[70]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[71]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[72]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[73]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[74]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[75]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[76]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[77]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[78]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[82]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[83]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[87]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[88]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[92]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[3]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[4]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[5]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[6]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[7]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[8]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[10]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[11]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[12]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[13]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[14]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[15]: sliceName must be a proper token [sliceName.empty() or

sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[16]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[17]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[18]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[19]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[20]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[21]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[22]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[23]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[24]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[25]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[26]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[27]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[28]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[29]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[30]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[31]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[32]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[33]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[34]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[35]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[36]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[37]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[38]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[39]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[40]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[41]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[42]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[43]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[44]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[45]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[46]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[47]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[48]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[49]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[50]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[51]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[52]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[53]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[54]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[55]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[56]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[57]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)

ContactPoint | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

Contract | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

Coverage | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

DataElement | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

DetectedIssue | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

Device | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

DeviceComponent | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

DeviceMetric | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

DeviceUseRequest | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

DeviceUseStatement | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

DiagnosticOrder | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

DiagnosticReport | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
DocumentManifest | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
DocumentReference | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
DomainResource | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ElementDefinition | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
EligibilityRequest | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
EligibilityResponse | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Encounter | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
EnrollmentRequest | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
EnrollmentResponse | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
EpisodeOfCare | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ExplanationOfBenefit | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Extension | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
FamilyMemberHistory | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Flag | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Goal | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Group | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
HealthcareService | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
HumanName | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Identifier | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ImagingObjectSelection | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ImagingStudy | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Immunization | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ImmunizationRecommendation | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ImplementationGuide | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
lipidprofile | ERROR @ StructureDefinition.snapshot.element[24]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[25]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[5]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[6]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
List | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Location | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Media | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Medication | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
MedicationAdministration | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
MedicationDispense | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
MedicationOrder | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
MedicationStatement | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
MessageHeader | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Meta | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

NamingSystem | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Narrative | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
NutritionOrder | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Observation | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
OperationDefinition | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
OperationOutcome | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Order | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
OrderResponse | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Organization | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Parameters | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Patient | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
PaymentNotice | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
PaymentReconciliation | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Period | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Person | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Practitioner | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Procedure | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ProcedureRequest | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ProcessRequest | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ProcessResponse | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Provenance | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
qicore-adverseevent | ERROR @ StructureDefinition: All element definitions must have unique ids (snapshot) ((ids: QICore-AdverseEvent,Basic.id,Basic.meta,Basic.implicitRules,Basic.language,Basic.text,Basic.contained,Basic.extension,type,discoveryDateTime,location,category,cause,Basic.extension.id,Basic.extension.extension,item,Basic.extension.extension.id,extension,Basic.extension.extension.url,Basic.extension.extension.value[x],certainty,Basic.extension.extension.id,extension,Basic.extension.extension.url,Basic.extension.extension.valueCodeableConcept,Basic.extension.url,Basic.extension.value[x],reaction,period,clinicalStudy,severity,Basic.modifierExtension,didNotOccur,Basic.identifier,Basic.code,Basic.subject,Basic.author,Basic.created)) [snapshot.element.id.trace('ids').isDistinct()] (src = InstanceValidator)
qicore-allergyintolerance | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[10]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[28]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[3]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[13]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-communication | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-communicationrequest | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-deviceuserequest | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-diagnosticorder | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[31]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[14]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-diagnosticreport | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[13]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[3]: sliceName must be a proper token [sliceName.empty() or

sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-encounter | ERROR @ StructureDefinition: All element definitions must have unique ids (snapshot) ((ids: QICore-Encounter,Encounter.id,Encounter.meta,Encounter.implicitRules,Encounter.language,Encounter.text,Encounter.contained,Encounter.extension,Encounter.relatedCondition,Encounter.extension.id,Encounter.extension.extension,condition,Encounter.extension.extension,Encounter.extension.extension,Encounter.extension.extension,role,Encounter.extension.extension,Encounter.extension.extension,Encounter.extension.extension,Encounter.extension.extension,Encounter.extension.extension,Encounter.extension.extension,Encounter.extension.url,Encounter.extension.valueReference,Encounter.reasonCancelled,Encounter.modifierExtension,Encounter.identifier,Encounter.status,Encounter.statusHistory,Encounter.statusHistory.id,Encounter.statusHistory.extension,Encounter.statusHistory.modifierExtension,Encounter.statusHistory.status,Encounter.statusHistory.period,Encounter.class,Encounter.type,Encounter.priority,Encounter.patient,Encounter.episodeOfCare,Encounter.incomingReferral,Encounter.participant,Encounter.participant.id,Encounter.participant.extension,Encounter.participant.modifierExtension,Encounter.participant.type,Encounter.participant.period,Encounter.participant.individual,Encounter.appointment,Encounter.period,Encounter.length,Encounter.reason,Encounter.indication,Encounter.hospitalization,Encounter.hospitalization.id,Encounter.hospitalization.extension,Encounter.hospitalization.modifierExtension,Encounter.hospitalization.preAdmissionIdentifier,Encounter.hospitalization.origin,Encounter.hospitalization.admitSource,Encounter.hospitalization.admittingDiagnosis,Encounter.hospitalization.reAdmission,Encounter.hospitalization.dietPreference,Encounter.hospitalization.specialCourtesy,Encounter.hospitalization.specialArrangement,Encounter.hospitalization.destination,Encounter.hospitalization.dischargeDisposition,Encounter.hospitalization.dischargeDiagnosis,Encounter.location,Encounter.location.id,Encounter.location.extension,Encounter.location.modifierExtension,Encounter.location.location,Encounter.location.status,Encounter.location.period,Encounter.serviceProvider,Encounter.partOf)) [snapshot.element.id.trace('ids').isDistinct()] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[26]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[3]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[4]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[5]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[6]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-familymemberhistory | ERROR @ StructureDefinition.snapshot.element[24]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[25]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[8]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-goal | ERROR @ StructureDefinition: All element definitions must have unique ids (snapshot) ((ids: QICore-Goal,Goal.id,Goal.meta,Goal.implicitRules,Goal.language,Goal.text,Goal.contained,Goal.extension,Goal.target,Goal.extension.id,Goal.extension.extension,measure,Goal.extension.extension,Goal.extension.extension,Goal.extension.extension,Goal.extension.extension,detail,Goal.extension.extension,Goal.extension.extension,Goal.extension.extension,Goal.extension.extension,Goal.extension.extension,Goal.extension.extension,Goal.extension.url,Goal.extension.value[x],Goal.reasonRejected,Goal.modifierExtension,Goal.identifier,Goal.subject,Goal.start[x],Goal.target[x],Goal.category,Goal.description,Goal.status,Goal.statusDate,Goal.statusReason,Goal.author,Goal.priority,Goal.addresses,Goal.note,Goal.outcome,Goal.outcome.id,Goal.outcome.extension,Goal.outcome.modifierExtension,Goal.outcome.result[x])) [snapshot.element.id.trace('ids').isDistinct()] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[26]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[3]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[4]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[5]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-imagingstudy | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[10]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[3]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-medication | ERROR @ StructureDefinition.snapshot.element[21]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[6]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-medicationdispense | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[10]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[29]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[30]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[31]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[32]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[33]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[34]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[35]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[36]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[3]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[15]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[16]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[17]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[18]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[19]: sliceName must be a

proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[20]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[21]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[22]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)

qicore-medicationorder | ERROR @ StructureDefinition.snapshot.element[24]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[25]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[26]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[27]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[28]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[29]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[30]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[31]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[10]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[11]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[12]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[13]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[14]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[15]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[16]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[17]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)

qicore-patient | ERROR @ StructureDefinition: All element definitions must have unique ids (snapshot) ((ids: QICore-Patient,Patient.id,Patient.meta,Patient.implicitRules,Patient.language,Patient.text,Patient.contained,Patient.extension,race,ethnicity,religion,birthPlace,disability,nationality,Patient.extension.id,Patient.extension.extension,code,Patient.extension.extension,Patient.extension.extension,Patient.extension.extension,Patient.extension.extension,period,Patient.extension.extension,Patient.extension.extension,Patient.extension.extension,Patient.extension.extension,Patient.extension.extension,Patient.extension.url,Patient.extension.value[x],cadavericDonor,militaryService,clinicalTrial,Patient.extension.id,Patient.extension.extension,NCT,Patient.extension.extension.id,extension,Patient.extension.extension.url,Patient.extension.extension.valueString,period,Patient.extension.extension.id,extension,Patient.extension.extension.url,Patient.extension.extension.valuePeriod,reason,Patient.extension.extension.id,extension,Patient.extension.extension.url,Patient.extension.extension.valueCodeableConcept,Patient.extension.url,Patient.extension.value[x],birthTime,Patient.modifierExtension,Patient.identifier,Patient.active,Patient.name,Patient.telecom,Patient.telecom.id,Patient.telecom.extension,telecom.preferred,Patient.telecom.system,Patient.telecom.value,Patient.telecom.use,Patient.telecom.rank,Patient.telecom.period,Patient.gender,Patient.birthDate,Patient.deceased[x],Patient.address,Patient.address.id,Patient.address.extension,address.preferred,Patient.address.use,Patient.address.type,Patient.address.text,Patient.address.line,Patient.address.city,Patient.address.district,Patient.address.state,Patient.address.postalCode,Patient.address.country,Patient.address.period,Patient.maritalStatus,Patient.multipleBirth[x],Patient.photo,Patient.contact,Patient.contact.id,Patient.contact.extension,Patient.contact.modifierExtension,Patient.contact.relationship,Patient.contact.name,Patient.contact.telecom,Patient.contact.address,Patient.contact.address.id,Patient.contact.address.extension,Patient.contact.address.use,Patient.contact.address.type,Patient.contact.address.text,Patient.contact.address.line,Patient.contact.address.city,Patient.contact.address.district,Patient.contact.address.state,Patient.contact.address.postalCode,Patient.contact.address.country,Patient.contact.address.period,Patient.contact.gender,Patient.contact.organization,Patient.contact.period,Patient.animal,Patient.animal.id,Patient.animal.extension,Patient.animal.modifierExtension,Patient.animal.species,Patient.animal.breed,Patient.animal.genderStatus,Patient.communication,Patient.communication.id,Patient.communication.extension,Patient.communication.modifierExtension,Patient.communication.language,Patient.communication.preferred,Patient.careProvider,Patient.managingOrganization,Patient.link,Patient.link.id,Patient.link.extension,Patient.link.modifierExtension,Patient.link.other,Patient.link.type)) [snapshot.element.id.trace('ids').isDistinct()] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[61]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[73]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[8]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[19]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[24]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)

qicore-practitioner | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[28]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[8]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)

qicore-procedure | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[10]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[3]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)

qicore-procedurerequest | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[10]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[11]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[3]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[4]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)

qicore-referralrequest | ERROR @ StructureDefinition.snapshot.element[9]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[2]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
qicore-specimen | ERROR @ StructureDefinition.snapshot.element[37]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[14]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
Quantity | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Questionnaire | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
QuestionnaireResponse | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Range | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Ratio | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Reference | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
ReferralRequest | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
RelatedPerson | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
RiskAssessment | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
SampledData | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
Schedule | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)
sdc-questionnaire | ERROR @ StructureDefinition: All element definitions must have unique ids (snapshot) ((ids: SDC-Questionnaire,Questionnaire.id,Questionnaire.meta,Questionnaire.implicitRules,Questionnaire.language,Questionnaire.text,Questionnaire.contained,Questionnaire.extension,submissionEndpoint,provenanceSignatureRequired,styleSensitive,category,Questionnaire.modifierExtension,Questionnaire.identifier,Questionnaire.version,Questionnaire.status,Questionnaire.date,Questionnaire.publisher,Questionnaire.telecom,Questionnaire.subjectType,Questionnaire.group,Questionnaire.group.id,Questionnaire.group.extension,groupLabel,Questionnaire.group.extension.id,extension,Questionnaire.group.extension.url,Questionnaire.group.extension.valueString,Questionnaire.group.extension.valueString.id,Questionnaire.group.extension.valueString.extension,groupLabel.renderingStyle,groupLabel.markup,Questionnaire.group.extension.valueString.value,groupInstruction,Questionnaire.group.extension.id,extension,Questionnaire.group.extension.url,Questionnaire.group.extension.valueString,Questionnaire.group.extension.valueString.id,Questionnaire.group.extension.valueString.extension,groupInstruction.renderingStyle,groupInstruction.markup,Questionnaire.group.extension.valueString.value,groupSecurity,Questionnaire.group.extension.id,extension,Questionnaire.group.extension.url,Questionnaire.group.extension.valueString,Questionnaire.group.extension.valueString.id,Questionnaire.group.extension.valueString.extension,groupSecurity.renderingStyle,groupSecurity.markup,Questionnaire.group.extension.valueString.value,groupHelp,Questionnaire.group.extension.id,extension,Questionnaire.group.extension.url,Questionnaire.group.extension.valueString,Questionnaire.group.extension.valueString.id,Questionnaire.group.extension.valueString.extension,groupHelp.renderingStyle,groupHelp.markup,Questionnaire.group.extension.valueString.value,groupTooltip,Questionnaire.group.extension.id,extension,Questionnaire.group.extension.url,Questionnaire.group.extension.valueString,Questionnaire.group.extension.valueString.id,Questionnaire.group.extension.valueString.extension,groupTooltip.renderingStyle,groupTooltip.markup,Questionnaire.group.extension.valueString.value,groupTrailing,Questionnaire.group.extension.id,extension,Questionnaire.group.extension.url,Questionnaire.group.extension.valueString,Questionnaire.group.extension.valueString.id,Questionnaire.group.extension.valueString.extension,groupTrailing.renderingStyle,groupTrailing.markup,Questionnaire.group.extension.valueString.value,groupOptionalDisplay,hidden,groupMinOccurs,groupMaxOccurs,specialGroup,Questionnaire.group.modifierExtension,Questionnaire.group.linkId,Questionnaire.group.title,Questionnaire.group.title.id,Questionnaire.group.title.extension,groupTitle.renderingStyle,groupTitle.markup,Questionnaire.group.title.value,Questionnaire.group.concept,Questionnaire.group.text,Questionnaire.group.text.id,Questionnaire.group.text.extension,groupText.renderingStyle,groupText.markup,Questionnaire.group.text.value,Questionnaire.group.required,Questionnaire.group.repeats,Questionnaire.group.group,Questionnaire.group.question,Questionnaire.group.question.id,Questionnaire.group.question.extension,questionMinOccurs,questionMaxOccurs,questionLabel,Questionnaire.group.question.extension.id,extension,Questionnaire.group.question.extension.url,Questionnaire.group.question.extension.valueString,Questionnaire.group.question.extension.valueString.id,Questionnaire.group.question.extension.valueString.extension,questionLabel.renderingStyle,questionLabel.markup,Questionnaire.group.question.extension.valueString.value,questionInstruction,Questionnaire.group.question.extension.id,extension,Questionnaire.group.question.extension.url,Questionnaire.group.question.extension.valueString,Questionnaire.group.question.extension.valueString.id,Questionnaire.group.question.extension.valueString.extension,questionInstruction.renderingStyle,questionInstruction.markup,Questionnaire.group.question.extension.valueString.value,questionSecurity,Questionnaire.group.question.extension.id,extension,Questionnaire.group.question.extension.url,Questionnaire.group.question.extension.valueString,Questionnaire.group.question.extension.valueString.id,Questionnaire.group.question.extension.valueString.extension,questionSecurity.renderingStyle,questionSecurity.markup,Questionnaire.group.question.extension.valueString.value,questionHelp,Questionnaire.group.question.extension.id,extension,Questionnaire.group.question.extension.url,Questionnaire.group.question.extension.valueString,Questionnaire.group.question.extension.valueString.id,Questionnaire.group.question.extension.valueString.extension,questionHelp.renderingStyle,questionHelp.markup,Questionnaire.group.question.extension.valueString.value,questionTooltip,Questionnaire.group.question.extension.id,extension,Questionnaire.group.question.extension.url,Questionnaire.group.question.extension.valueString,Questionnaire.group.question.extension.valueString.id,Questionnaire.group.question.extension.valueString.extension,questionTooltip.renderingStyle,questionTooltip.markup,Questionnaire.group.question.extension.valueString.value,questionTrailing,Questionnaire.group.question.extension.id,extension,Questionnaire.group.question.extension.url,Questionnaire.group.question.extension.valueString,Questionnaire.group.question.extension.valueString.id,Questionnaire.group.question.extension.valueString.extension,questionTrailing.renderingStyle,questionTrailing.markup,Questionnaire.group.question.extension.valueString.value,questionUnits,Questionnaire.group.question.extension.id,extension,Questionnaire.group.question.extension.url,Questionnaire.group.question.extension.valueString,Questionnaire.group.question.extension.valueString.id,Questionnaire.group.question.extension.valueString.extension,questionUnits.renderingStyle,questionUnits.markup,Questionnaire.group.question.extension.valueString.value,questionEnableWhen,Questionnaire.group.question.extension.id,Questionnaire.group.question.extension.extension,question,Questionnaire.group.question.extension.extension.id,extension,Questionnaire.group.question.extension.extension.url,Questionnaire.group.question.extension.extension.valueString,answered,Questionnaire.group.question.extension.extension.id,extension,Questionnaire.group.question.extension.extension.url,Questionnaire.group.question.extension.extension.valueBoolean,answer,Questionnaire.group.question.extension.extension.id,extension,Questionnaire.group.question.extension.extension.url,Questionnaire.group.question.extension.extension.value[x],Questionnaire.group.question.extension.url,Questionnaire.group.question.extension.value[x],questionOptionalDisplay,hidden,questionControl,choiceOrientation,defaultValue,readOnly,minLength,maxLength,regex,entryFormat,minValue,maxValue,maxDecimalPlaces,deReference,Questionnaire.group.question.modifierExtension,Questionnaire.group.question.linkId,Questionnaire.group.question.concept,Questionnaire.group.question.text,Questionnaire.group.question.text.id,Questionnaire.group.question.text.extension,questionText.render

ingStyle,questionText.markup,Questionnaire.group.question.text.value,Questionnaire.group.question.type,Questionnaire.group.question.required,Questionnaire.group.question.repeats,Questionnaire.group.question.options,Questionnaire.group.question.option,Questionnaire.group.question.group)) [snapshot.element.id.trace('ids').isDistinct()] (src = InstanceValidator) ERROR @ StructureDefinition: All element definitions must have unique ids (diff) ((ids: SDC-Questionnaire,submissionEndpoint,provenanceSignatureRequired,styleSensitive,category,Questionnaire.identifier,Questionnaire.version,Questionnaire.status,Questionnaire.date,Questionnaire.publisher,Questionnaire.subjectType,Questionnaire.group,groupLabel,Questionnaire.group.extension.valueString,groupLabel.renderingStyle,groupLabel.markup,groupInstruction,Questionnaire.group.extension.valueString,groupInstruction.renderingStyle,groupInstruction.markup,groupSecurity,Questionnaire.group.extension.valueString,groupSecurity.renderingStyle,groupSecurity.markup,groupHelp,Questionnaire.group.extension.valueString,groupHelp.renderingStyle,groupHelp.markup,groupTooltip,Questionnaire.group.extension.valueString,groupTooltip.renderingStyle,groupTooltip.markup,groupTrailing,Questionnaire.group.extension.valueString,groupTrailing.renderingStyle,groupTrailing.markup,groupOptionalDisplay,hidden,groupMinOccurs,groupMaxOccurs,specialGroup,Questionnaire.group.linkId,Questionnaire.group.title,groupTitle.renderingStyle,groupTitle.markup,Questionnaire.group.concept,Questionnaire.group.text,groupText.renderingStyle,groupText.markup,Questionnaire.group.required,Questionnaire.group.repeats,Questionnaire.group.group,Questionnaire.group.question,questionMinOccurs,questionMaxOccurs,questionLabel,Questionnaire.group.question.extension.valueString,questionLabel.renderingStyle,questionLabel.markup,questionInstruction,Questionnaire.group.question.extension.valueString,questionInstruction.renderingStyle,questionInstruction.markup,questionSecurity,Questionnaire.group.question.extension.valueString,questionSecurity.renderingStyle,questionSecurity.markup,questionHelp,Questionnaire.group.question.extension.valueString,questionHelp.renderingStyle,questionHelp.markup,questionTooltip,Questionnaire.group.question.extension.valueString,questionTooltip.renderingStyle,questionTooltip.markup,questionTrailing,Questionnaire.group.question.extension.valueString,questionTrailing.renderingStyle,questionTrailing.markup,questionUnits,Questionnaire.group.question.extension.valueString,questionUnits.renderingStyle,questionUnits.markup,questionEnableWhen,question,answered,answer,questionOptionalDisplay,hidden,questionControl,choiceOrientation,defaultValue,readOnly,minLength,maxLength,regex,entryFormat,minValue,maxValue,maxDecimalPlaces,deReference,Questionnaire.group.question.linkId,Questionnaire.group.question.concept,Questionnaire.group.question.text,questionText.renderingStyle,questionText.markup,Questionnaire.group.question.type,Questionnaire.group.question.required,Questionnaire.group.question.repeats,Questionnaire.group.question.options,Questionnaire.group.question.option,Questionnaire.group.question.group)) [differential.element.id.trace('ids').isDistinct()] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[31]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[32]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[41]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[42]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[51]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[52]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[61]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[62]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[71]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[72]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[81]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[82]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[94]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[95]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[101]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[102]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[119]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[120]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[129]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[130]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[139]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[140]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[149]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[150]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[159]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[160]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[169]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[170]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[179]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[180]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[222]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[223]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[15]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[16]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[19]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[20]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[23]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[24]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[27]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[28]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[31]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[32]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[35]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src

= InstanceValidator) ERROR @ StructureDefinition.differential.element[36]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[44]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[45]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[48]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[49]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[58]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[59]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[62]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[63]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[66]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[67]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[70]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[71]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[74]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[75]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[78]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[79]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[82]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[83]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[105]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[106]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)

sdc-valueset | ERROR @ StructureDefinition: All element definitions must have unique ids (snapshot) ((ids: SDC-ValueSet,ValueSet.id,ValueSet.meta,ValueSet.implicitRules,ValueSet.language,ValueSet.text,ValueSet.contained,ValueSet.extension,styleSensitive,ValueSet.modifierExtension,ValueSet.url,ValueSet.identifier,ValueSet.version,ValueSet.name,ValueSet.status,ValueSet.experimental,ValueSet.publisher,ValueSet.contact,ValueSet.contact.id,ValueSet.contact.extension,ValueSet.contact.modifierExtension,ValueSet.contact.name,ValueSet.contact.telecom,ValueSet.date,ValueSet.lockedDate,ValueSet.description,ValueSet.useContext,ValueSet.immutable,ValueSet.requirements,ValueSet.copyright,ValueSet.extensible,ValueSet.codeSystem,ValueSet.codeSystem.id,ValueSet.codeSystem.extension,ValueSet.codeSystem.modifierExtension,ValueSet.codeSystem.system,ValueSet.codeSystem.version,ValueSet.codeSystem.caseSensitive,ValueSet.codeSystem.concept,ValueSet.codeSystem.concept.id,ValueSet.codeSystem.concept.extension,conceptLabel,ValueSet.codeSystem.concept.extension.id,extension,ValueSet.codeSystem.concept.extension.url,ValueSet.codeSystem.concept.extension.valueString,ValueSet.codeSystem.concept.extension.valueString.id,ValueSet.codeSystem.concept.extension.valueString.extension,conceptLabel.renderingStyle,conceptLabel.markup,ValueSet.codeSystem.concept.extension.valueString.value,conceptConceptOrder,ValueSet.codeSystem.concept.modifierExtension,ValueSet.codeSystem.concept.code,ValueSet.codeSystem.concept.abstract,ValueSet.codeSystem.concept.display,ValueSet.codeSystem.concept.display.id,ValueSet.codeSystem.concept.display.extension,conceptDisplayLabel.renderingStyle,conceptDisplayLabel.markup,ValueSet.codeSystem.concept.display.value,ValueSet.codeSystem.concept.definition,designation,ValueSet.codeSystem.concept.designation.id,ValueSet.codeSystem.concept.designation.extension,ValueSet.codeSystem.concept.designation.modifierExtension,ValueSet.codeSystem.concept.designation.language,ValueSet.codeSystem.concept.designation.use,ValueSet.codeSystem.concept.designation.value,ValueSet.codeSystem.concept.concept,ValueSet.compose,ValueSet.compose.id,ValueSet.compose.extension,ValueSet.compose.modifierExtension,ValueSet.compose.import,ValueSet.compose.include,ValueSet.compose.include.id,ValueSet.compose.include.extension,ValueSet.compose.include.modifierExtension,ValueSet.compose.include.system,ValueSet.compose.include.version,ValueSet.compose.include.concept,ValueSet.compose.include.concept.id,ValueSet.compose.include.concept.extension,conceptLabel,ValueSet.compose.include.concept.extension.id,extension,ValueSet.compose.include.concept.extension.url,ValueSet.compose.include.concept.extension.valueString,ValueSet.compose.include.concept.extension.valueString.id,ValueSet.compose.include.concept.extension.valueString.extension,code.renderingStyle,code.markup,ValueSet.compose.include.concept.extension.valueString.value,conceptConceptOrder,ValueSet.compose.include.concept.modifierExtension,ValueSet.compose.include.concept.code,ValueSet.compose.include.concept.display,ValueSet.compose.include.concept.display.id,ValueSet.compose.include.concept.display.extension,codeDisplay.renderingStyle,codeDisplay.markup,ValueSet.compose.include.concept.display.value,ValueSet.compose.include.concept.designation,ValueSet.compose.include.filter,ValueSet.compose.include.filter.id,ValueSet.compose.include.filter.extension,ValueSet.compose.include.filter.modifierExtension,ValueSet.compose.include.filter.property,ValueSet.compose.include.filter.op,ValueSet.compose.include.filter.value,ValueSet.compose.exclude,ValueSet.expansion,ValueSet.expansion.id,ValueSet.expansion.extension,ValueSet.expansion.modifierExtension,ValueSet.expansion.identifier,ValueSet.expansion.timestamp,ValueSet.expansion.total,ValueSet.expansion.offset,ValueSet.expansion.parameter,ValueSet.expansion.parameter.id,ValueSet.expansion.parameter.extension,ValueSet.expansion.parameter.modifierExtension,ValueSet.expansion.parameter.name,ValueSet.expansion.parameter.value[x],ValueSet.expansion.contains,ValueSet.expansion.contains.id,ValueSet.expansion.contains.extension,expansionLabel,ValueSet.expansion.contains.extension.id,extension,ValueSet.expansion.contains.extension.url,ValueSet.expansion.contains.extension.valueString,ValueSet.expansion.contains.extension.valueString.id,ValueSet.expansion.contains.extension.valueString.extension,expansion.renderingStyle,expansion.markup,ValueSet.expansion.contains.extension.valueString.value,conceptConceptOrder,ValueSet.expansion.contains.modifierExtension,ValueSet.expansion.contains.system,ValueSet.expansion.contains.abstract,ValueSet.expansion.contains.version,ValueSet.expansion.contains.code,ValueSet.expansion.contains.display,ValueSet.expansion.contains.display.id,ValueSet.expansion.contains.display.extension,expansionDisplay.renderingStyle,expansionDisplay.markup,ValueSet.expansion.contains.display.value,ValueSet.expansion.contains.contains)) [snapshot.element.id.trace('ids').isDistinct()] (src = InstanceValidator) ERROR @ StructureDefinition: All element definitions must have unique ids (diff) ((ids: SDC-ValueSet,styleSensitive,ValueSet.url,ValueSet.name,ValueSet.status,ValueSet.experimental,ValueSet.date,ValueSet.lockedDate,ValueSet.description,ValueSet.useContext,ValueSet.immutable,ValueSet.extensible,ValueSet.codeSystem,ValueSet.codeSystem.system,ValueSet.codeSystem.version,ValueSet.codeSystem.caseSensitive,ValueSet.codeSystem.concept,conceptLabel,ValueSet.codeSystem.concept.extension.valueString,conceptLabel.renderingStyle,conceptLabel.markup,conceptConceptOrder,ValueSet.codeSystem.concept.code,ValueSet.codeSystem.concept.abstract,ValueSet.codeSystem.concept.display,conceptDisplayLabel.renderingStyle,conceptDisplayLabel.markup,ValueSet.codeSystem.concept.definition,ValueSet.codeSystem.concept.concept,ValueSet.compose,ValueSet.compose.import,ValueSet.compose.include,ValueSet.compose.include.system,ValueSet.compose.include.version,ValueSet.compose.include.concept,conceptLabel,ValueSet.compose.include.concept.extension.valueString,code.renderingStyle,code.markup,conceptConceptOrder,ValueSet.compose.include.concept.code,ValueSet.compose.include.concept.display,codeDisplay.renderingStyle,codeDisplay.markup,ValueSet.compose.include.filter,ValueSet.compose.include.filter.property,ValueSet.compose.include.filter.op,ValueSet.compose.include.filter.value,ValueSet.compose.exclude,ValueSet.expansion,ValueSet.expansion.identifier,ValueSet.expansion.timestamp,ValueSet.expansion.contains,expansionLabel,ValueSet.expansion.contains.extension.valueString,expansion.renderingStyle,expansion.markup,conceptConceptOrder,ValueSet.expansion.contains.system,ValueSet.expansion.contains.abstract,ValueSet.expansion.contains.version,ValueSet.expansion.contains.code,ValueSet.expansion.contains.display,expansionDisplay.renderingStyle,expansionDisplay.markup,ValueSet.expansion.contains.contains)) [differential.element.id.trace('ids').isDistinct()] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[49]:

sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[50]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[59]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[60]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[92]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[93]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[101]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[102]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[137]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[138]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[149]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[150]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[20]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[21]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[26]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[27]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[38]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[39]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[43]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[44]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[56]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[57]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[64]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.differential.element[65]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)
sdcde-dataelement | ERROR @ StructureDefinition: All element definitions must have unique ids (snapshot) ((ids: SDCDE-DataElement,DataElement.id,DataElement.meta,DataElement.implicitRules,DataElement.language,DataElement.text,DataElement.contained,DataElement.extension,effectivePeriod,documentReference,administrativeStatus,changeDescription,submitterOrganization,registryOrganization,DataElement.modifierExtension,DataElement.url,DataElement.identifier,DataElement.version,DataElement.name,DataElement.status,DataElement.experimental,DataElement.publisher,DataElement.contact,DataElement.contact.id,DataElement.contact.extension,contactAddress,DataElement.contact.modifierExtension,DataElement.contact.name,DataElement.contact.telecom,DataElement.date,DataElement.useContext,DataElement.useContext.id,DataElement.useContext.extension,classificationOrContext,DataElement.useContext.coding,DataElement.useContext.text,DataElement.copyright,DataElement.stringency,DataElement.mapping,ObjectClassPropertyMapping,DataElement.mapping.id,DataElement.mapping.extension,DataElement.mapping.modifierExtension,DataElement.mapping.identity,DataElement.mapping.uri,DataElement.mapping.name,DataElement.mapping.comments,OtherMapping,DataElement.mapping.id,DataElement.mapping.extension,DataElement.mapping.modifierExtension,DataElement.mapping.identity,DataElement.mapping.uri,DataElement.mapping.name,DataElement.mapping.comments,DataElement.element,DataElement.element.id,DataElement.element.extension,question,allowedUnits,minLength,regex,entryFormat,maxDecimalPlaces,DataElement.element.path,DataElement.element.representation,DataElement.element.name,DataElement.element.label,DataElement.element.code,DataElement.element.code.id,DataElement.element.code.extension,isDataElementConcept,DataElement.element.code.system,DataElement.element.code.version,DataElement.element.code.code,DataElement.element.code.display,DataElement.element.code.userSelected,DataElement.element.slicing,DataElement.element.slicing.id,DataElement.element.slicing.extension,DataElement.element.slicing.discriminator,DataElement.element.slicing.description,DataElement.element.slicing.ordered,DataElement.element.slicing.rules,DataElement.element.short,DataElement.element.definition,DataElement.element.comments,DataElement.element.requirements,DataElement.element.alias,DataElement.element.min,DataElement.element.max,DataElement.element.base,DataElement.element.base.id,DataElement.element.base.extension,DataElement.element.base.path,DataElement.element.base.min,DataElement.element.base.max,DataElement.element.type,DataElement.element.type.id,DataElement.element.type.extension,DataElement.element.type.code,DataElement.element.type.profile,DataElement.element.type.aggregation,DataElement.element.nameReference,DataElement.element.defaultValue[x],DataElement.element.meaningWhenMissing,DataElement.element.fixed[x],DataElement.element.pattern[x],DataElement.element.example[x],DataElement.element.minValue[x],l,DataElement.element.maxLength,DataElement.element.condition,DataElement.element.constraint,DataElement.element.constraint.id,DataElement.element.constraint.extension,DataElement.element.constraint.key,DataElement.element.constraint.requirements,DataElement.element.constraint.severity,DataElement.element.constraint.human,DataElement.element.constraint.xpath,DataElement.element.mustSupport,DataElement.element.isModifier,DataElement.element.isSummary,DataElement.element.binding,DataElement.element.binding.id,DataElement.element.binding.extension,DataElement.element.binding.strength,DataElement.element.binding.description,DataElement.element.binding.valueSetReference,DataElement.element.binding.valueSetReference.id,DataElement.element.binding.valueSetReference.extension,permittedValueValueset,permittedValueConcepmap,DataElement.element.binding.valueSetReference.reference,DataElement.element.binding.valueSetReference.display,DataElement.element.mapping,ClassPropertyMapping,DataElement.element.mapping.id,DataElement.element.mapping.extension,objectClass,DataElement.element.mapping.extension.id,extension,DataElement.element.mapping.extension.url,DataElement.element.mapping.extension.valueCoding,property,DataElement.element.mapping.extension.id,extension,DataElement.element.mapping.extension.url,DataElement.element.mapping.extension.valueCoding,DataElement.element.mapping.identity,DataElement.element.mapping.language,DataElement.element.mapping.map,OtherElementMapping,DataElement.element.mapping.id,MappingEquivalence,DataElement.element.mapping.identity,DataElement.element.mapping.language,DataElement.element.mapping.map))
[snapshot.element.id.trace('ids').isDistinct()] (src = InstanceValidator) ERROR @ StructureDefinition: All element definitions must have unique ids (diff) ((ids: SDCDE-DataElement,effectivePeriod,documentReference,administrativeStatus,changeDescription,submitterOrganization,registryOrganization,DataElement.url,DataElement.identifier,DataElement.version,DataElement.name,DataElement.status,DataElement.experimental,DataElement.publisher,DataElement.contact,contactAddress,DataElement.contact.name,DataElement.contact.telecom,DataElement.date,DataElement.useContext,classificationOrContext,DataElement.copyright,DataElement.stringency,ObjectClassPropertyMapping,DataElement.mapping.identity,DataElement.mapping.uri,DataElement.mapping.name,DataElement.mapping.comments,OtherMapping,DataElement.mapping.identity,DataElement.mapping.uri,DataElement.mapping.name,DataElement.mapping.comments,DataElement.element,question,allowedUnits,minLength,regex,entryFormat,maxDecimalPlaces,DataElement.element.path,DataElement.element.name,DataElement.element.label,DataElement.element.code,isDataElementConcept,DataElement.element.definition,DataElement.element.comments,DataElement.element.requirements,DataElement.element.alias,DataElement.element.min,DataElement.element.max,DataElement.element.type,DataElement.element.type.code,DataElement.element.defaultValue[x],DataElement.element.meaningWhenMissing,DataElement.element.example[x],DataElement.element.minValue[x],l,DataElement.element.maxLength,DataElement.element.condition,DataElement.element.constraint,DataElement.element.constraint.key,DataElement.element.constraint.requirements,DataElement.element.constraint.severity,DataElement.element.constraint.human,DataElement.element.constraint.xpath,DataElement.element.mustSupport,DataElement.element.binding,DataElement.element.binding.strength,DataElement.element.binding.description,DataElement.element.binding.valueSetR

eference,permittedValueValueset,permittedValueConcepmap,ClassPropertyMapping,objectClass,DataElement.element.mapping.extension.valueCoding,property,DataElement.element.mapping.extension.valueCoding,DataElement.element.mapping.identity,DataElement.element.mapping.map,OtherElementMapping,MappingEquivalence,DataElement.element.mapping.identity,DataElement.element.mapping.language,DataElement.element.mapping.map)) [differential.element.id.trace('ids').isDistinct()] (src = InstanceValidator)

sdcde-valueset | ERROR @ StructureDefinition: All element definitions must have unique ids (snapshot) ((ids: SDCDE-ValueSet,ValueSet.id,ValueSet.meta,ValueSet.implicitRules,ValueSet.language,ValueSet.text,ValueSet.contained,ValueSet.extension,styleSensitive,trustedExpansion,effectiveDate,expirationDate,ValueSet.modifierExtension,ValueSet.url,ValueSet.identifier,ValueSet.version,ValueSet.name,ValueSet.status,ValueSet.experimental,ValueSet.publisher,ValueSet.contact,ValueSet.contact.id,ValueSet.contact.extension,ValueSet.contact.modifierExtension,ValueSet.contact.name,ValueSet.contact.telecom,ValueSet.date,ValueSet.lockedDate,ValueSet.description,ValueSet.useContext,ValueSet.immutable,ValueSet.requirements,ValueSet.copyright,ValueSet.extensible,ValueSet.codeSystem,ValueSet.codeSystem.id,ValueSet.codeSystem.extension,ValueSet.codeSystem.modifierExtension,ValueSet.codeSystem.system,ValueSet.codeSystem.version,ValueSet.codeSystem.caseSensitive,ValueSet.codeSystem.concept,ValueSet.codeSystem.concept.id,ValueSet.codeSystem.concept.extension,conceptConceptOrder,ValueSet.codeSystem.concept.modifierExtension,ValueSet.codeSystem.concept.code,ValueSet.codeSystem.concept.abstract,ValueSet.codeSystem.concept.display,ValueSet.codeSystem.concept.definition,designation,ValueSet.codeSystem.concept.designation.id,ValueSet.codeSystem.concept.designation.extension,ValueSet.codeSystem.concept.designation.modifierExtension,ValueSet.codeSystem.concept.designation.language,ValueSet.codeSystem.concept.designation.use,ValueSet.codeSystem.concept.designation.value,ValueSet.codeSystem.concept.concept,ValueSet.compose,ValueSet.compose.id,ValueSet.compose.extension,ValueSet.compose.modifierExtension,ValueSet.compose.import,ValueSet.compose.include,ValueSet.compose.include.id,ValueSet.compose.include.extension,ValueSet.compose.include.modifierExtension,ValueSet.compose.include.system,ValueSet.compose.include.version,ValueSet.compose.include.concept,ValueSet.compose.include.concept.id,ValueSet.compose.include.concept.extension,conceptConceptOrder,ValueSet.compose.include.concept.modifierExtension,ValueSet.compose.include.concept.code,ValueSet.compose.include.concept.display,ValueSet.compose.include.concept.designation,ValueSet.compose.include.filter,ValueSet.compose.include.filter.id,ValueSet.compose.include.filter.extension,ValueSet.compose.include.filter.modifierExtension,ValueSet.compose.include.filter.property,ValueSet.compose.include.filter.op,ValueSet.compose.include.filter.value,ValueSet.compose.exclude,ValueSet.expansion,ValueSet.expansion.id,ValueSet.expansion.extension,ValueSet.expansion.modifierExtension,ValueSet.expansion.identifier,ValueSet.expansion.timestamp,ValueSet.expansion.total,ValueSet.expansion.offset,ValueSet.expansion.parameter,ValueSet.expansion.parameter.id,ValueSet.expansion.parameter.extension,ValueSet.expansion.parameter.modifierExtension,ValueSet.expansion.parameter.name,ValueSet.expansion.parameter.value[x],ValueSet.expansion.contains,ValueSet.expansion.contains.id,ValueSet.expansion.contains.extension,conceptConceptOrder,definition,systemName,ValueSet.expansion.contains.modifierExtension,ValueSet.expansion.contains.system,ValueSet.expansion.contains.abstract,ValueSet.expansion.contains.version,ValueSet.expansion.contains.code,ValueSet.expansion.contains.display,ValueSet.expansion.contains.contains)) [snapshot.element.id.trace('ids').isDistinct()] (src = InstanceValidator) ERROR @ StructureDefinition: All element definitions must have unique ids (diff) ((ids: SDCDE-ValueSet,styleSensitive,trustedExpansion,effectiveDate,expirationDate,ValueSet.url,ValueSet.identifier,ValueSet.version,ValueSet.name,ValueSet.status,ValueSet.experimental,ValueSet.publisher,ValueSet.date,ValueSet.lockedDate,ValueSet.description,ValueSet.useContext,ValueSet.immutable,ValueSet.extensible,ValueSet.codeSystem,ValueSet.codeSystem.system,ValueSet.codeSystem.version,ValueSet.codeSystem.caseSensitive,ValueSet.codeSystem.concept,conceptConceptOrder,ValueSet.codeSystem.concept.code,ValueSet.codeSystem.concept.abstract,ValueSet.codeSystem.concept.display,ValueSet.codeSystem.concept.definition,ValueSet.codeSystem.concept.concept,ValueSet.compose,ValueSet.compose.import,ValueSet.compose.include,ValueSet.compose.include.system,ValueSet.compose.include.version,ValueSet.compose.include.concept,conceptConceptOrder,ValueSet.compose.include.concept.code,ValueSet.compose.include.concept.display,ValueSet.compose.include.filter,ValueSet.compose.include.filter.property,ValueSet.compose.include.filter.op,ValueSet.compose.include.filter.value,ValueSet.compose.exclude,ValueSet.expansion,ValueSet.expansion.identifier,ValueSet.expansion.timestamp,ValueSet.expansion.contains,conceptConceptOrder,definition,systemName,ValueSet.expansion.contains.system,ValueSet.expansion.contains.abstract,ValueSet.expansion.contains.version,ValueSet.expansion.contains.code,ValueSet.expansion.contains.display,ValueSet.expansion.contains.contains))
[differential.element.id.trace('ids').isDistinct()] (src = InstanceValidator)

SearchParameter | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

Signature | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

Slot | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

Specimen | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

example | ERROR @ StructureDefinition.snapshot.element[16]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator) ERROR @ StructureDefinition.snapshot.element[17]: sliceName must be a proper token [sliceName.empty() or sliceName.matches('^[a-zA-Z0-9\\-_]+$')] (src = InstanceValidator)

StructureDefinition | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

Subscription | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

Substance | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

SupplyDelivery | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

SupplyRequest | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

TestScript | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

Timing | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

uslab-patient | ERROR @ StructureDefinition: All element definitions must have unique ids (snapshot) ((ids: USLab-Patient,Patient.id,Patient.meta,Patient.implicitRules,Patient.language,Patient.text,Patient.contained,Patient.extension,race,Patient.extension.id,extension,Patient.extension.url,Patient.extension.valueCodeableConcept,Patient.extension.valueCodeableConcept.id,Patient.extension.valueCodeableConcept.extension,Patient.extension.valueCodeableConcept.coding,Patient.extension.valueCodeableConcept.coding.id,Patient.extension.valueCodeableConcept.coding.extension,Patient.extension.valueCodeableConcept.coding.system,Patient.extension.valueCodeableConcept.coding.version,Patient.extension.valueCodeableConcept.coding.code,Patient.extension.valueCodeableConcept.coding.display,Patient.extension.valueCodeableConcept.coding.userSelected,Patient.extension.valueCodeableConcept.text,ethnicity,Patient.extension.id,extension,Patient.extension.url,Patient.extension.valueCodeableConcept,Patient.extension.valueCodeableConcept.id,Patient.extension.valueCodeableConcept.extension,Patient.extension.valueCodeableConcept.coding,Patient.extension.valueCodeableConcept.coding.id,Patient.extension.valueCodeableConcept.coding.extension,Patient.extension.valueCodeableConcept.coding.system,Patient.extension.valueCodeableConcept.coding.version,Patient.extension.valueCodeableConcept.coding.code,Patient.extension.valueCodeableConcept.coding.display,Patient.extension.valueCodeableConcept.coding.userSelected,Patient.extension.valueCodeableConcept.text,Patient.modifierExtension,Patient.identifier,Patient.identifier.id,Patient.identifier.extension,Patient.identifier.use,Patient.identifier.type,Patient.identifier.

system,Patient.identifier.value,Patient.identifier.period,Patient.identifier.assigner,Patient.active,Patient.name,Patient.name.id,Patient.name.extension,Patient.name.use,Patient.name.text,Patient.name.family,Patient.name.given,Patient.name.prefix,Patient.name.suffix,Patient.name.period,Patient.telecom,Patient.gender,Patient.birthDate,Patient.deceased[x],Patient.address,Patient.address.id,Patient.address.extension,Patient.address.use,Patient.address.type,Patient.address.text,Patient.address.line,Patient.address.city,Patient.address.district,Patient.address.state,Patient.address.postalCode,Patient.address.country,Patient.address.period,Patient.maritalStatus,Patient.multipleBirth[x],Patient.photo,Patient.contact,Patient.contact.id,Patient.contact.extension,Patient.contact.modifierExtension,Patient.contact.relationship,Patient.contact.name,Patient.contact.telecom,Patient.contact.address,Patient.contact.gender,Patient.contact.organization,Patient.contact.period,Patient.animal,Patient.animal.id,Patient.animal.extension,Patient.animal.modifierExtension,Patient.animal.species,Patient.animal.breed,Patient.animal.genderStatus,Patient.communication,Patient.communication.id,Patient.communication.extension,Patient.communication.modifierExtension,Patient.communication.language,Patient.communication.preferred,Patient.careProvider,Patient.managingOrganization,Patient.link,Patient.link.id,Patient.link.extension,Patient.link.modifierExtension,Patient.link.other,Patient.link.type)) [snapshot.element.id.trace('ids').isDistinct()] (src = InstanceValidator) ERROR @ StructureDefinition: All element definitions must have unique ids (diff) ((ids: USLab-Patient,race,Patient.extension.valueCodeableConcept,Patient.extension.valueCodeableConcept.coding,Patient.extension.valueCodeableConcept.coding.system,Patient.extension.valueCodeableConcept.coding.code,Patient.extension.valueCodeableConcept.coding.display,ethnicity,Patient.extension.valueCodeableConcept,Patient.extension.valueCodeableConcept.coding,Patient.extension.valueCodeableConcept.coding.system,Patient.extension.valueCodeableConcept.coding.code,Patient.extension.valueCodeableConcept.coding.display,Patient.identifier,Patient.identifier.use,Patient.identifier.system,Patient.identifier.value,Patient.name,Patient.name.use,Patient.name.family,Patient.name.given,Patient.name.suffix,Patient.gender,Patient.birthDate,Patient.deceased[x],Patient.address,Patient.address.use,Patient.address.line,Patient.address.city,Patient.address.state,Patient.address.postalCode,Patient.address.country)) [differential.element.id.trace('ids').isDistinct()] (src = InstanceValidator)

uslab-phpatient | ERROR @ StructureDefinition: All element definitions must have unique ids (snapshot) ((ids: USLab-PHPatient,Patient.id,Patient.meta,Patient.implicitRules,Patient.language,Patient.text,Patient.contained,Patient.extension,DAFRace,Patient.extension.id,extension,Patient.extension.url,Patient.extension.valueCodeableConcept,Patient.extension.valueCodeableConcept.id,Patient.extension.valueCodeableConcept.extension,Patient.extension.valueCodeableConcept.coding,Patient.extension.valueCodeableConcept.coding.id,Patient.extension.valueCodeableConcept.coding.extension,Patient.extension.valueCodeableConcept.coding.system,Patient.extension.valueCodeableConcept.coding.version,Patient.extension.valueCodeableConcept.coding.code,Patient.extension.valueCodeableConcept.coding.display,Patient.extension.valueCodeableConcept.coding.userSelected,Patient.extension.valueCodeableConcept.text,DAFethnicity,Patient.extension.id,extension,Patient.extension.url,Patient.extension.valueCodeableConcept,Patient.extension.valueCodeableConcept.id,Patient.extension.valueCodeableConcept.extension,Patient.extension.valueCodeableConcept.coding,Patient.extension.valueCodeableConcept.coding.id,Patient.extension.valueCodeableConcept.coding.extension,Patient.extension.valueCodeableConcept.coding.system,Patient.extension.valueCodeableConcept.coding.version,Patient.extension.valueCodeableConcept.coding.code,Patient.extension.valueCodeableConcept.coding.display,Patient.extension.valueCodeableConcept.coding.userSelected,Patient.extension.valueCodeableConcept.text,Patient.modifierExtension,Patient.identifier,Patient.identifier.id,Patient.identifier.extension,Patient.identifier.use,Patient.identifier.type,Patient.identifier.system,Patient.identifier.value,Patient.identifier.period,Patient.identifier.assigner,Patient.active,Patient.name,Patient.name.id,Patient.name.extension,Patient.name.use,Patient.name.text,Patient.name.family,Patient.name.given,Patient.name.prefix,Patient.name.suffix,Patient.name.period,Patient.telecom,USLabHomePhone,Patient.telecom.id,Patient.telecom.extension,Patient.telecom.system,Patient.telecom.value,Patient.telecom.use,Patient.telecom.rank,Patient.telecom.period,USLabWorkPhone,Patient.telecom.id,Patient.telecom.extension,Patient.telecom.system,Patient.telecom.value,Patient.telecom.use,Patient.telecom.rank,Patient.telecom.period,Patient.gender,Patient.birthDate,Patient.deceased[x],Patient.address,Patient.address.id,Patient.address.extension,USLabPHCountry,Patient.address.use,Patient.address.type,Patient.address.text,Patient.address.line,Patient.address.city,Patient.address.district,Patient.address.state,Patient.address.postalCode,Patient.address.country,Patient.address.period,Patient.maritalStatus,Patient.multipleBirth[x],Patient.photo,Patient.contact,Patient.contact.id,Patient.contact.extension,Patient.contact.modifierExtension,Patient.contact.relationship,Patient.contact.relationship.id,Patient.contact.relationship.extension,Patient.contact.relationship.coding,Patient.contact.relationship.coding.id,Patient.contact.relationship.coding.extension,Patient.contact.relationship.coding.system,Patient.contact.relationship.coding.version,Patient.contact.relationship.coding.code,Patient.contact.relationship.coding.display,Patient.contact.relationship.coding.userSelected,Patient.contact.relationship.text,Patient.contact.name,Patient.contact.name.id,Patient.contact.name.extension,Patient.contact.name.use,Patient.contact.name.text,Patient.contact.name.family,Patient.contact.name.given,Patient.contact.name.prefix,Patient.contact.name.suffix,Patient.contact.name.period,Patient.contact.telecom,Patient.contact.telecom.id,Patient.contact.telecom.extension,Patient.contact.telecom.system,Patient.contact.telecom.value,Patient.contact.telecom.use,Patient.contact.telecom.rank,Patient.contact.telecom.period,Patient.contact.address,Patient.contact.address.id,Patient.contact.address.extension,CountyCountact,Patient.contact.address.use,Patient.contact.address.type,Patient.contact.address.text,Patient.contact.address.line,Patient.contact.address.city,Patient.contact.address.district,Patient.contact.address.state,Patient.contact.address.postalCode,Patient.contact.address.country,Patient.contact.address.period,Patient.contact.gender,Patient.contact.organization,Patient.contact.period,Patient.animal,Patient.animal.id,Patient.animal.extension,Patient.animal.modifierExtension,Patient.animal.species,Patient.animal.breed,Patient.animal.genderStatus,Patient.communication,Patient.communication.id,Patient.communication.extension,Patient.communication.modifierExtension,Patient.communication.language,Patient.communication.preferred,Patient.careProvider,Patient.managingOrganization,Patient.link,Patient.link.id,Patient.link.extension,Patient.link.modifierExtension,Patient.link.other,Patient.link.type)) [snapshot.element.id.trace('ids').isDistinct()] (src = InstanceValidator) ERROR @ StructureDefinition: All element definitions must have unique ids (diff) ((ids: USLab-PHPatient,DAFRace,Patient.extension.valueCodeableConcept,Patient.extension.valueCodeableConcept.coding,Patient.extension.valueCodeableConcept.coding.system,Patient.extension.valueCodeableConcept.coding.code,Patient.extension.valueCodeableConcept.coding.display,DAFethnicity,Patient.extension.valueCodeableConcept,Patient.extension.valueCodeableConcept.coding,Patient.extension.valueCodeableConcept.coding.system,Patient.extension.valueCodeableConcept.coding.code,Patient.extension.valueCodeableConcept.coding.display,Patient.identifier,Patient.identifier.use,Patient.identifier.system,Patient.identifier.value,Patient.name,Patient.name.use,Patient.name.family,Patient.name.given,Patient.name.prefix,Patient.name.suffix,USLabHomePhone,Patient.telecom.system,Patient.telecom.value,Patient.telecom.use,USLabWorkPhone,Patient.telecom.system,Patient.telecom.value,Patient.telecom.use,Patient.gender,Patient.birthDate,Patient.deceased[x],Patient.address,USLabPHCountry,Patient.address.use,Patient.address.line,Patient.address.city,Patient.address.state,Patient.address.postalCode,Patient.address.country,Patient.contact,Patient.contact.relationship,Patient.contact.relationship.coding,Patient.contact.relationship.coding.code,Patient.contact.name,Patient.contact.name.family,Patient.contact.name.given,Patient.contact.name.suffix,Patient.contact.telecom,Patient.contact.telecom.system,Patient.contact.telecom.value,Patient.contact.address,CountyCountact,Patient.contact.address.use,Patient.contact.address.line,Patient.contact.address.city,Patient.contact.address.state,Patient.contact.address.postalCode,Patient.contact.address.country)) [differential.element.id.trace('ids').isDistinct()] (src = InstanceValidator)

ValueSet | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

VisionPrescription | ERROR @ StructureDefinition: element.base cannot appear if there is no base on the structure definition [snapshot.exists() implies (snapshot.element.base.exists() = baseDefinition.exists())] (src = InstanceValidator)

5.4 Resource OperationDefinition - Content

URL: https://hl7.org/fhir/STU3/operationdefinition.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 4 | Trial Use | Compartments: Not linked to any defined compartments

A formal computable definition of an operation (on the RESTful interface) or a named query (using the search interaction).

5.4.1 Scope and Usage

The OperationDefinition resource provides a formal computable definition of an operation or a named query. The OperationDefinition serves two principal purposes:

* To allow for automatic determination of system compatibility

* To allow for dynamic generation of forms to drive the operations

See below for further information about these, and about how operations and named queries are executed.

5.4.2 Boundaries and Relationships

OperationDefinitions are published to define operations that servers can implement in a common fashion. The FHIR specification itself describes some (see below), and other organizations (including IHE, national programs, jurisdictions and vendors) are able to publish additional OperationDefinitions.

OperationDefinition resources are referred to from two different places:

* From a Capability Statement, to declare what operations a system does or should implement

* From another OperationDefinition resource. This allows for a server to describe a limited implementation of a standard operation, or to allow traceability if the server has to rename the operation due to a name clash

This resource is referenced by CapabilityStatement and ServiceDefinition

5.4.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
OperationDefinition | | | DomainResource | Definition of an operation or a named query Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this operation definition (globally unique)
version | ? | 0..1 | string | Business version of the operation definition
name | ? | 1..1 | string | Name for this operation definition (computer friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
kind | | 1..1 | code | operation | query OperationKind (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the operation definition
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for operation definition (if applicable) Jurisdiction ValueSet (Extensible)
purpose | | 0..1 | markdown | Why this operation definition is defined
idempotent | ? | 0..1 | boolean | Whether content is unchanged by the operation
code | ? | 1..1 | code | Name used to invoke the operation
comment | | 0..1 | string | Additional information about use
base | ? | 0..1 | Reference(OperationDefinition) | Marks this as a profile of the base
resource | ? | 0..* | code | Types this operation applies to ResourceType (Required)
system | ? | 1..1 | boolean | Invoke at the system level?
type | ? | 1..1 | boolean | Invole at the type level?
instance | ? | 1..1 | boolean | Invoke on an instance?
parameter | I | 0..* | BackboneElement | Parameters for the operation/query + Either a type must be provided, or parts + A search type can only be specified for parameters of type string
name | | 1..1 | code | Name in Parameters.parameter.name or in URL
use | | 1..1 | code | in | out OperationParameterUse (Required)
min | | 1..1 | integer | Minimum Cardinality
max | | 1..1 | string | Maximum Cardinality (a number or *)
documentation | | 0..1 | string | Description of meaning/use
type | I | 0..1 | code | What type this parameter has FHIRAllTypes (Required)
searchType | I | 0..1 | code | number | date | string | token | reference | composite | quantity | uri SearchParamType (Required)
profile | | 0..1 | Reference(StructureDefinition) | Profile on the type
binding | | 0..1 | BackboneElement | ValueSet details if this is coded
strength | | 1..1 | code | required | extensible | preferred | example BindingStrength (Required)
valueSet[x] | | 1..1 | | Source of value set
valueSetUri | | | uri

valueSetReference | | | Reference(ValueSet)
part | I | 0..* | see parameter | Parts of a nested Parameter
overload | | 0..* | BackboneElement | Define overloaded variants for when generating code
parameterName | | 0..* | string | Name of parameter to include in overload
comment | | 0..1 | string | Comments to go on overload
Documentation for this format

UML Diagram (Legend)

XML Template

```
<OperationDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this operation definition (globally unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the operation definition -->
 <name value="[string]"/><!-- 1..1 Name for this operation definition (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <kind value="[code]"/><!-- 1..1 operation | query -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the operation definition -->
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for operation definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this operation definition is defined -->
 <idempotent value="[boolean]"/><!-- 0..1 Whether content is unchanged by the operation -->
 <code value="[code]"/><!-- 1..1 Name used to invoke the operation -->
 <comment value="[string]"/><!-- 0..1 Additional information about use -->
 <base><!-- 0..1 Reference(OperationDefinition) Marks this as a profile of the base --></base>
 <resource value="[code]"/><!-- 0..* Types this operation applies to -->
 <system value="[boolean]"/><!-- 1..1 Invoke at the system level? -->
 <type value="[boolean]"/><!-- 1..1 Invole at the type level? -->
 <instance value="[boolean]"/><!-- 1..1 Invoke on an instance? -->
 <parameter>  <!-- 0..* Parameters for the operation/query -->
  <name value="[code]"/><!-- 1..1 Name in Parameters.parameter.name or in URL -->
  <use value="[code]"/><!-- 1..1 in | out -->
  <min value="[integer]"/><!-- 1..1 Minimum Cardinality -->
  <max value="[string]"/><!-- 1..1 Maximum Cardinality (a number or *) -->
  <documentation value="[string]"/><!-- 0..1 Description of meaning/use -->
  <type value="[code]"/><!--  0..1 What type this parameter has -->
  <searchType value="[code]"/><!--  0..1 number | date | string | token | reference | composite | quantity | uri -->
  <profile><!-- 0..1 Reference(StructureDefinition) Profile on the type --></profile>
  <binding>  <!-- 0..1 ValueSet details if this is coded -->
   <strength value="[code]"/><!-- 1..1 required | extensible | preferred | example -->
   <valueSet[x]><!-- 1..1 uri|Reference(ValueSet) Source of value set --></valueSet[x]>
  </binding>
  <part><!--  0..* Content as for OperationDefinition.parameter Parts of a nested Parameter --></part>
 </parameter>
 <overload>  <!-- 0..* Define overloaded variants for when  generating code -->
  <parameterName value="[string]"/><!-- 0..* Name of parameter to include in overload -->
  <comment value="[string]"/><!-- 0..1 Comments to go on overload -->
 </overload>
</OperationDefinition>
```

JSON Template

```
{
  "resourceType" : "OperationDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this operation definition (globally unique)
  "version" : "<string>", // Business version of the operation definition
  "name" : "<string>", // R!  Name for this operation definition (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "kind" : "<code>", // R!  operation | query
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the operation definition
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for operation definition (if applicable)
  "purpose" : "<markdown>", // Why this operation definition is defined
  "idempotent" : <boolean>, // Whether content is unchanged by the operation
  "code" : "<code>", // R!  Name used to invoke the operation
  "comment" : "<string>", // Additional information about use
  "base" : { Reference(OperationDefinition) }, // Marks this as a profile of the base
  "resource" : ["<code>"], // Types this operation applies to
  "system" : <boolean>, // R!  Invoke at the system level?
  "type" : <boolean>, // R!  Invole at the type level?
  "instance" : <boolean>, // R!  Invoke on an instance?
  "parameter" : [{ // Parameters for the operation/query
    "name" : "<code>", // R!  Name in Parameters.parameter.name or in URL
    "use" : "<code>", // R!  in | out
    "min" : <integer>, // R!  Minimum Cardinality
    "max" : "<string>", // R!  Maximum Cardinality (a number or *)
    "documentation" : "<string>", // Description of meaning/use
```

```
    "type" : "<code>", // C? What type this parameter has
    "searchType" : "<code>", // C? number | date | string | token | reference | composite | quantity | uri
    "profile" : { Reference(StructureDefinition) }, // Profile on the type
    "binding" : { // ValueSet details if this is coded
      "strength" : "<code>", // R!  required | extensible | preferred | example
      // valueSet[x]: Source of value set. One of these 2:
      "valueSetUri" : "<uri>"
      "valueSetReference" : { Reference(ValueSet) }
    },
    "part" : [{ Content as for OperationDefinition.parameter }] // C? Parts of a nested Parameter
  }],
  "overload" : [{ // Define overloaded variants for when  generating code
    "parameterName" : ["<string>"], // Name of parameter to include in overload
    "comment" : "<string>" // Comments to go on overload
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:OperationDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:OperationDefinition.url [ uri ]; # 0..1 Logical URI to reference this operation definition (globally unique)
  fhir:OperationDefinition.version [ string ]; # 0..1 Business version of the operation definition
  fhir:OperationDefinition.name [ string ]; # 1..1 Name for this operation definition (computer friendly)
  fhir:OperationDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:OperationDefinition.kind [ code ]; # 1..1 operation | query
  fhir:OperationDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:OperationDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:OperationDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:OperationDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:OperationDefinition.description [ markdown ]; # 0..1 Natural language description of the operation definition
  fhir:OperationDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:OperationDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for operation definition
     (if applicable)
  fhir:OperationDefinition.purpose [ markdown ]; # 0..1 Why this operation definition is defined
  fhir:OperationDefinition.idempotent [ boolean ]; # 0..1 Whether content is unchanged by the operation
  fhir:OperationDefinition.code [ code ]; # 1..1 Name used to invoke the operation
  fhir:OperationDefinition.comment [ string ]; # 0..1 Additional information about use
  fhir:OperationDefinition.base [ Reference(OperationDefinition) ]; # 0..1 Marks this as a profile of the base
  fhir:OperationDefinition.resource [ code ], ... ; # 0..* Types this operation applies to
  fhir:OperationDefinition.system [ boolean ]; # 1..1 Invoke at the system level?
  fhir:OperationDefinition.type [ boolean ]; # 1..1 Invole at the type level?
  fhir:OperationDefinition.instance [ boolean ]; # 1..1 Invoke on an instance?
  fhir:OperationDefinition.parameter [ # 0..* Parameters for the operation/query
    fhir:OperationDefinition.parameter.name [ code ]; # 1..1 Name in Parameters.parameter.name or in URL
    fhir:OperationDefinition.parameter.use [ code ]; # 1..1 in | out
    fhir:OperationDefinition.parameter.min [ integer ]; # 1..1 Minimum Cardinality
    fhir:OperationDefinition.parameter.max [ string ]; # 1..1 Maximum Cardinality (a number or *)
    fhir:OperationDefinition.parameter.documentation [ string ]; # 0..1 Description of meaning/use
    fhir:OperationDefinition.parameter.type [ code ]; # 0..1 What type this parameter has
    fhir:OperationDefinition.parameter.searchType [ code ]; # 0..1 number | date | string | token | reference |
    composite | quantity | uri
    fhir:OperationDefinition.parameter.profile [ Reference(StructureDefinition) ]; # 0..1 Profile on the type
    fhir:OperationDefinition.parameter.binding [ # 0..1 ValueSet details if this is coded
      fhir:OperationDefinition.parameter.binding.strength [ code ]; # 1..1 required | extensible | preferred | example
      # OperationDefinition.parameter.binding.valueSet[x] : 1..1 Source of value set. One of these 2
        fhir:OperationDefinition.parameter.binding.valueSetUri [ uri ]
        fhir:OperationDefinition.parameter.binding.valueSetReference [ Reference(ValueSet) ]
    ];
    fhir:OperationDefinition.parameter.part [ See OperationDefinition.parameter ], ... ; # 0..* Parts of a nested
    Parameter
  ], ...;
  fhir:OperationDefinition.overload [ # 0..* Define overloaded variants for when  generating code
    fhir:OperationDefinition.overload.parameterName [ string ], ... ; # 0..* Name of parameter to include in overload
    fhir:OperationDefinition.overload.comment [ string ]; # 0..1 Comments to go on overload
  ], ...;
]
```

Changes since DSTU2

OperationDefinition

OperationDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status

OperationDefinition.experimental | Now marked as Modifier

OperationDefinition.contact | Type changed from BackboneElement to ContactDetail

OperationDefinition.description | Type changed from string to markdown

OperationDefinition.useContext | Added Element

OperationDefinition.jurisdiction | Added Element

OperationDefinition.purpose | Renamed from requirements to purpose Type changed from string to markdown

OperationDefinition.comment | Renamed from notes to comment

OperationDefinition.resource | Renamed from type to resource

OperationDefinition.type | Min Cardinality changed from 0 to 1 Max Cardinality changed from * to 1 Type changed from code to boolean Remove Binding http://hl7.org/fhir/ValueSet/resource-types (required)
OperationDefinition.parameter.type | Change value set from http://hl7.org/fhir/ValueSet/operation-parameter-type to http://hl7.org/fhir/ValueSet/all-types
OperationDefinition.parameter.searchType | Added Element
OperationDefinition.overload | Added Element
OperationDefinition.overload.parameterName | Added Element
OperationDefinition.overload.comment | Added Element
OperationDefinition.contact.name | deleted
OperationDefinition.contact.telecom | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 18 tests that all execute ok. 1 fail round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
OperationDefinition | | | DomainResource | Definition of an operation or a named query Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this operation definition (globally unique)
version | ? | 0..1 | string | Business version of the operation definition
name | ? | 1..1 | string | Name for this operation definition (computer friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
kind | | 1..1 | code | operation | query OperationKind (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the operation definition
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for operation definition (if applicable) Jurisdiction ValueSet (Extensible)
purpose | | 0..1 | markdown | Why this operation definition is defined
idempotent | ? | 0..1 | boolean | Whether content is unchanged by the operation
code | ? | 1..1 | code | Name used to invoke the operation
comment | | 0..1 | string | Additional information about use
base | ? | 0..1 | Reference(OperationDefinition) | Marks this as a profile of the base
resource | ? | 0..* | code | Types this operation applies to ResourceType (Required)
system | ? | 1..1 | boolean | Invoke at the system level?
type | ? | 1..1 | boolean | Invole at the type level?
instance | ? | 1..1 | boolean | Invoke on an instance?
parameter | I | 0..* | BackboneElement | Parameters for the operation/query + Either a type must be provided, or parts + A search type can only be specified for parameters of type string
name | | 1..1 | code | Name in Parameters.parameter.name or in URL
use | | 1..1 | code | in | out OperationParameterUse (Required)
min | | 1..1 | integer | Minimum Cardinality
max | | 1..1 | string | Maximum Cardinality (a number or *)
documentation | | 0..1 | string | Description of meaning/use
type | I | 0..1 | code | What type this parameter has FHIRAllTypes (Required)
searchType | I | 0..1 | code | number | date | string | token | reference | composite | quantity | uri SearchParamType (Required)
profile | | 0..1 | Reference(StructureDefinition) | Profile on the type
binding | | 0..1 | BackboneElement | ValueSet details if this is coded
strength | | 1..1 | code | required | extensible | preferred | example BindingStrength (Required)
valueSet[x] | | 1..1 | | Source of value set
valueSetUri | | | uri
valueSetReference | | | Reference(ValueSet)
part | I | 0..* | see parameter | Parts of a nested Parameter
overload | | 0..* | BackboneElement | Define overloaded variants for when generating code
parameterName | | 0..* | string | Name of parameter to include in overload
comment | | 0..1 | string | Comments to go on overload
Documentation for this format

UML Diagram (Legend)

XML Template

```
<OperationDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this operation definition (globally unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the operation definition -->
 <name value="[string]"/><!-- 1..1 Name for this operation definition (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <kind value="[code]"/><!-- 1..1 operation | query -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
```

```
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the operation definition -->
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for operation definition (if applicable)
    --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this operation definition is defined -->
 <idempotent value="[boolean]"/><!-- 0..1 Whether content is unchanged by the operation -->
 <code value="[code]"/><!-- 1..1 Name used to invoke the operation -->
 <comment value="[string]"/><!-- 0..1 Additional information about use -->
 <base><!-- 0..1 Reference(OperationDefinition) Marks this as a profile of the base --></base>
 <resource value="[code]"/><!-- 0..* Types this operation applies to -->
 <system value="[boolean]"/><!-- 1..1 Invoke at the system level? -->
 <type value="[boolean]"/><!-- 1..1 Invole at the type level? -->
 <instance value="[boolean]"/><!-- 1..1 Invoke on an instance? -->
 <parameter>  <!-- 0..* Parameters for the operation/query -->
  <name value="[code]"/><!-- 1..1 Name in Parameters.parameter.name or in URL -->
  <use value="[code]"/><!-- 1..1 in | out -->
  <min value="[integer]"/><!-- 1..1 Minimum Cardinality -->
  <max value="[string]"/><!-- 1..1 Maximum Cardinality (a number or *) -->
  <documentation value="[string]"/><!-- 0..1 Description of meaning/use -->
  <type value="[code]"/><!--  0..1 What type this parameter has -->
  <searchType value="[code]"/><!--  0..1 number | date | string | token | reference | composite | quantity | uri -->
  <profile><!-- 0..1 Reference(StructureDefinition) Profile on the type --></profile>
  <binding>  <!-- 0..1 ValueSet details if this is coded -->
   <strength value="[code]"/><!-- 1..1 required | extensible | preferred | example -->
   <valueSet[x]><!-- 1..1 uri|Reference(ValueSet) Source of value set --></valueSet[x]>
  </binding>
  <part><!--  0..* Content as for OperationDefinition.parameter Parts of a nested Parameter --></part>
 </parameter>
 <overload>  <!-- 0..* Define overloaded variants for when  generating code -->
  <parameterName value="[string]"/><!-- 0..* Name of parameter to include in overload -->
  <comment value="[string]"/><!-- 0..1 Comments to go on overload -->
 </overload>
</OperationDefinition>
```

JSON Template

```
{
  "resourceType" : "OperationDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this operation definition (globally unique)
  "version" : "<string>", // Business version of the operation definition
  "name" : "<string>", // R!  Name for this operation definition (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "kind" : "<code>", // R!  operation | query
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the operation definition
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for operation definition (if applicable)
  "purpose" : "<markdown>", // Why this operation definition is defined
  "idempotent" : <boolean>, // Whether content is unchanged by the operation
  "code" : "<code>", // R!  Name used to invoke the operation
  "comment" : "<string>", // Additional information about use
  "base" : { Reference(OperationDefinition) }, // Marks this as a profile of the base
  "resource" : ["<code>"], // Types this operation applies to
  "system" : <boolean>, // R!  Invoke at the system level?
  "type" : <boolean>, // R!  Invole at the type level?
  "instance" : <boolean>, // R!  Invoke on an instance?
  "parameter" : [{ // Parameters for the operation/query
    "name" : "<code>", // R!  Name in Parameters.parameter.name or in URL
    "use" : "<code>", // R!  in | out
    "min" : <integer>, // R!  Minimum Cardinality
    "max" : "<string>", // R!  Maximum Cardinality (a number or *)
    "documentation" : "<string>", // Description of meaning/use
    "type" : "<code>", // C? What type this parameter has
    "searchType" : "<code>", // C? number | date | string | token | reference | composite | quantity | uri
    "profile" : { Reference(StructureDefinition) }, // Profile on the type
    "binding" : { // ValueSet details if this is coded
      "strength" : "<code>", // R!  required | extensible | preferred | example
      // valueSet[x]: Source of value set. One of these 2:
      "valueSetUri" : "<uri>"
      "valueSetReference" : { Reference(ValueSet) }
    },
    "part" : [{ Content as for OperationDefinition.parameter }] // C? Parts of a nested Parameter
  }],
  "overload" : [{ // Define overloaded variants for when  generating code
    "parameterName" : ["<string>"], // Name of parameter to include in overload
    "comment" : "<string>" // Comments to go on overload
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:OperationDefinition;
```

```
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:OperationDefinition.url [ uri ]; # 0..1 Logical URI to reference this operation definition (globally unique)
  fhir:OperationDefinition.version [ string ]; # 0..1 Business version of the operation definition
  fhir:OperationDefinition.name [ string ]; # 1..1 Name for this operation definition (computer friendly)
  fhir:OperationDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:OperationDefinition.kind [ code ]; # 1..1 operation | query
  fhir:OperationDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:OperationDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:OperationDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:OperationDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:OperationDefinition.description [ markdown ]; # 0..1 Natural language description of the operation definition
  fhir:OperationDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:OperationDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for operation definition
     (if applicable)
  fhir:OperationDefinition.purpose [ markdown ]; # 0..1 Why this operation definition is defined
  fhir:OperationDefinition.idempotent [ boolean ]; # 0..1 Whether content is unchanged by the operation
  fhir:OperationDefinition.code [ code ]; # 1..1 Name used to invoke the operation
  fhir:OperationDefinition.comment [ string ]; # 0..1 Additional information about use
  fhir:OperationDefinition.base [ Reference(OperationDefinition) ]; # 0..1 Marks this as a profile of the base
  fhir:OperationDefinition.resource [ code ], ... ; # 0..* Types this operation applies to
  fhir:OperationDefinition.system [ boolean ]; # 1..1 Invoke at the system level?
  fhir:OperationDefinition.type [ boolean ]; # 1..1 Invole at the type level?
  fhir:OperationDefinition.instance [ boolean ]; # 1..1 Invoke on an instance?
  fhir:OperationDefinition.parameter [ # 0..* Parameters for the operation/query
    fhir:OperationDefinition.parameter.name [ code ]; # 1..1 Name in Parameters.parameter.name or in URL
    fhir:OperationDefinition.parameter.use [ code ]; # 1..1 in | out
    fhir:OperationDefinition.parameter.min [ integer ]; # 1..1 Minimum Cardinality
    fhir:OperationDefinition.parameter.max [ string ]; # 1..1 Maximum Cardinality (a number or *)
    fhir:OperationDefinition.parameter.documentation [ string ]; # 0..1 Description of meaning/use
    fhir:OperationDefinition.parameter.type [ code ]; # 0..1 What type this parameter has
    fhir:OperationDefinition.parameter.searchType [ code ]; # 0..1 number | date | string | token | reference |
    composite | quantity | uri
    fhir:OperationDefinition.parameter.profile [ Reference(StructureDefinition) ]; # 0..1 Profile on the type
    fhir:OperationDefinition.parameter.binding [ # 0..1 ValueSet details if this is coded
      fhir:OperationDefinition.parameter.binding.strength [ code ]; # 1..1 required | extensible | preferred | example
      # OperationDefinition.parameter.binding.valueSet[x] : 1..1 Source of value set. One of these 2
        fhir:OperationDefinition.parameter.binding.valueSetUri [ uri ]
        fhir:OperationDefinition.parameter.binding.valueSetReference [ Reference(ValueSet) ]
    ];
    fhir:OperationDefinition.parameter.part [ See OperationDefinition.parameter ], ... ; # 0..* Parts of a nested
    Parameter
  ], ...;
  fhir:OperationDefinition.overload [ # 0..* Define overloaded variants for when  generating code
    fhir:OperationDefinition.overload.parameterName [ string ], ... ; # 0..* Name of parameter to include in overload
    fhir:OperationDefinition.overload.comment [ string ]; # 0..1 Comments to go on overload
  ], ...;
]
```

Changes since DSTU2

OperationDefinition

OperationDefinition.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status

OperationDefinition.experimental | Now marked as Modifier

OperationDefinition.contact | Type changed from BackboneElement to ContactDetail

OperationDefinition.description | Type changed from string to markdown

OperationDefinition.useContext | Added Element

OperationDefinition.jurisdiction | Added Element

OperationDefinition.purpose | Renamed from requirements to purpose Type changed from string to markdown

OperationDefinition.comment | Renamed from notes to comment

OperationDefinition.resource | Renamed from type to resource

OperationDefinition.type | Min Cardinality changed from 0 to 1 Max Cardinality changed from * to 1 Type changed from code to boolean Remove Binding http://hl7.org/fhir/ValueSet/resource-types (required)

OperationDefinition.parameter.type | Change value set from http://hl7.org/fhir/ValueSet/operation-parameter-type to http://hl7.org/fhir/ValueSet/all-types

OperationDefinition.parameter.searchType | Added Element

OperationDefinition.overload | Added Element

OperationDefinition.overload.parameterName | Added Element

OperationDefinition.overload.comment | Added Element

OperationDefinition.contact.name | deleted

OperationDefinition.contact.telecom | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 18 tests that all execute ok. 1 fail round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

5.4.3.1 Terminology Bindings

Path | Definition | Type | Reference

OperationDefinition.status | The lifecycle status of a Value Set or Concept Map. | Required | PublicationStatus
OperationDefinition.kind | Whether an operation is a normal operation or a query. | Required | OperationKind
OperationDefinition.jurisdiction | Countries and regions within which this artifact is targeted for use | Extensible | Jurisdiction ValueSet
OperationDefinition.resource | One of the resource types defined as part of FHIR. | Required | Resource Types
OperationDefinition.parameter.use | Whether an operation parameter is an input or an output parameter. | Required | OperationParameterUse
OperationDefinition.parameter.type | Either an abstract type, a resource or a data type. | Required | All Types
OperationDefinition.parameter.searchType | Data types allowed to be used for search parameters. | Required | SearchParamType
OperationDefinition.parameter.binding.strength | Indication of the degree of conformance expectations associated with a binding. | Required | BindingStrength

5.4.3.2 Constraints

* opd-1: On OperationDefinition.parameter: Either a type must be provided, or parts (expression on OperationDefinition.parameter: type.exists() or part.exists())

* opd-2: On OperationDefinition.parameter: A search type can only be specified for parameters of type string (expression on OperationDefinition.parameter: searchType implies type = 'string')

5.4.3.3 Operations defined as part of this Specification

Base Operations (All resource types)
Validate a resource | [base]/[Resource]/$validate | [base]/[Resource]/[id]/$validate
Access a list of profiles, tags, and security labels | [base]/$meta | [base]/[Resource]/$meta | [base]/[Resource]/[id]/$meta
Add profiles, tags, and security labels to a resource | [base]/[Resource]/[id]/$meta-add
Delete profiles, tags, and security labels for a resource | [base]/[Resource]/[id]/$meta-delete
Operations Defined by Resource Types
Apply | [base]/ActivityDefinition/[id]/$apply
Data Requirements | [base]/ActivityDefinition/[id]/$data-requirements
Fetch a subset of the CapabilityStatement resource | [base]/CapabilityStatement/$subset | [base]/CapabilityStatement/[id]/$subset
Test if a server implements a client's required operations | [base]/CapabilityStatement/$implements | [base]/CapabilityStatement/[id]/$implements
Test if a server implements a client's required operations | [base]/CapabilityStatement/$conforms
Concept Look Up & Decomposition | [base]/CodeSystem/$lookup
Subsumption Testing | [base]/CodeSystem/$subsumes | [base]/CodeSystem/[id]/$subsumes
Code Composition based on supplied properties | [base]/CodeSystem/$compose | [base]/CodeSystem/[id]/$compose
Generate a Document | [base]/Composition/$document | [base]/Composition/[id]/$document
Concept Translation | [base]/ConceptMap/$translate | [base]/ConceptMap/[id]/$translate
Closure Table Maintenance | [base]/$closure
Fetch Encounter Record | [base]/Encounter/[id]/$everything
Data Requirements | [base]/$data-requirements | [base]/Library/[id]/$data-requirements
Find a functional list | [base]/List/$find
Evaluate Measure | [base]/Measure/$evaluate-measure | [base]/Measure/[id]/$evaluate-measure
Data Requirements | [base]/Measure/[id]/$data-requirements
Process Message | [base]/$process-message
Observation Statistics | [base]/Observation/$stats
Last N Observations Query | [base]/Observation/$lastn
Find patient matches using MPI based logic | [base]/Patient/$match
Fetch Patient Record | [base]/Patient/$everything | [base]/Patient/[id]/$everything
Apply | [base]/PlanDefinition/[id]/$apply
Data Requirements | [base]/PlanDefinition/[id]/$data-requirements
Populate Questionnaire | [base]/Questionnaire/$populate | [base]/Questionnaire/[id]/$populate
Generate HTML for Questionnaire | [base]/Questionnaire/$populatehtml | [base]/Questionnaire/[id]/$populatehtml
Generate a link to a Questionnaire completion webpage | [base]/Questionnaire/$populatelink | [base]/Questionnaire/[id]/$populatelink
Evaluate | [base]/ServiceDefinition/[id]/$evaluate
Data Requirements | [base]/ServiceDefinition/[id]/$data-requirements
Build Questionnaire | [base]/StructureDefinition/$questionnaire | [base]/StructureDefinition/[id]/$questionnaire
Model Instance Transformation | [base]/StructureMap/$transform | [base]/StructureMap/[id]/$transform
Value Set Expansion | [base]/ValueSet/$expand | [base]/ValueSet/[id]/$expand
Value Set based Validation | [base]/ValueSet/$validate-code | [base]/ValueSet/[id]/$validate-code

5.4.3.4 Executing Operations and Named Queries

Operations are executed by POSTing to a URL that is defined by the operation definition. Named queries are executed by performing a search with the value of the search parameter "_query" set to the name provided in the definition.

If the named query is to be performed over the RESTful API, all the parameters must be simple search parameters, so that they can be represented directly in the URL without tricky encoding issues. Named queries always return a bundle containing a set of resources, so all the out parameters must be resources, not data types.

5.4.3.5 Passing Resources to Operations

There are two ways to pass resources to an operation: directly or by reference. The definition of an operation distinguishes between these two, since they have very different behaviors and consequences.

As an example, take the Questionnaire.$populate operation. This operation takes a questionnaire as a direct parameter. The type of the parameter is defined as 'Questionnaire'. In a Parameters resource, it would be represented like this:

```
<parameter>
```

```
    <name value="questionnaire"/>
    <resource>
      <Questionnaire>
        <!-- Questionnaire contents -->
      </Questionnaire>
    </resource>
  </parameter>
```

or, in JSON:

```
 "parameter": [
    {
      "name": "questionnaire",
      "resource": {
        "resourceType": "Questionnaire",
        // Questionnaire contents
      }
    }
  ]
```

Other parameters are passed by reference. For example, $populate takes a set of references to other resources that should be used to pre-populate the questionnaire. These are passed by reference, and the type of the parameter is Reference(Any) - a reference to any kind of resource. In a parameters resource, it would be represented like this:

```
  <parameter>
    <name value="content"/>
    <valueReference>
      <reference value="Patient/123">
      </Questionnaire>
    </valueReference>
  </parameter>
```

or, in JSON:

```
 "parameter": [
    {
      "name": "questionnaire",
      "valueReference" : {
        "reference" : "Patient/123"
      }
    }
  ]
```

Some operations can take either form; in that case, two distinct parameters must be defined, one for a resource as a direct parameter, and one for a reference.

5.4.3.6 Renaming OperationDefinition.name

It is possible for two different organizations to create different operation definitions with the same name or, perhaps more likely, to define equivalent operations that have the same name but incompatible approaches in their parameter lists.

It is also possible, though unlikely, that a server will be required to support both of these operations. If this is the case, the server is able to do this by giving one of them a new name and referring to it by definition in the capability statement. To illustrate this, assume that two different organizations, "orgA" and "orgB", both define an operation called "dothis", and the definitions are incompatible. OrgA publishes its operation definition at http://orga.com/fhir/dothis.xml, and OrgB publishes its operation at http://fhir.orgb.com/meta/OperationDefinition/dothis. The server is able to implement both. Its capability statement will say:

```
<CapabilityStatement xmlns="http://hl7.org/fhir">
  <!-- snip -->
  <rest>
    <!-- snip -->
    <operation>
      <name value="dothis"/>
      <definition>
        <reference value="http://orga.com/fhir/dothis.xml"/>
      </definition>
    </operation>
    <operation>
      <name value="dothis2"/>
      <definition>
        <reference value="http://fhir.orgb.com/meta/OperationDefinition/dothis"/>
      </definition>
    </operation>
    <!-- snip -->
  </rest>
  <!-- snip -->
</CapabilityStatement>
```

If a general purpose cross server client is looking for the implementation of the http://fhir.orgb.com/meta/OperationDefinition/dothis operation and wants to be robust against this name clash problem, instead of simply executing the $dothis operation, it can look at the server's CapabilityStatement for the underlying definition URI and then execute the name given in the capability statement.

5.4.3.7 Determining System Compatibility

A client can determine the compatibility of the server by processing its capability statement and ensuring that the server implements the specific operation definitions and parameters required by the client. The client can then report a useful error

to the user rather than allowing mystifying operational errors to occur.

However, there are fundamental limitations to this approach because there are many aspects of these operations that are not (or cannot be) defined in a formal fashion using OperationDefinition, for example, co-occurrence constraints among parameters.

In the same sense, a 3rd party tool can examine a server's CapabilityStatement and a client's definition of an acceptable server to determine whether those two system are interoperable or not.

5.4.3.8 Dynamically Generating Forms

Finally, it is possible to generate user interface forms automatically from the OperationDefinition. The documentation in the OperationDefinition.description and OperationDefinition.parameter.documentation should be sufficient to allow moderately technical users to guess at the correct content of the form. For this reason, highly technical documentation should go in OperationDefinition.comment.

It is anticipated that this would be used to automate development processes, rather than to generate end-user forms: such users will usually need more support than can be offered in a generated form.

5.4.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
base | reference | Marks this as a profile of the base | OperationDefinition.base (OperationDefinition)
code | token | Name used to invoke the operation | OperationDefinition.code
date | date | The operation definition publication date | OperationDefinition.date
description | string | The description of the operation definition | OperationDefinition.description
instance | token | Invoke on an instance? | OperationDefinition.instance
jurisdiction | token | Intended jurisdiction for the operation definition | OperationDefinition.jurisdiction
kind | token | operation | query | OperationDefinition.kind
name | string | Computationally friendly name of the operation definition | OperationDefinition.name
param-profile | reference | Profile on the type | OperationDefinition.parameter.profile (StructureDefinition)
publisher | string | Name of the publisher of the operation definition | OperationDefinition.publisher
status | token | The current status of the operation definition | OperationDefinition.status
system | token | Invoke at the system level? | OperationDefinition.system
type | token | Invole at the type level? | OperationDefinition.type
url | uri | The uri that identifies the operation definition | OperationDefinition.url
version | token | The business version of the operation definition | OperationDefinition.version

5.4.5 Resource OperationDefinition - Examples

URL: https://hl7.org/fhir/STU3/operationdefinition-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.4.5 Resource OperationDefinition - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

OperationDefinition example | example | XML | JSON | Turtle
ValueSet Expansion | ValueSet-expand | XML | JSON | Turtle
ValueSet Validation | ValueSet-validate-code | XML | JSON | Turtle
ValueSet Lookup | CodeSystem-lookup | XML | JSON | Turtle
ConceptMap Translate | ConceptMap-translate | XML | JSON | Turtle
ConceptMap Closure | ConceptMap-closure | XML | JSON | Turtle
Questionnaire Populate | Questionnaire-populate | XML | JSON | Turtle
Library data requirements | Library-data-requirements | XML | JSON | Turtle
Measure data requirements | Measure-data-requirements | XML | JSON | Turtle
Measure evaluate measure | Measure-evaluate-measure | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.4.6 Resource OperationDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/operationdefinition-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.4.6 Resource OperationDefinition - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 4 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the OperationDefinition resource.

OperationDefinition

Definition | A formal computable definition of an operation (on the RESTful interface) or a named query (using the search interaction).

Control | 1..1

OperationDefinition.url

Definition | An absolute URI that is used to identify this operation definition when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this operation definition is (or will be) published. The URL SHOULD include the major version of the operation definition. For more information see Technical and Business Versions.

Control | 0..1

Type | uri

Requirements | Allows the operation definition to be referenced by a single globally unique identifier.

Summary | true

Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.

OperationDefinition.version

Definition | The identifier that is used to identify this version of the operation definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the operation definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.

Note | This is a business versionId, not a resource version id (see discussion)

Control | 0..1

Type | string

Summary | true

Comments | There may be different operation definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the operation definition with the format [url]|[version].

OperationDefinition.name

Definition | A natural language name identifying the operation definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.

Control | 1..1

Type | string

Requirements | Support human navigation and code generation.

Summary | true

Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.

OperationDefinition.status

Definition | The status of this operation definition. Enables tracking the life-cycle of the content.

Control | 1..1

Terminology Binding | PublicationStatus (Required)

Type | code

Is Modifier | true

Summary | true

Comments | Allows filtering of operation definitions that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired operation definition without due consideration.

OperationDefinition.kind

Definition | Whether this is an operation or a named query.

Control | 1..1

Terminology Binding | OperationKind (Required)

Type | code

Comments | Named queries are invoked differently, and have different capabilities.

OperationDefinition.experimental

Definition | A boolean value to indicate that this operation definition is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.

Control | 0..1

Type | boolean

Is Modifier | true

Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level operation definition.

Summary | true

Comments | Allows filtering of operation definition that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental operation definition in production.

OperationDefinition.date

Definition | The date (and optionally time) when the operation definition was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the operation definition changes.

Control | 0..1

Type | dateTime

Alternate Names | Revision Date

Summary | true

Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the operation definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.

OperationDefinition.publisher

Definition | The name of the individual or organization that published the operation definition.

Control | 0..1

Type | string

Requirements | Helps establish the "authority/credibility" of the operation definition. May also allow for contact.
Summary | true
Comments | Usually an organization, but may be an individual. The publisher (or steward) of the operation definition is the organization or individual primarily responsible for the maintenance and upkeep of the operation definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the operation definition. This item SHOULD be populated unless the information is available from context.
OperationDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
OperationDefinition.description
Definition | A free text natural language description of the operation definition from a consumer's perspective.
Control | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the operation definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the operation definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created).
OperationDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate operation definition instances.
Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
OperationDefinition.jurisdiction
Definition | A legal or geographic region in which the operation definition is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the operation definition to be used in jurisdictions other than those for which it was originally designed or intended.
OperationDefinition.purpose
Definition | Explaination of why this operation definition is needed and why it has been designed as it has.
Control | 0..1
Type | markdown
Comments | This element does not describe the usage of the operation definition Instead it provides traceability of "why" the resource is either needed or "why" it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this operation definition.
OperationDefinition.idempotent
Definition | Operations that are idempotent (see HTTP specification definition of idempotent) may be invoked by performing an HTTP GET operation instead of a POST.
Control | 0..1
Type | boolean
Summary | true
OperationDefinition.code
Definition | The name used to invoke the operation.
Control | 1..1
Type | code
Summary | true
Comments | The code does not include the '$' prefix that is always included in the URL when the operation is invoked.
OperationDefinition.comment
Definition | Additional information about how to use this operation or named query.
Control | 0..1
Type | string
OperationDefinition.base
Definition | Indicates that this operation definition is a constraining profile on the base.
Control | 0..1
Type | Reference(OperationDefinition)
Summary | true
Comments | A constrained profile can make optional parameters required or not used and clarify documentation.
OperationDefinition.resource
Definition | The types on which this operation can be executed.
Control | 0..*
Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
Comments | If the type is an abstract resource ("Resource" or "DomainResource") then the operation can be invoked on any concrete specialization.
OperationDefinition.system

Definition | Indicates whether this operation or named query can be invoked at the system level (e.g. without needing to choose a resource type for the context).
Control | 1..1
Type | boolean
Summary | true
OperationDefinition.type
Definition | Indicates whether this operation or named query can be invoked at the resource type level for any given resource type level (e.g. without needing to choose a specific resource id for the context).
Control | 1..1
Type | boolean
Summary | true
OperationDefinition.instance
Definition | Indicates whether this operation can be invoked on a particular instance of one of the given types.
Control | 1..1
Type | boolean
Summary | true
OperationDefinition.parameter
Definition | The parameters for the operation/query.
Control | 0..*
Comments | Query Definitions only have one output parameter, named "result". This may not be described, but can be to allow a profile to be defined.
Invariants | Defined on this element opd-1: Either a type must be provided, or parts (expression : type.exists() or part.exists(), xpath: exists(f:type) or exists(f:part)) opd-2: A search type can only be specified for parameters of type string (expression : searchType implies type = 'string', xpath: not(exists(f:searchType)) or (f:type/@value = 'string'))
OperationDefinition.parameter.name
Definition | The name of used to identify the parameter.
Control | 1..1
Type | code
Comments | This name must be a token (start with a letter in a..z, and only contain letters, numerals, and underscore. Note that for search parameters (type = string, with a search type), the name may be altered by the search modifiers.
OperationDefinition.parameter.use
Definition | Whether this is an input or an output parameter.
Control | 1..1
Terminology Binding | OperationParameterUse (Required)
Type | code
Comments | If a parameter name is used for both an input and an output parameter, the parameter should be defined twice.
OperationDefinition.parameter.min
Definition | The minimum number of times this parameter SHALL appear in the request or response.
Control | 1..1
Type | integer
OperationDefinition.parameter.max
Definition | The maximum number of times this element is permitted to appear in the request or response.
Control | 1..1
Type | string
OperationDefinition.parameter.documentation
Definition | Describes the meaning or use of this parameter.
Control | 0..1
Type | string
OperationDefinition.parameter.type
Definition | The type for this parameter.
Control | 0..1
Terminology Binding | Any defined Resource or Data Type name (including "Any" and "Type")
Type | code
Comments | if there is no stated parameter, then the parameter is a multi-part parameter; type and must have at least one part defined.
Invariants | Affect this element opd-1: Either a type must be provided, or parts (expression : type.exists() or part.exists(), xpath: exists(f:type) or exists(f:part))
OperationDefinition.parameter.searchType
Definition | How the parameter is understood as a search parameter. This is only used if the parameter type is 'string'.
Control | 0..1
Terminology Binding | SearchParamType (Required)
Type | code
Invariants | Affect this element opd-2: A search type can only be specified for parameters of type string (expression : searchType implies type = 'string', xpath: not(exists(f:searchType)) or (f:type/@value = 'string'))
OperationDefinition.parameter.profile
Definition | A profile the specifies the rules that this parameter must conform to.
Control | 0..1
Type | Reference(StructureDefinition)
OperationDefinition.parameter.binding
Definition | Binds to a value set if this parameter is coded (code, Coding, CodeableConcept).
Control | 0..1
OperationDefinition.parameter.binding.strength
Definition | Indicates the degree of conformance expectations associated with this binding - that is, the degree to which the provided value set must be adhered to in the instances.

Control | 1..1
Terminology Binding | BindingStrength (Required)
Type | code
Alternate Names | conformance; extensibility
Comments | For further discussion, see Using Terminologies.
OperationDefinition.parameter.binding.valueSet[x]
Definition | Points to the value set or external definition (e.g. implicit value set) that identifies the set of codes to be used.
Control | 1..1
Type | uri|Reference(ValueSet)
[x] Note | See Choice of Data Types for further information about how to use [x]
Comments | For value sets with a referenceResource, the display can contain the value set description. The reference may be version-specific or not.
OperationDefinition.parameter.part
Definition | The parts of a nested Parameter.
Control | 0..*
Type | See OperationDefinition.parameter
Comments | Query Definitions only have one output parameter, named "result". This may not be described, but can be to allow a profile to be defined.
Invariants | Affect this element opd-1: Either a type must be provided, or parts (expression : type.exists() or part.exists(), xpath: exists(f:type) or exists(f:part))
OperationDefinition.overload
Definition | Defines an appropriate combination of parameters to use when invoking this operation, to help code generators when generating overloaded parameter sets for this operation.
Control | 0..*
Comments | The combinations are suggestions as to which sets of parameters to use together, but the combinations are not intended to be authoritative.
OperationDefinition.overload.parameterName
Definition | Name of parameter to include in overload.
Control | 0..*
Type | string
OperationDefinition.overload.comment
Definition | Comments to go on overload.
Control | 0..1
Type | string

5.4.7 Resource OperationDefinition - Mappings

URL: https://hl7.org/fhir/STU3/operationdefinition-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.4.7 Resource OperationDefinition - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Workflow Mapping

Mappings for the operationdefinition resource.

5.4.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

OperationDefinition | n/a

5.4.7.2 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

OperationDefinition | Definition
url | Definition.url
version | Definition.version
name | Definition.title
status | Definition.status (different ValueSet)
kind
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
idempotent
code
comment
base | Definition.definition

resource | Definition.subjectCodableConcept
system
type
instance
parameter
name
use
min
max
documentation
type
searchType
profile
binding
strength
valueSet[x]
part
overload
parameterName
comment

5.4.8 Resource OperationDefinition - Design Notes

URL: https://hl7.org/fhir/STU3/operationdefinition-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.4.8 Resource OperationDefinition - Design Notes

to do

5.4.9 Resource OperationDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/operationdefinition-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.4.9 Resource OperationDefinition - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the OperationDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

5.4.9.1 Profiles

No Profiles defined for this resource

5.4.9.2 Extensions

resource-approvalDate | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | lastReviewDate | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

5.4.9.3 Search Extensions

No Search Extensions defined for this resource

5.4.10 Resource OperationDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/operationdefinition-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.4.10 Resource OperationDefinition - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for OperationDefinition.

Functional status for this map: 18 tests that all execute ok. 1 fail round-trip testing and all r3 resources are valid. (see documentation)

5.4.10.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/OperationDefinition2to3" = "R2 to R3 Conversions for OperationDefinition"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/OperationDefinition" alias OperationDefinitionR2 as source
uses "http://hl7.org/fhir/StructureDefinition/OperationDefinition" alias OperationDefinition as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  OperationDefinition extends DomainResource
  input src : OperationDefinitionR2 as source
  input tgt : OperationDefinition as target

  "OperationDefinition-url" : for src.url make tgt.url
  "OperationDefinition-version" : for src.version make tgt.version
  "OperationDefinition-name" : for src.name make tgt.name
  "OperationDefinition-status" : for src.status make tgt.status
  "OperationDefinition-kind" : for src.kind make tgt.kind
  "OperationDefinition-experimental" : for src.experimental make tgt.experimental
  "OperationDefinition-date" : for src.date make tgt.date
  "OperationDefinition-publisher" : for src.publisher make tgt.publisher
  "OperationDefinition-contact" : for src.contact as vs0 make tgt.contact as vt0 then opdContact(vs0, vt0)
  "OperationDefinition-idempotent" : for src.idempotent make tgt.idempotent
  "OperationDefinition-code" : for src.code make tgt.code
  "OperationDefinition-comment" : for src.notes make tgt.comment
  "OperationDefinition-requirements" : for src.requirements make tgt.purpose
  "OperationDefinition-description" : for src.description make tgt.description
  "OperationDefinition-base" : for src.base make tgt.base
  "OperationDefinition-system" : for src.system make tgt.system
  "OperationDefinition-type1" : for src.type make tgt.resource
  "OperationDefinition-type2" : for src where type.exists() make tgt.type = true
  "OperationDefinition-type2" : for src where type.empty() make tgt.type = false
  "OperationDefinition-instance" : for src.instance make tgt.instance
  "OperationDefinition-parameter" : for src.parameter as vs0 make tgt.parameter as vt0 then opParam(vs0, vt0)
endgroup

group opdContact extends Element
  input src as source
  input tgt as target

  "OperationDefinition.contact-name" : for src.name make tgt.name
  "OperationDefinition.contact-telecom" : for src.telecom make tgt.telecom
endgroup

group opParam extends BackboneElement
  input src as source
  input tgt as target

  "OperationDefinition.parameter-name" : for src.name make tgt.name
  "OperationDefinition.parameter-use" : for src.use make tgt.use
  "OperationDefinition.parameter-min" : for src.min make tgt.min
  "OperationDefinition.parameter-max" : for src.max make tgt.max
  "OperationDefinition.parameter-documentation" : for src.documentation make tgt.documentation
  "OperationDefinition.parameter-type1" : for src.type as v where ($this in ('number' | 'date' | 'string' | 'token' |
    'reference' | 'composite' | 'quantity' | 'uri')).not() make tgt.type = v
  "OperationDefinition.parameter-type2" : for src.type as v where ($this in ('number' | 'date' | 'string' | 'token' |
    'reference' | 'composite' | 'quantity' | 'uri')) make tgt.searchType = v, tgt.type = 'string'
  "OperationDefinition.parameter-profile" : for src.profile make tgt.profile
  "OperationDefinition.parameter-part" : for src.part as ps0 make tgt.part as pt0 then opPart(ps0, pt0)
endgroup

group opPart extends BackboneElement
  input src as source
  input tgt as target

  "OperationDefinition.parameter-part-name" : for src.name make tgt.name
  "OperationDefinition.parameter-part-use" : for src.use make tgt.use
  "OperationDefinition.parameter-part-min" : for src.min make tgt.min
  "OperationDefinition.parameter-part-max" : for src.max make tgt.max
  "OperationDefinition.parameter-part-documentation" : for src.documentation make tgt.documentation
  "OperationDefinition.parameter-part-type" : for src.type make tgt.type
  "OperationDefinition.parameter-part-profile" : for src.profile make tgt.profile
  "OperationDefinition.parameter-part-part" : for src.part as psrc make tgt.part as ptgt then opPart(psrc, ptgt)
endgroup
```

5.4.10.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/OperationDefinition3to2" = "R3 to R2 Conversion for OperationDefinition"

uses "http://hl7.org/fhir/StructureDefinition/OperationDefinition" alias OperationDefinition as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/OperationDefinition" alias OperationDefinitionR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  OperationDefinition extends DomainResource
  input src : OperationDefinition as source
  input tgt : OperationDefinitionR2 as target

  "OperationDefinition-url" : for src.url make tgt.url
  "OperationDefinition-version" : for src.version make tgt.version
```

```
  "OperationDefinition-name" : for src.name make tgt.name
  "OperationDefinition-status" : for src.status make tgt.status
  "OperationDefinition-kind" : for src.kind make tgt.kind
  "OperationDefinition-experimental" : for src.experimental make tgt.experimental
  "OperationDefinition-publisher" : for src.publisher make tgt.publisher
  "OperationDefinition-contact" : for src.contact as vs0 make tgt.contact as vt0 then opdContact(vs0, vt0)
  "OperationDefinition-date" : for src.date make tgt.date
  "OperationDefinition-requirements" : for src.purpose make tgt.requirements
  "OperationDefinition-description" : for src.description make tgt.description
  "OperationDefinition-idempotent" : for src.idempotent make tgt.idempotent
  "OperationDefinition-code" : for src.code make tgt.code
  "OperationDefinition-notes" : for src.comment make tgt.notes
  "OperationDefinition-base" : for src.base make tgt.base
  "OperationDefinition-system" : for src.system make tgt.system
  "OperationDefinition-type" : for src.resource make tgt.type
  "OperationDefinition-instance" : for src.instance make tgt.instance
  "OperationDefinition-parameter" : for src.parameter as vs0 make tgt.parameter as vt0 then opdParameter(vs0, vt0)
endgroup

group opdContact extends Element
  input src as source
  input tgt as target

  "OperationDefinition.contact-name" : for src.name make tgt.name
  "OperationDefinition.contact-telecom" : for src.telecom make tgt.telecom
endgroup

group opdParameter extends BackboneElement
  input src as source
  input tgt as target

  "OperationDefinition.parameter-name" : for src.name make tgt.name
  "OperationDefinition.parameter-use" : for src.use make tgt.use
  "OperationDefinition.parameter-min" : for src.min make tgt.min
  "OperationDefinition.parameter-max" : for src.max make tgt.max
  "OperationDefinition.parameter-documentation" : for src.documentation make tgt.documentation
  "OperationDefinition.parameter-type" : for src.type where src.searchType.empty() make tgt.type
  "OperationDefinition.parameter-type" : for src.searchType make tgt.type
  "OperationDefinition.parameter-profile" : for src.profile make tgt.profile
  "OperationDefinition.parameter-part" : for src.part as ps0 make tgt.part as pt0 then opdPart(ps0, pt0)
endgroup

group opdPart extends BackboneElement
  input src as source
  input tgt as target

  "OperationDefinition.parameter-part-name" : for src.name make tgt.name
  "OperationDefinition.parameter-part-use" : for src.use make tgt.use
  "OperationDefinition.parameter-part-min" : for src.min make tgt.min
  "OperationDefinition.parameter-part-max" : for src.max make tgt.max
  "OperationDefinition.parameter-part-documentation" : for src.documentation make tgt.documentation
  "OperationDefinition.parameter-part-type" : for src.type make tgt.type
  "OperationDefinition.parameter-part-profile" : for src.profile make tgt.profile
  "OperationDefinition.parameter-part-part" : for src.part as psrc make tgt.part as ptgt then opdPart(psrc, ptgt)
endgroup
```

5.4.10.3 R3 Validation Errors

5.5 Resource SearchParameter - Content

URL: https://hl7.org/fhir/STU3/searchparameter.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

A search parameter that defines a named search item that can be used to search/filter on a resource.

5.5.1 Scope and Usage

A SearchParameter resource specifies a search parameter that may be used on the RESTful API to search or filter on a resource. The SearchParameter resource declares:

* how to refer to the search parameter from a client

* how the search parameter is to be understood by the server

* where in the source resource the parameter matches

5.5.2 Boundaries and Relationships

* Search Parameters are referred to by CapabilityStatement resources via the canonical URL for a search parameter (CapabilityStatement.rest.resource.searchParam.definition)

* Search Parameters can appear in an Implementation Guide to specify how resources can be found

5.5.3 Background and Context

Implementers should be familiar with the background and concepts described in Search on the RESTful API before working with this resource.

There is a registry of all Search Parameters.

5.5.4 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
SearchParameter | I | | DomainResource | Search Parameter for a resource + If an xpath is present, there SHALL be an xpathUsage + Search parameters can only have chain names when the search parameter type is 'reference' Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 1..1 | uri | Logical URI to reference this search parameter (globally unique)
version | ? | 0..1 | string | Business version of the search parameter
name | ? | 1..1 | string | Name for this search parameter (computer friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for search parameter (if applicable) Jurisdiction ValueSet (Extensible)
purpose | | 0..1 | markdown | Why this search parameter is defined
code | ? | 1..1 | code | Code used in URL
base | ? | 1..* | code | The resource type(s) this search parameter applies to ResourceType (Required)
type | ? | 1..1 | code | number | date | string | token | reference | composite | quantity | uri SearchParamType (Required)
derivedFrom | | 0..1 | uri | Original Definition for the search parameter
description | ? | 1..1 | markdown | Natural language description of the search parameter
expression | I | 0..1 | string | FHIRPath expression that extracts the values
xpath | I | 0..1 | string | XPath that extracts the values
xpathUsage | I | 0..1 | code | normal | phonetic | nearby | distance | other XPathUsageType (Required)
target | | 0..* | code | Types of resource (if a resource reference) ResourceType (Required)
comparator | | 0..* | code | eq | ne | gt | lt | ge | le | sa | eb | ap SearchComparator (Required)
modifier | | 0..* | code | missing | exact | contains | not | text | in | not-in | below | above | type SearchModifierCode (Required)
chain | | 0..* | string | Chained names supported
component | | 0..* | BackboneElement | For Composite resources to define the parts
definition | | 1..1 | Reference(SearchParameter) | Defines how the part works
expression | | 1..1 | string | Subexpression relative to main expression
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SearchParameter xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Logical URI to reference this search parameter (globally unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the search parameter -->
 <name value="[string]"/><!-- 1..1 Name for this search parameter (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for search parameter (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this search parameter is defined -->
 <code value="[code]"/><!-- 1..1 Code used in URL -->
 <base value="[code]"/><!-- 1..* The resource type(s) this search parameter applies to -->
 <type value="[code]"/><!-- 1..1 number | date | string | token | reference | composite | quantity | uri -->
 <derivedFrom value="[uri]"/><!-- 0..1 Original Definition for the search parameter -->
 <description value="[markdown]"/><!-- 1..1 Natural language description of the search parameter -->
 <expression value="[string]"/><!--  0..1 FHIRPath expression that extracts the values -->
 <xpath value="[string]"/><!--  0..1 XPath that extracts the values -->
 <xpathUsage value="[code]"/><!--  0..1 normal | phonetic | nearby | distance | other -->
 <target value="[code]"/><!-- 0..* Types of resource (if a resource reference) -->
 <comparator value="[code]"/><!-- 0..* eq | ne | gt | lt | ge | le | sa | eb | ap -->
 <modifier value="[code]"/><!-- 0..* missing | exact | contains | not | text | in | not-in | below | above | type -->
 <chain value="[string]"/><!-- 0..* Chained names supported -->
```

```
 <component>  <!-- 0..* For Composite resources to define the parts -->
  <definition><!-- 1..1 Reference(SearchParameter) Defines how the part works --></definition>
  <expression value="[string]"/><!-- 1..1 Subexpression relative to main expression -->
 </component>
</SearchParameter>
```

JSON Template

```
{
  "resourceType" : "SearchParameter",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Logical URI to reference this search parameter (globally unique)
  "version" : "<string>", // Business version of the search parameter
  "name" : "<string>", // R!  Name for this search parameter (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for search parameter (if applicable)
  "purpose" : "<markdown>", // Why this search parameter is defined
  "code" : "<code>", // R!  Code used in URL
  "base" : ["<code>"], // R!  The resource type(s) this search parameter applies to
  "type" : "<code>", // R!  number | date | string | token | reference | composite | quantity | uri
  "derivedFrom" : "<uri>", // Original Definition for the search parameter
  "description" : "<markdown>", // R!  Natural language description of the search parameter
  "expression" : "<string>", // C? FHIRPath expression that extracts the values
  "xpath" : "<string>", // C? XPath that extracts the values
  "xpathUsage" : "<code>", // C? normal | phonetic | nearby | distance | other
  "target" : ["<code>"], // Types of resource (if a resource reference)
  "comparator" : ["<code>"], // eq | ne | gt | lt | ge | le | sa | eb | ap
  "modifier" : ["<code>"], // missing | exact | contains | not | text | in | not-in | below | above | type
  "chain" : ["<string>"], // Chained names supported
  "component" : [{ // For Composite resources to define the parts
    "definition" : { Reference(SearchParameter) }, // R!  Defines how the part works
    "expression" : "<string>" // R!  Subexpression relative to main expression
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SearchParameter;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SearchParameter.url [ uri ]; # 1..1 Logical URI to reference this search parameter (globally unique)
  fhir:SearchParameter.version [ string ]; # 0..1 Business version of the search parameter
  fhir:SearchParameter.name [ string ]; # 1..1 Name for this search parameter (computer friendly)
  fhir:SearchParameter.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:SearchParameter.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:SearchParameter.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:SearchParameter.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:SearchParameter.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:SearchParameter.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:SearchParameter.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for search parameter (if
    applicable)
  fhir:SearchParameter.purpose [ markdown ]; # 0..1 Why this search parameter is defined
  fhir:SearchParameter.code [ code ]; # 1..1 Code used in URL
  fhir:SearchParameter.base [ code ], ... ; # 1..* The resource type(s) this search parameter applies to
  fhir:SearchParameter.type [ code ]; # 1..1 number | date | string | token | reference | composite | quantity | uri
  fhir:SearchParameter.derivedFrom [ uri ]; # 0..1 Original Definition for the search parameter
  fhir:SearchParameter.description [ markdown ]; # 1..1 Natural language description of the search parameter
  fhir:SearchParameter.expression [ string ]; # 0..1 FHIRPath expression that extracts the values
  fhir:SearchParameter.xpath [ string ]; # 0..1 XPath that extracts the values
  fhir:SearchParameter.xpathUsage [ code ]; # 0..1 normal | phonetic | nearby | distance | other
  fhir:SearchParameter.target [ code ], ... ; # 0..* Types of resource (if a resource reference)
  fhir:SearchParameter.comparator [ code ], ... ; # 0..* eq | ne | gt | lt | ge | le | sa | eb | ap
  fhir:SearchParameter.modifier [ code ], ... ; # 0..* missing | exact | contains | not | text | in | not-in | below |
    above | type
  fhir:SearchParameter.chain [ string ], ... ; # 0..* Chained names supported
  fhir:SearchParameter.component [ # 0..* For Composite resources to define the parts
    fhir:SearchParameter.component.definition [ Reference(SearchParameter) ]; # 1..1 Defines how the part works
    fhir:SearchParameter.component.expression [ string ]; # 1..1 Subexpression relative to main expression
  ], ...;
]
```

Changes since DSTU2

SearchParameter

SearchParameter.version | Added Element

SearchParameter.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status

SearchParameter.experimental | Now marked as Modifier

SearchParameter.contact | Type changed from BackboneElement to ContactDetail

SearchParameter.useContext | Added Element
SearchParameter.jurisdiction | Added Element
SearchParameter.purpose | Renamed from requirements to purpose Type changed from string to markdown
SearchParameter.base | Max Cardinality changed from 1 to *
SearchParameter.derivedFrom | Added Element
SearchParameter.description | Type changed from string to markdown
SearchParameter.expression | Added Element
SearchParameter.comparator | Added Element
SearchParameter.modifier | Added Element
SearchParameter.chain | Added Element
SearchParameter.component | Added Element
SearchParameter.component.definition | Added Element
SearchParameter.component.expression | Added Element
SearchParameter.contact.name | deleted
SearchParameter.contact.telecom | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
SearchParameter | I | | DomainResource | Search Parameter for a resource + If an xpath is present, there SHALL be an xpathUsage + Search parameters can only have chain names when the search parameter type is 'reference' Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 1..1 | uri | Logical URI to reference this search parameter (globally unique)
version | ? | 0..1 | string | Business version of the search parameter
name | ? | 1..1 | string | Name for this search parameter (computer friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for search parameter (if applicable) Jurisdiction ValueSet (Extensible)
purpose | | 0..1 | markdown | Why this search parameter is defined
code | ? | 1..1 | code | Code used in URL
base | ? | 1..* | code | The resource type(s) this search parameter applies to ResourceType (Required)
type | ? | 1..1 | code | number | date | string | token | reference | composite | quantity | uri SearchParamType (Required)
derivedFrom | | 0..1 | uri | Original Definition for the search parameter
description | ? | 1..1 | markdown | Natural language description of the search parameter
expression | I | 0..1 | string | FHIRPath expression that extracts the values
xpath | I | 0..1 | string | XPath that extracts the values
xpathUsage | I | 0..1 | code | normal | phonetic | nearby | distance | other XPathUsageType (Required)
target | | 0..* | code | Types of resource (if a resource reference) ResourceType (Required)
comparator | | 0..* | code | eq | ne | gt | lt | ge | le | sa | eb | ap SearchComparator (Required)
modifier | | 0..* | code | missing | exact | contains | not | text | in | not-in | below | above | type SearchModifierCode (Required)
chain | | 0..* | string | Chained names supported
component | | 0..* | BackboneElement | For Composite resources to define the parts
definition | | 1..1 | Reference(SearchParameter) | Defines how the part works
expression | | 1..1 | string | Subexpression relative to main expression
Documentation for this format

UML Diagram (Legend)

XML Template

```
<SearchParameter xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Logical URI to reference this search parameter (globally unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the search parameter -->
 <name value="[string]"/><!-- 1..1 Name for this search parameter (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for search parameter (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this search parameter is defined -->
 <code value="[code]"/><!-- 1..1 Code used in URL -->
 <base value="[code]"/><!-- 1..* The resource type(s) this search parameter applies to -->
 <type value="[code]"/><!-- 1..1 number | date | string | token | reference | composite | quantity | uri -->
 <derivedFrom value="[uri]"/><!-- 0..1 Original Definition for the search parameter -->
 <description value="[markdown]"/><!-- 1..1 Natural language description of the search parameter -->
 <expression value="[string]"/><!--  0..1 FHIRPath expression that extracts the values -->
```

```
 <xpath value="[string]"/><!--  0..1 XPath that extracts the values -->
 <xpathUsage value="[code]"/><!--  0..1 normal | phonetic | nearby | distance | other -->
 <target value="[code]"/><!-- 0..* Types of resource (if a resource reference) -->
 <comparator value="[code]"/><!-- 0..* eq | ne | gt | lt | ge | le | sa | eb | ap -->
 <modifier value="[code]"/><!-- 0..* missing | exact | contains | not | text | in | not-in | below | above | type -->
 <chain value="[string]"/><!-- 0..* Chained names supported -->
 <component>  <!-- 0..* For Composite resources to define the parts -->
  <definition><!-- 1..1 Reference(SearchParameter) Defines how the part works --></definition>
  <expression value="[string]"/><!-- 1..1 Subexpression relative to main expression -->
 </component>
</SearchParameter>
```

JSON Template

```
{
  "resourceType" : "SearchParameter",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Logical URI to reference this search parameter (globally unique)
  "version" : "<string>", // Business version of the search parameter
  "name" : "<string>", // R!  Name for this search parameter (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for search parameter (if applicable)
  "purpose" : "<markdown>", // Why this search parameter is defined
  "code" : "<code>", // R!  Code used in URL
  "base" : ["<code>"], // R!  The resource type(s) this search parameter applies to
  "type" : "<code>", // R!  number | date | string | token | reference | composite | quantity | uri
  "derivedFrom" : "<uri>", // Original Definition for the search parameter
  "description" : "<markdown>", // R!  Natural language description of the search parameter
  "expression" : "<string>", // C? FHIRPath expression that extracts the values
  "xpath" : "<string>", // C? XPath that extracts the values
  "xpathUsage" : "<code>", // C? normal | phonetic | nearby | distance | other
  "target" : ["<code>"], // Types of resource (if a resource reference)
  "comparator" : ["<code>"], // eq | ne | gt | lt | ge | le | sa | eb | ap
  "modifier" : ["<code>"], // missing | exact | contains | not | text | in | not-in | below | above | type
  "chain" : ["<string>"], // Chained names supported
  "component" : [{ // For Composite resources to define the parts
    "definition" : { Reference(SearchParameter) }, // R!  Defines how the part works
    "expression" : "<string>" // R!  Subexpression relative to main expression
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:SearchParameter;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:SearchParameter.url [ uri ]; # 1..1 Logical URI to reference this search parameter (globally unique)
  fhir:SearchParameter.version [ string ]; # 0..1 Business version of the search parameter
  fhir:SearchParameter.name [ string ]; # 1..1 Name for this search parameter (computer friendly)
  fhir:SearchParameter.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:SearchParameter.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:SearchParameter.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:SearchParameter.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:SearchParameter.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:SearchParameter.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:SearchParameter.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for search parameter (if
    applicable)
  fhir:SearchParameter.purpose [ markdown ]; # 0..1 Why this search parameter is defined
  fhir:SearchParameter.code [ code ]; # 1..1 Code used in URL
  fhir:SearchParameter.base [ code ], ... ; # 1..* The resource type(s) this search parameter applies to
  fhir:SearchParameter.type [ code ]; # 1..1 number | date | string | token | reference | composite | quantity | uri
  fhir:SearchParameter.derivedFrom [ uri ]; # 0..1 Original Definition for the search parameter
  fhir:SearchParameter.description [ markdown ]; # 1..1 Natural language description of the search parameter
  fhir:SearchParameter.expression [ string ]; # 0..1 FHIRPath expression that extracts the values
  fhir:SearchParameter.xpath [ string ]; # 0..1 XPath that extracts the values
  fhir:SearchParameter.xpathUsage [ code ]; # 0..1 normal | phonetic | nearby | distance | other
  fhir:SearchParameter.target [ code ], ... ; # 0..* Types of resource (if a resource reference)
  fhir:SearchParameter.comparator [ code ], ... ; # 0..* eq | ne | gt | lt | ge | le | sa | eb | ap
  fhir:SearchParameter.modifier [ code ], ... ; # 0..* missing | exact | contains | not | text | in | not-in | below |
    above | type
  fhir:SearchParameter.chain [ string ], ... ; # 0..* Chained names supported
  fhir:SearchParameter.component [ # 0..* For Composite resources to define the parts
    fhir:SearchParameter.component.definition [ Reference(SearchParameter) ]; # 1..1 Defines how the part works
    fhir:SearchParameter.component.expression [ string ]; # 1..1 Subexpression relative to main expression
  ], ...;
]
```

Changes since DSTU2

SearchParameter

SearchParameter.version | Added Element
SearchParameter.status | Min Cardinality changed from 0 to 1 Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status
SearchParameter.experimental | Now marked as Modifier
SearchParameter.contact | Type changed from BackboneElement to ContactDetail
SearchParameter.useContext | Added Element
SearchParameter.jurisdiction | Added Element
SearchParameter.purpose | Renamed from requirements to purpose Type changed from string to markdown
SearchParameter.base | Max Cardinality changed from 1 to *
SearchParameter.derivedFrom | Added Element
SearchParameter.description | Type changed from string to markdown
SearchParameter.expression | Added Element
SearchParameter.comparator | Added Element
SearchParameter.modifier | Added Element
SearchParameter.chain | Added Element
SearchParameter.component | Added Element
SearchParameter.component.definition | Added Element
SearchParameter.component.expression | Added Element
SearchParameter.contact.name | deleted
SearchParameter.contact.telecom | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

5.5.4.1 Terminology Bindings

Path | Definition | Type | Reference
SearchParameter.status | The lifecycle status of a Value Set or Concept Map. | Required | PublicationStatus
SearchParameter.jurisdiction | Countries and regions within which this artifact is targeted for use | Extensible | Jurisdiction ValueSet
SearchParameter.base SearchParameter.target | One of the resource types defined as part of FHIR. | Required | Resource Types
SearchParameter.type | Data types allowed to be used for search parameters. | Required | SearchParamType
SearchParameter.xpathUsage | How a search parameter relates to the set of elements returned by evaluating its xpath query. | Required | XPathUsageType
SearchParameter.comparator | What Search Comparator Codes are supported in search | Required | SearchComparator
SearchParameter.modifier | A supported modifier for a search parameter. | Required | SearchModifierCode

5.5.4.2 Constraints

* spd-1: If an xpath is present, there SHALL be an xpathUsage (expression : xpath.empty() or xpathUsage.exists())

* spd-2: Search parameters can only have chain names when the search parameter type is 'reference' (expression : chain.empty() or type = 'reference')

5.5.5 SearchParameters on Extensions

The SearchParameter resource may be used to define searches on extensions. Depending on the context of the extension (as defined in the StructureDefinition), this may require multiple instances of the SearchParameter.xpath element like so:

* If the extension's context is scoped by a single resource, SearchParameter.base must be set to that resource, and there may be one or more xpath elements corresponding to the elements specified in the extension's context, e.g. "f:Patient/f:name/f:extension[@url='extension url']"

* If the extension's context specifies multiple resources, SearchParameter.base must be set to "DomainResource".

* You may use a single xpath element with the full URL of the Extension to indicate that the SearchParameter would search in all elements of the given SearchParameter.base resource for the given extension.

5.5.6 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
base | token | The resource type(s) this search parameter applies to | SearchParameter.base
code | token | Code used in URL | SearchParameter.code
component | reference | Defines how the part works | SearchParameter.component.definition (SearchParameter)
date | date | The search parameter publication date | SearchParameter.date
derived-from | uri | Original Definition for the search parameter | SearchParameter.derivedFrom
description | string | The description of the search parameter | SearchParameter.description
jurisdiction | token | Intended jurisdiction for the search parameter | SearchParameter.jurisdiction
name | string | Computationally friendly name of the search parameter | SearchParameter.name
publisher | string | Name of the publisher of the search parameter | SearchParameter.publisher
status | token | The current status of the search parameter | SearchParameter.status
target | token | Types of resource (if a resource reference) | SearchParameter.target

type | token | number | date | string | token | reference | composite | quantity | uri | SearchParameter.type
url | uri | The uri that identifies the search parameter | SearchParameter.url
version | token | The business version of the search parameter | SearchParameter.version

5.5.7 Resource SearchParameter - Examples

URL: https://hl7.org/fhir/STU3/searchparameter-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.5.7 Resource SearchParameter - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

General example | example | XML | JSON | Turtle
Search parameter for an extension example | example-extension | XML | JSON | Turtle
Search parameter for a reference example | example-reference | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.5.8 Resource SearchParameter - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/searchparameter-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.5.8 Resource SearchParameter - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 3 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the SearchParameter resource.

SearchParameter
Definition | A search parameter that defines a named search item that can be used to search/filter on a resource.
Control | 1..1
Comments | In FHIR, search is not performed directly on a resource (by XML or JSON path), but on a named parameter that maps into the resource content.
Invariants | Defined on this element spd-1: If an xpath is present, there SHALL be an xpathUsage (expression : xpath.empty() or xpathUsage.exists(), xpath: not(exists(f:xpath)) or exists(f:xpathUsage)) spd-2: Search parameters can only have chain names when the search parameter type is 'reference' (expression : chain.empty() or type = 'reference', xpath: not(exists(f:chain)) or (f:type/@value = 'reference'))
SearchParameter.url
Definition | An absolute URI that is used to identify this search parameter when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this search parameter is (or will be) published. The URL SHOULD include the major version of the search parameter. For more information see Technical and Business Versions.
Control | 1..1
Type | uri
Requirements | Allows the search parameter to be referenced by a single globally unique identifier. To allow referencing and reusing search parameter definitions under other names on operational systems (in case of name clash).
Summary | true
Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.
SearchParameter.version
Definition | The identifier that is used to identify this version of the search parameter when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the search parameter author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Control | 0..1
Type | string
Summary | true
Comments | There may be different search parameter instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the search parameter with the format [url]|[version].
SearchParameter.name
Definition | A natural language name identifying the search parameter. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Control | 1..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.
SearchParameter.status
Definition | The status of this search parameter. Enables tracking the life-cycle of the content.
Control | 1..1

Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Allows filtering of search parameters that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired search parameter without due consideration.
SearchParameter.experimental
Definition | A boolean value to indicate that this search parameter is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.
Control | 0..1
Type | boolean
Is Modifier | true
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level search parameter.
Summary | true
Comments | Allows filtering of search parameter that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental search parameter in production.
SearchParameter.date
Definition | The date (and optionally time) when the search parameter was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the search parameter changes.
Control | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the search parameter. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
SearchParameter.publisher
Definition | The name of the individual or organization that published the search parameter.
Control | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the search parameter. May also allow for contact.
Summary | true
Comments | Usually an organization, but may be an individual. The publisher (or steward) of the search parameter is the organization or individual primarily responsible for the maintenance and upkeep of the search parameter. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the search parameter. This item SHOULD be populated unless the information is available from context.
SearchParameter.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
SearchParameter.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate search parameter instances.
Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
SearchParameter.jurisdiction
Definition | A legal or geographic region in which the search parameter is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the search parameter to be used in jurisdictions other than those for which it was originally designed or intended.
SearchParameter.purpose
Definition | Explaination of why this search parameter is needed and why it has been designed as it has.
Control | 0..1
Type | markdown
Comments | This element does not describe the usage of the search parameter Instead it provides traceability of "why" the resource is either needed or "why" it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this search parameter.
SearchParameter.code
Definition | The code used in the URL or the parameter name in a parameters resource for this search parameter.
Control | 1..1
Type | code
Summary | true
Comments | For maximum compatibility, use only lowercase ASCII characters.
SearchParameter.base
Definition | The base resource type(s) that this search parameter can be used against.
Control | 1..*

Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
Comments | A search parameter must always have at least one type. When search parameters have more than one resource, they can be used against any of the listed resource types, or in a cross-type search (see Cross Resource Search).
SearchParameter.type
Definition | The type of value a search parameter refers to, and how the content is interpreted.
Control | 1..1
Terminology Binding | SearchParamType (Required)
Type | code
Summary | true
SearchParameter.derivedFrom
Definition | Where this search parameter is originally defined. If a derivedFrom is provided, then the details in the search parameter must be consistent with the definition from which it is defined. I.e. the parameter should have the same meaning, and (usually) the functionality should be a proper subset of the underlying search parameter.
Control | 0..1
Type | uri
Comments | The intent of this is that a server can designate that it provides support for a search parameter defined in the specification itself (e.g. value-quantity , but detail how it is supported by the server.
SearchParameter.description
Definition | A free text natural language description of the search parameter from a consumer's perspective. and how it used.
Control | 1..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the search parameter was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the search parameter as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created).
SearchParameter.expression
Definition | A FHIRPath expression that returns a set of elements for the search parameter.
Control | 0..1
Type | string
Comments | Note that the elements returned by the expression are sometimes complex elements where logic is required to determine quite how to handle them; e.g. CodeableConcepts may contain text and/or multiple codings, where the codings themselves contain a code and a system. For composite search parameters, the outcome of the expression must a collection of base elements from which the composites are derived.
Invariants | Affect this element spd-1: If an xpath is present, there SHALL be an xpathUsage (expression : xpath.empty() or xpathUsage.exists(), xpath: not(exists(f:xpath)) or exists(f:xpathUsage))
SearchParameter.xpath
Definition | An XPath expression that returns a set of elements for the search parameter.
Control | 0..1
Type | string
Comments | Note that the elements returned by the XPath are sometimes complex elements where logic is required to determine quite how to handle them; e.g. CodeableConcepts may contain text and/or multiple codings, where the codings themselves contain a code and a system.
Invariants | Affect this element spd-1: If an xpath is present, there SHALL be an xpathUsage (expression : xpath.empty() or xpathUsage.exists(), xpath: not(exists(f:xpath)) or exists(f:xpathUsage))
SearchParameter.xpathUsage
Definition | How the search parameter relates to the set of elements returned by evaluating the xpath query.
Control | 0..1
Terminology Binding | XPathUsageType (Required)
Type | code
Invariants | Affect this element spd-1: If an xpath is present, there SHALL be an xpathUsage (expression : xpath.empty() or xpathUsage.exists(), xpath: not(exists(f:xpath)) or exists(f:xpathUsage))
SearchParameter.target
Definition | Types of resource (if a resource is referenced).
Control | 0..*
Terminology Binding | Any defined Resource Type name
Type | code
SearchParameter.comparator
Definition | Comparators supported for the search parameter.
Control | 0..*
Terminology Binding | SearchComparator (Required)
Type | code
Comments | If no comparators are listed, clients should not expect servers to support any comparators.
SearchParameter.modifier
Definition | A modifier supported for the search parameter.
Control | 0..*
Terminology Binding | SearchModifierCode (Required)
Type | code
SearchParameter.chain
Definition | Contains the names of any search parameters which may be chained to the containing search parameter. Chained parameters may be added to search parameters of type reference, and specify that resources will only be returned if they contain a reference to a resource which matches the chained parameter value. Values for this field should be drawn from SearchParameter.code for a parameter on the target resource type.
Control | 0..*
Type | string

Comments | Systems are not required to list all the chain names they support, but if they don't list them, clients may not know to use them.
SearchParameter.component
Definition | Used to define the parts of a composite search parameter.
Control | 0..*
SearchParameter.component.definition
Definition | The definition of the search parameter that describes this part.
Control | 1..1
Type | Reference(SearchParameter)
SearchParameter.component.expression
Definition | A sub-expression that defines how to extract values for this component from the output of the main SearchParameter.expression.
Control | 1..1
Type | string
Comments | This expression overrides the expression in the definition, and extracts the index values from the outcome of the composite expression.

5.5.9 Resource SearchParameter - Mappings

URL: https://hl7.org/fhir/STU3/searchparameter-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.5.9 Resource SearchParameter - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the searchparameter resource.

5.5.9.1 Mappings for RIM Mapping (http://hl7.org/v3)

SearchParameter | n/a

5.5.9.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

SearchParameter | n/a

5.5.10 Resource SearchParameter - Design Notes

URL: https://hl7.org/fhir/STU3/searchparameter-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.5.10 Resource SearchParameter - Design Notes

to do

5.5.11 Resource SearchParameter - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/searchparameter-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.5.11 Resource SearchParameter - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the SearchParameter resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

5.5.11.1 Profiles

No Profiles defined for this resource

5.5.11.2 Extensions

resource-approvalDate | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | lastReviewDate | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

5.5.11.3 Search Extensions

No Search Extensions defined for this resource

5.5.12 Standard Search Parameter Extensions

URL: https://hl7.org/fhir/STU3/searchparameter-extensions.html

This profile describes common extensions that are used with Search Parameters

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.5.12 Standard Search Parameter Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

5.5.12.1 Content

5.5.13 Resource SearchParameter - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/searchparameter-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.5.13 Resource SearchParameter - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for SearchParameter.

Functional status for this map: 2 tests that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

5.5.13.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/SearchParameter2to3" = "R2 to R3 Conversions for SearchParameter"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/SearchParameter" alias SearchParameterR2 as source
uses "http://hl7.org/fhir/StructureDefinition/SearchParameter" alias SearchParameter as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/DomainResource2to3"
imports "http://hl7.org/fhir/StructureMap/ContactPoint2to3"

group for type+types  SearchParameter extends DomainResource
  input src : SearchParameterR2 as source
  input tgt : SearchParameter as target

  "SearchParameter-url" : for src.url make tgt.url
  "SearchParameter-name" : for src.name make tgt.name
  "SearchParameter-status" : for src.status default "draft" make tgt.status
  "SearchParameter-experimental" : for src.experimental make tgt.experimental
  "SearchParameter-date" : for src.date make tgt.date
  "SearchParameter-publisher" : for src.publisher make tgt.publisher
  "SearchParameter-contact" : for src.contact as vs0 make tgt.contact as vt0 then {
    "ContactDetail-name" : for vs0.name make vt0.name
    "ContactDetail-telecom" : for vs0.telecom make vt0.telecom
  }

  "SearchParameter-purpose" : for src.requirements make tgt.purpose
  "SearchParameter-code" : for src.code make tgt.code
  "SearchParameter-base" : for src.base make tgt.base
  "SearchParameter-type" : for src.type make tgt.type
  "SearchParameter-description" : for src.description make tgt.description
  "SearchParameter-xpath" : for src.xpath make tgt.xpath
  "SearchParameter-xpathUsage" : for src.xpathUsage make tgt.xpathUsage
  "SearchParameter-target" : for src.target make tgt.target

endgroup
```

5.5.13.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/SearchParameter3to2" = "R3 to R2 Conversion for SearchParameter"

uses "http://hl7.org/fhir/StructureDefinition/SearchParameter" alias SearchParameter as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/SearchParameter" alias SearchParameterR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/DomainResource3to2"
imports "http://hl7.org/fhir/StructureMap/ContactPoint3to2"

group for type+types  SearchParameter extends DomainResource
  input src : SearchParameter as source
  input tgt : SearchParameterR2 as target

  "SearchParameter-url" : for src.url make tgt.url
  "SearchParameter-name" : for src.name make tgt.name
```

```
  "SearchParameter-status" : for src.status make tgt.status
  "SearchParameter-experimental" : for src.experimental make tgt.experimental
  "SearchParameter-publisher" : for src.publisher make tgt.publisher
  "SearchParameter-contact" : for src.contact as vs0 make tgt.contact as vt0 then {
    "SearchParameter.contact-name" : for vs0.name make vt0.name
    "SearchParameter.contact-telecom" : for vs0.telecom make vt0.telecom
  }

  "SearchParameter-date" : for src.date make tgt.date
  "SearchParameter-requirements" : for src.purpose make tgt.requirements
  "SearchParameter-code" : for src.code make tgt.code
  "SearchParameter-base" : for src.base only_one make tgt.base
  "SearchParameter-type" : for src.type make tgt.type
  "SearchParameter-description" : for src.description make tgt.description
  "SearchParameter-xpath" : for src.xpath make tgt.xpath
  "SearchParameter-xpathUsage" : for src.xpathUsage make tgt.xpathUsage
  "SearchParameter-target" : for src.target make tgt.target

endgroup
```

5.5.13.3 R3 Validation Errors

5.6 Resource CompartmentDefinition - Content

URL: https://hl7.org/fhir/STU3/compartmentdefinition.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

A compartment definition that defines how resources are accessed on a server.

5.6.1 Scope and Usage

Each resource may belong to one or more logical compartments. A compartment is a logical grouping of resources which share a common property. Compartments have two principal roles:

* Function as an access mechanism for finding a set of related resources quickly

* Provide a definitional basis for applying access control to resources quickly

Note:

At present, CompartmentDefinitions can only be defined by HL7 International. This is because their existence creates significant impact on the behavior of servers.

5.6.2 Boundaries and Relationships

Compartment definitions describe how particular compartment instances are named and identified, and how systems know which resources are in the compartment. The following compartments are defined by this specification:

Title	Description	Identity	Membership
Patient	The set of resources associated with a particular patient	There is an instance of the patient compartment for each patient resource, and the identity of the compartment is the same as the patient. When a patient is linked to another patient, all the records associated with the linked patient are in the compartment associated with the target of the link.	The patient compartment includes any resources where the subject of the resource is the patient, and some other resources that are directly linked to resources in the patient compartment
Encounter	The set of resources associated with a particular encounter	There is an instance of the encounter compartment for each encounter resource, and the identity of the compartment is the same as the encounter	The encounter compartment includes any resources where the resource has an explicitly nominated encounter, and some other resources that themselves link to resources in the encounter compartment. Note that for many resources, the exact nature of the link to encounter can be ambiguous (e.g. for a DiagnosticReport, is it the encounter when it was initiated, or when it was reported?)
RelatedPerson	The set of resources associated with a particular 'related person'	There is an instance of the relatedPerson compartment for each relatedPerson resource, and the identity of the compartment is the same as the relatedPerson	The relatedPerson compartment includes any resources where the resource is explicitly linked to relatedPerson (usually as author)
Practitioner	The set of resources associated with a particular practitioner	There is an instance of the practitioner compartment for each Practitioner resource, and the identity of the compartment is the same as the Practitioner	The practitioner compartment includes any resources where the resource is explicitly linked to a Practitioner (usually as author, but other kinds of linkage exist)
Device	The set of resources associated with a particular device	There is an instance of the practitioner compartment for each Device resource, and the identity of the compartment is the same as the Device	The device compartment includes any resources where the resource is explicitly linked to a Device (mostly subject or performer)

The full definitions of these compartments are published as CompartmentDefinition resources. Servers typically do not support the full definition of a compartment, and are not required to. Systems may publish CompartmentDefinition resources so that other systems may make use of compartments properly.

* CompartmentDefinitions are used by CapabilityStatement instances for specifying how resources are accessed

5.6.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
CompartmentDefinition | | | DomainResource | Compartment Definition for a resource Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 1..1 | uri | Logical URI to reference this compartment definition (globally unique)
name | ? | 1..1 | string | Name for this compartment definition (computer friendly)
title | ? | 0..1 | string | Name for this compartment definition (human friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the compartment definition
purpose | | 0..1 | markdown | Why this compartment definition is defined
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for compartment definition (if applicable) Jurisdiction ValueSet (Extensible)
code | ? | 1..1 | code | Patient | Encounter | RelatedPerson | Practitioner | Device CompartmentType (Required)
search | ? | 1..1 | boolean | Whether the search syntax is supported
resource | ? | 0..* | BackboneElement | How a resource is related to the compartment
code | ? | 1..1 | code | Name of resource type ResourceType (Required)
param | ? | 0..* | string | Search Parameter Name, or chained parameters
documentation | | 0..1 | string | Additional documentation about the resource and compartment
Documentation for this format

UML Diagram (Legend)

XML Template

```
<CompartmentDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Logical URI to reference this compartment definition (globally unique) -->
 <name value="[string]"/><!-- 1..1 Name for this compartment definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this compartment definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the compartment definition -->
 <purpose value="[markdown]"/><!-- 0..1 Why this compartment definition is defined -->
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for compartment definition (if applicable)
    --></jurisdiction>
 <code value="[code]"/><!-- 1..1 Patient | Encounter | RelatedPerson | Practitioner | Device -->
 <search value="[boolean]"/><!-- 1..1 Whether the search syntax is supported -->
 <resource>  <!-- 0..* How a resource is related to the compartment -->
  <code value="[code]"/><!-- 1..1 Name of resource type -->
  <param value="[string]"/><!-- 0..* Search Parameter Name, or chained parameters -->
  <documentation value="[string]"/><!-- 0..1 Additional documentation about the resource and compartment -->
 </resource>
</CompartmentDefinition>
```

JSON Template

```
{
  "resourceType" : "CompartmentDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Logical URI to reference this compartment definition (globally unique)
  "name" : "<string>", // R!  Name for this compartment definition (computer friendly)
  "title" : "<string>", // Name for this compartment definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the compartment definition
  "purpose" : "<markdown>", // Why this compartment definition is defined
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for compartment definition (if applicable)
  "code" : "<code>", // R!  Patient | Encounter | RelatedPerson | Practitioner | Device
  "search" : <boolean>, // R!  Whether the search syntax is supported
  "resource" : [{ // How a resource is related to the compartment
    "code" : "<code>", // R!  Name of resource type
    "param" : ["<string>"], // Search Parameter Name, or chained parameters
    "documentation" : "<string>" // Additional documentation about the resource and compartment
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CompartmentDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CompartmentDefinition.url [ uri ]; # 1..1 Logical URI to reference this compartment definition (globally unique)
  fhir:CompartmentDefinition.name [ string ]; # 1..1 Name for this compartment definition (computer friendly)
  fhir:CompartmentDefinition.title [ string ]; # 0..1 Name for this compartment definition (human friendly)
  fhir:CompartmentDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:CompartmentDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:CompartmentDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:CompartmentDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:CompartmentDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:CompartmentDefinition.description [ markdown ]; # 0..1 Natural language description of the compartment definition
  fhir:CompartmentDefinition.purpose [ markdown ]; # 0..1 Why this compartment definition is defined
  fhir:CompartmentDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:CompartmentDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for compartment
    definition (if applicable)
  fhir:CompartmentDefinition.code [ code ]; # 1..1 Patient | Encounter | RelatedPerson | Practitioner | Device
  fhir:CompartmentDefinition.search [ boolean ]; # 1..1 Whether the search syntax is supported
  fhir:CompartmentDefinition.resource [ # 0..* How a resource is related to the compartment
    fhir:CompartmentDefinition.resource.code [ code ]; # 1..1 Name of resource type
    fhir:CompartmentDefinition.resource.param [ string ], ... ; # 0..* Search Parameter Name, or chained parameters
    fhir:CompartmentDefinition.resource.documentation [ string ]; # 0..1 Additional documentation about the resource and
     compartment
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

CompartmentDefinition | | | DomainResource | Compartment Definition for a resource Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | ? | 1..1 | uri | Logical URI to reference this compartment definition (globally unique)

name | ? | 1..1 | string | Name for this compartment definition (computer friendly)

title | ? | 0..1 | string | Name for this compartment definition (human friendly)

status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage

date | ? | 0..1 | dateTime | Date this was last changed

publisher | ? | 0..1 | string | Name of the publisher (organization or individual)

contact | ? | 0..* | ContactDetail | Contact details for the publisher

description | | 0..1 | markdown | Natural language description of the compartment definition

purpose | | 0..1 | markdown | Why this compartment definition is defined

useContext | ? | 0..* | UsageContext | Context the content is intended to support

jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for compartment definition (if applicable) Jurisdiction ValueSet (Extensible)

code | ? | 1..1 | code | Patient | Encounter | RelatedPerson | Practitioner | Device CompartmentType (Required)

search | ? | 1..1 | boolean | Whether the search syntax is supported

resource | ? | 0..* | BackboneElement | How a resource is related to the compartment

code | ? | 1..1 | code | Name of resource type ResourceType (Required)

param | ? | 0..* | string | Search Parameter Name, or chained parameters

documentation | | 0..1 | string | Additional documentation about the resource and compartment

Documentation for this format

UML Diagram (Legend)

XML Template

```
<CompartmentDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Logical URI to reference this compartment definition (globally unique) -->
 <name value="[string]"/><!-- 1..1 Name for this compartment definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this compartment definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the compartment definition -->
 <purpose value="[markdown]"/><!-- 0..1 Why this compartment definition is defined -->
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for compartment definition (if applicable)
    --></jurisdiction>
 <code value="[code]"/><!-- 1..1 Patient | Encounter | RelatedPerson | Practitioner | Device -->
 <search value="[boolean]"/><!-- 1..1 Whether the search syntax is supported -->
 <resource>  <!-- 0..* How a resource is related to the compartment -->
```

```
  <code value="[code]"/><!-- 1..1 Name of resource type -->
  <param value="[string]"/><!-- 0..* Search Parameter Name, or chained parameters -->
  <documentation value="[string]"/><!-- 0..1 Additional documentation about the resource and compartment -->
 </resource>
</CompartmentDefinition>
```

JSON Template

```
{
  "resourceType" : "CompartmentDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Logical URI to reference this compartment definition (globally unique)
  "name" : "<string>", // R!  Name for this compartment definition (computer friendly)
  "title" : "<string>", // Name for this compartment definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the compartment definition
  "purpose" : "<markdown>", // Why this compartment definition is defined
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for compartment definition (if applicable)
  "code" : "<code>", // R!  Patient | Encounter | RelatedPerson | Practitioner | Device
  "search" : <boolean>, // R!  Whether the search syntax is supported
  "resource" : [{ // How a resource is related to the compartment
    "code" : "<code>", // R!  Name of resource type
    "param" : ["<string>"], // Search Parameter Name, or chained parameters
    "documentation" : "<string>" // Additional documentation about the resource and compartment
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:CompartmentDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:CompartmentDefinition.url [ uri ]; # 1..1 Logical URI to reference this compartment definition (globally unique)
  fhir:CompartmentDefinition.name [ string ]; # 1..1 Name for this compartment definition (computer friendly)
  fhir:CompartmentDefinition.title [ string ]; # 0..1 Name for this compartment definition (human friendly)
  fhir:CompartmentDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:CompartmentDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:CompartmentDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:CompartmentDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:CompartmentDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:CompartmentDefinition.description [ markdown ]; # 0..1 Natural language description of the compartment definition
  fhir:CompartmentDefinition.purpose [ markdown ]; # 0..1 Why this compartment definition is defined
  fhir:CompartmentDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:CompartmentDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for compartment
    definition (if applicable)
  fhir:CompartmentDefinition.code [ code ]; # 1..1 Patient | Encounter | RelatedPerson | Practitioner | Device
  fhir:CompartmentDefinition.search [ boolean ]; # 1..1 Whether the search syntax is supported
  fhir:CompartmentDefinition.resource [ # 0..* How a resource is related to the compartment
    fhir:CompartmentDefinition.resource.code [ code ]; # 1..1 Name of resource type
    fhir:CompartmentDefinition.resource.param [ string ], ... ; # 0..* Search Parameter Name, or chained parameters
    fhir:CompartmentDefinition.resource.documentation [ string ]; # 0..1 Additional documentation about the resource and
     compartment
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

5.6.3.1 Terminology Bindings

Path	Definition	Type	Reference
CompartmentDefinition.status	The lifecycle status of a Value Set or Concept Map.	Required	PublicationStatus
CompartmentDefinition.jurisdiction	Countries and regions within which this artifact is targeted for use	Extensible	Jurisdiction ValueSet
CompartmentDefinition.code	Which compartment a compartment definition describes	Required	CompartmentType
CompartmentDefinition.resource.code	One of the resource types defined as part of FHIR.	Required	Resource Types

5.6.3.2 Using Compartments

As an example of compartment usage, to retrieve a list of a patient's conditions, use the URL:

```
GET [base]/Patient/[id]/Condition
```

Additional search parameters can be defined, such as this hypothetical search for acute conditions:

```
GET [base]/Patient/[id]/Condition?code:in=http://hspc.org/ValueSet/acute-concerns
```

Note that as searches, these are syntactic variations on these two search URLs respectively:

```
GET [base]/Condition?patient=[id]
GET [base]/Condition?patient=[id]&code:in=http://hspc.org/ValueSet/acute-concerns
```

Note that the outcome of a compartment search is the same as the equivalent normal search. As an example, both these searches return the same outcome if there is no patient 333:

```
GET [base]/Patient/333/Condition
GET [base]/Condition?patient=333
```

Whether the patient doesn't exist, or the user has no access to the patient, both these searches return an empty bundle with no matches. Some systems will include an operation outcome warning that there is no matching patient.

However, there is a key difference in functionality between compartment based searches and direct searches with parameters. Consider this search:

```
GET [base]/Patient/[id]/Communication
```

Because the definition of the patient compartment for Communication says that a Communication resource is in the patient compartment if the subject, sender, or recipient is the patient, the compartment search is actually the same as the union of these 3 searches:

```
GET [base]/Communication?subject=[id]
GET [base]/Communication?sender=[id]
GET [base]/Communication?recipient=[id]
```

There is no way to do this as a single search, except by using the _filter:

```
GET [base]/Communication?_filter=subject re [id] or sender re [id] or recipient re [id]
```

Further details of searching by compartment are described under the search operation. As a search related operation, the assignment of resources to compartments is only based on the current version of any of the resources involved. Note that contained patient resources cannot create a patient compartment of their own.

Compartments may be used explicitly, like this, but can also be used implicitly. For instance, if a FHIR server is providing a patient view of a record, the authorized user associated with use of the FHIR RESTful API may be limited to accessing records from the compartment instance(s) logically associated with their identity. Irrespective of whether compartments are being used explicitly or implicitly, servers will need to make arrangements to make some resources with no direct link to a patient available to the client (medications, substances, etc.).

Note that resources may cross between compartments, or interlink them. Examples of this would be where a Diagnostic Report identifies a subject, but an Observation it references identifies a different subject, or where a List resource references items that identify different subjects. Such cross-linking may arise for many valid reasons, including:

* Cases where subject records are inter-linked - Transplants, Perinatal care, family therapy etc.

* Workflow management where action lists link multiple patients and/or practitioners

Given the wide variety of use cases and contexts in which FHIR is used, compartments do not define how cross-linking is handled. Systems may reject resources, remove them from both compartments, or place them in both, or act in some other fashion.

It is at the discretion of the server whether to include resources in a compartment when the reference to the resource that establishes the compartment is in an extension.

Some resources are not in any compartment, e.g. Medication, Substance, Location. These resources are not directly to a patient or authored record, and are sometimes called 'master files'. Servers will need to make arrangements to make these resources available to the clients that are limited to particular compartments. For example, a Medication resource describes a medication itself and does not link to a patient; however, a resource such as MedicationAdministration connects the Medication (details of what was administered) to the patient (for whom was it administered), and so is required to interpret the administration.

5.6.4 Defining New Compartments

Compartments are defined and added the list above when implementer communities identify them as common access points for data. As described below, compartments have both syntactical and logical consequences, and both these aspects of their functionality are evaluated when deciding whether to define compartments.

5.6.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
code | token | Patient | Encounter | RelatedPerson | Practitioner | Device | CompartmentDefinition.code
date | date | The compartment definition publication date | CompartmentDefinition.date
description | string | The description of the compartment definition | CompartmentDefinition.description
jurisdiction | token | Intended jurisdiction for the compartment definition | CompartmentDefinition.jurisdiction
name | string | Computationally friendly name of the compartment definition | CompartmentDefinition.name
publisher | string | Name of the publisher of the compartment definition | CompartmentDefinition.publisher
resource | token | Name of resource type | CompartmentDefinition.resource.code
status | token | The current status of the compartment definition | CompartmentDefinition.status
title | string | The human-friendly name of the compartment definition | CompartmentDefinition.title
url | uri | The uri that identifies the compartment definition | CompartmentDefinition.url

5.6.6 Resource CompartmentDefinition - Examples

URL: https://hl7.org/fhir/STU3/compartmentdefinition-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.6.6 Resource CompartmentDefinition - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Full Patient Compartment | patient | XML | JSON | Turtle
Full Practitioner Compartment | practitioner | XML | JSON | Turtle
Full RelatedPerson Compartment | relatedperson | XML | JSON | Turtle
Full Device Compartment | device | XML | JSON | Turtle
Full Encounter Compartment | encounter | XML | JSON | Turtle
Example Compartment | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.6.7 Resource CompartmentDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/compartmentdefinition-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.6.7 Resource CompartmentDefinition - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the CompartmentDefinition resource.

CompartmentDefinition
Definition | A compartment definition that defines how resources are accessed on a server.
Control | 1..1
Comments | In FHIR, search is not performed directly on a resource (by XML or JSON path), but on a named parameter that maps into the resource content.
CompartmentDefinition.url
Definition | An absolute URI that is used to identify this compartment definition when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this compartment definition is (or will be) published. The URL SHOULD include the major version of the compartment definition. For more information see Technical and Business Versions.
Control | 1..1
Type | uri
Requirements | Allows the compartment definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.
CompartmentDefinition.name
Definition | A natural language name identifying the compartment definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Control | 1..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly. This is often the same as the code for the parameter, but does not need to be.
CompartmentDefinition.title
Definition | A short, descriptive, user-friendly title for the compartment definition.
Control | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
CompartmentDefinition.status
Definition | The status of this compartment definition. Enables tracking the life-cycle of the content.
Control | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Allows filtering of compartment definitions that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired compartment definition without due consideration.
CompartmentDefinition.experimental
Definition | A boolean value to indicate that this compartment definition is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.

Control | 0..1
Type | boolean
Is Modifier | true
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level compartment definition.
Summary | true
Comments | Allows filtering of compartment definition that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental compartment definition in production.
CompartmentDefinition.date
Definition | The date (and optionally time) when the compartment definition was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the compartment definition changes.
Control | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the compartment definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
CompartmentDefinition.publisher
Definition | The name of the individual or organization that published the compartment definition.
Control | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the compartment definition. May also allow for contact.
Summary | true
Comments | Usually an organization, but may be an individual. The publisher (or steward) of the compartment definition is the organization or individual primarily responsible for the maintenance and upkeep of the compartment definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the compartment definition. This item SHOULD be populated unless the information is available from context.
CompartmentDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
CompartmentDefinition.description
Definition | A free text natural language description of the compartment definition from a consumer's perspective.
Control | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the compartment definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the compartment definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created).
CompartmentDefinition.purpose
Definition | Explaination of why this compartment definition is needed and why it has been designed as it has.
Control | 0..1
Type | markdown
Comments | This element does not describe the usage of the compartment definition Instead it provides traceability of "why" the resource is either needed or "why" it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this compartment definition.
CompartmentDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate compartment definition instances.
Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
CompartmentDefinition.jurisdiction
Definition | A legal or geographic region in which the compartment definition is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the compartment definition to be used in jurisdictions other than those for which it was originally designed or intended.
CompartmentDefinition.code
Definition | Which compartment this definition describes.
Control | 1..1
Terminology Binding | CompartmentType (Required)
Type | code
Summary | true
Comments | Only the specification can define the compartments that can exist. Servers can choose to support them.
CompartmentDefinition.search

Definition | Whether the search syntax is supported,.
Control | 1..1
Type | boolean
Summary | true
Comments | Servers may define and use compartments to manage logical access without implementing the compartment related syntax.
CompartmentDefinition.resource
Definition | Information about how a resource is related to the compartment.
Control | 0..*
Summary | true
CompartmentDefinition.resource.code
Definition | The name of a resource supported by the server.
Control | 1..1
Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
CompartmentDefinition.resource.param
Definition | The name of a search parameter that represents the link to the compartment. More than one may be listed because a resource may be linked to a compartment in more than one way,.
Control | 0..*
Type | string
Summary | true
Comments | If no search parameters are listed, then the resource is not linked to the compartment.
CompartmentDefinition.resource.documentation
Definition | Additional documentation about the resource and compartment.
Control | 0..1
Type | string

5.6.8 Resource CompartmentDefinition - Mappings

URL: https://hl7.org/fhir/STU3/compartmentdefinition-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.6.8 Resource CompartmentDefinition - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

HL7 v2 Mapping

Mappings for the compartmentdefinition resource.

5.6.8.1 Mappings for RIM Mapping (http://hl7.org/v3)

CompartmentDefinition | n/a

5.6.8.2 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

CompartmentDefinition | n/a

5.6.9 Resource CompartmentDefinition - Design Notes

URL: https://hl7.org/fhir/STU3/compartmentdefinition-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.6.9 Resource CompartmentDefinition - Design Notes

to do

5.6.10 Resource CompartmentDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/compartmentdefinition-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.6.10 Resource CompartmentDefinition - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the CompartmentDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

5.6.10.1 Profiles

No Profiles defined for this resource

5.6.10.2 Extensions

resource-approvalDate | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | lastReviewDate | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

5.6.10.3 Search Extensions

No Search Extensions defined for this resource

5.7 Resource DataElement - Content

URL: https://hl7.org/fhir/STU3/dataelement.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

The formal description of a single piece of information that can be gathered and reported.

5.7.1 Scope and Usage

DataElement is an infrastructure resource that supports the defining individual pieces of data that might be collected or stored. While these definitions might apply to elements found in FHIR resources and profiles, they can also apply to questionnaire questions, elements in data stores, and non-FHIR specification (HL7 v2 , CDA , CDISC, etc.) I.e. the definitions aren't FHIR-specific.

This resource covers two major use-cases:

* Definitions of types of measurements or observations that may be requested or performed. In HL7, these are sometimes referred to as service, test or observation "master files"

* Definitions of "data elements" (DEs) that may be used in questionnaires (survey instruments and data collection forms) and profiles and potentially mapped to elements in other resources and profiles

The purpose of the first use-case is to allow systems to identify what types of lab orders, diagnostic studies and other types of observations may be requested or performed within a particular organization or other context. An ordering practitioner can query for a list of data elements by category name or other criteria and identify which, within a set of similar tests, they wish to request to be performed.

The focus of the second use-case is standardizing data capture and reporting. By defining standard names, data capture constraints, questions and other characteristics, the data gathered within and across organizations via questionnaires, as part of clinical studies, etc. can be made more consistent. When designing clinical studies, constructing questionnaires, building profiles or performing other tasks that involve determining what data will be captured or exchanged and how, designers can query to find pre-defined data element definitions they can leverage or map to. By encouraging consistency around data element definitions, data types, value sets, string lengths and other constraints, data becomes more easily exchangeable and comparable across systems. This benefits interoperability and clinical research. (For more discussion, see the section on standardization below.)

The scope covers base capabilities of the ISO 11179 Metadata Registries specification which defines DEs. It also covers observation definitions by ontologies such as LOINC. The term "observation" is interpreted in its broadest sense as "any element that might be thought of as the 'value' in a name-value pair". So it includes such concepts as patient gender, practitioner address and other data elements that would not typically be captured using the Observation resource.

5.7.2 Boundaries and Relationships

This resource has significant overlap with StructureDefinition and Questionnaire.

StructureDefinition also defines data elements. However, it does so only in the context of FHIR-specific artifacts (FHIR resources, data types and profiles). As well, StructureDefinition typically identifies a number of data elements together in context. DataElement defines only a single data element and it does so in a manner that is not directly tied to FHIR. Data elements might define the value of an Observation, constrain the allowed answer to a question in a Questionnaire (including providing a list of permitted answers), describe the permitted value captured of an element in some other resource (Patient, FamilyMemberHistory, etc.) or even used outside FHIR entirely in a CDA document or HL7 v2 specification.

Questionnaire is to both define forms, surveys and other structures that can be filled out. Questionnaire also defines data elements. However, it does so only in the context of a particular questionnaire design. In contrast, DataElement is focused on defining data elements independent of their use in questionnaires and other structures. A single DataElement might be referenced in numerous Questionnaires, or even potentially in multiple places within a single Questionnaire. This reference might either be implicit or may be explicit through an extension. (For implementability reasons, the data constraints should still be explicitly exposed within the Questionnaire rather than being included "by reference" to the DataElement.)

Note: Extensions on DataElement that define the characteristics of a data element will typically also be applicable to Profile's ElementDefinition and Questionnaire's Question data element as both are also used to define the characteristics of a data element.

DataElement differs from Observation in that it describes what kind of observations can occur, while Observation focuses on a specific observation of a specific subject at a particular time that has occurred.

ActivityDefinition defines actions that can occur. DataElement defines individual pieces of data that can be captured. An ActivityDefinition that is focused on capturing data might be linked to the DataElement that defines the data to be captured.

5.7.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
DataElement | | | DomainResource | Resource data element Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this data element (globally unique)
identifier | ? | 0..* | Identifier | Additional identifier for the data element
version | ? | 0..1 | string | Business version of the data element
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
name | ? | 0..1 | string | Name for this data element (computer friendly)
title | ? | 0..1 | string | Name for this data element (human friendly)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for data element (if applicable) Jurisdiction ValueSet (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
stringency | ? | 0..1 | code | comparable | fully-specified | equivalent | convertable | scaleable | flexible DataElementStringency (Required)
mapping | I | 0..* | BackboneElement | External specification mapped to + At least one of name or uri SHALL be present
identity | | 1..1 | id | Internal id when this mapping is used
uri | | 0..1 | uri | Identifies what this mapping refers to
name | | 0..1 | string | Names what this mapping refers to
comment | | 0..1 | string | Versions, issues, scope limitations, etc.
element | ?I | 1..* | ElementDefinition | Definition of element + No base allowed + No slicing allowed
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DataElement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this data element (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the data element --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the data element -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <name value="[string]"/><!-- 0..1 Name for this data element (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this data element (human friendly) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for data element (if applicable) --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <stringency value="[code]"/><!-- 0..1 comparable | fully-specified | equivalent | convertable | scaleable | flexible
    -->
 <mapping>  <!--  0..* External specification mapped to -->
  <identity value="[id]"/><!-- 1..1 Internal id when this mapping is used -->
  <uri value="[uri]"/><!-- 0..1 Identifies what this mapping refers to -->
  <name value="[string]"/><!-- 0..1 Names what this mapping refers to -->
  <comment value="[string]"/><!-- 0..1 Versions, issues, scope limitations, etc. -->
 </mapping>
 <element><!-- 1..* ElementDefinition Definition of element --></element>
</DataElement>
```

JSON Template

```
{
```

```
  "resourceType" : "DataElement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this data element (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the data element
  "version" : "<string>", // Business version of the data element
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "name" : "<string>", // Name for this data element (computer friendly)
  "title" : "<string>", // Name for this data element (human friendly)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for data element (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "stringency" : "<code>", // comparable | fully-specified | equivalent | convertable | scaleable | flexible
  "mapping" : [{ // C? External specification mapped to
    "identity" : "<id>", // R!  Internal id when this mapping is used
    "uri" : "<uri>", // Identifies what this mapping refers to
    "name" : "<string>", // Names what this mapping refers to
    "comment" : "<string>" // Versions, issues, scope limitations, etc.
  }],
  "element" : [{ ElementDefinition }] // R!  Definition of element
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DataElement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DataElement.url [ uri ]; # 0..1 Logical URI to reference this data element (globally unique)
  fhir:DataElement.identifier [ Identifier ], ... ; # 0..* Additional identifier for the data element
  fhir:DataElement.version [ string ]; # 0..1 Business version of the data element
  fhir:DataElement.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:DataElement.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:DataElement.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:DataElement.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:DataElement.name [ string ]; # 0..1 Name for this data element (computer friendly)
  fhir:DataElement.title [ string ]; # 0..1 Name for this data element (human friendly)
  fhir:DataElement.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:DataElement.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:DataElement.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for data element (if applicable)
  fhir:DataElement.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:DataElement.stringency [ code ]; # 0..1 comparable | fully-specified | equivalent | convertable | scaleable |
    flexible
  fhir:DataElement.mapping [ # 0..* External specification mapped to
    fhir:DataElement.mapping.identity [ id ]; # 1..1 Internal id when this mapping is used
    fhir:DataElement.mapping.uri [ uri ]; # 0..1 Identifies what this mapping refers to
    fhir:DataElement.mapping.name [ string ]; # 0..1 Names what this mapping refers to
    fhir:DataElement.mapping.comment [ string ]; # 0..1 Versions, issues, scope limitations, etc.
  ], ...;
  fhir:DataElement.element [ ElementDefinition ], ... ; # 1..* Definition of element
]
```

Changes since DSTU2

DataElement

DataElement.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status

DataElement.experimental | Now marked as Modifier

DataElement.title | Added Element

DataElement.contact | Type changed from BackboneElement to ContactDetail

DataElement.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)

DataElement.jurisdiction | Added Element

DataElement.copyright | Type changed from string to markdown

DataElement.mapping.comment | Renamed from comments to comment

DataElement.contact.name | deleted

DataElement.contact.telecom | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 4 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (4 errors).).

Structure

Name | Flags | Card. | Type | Description & Constraints

DataElement | | | DomainResource | Resource data element Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | ? | 0..1 | uri | Logical URI to reference this data element (globally unique)

identifier | ? | 0..* | Identifier | Additional identifier for the data element

version | ? | 0..1 | string | Business version of the data element
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
name | ? | 0..1 | string | Name for this data element (computer friendly)
title | ? | 0..1 | string | Name for this data element (human friendly)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for data element (if applicable) Jurisdiction ValueSet (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
stringency | ? | 0..1 | code | comparable | fully-specified | equivalent | convertable | scaleable | flexible DataElementStringency (Required)
mapping | I | 0..* | BackboneElement | External specification mapped to + At least one of name or uri SHALL be present
identity | | 1..1 | id | Internal id when this mapping is used
uri | | 0..1 | uri | Identifies what this mapping refers to
name | | 0..1 | string | Names what this mapping refers to
comment | | 0..1 | string | Versions, issues, scope limitations, etc.
element | ?I | 1..* | ElementDefinition | Definition of element + No base allowed + No slicing allowed
Documentation for this format

UML Diagram (Legend)

XML Template

```
<DataElement xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this data element (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the data element --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the data element -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <name value="[string]"/><!-- 0..1 Name for this data element (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this data element (human friendly) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for data element (if applicable) --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <stringency value="[code]"/><!-- 0..1 comparable | fully-specified | equivalent | convertable | scaleable | flexible
    -->
 <mapping>  <!--  0..* External specification mapped to -->
  <identity value="[id]"/><!-- 1..1 Internal id when this mapping is used -->
  <uri value="[uri]"/><!-- 0..1 Identifies what this mapping refers to -->
  <name value="[string]"/><!-- 0..1 Names what this mapping refers to -->
  <comment value="[string]"/><!-- 0..1 Versions, issues, scope limitations, etc. -->
 </mapping>
 <element><!-- 1..* ElementDefinition Definition of element --></element>
</DataElement>
```

JSON Template

```
{
  "resourceType" : "DataElement",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this data element (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the data element
  "version" : "<string>", // Business version of the data element
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "name" : "<string>", // Name for this data element (computer friendly)
  "title" : "<string>", // Name for this data element (human friendly)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for data element (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "stringency" : "<code>", // comparable | fully-specified | equivalent | convertable | scaleable | flexible
  "mapping" : [{ // C? External specification mapped to
    "identity" : "<id>", // R!  Internal id when this mapping is used
    "uri" : "<uri>", // Identifies what this mapping refers to
    "name" : "<string>", // Names what this mapping refers to
    "comment" : "<string>" // Versions, issues, scope limitations, etc.
  }],
  "element" : [{ ElementDefinition }] // R!  Definition of element
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:DataElement;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root
```

```
  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:DataElement.url [ uri ]; # 0..1 Logical URI to reference this data element (globally unique)
  fhir:DataElement.identifier [ Identifier ], ... ; # 0..* Additional identifier for the data element
  fhir:DataElement.version [ string ]; # 0..1 Business version of the data element
  fhir:DataElement.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:DataElement.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:DataElement.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:DataElement.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:DataElement.name [ string ]; # 0..1 Name for this data element (computer friendly)
  fhir:DataElement.title [ string ]; # 0..1 Name for this data element (human friendly)
  fhir:DataElement.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:DataElement.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:DataElement.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for data element (if applicable)
  fhir:DataElement.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:DataElement.stringency [ code ]; # 0..1 comparable | fully-specified | equivalent | convertable | scaleable |
    flexible
  fhir:DataElement.mapping [ # 0..* External specification mapped to
    fhir:DataElement.mapping.identity [ id ]; # 1..1 Internal id when this mapping is used
    fhir:DataElement.mapping.uri [ uri ]; # 0..1 Identifies what this mapping refers to
    fhir:DataElement.mapping.name [ string ]; # 0..1 Names what this mapping refers to
    fhir:DataElement.mapping.comment [ string ]; # 0..1 Versions, issues, scope limitations, etc.
  ], ...;
  fhir:DataElement.element [ ElementDefinition ], ... ; # 1..* Definition of element
]
```

Changes since DSTU2

DataElement

DataElement.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status

DataElement.experimental | Now marked as Modifier

DataElement.title | Added Element

DataElement.contact | Type changed from BackboneElement to ContactDetail

DataElement.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)

DataElement.jurisdiction | Added Element

DataElement.copyright | Type changed from string to markdown

DataElement.mapping.comment | Renamed from comments to comment

DataElement.contact.name | deleted

DataElement.contact.telecom | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 4 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (4 errors).).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

5.7.3.1 Terminology Bindings

Path | Definition | Type | Reference

DataElement.status | The lifecycle status of a Value Set or Concept Map. | Required | PublicationStatus

DataElement.jurisdiction | Countries and regions within which this artifact is targeted for use | Extensible | Jurisdiction ValueSet

DataElement.stringency | Indicates the degree of precision of the data element definition. | Required | DataElementStringency

5.7.3.2 Constraints

* dae-1: On DataElement.element: No base allowed (expression on DataElement.element: base.empty())

* dae-2: On DataElement.element: No slicing allowed (expression on DataElement.element: slicing.empty())

* dae-3: On DataElement.mapping: At least one of name or uri SHALL be present (expression on DataElement.mapping: uri.exists() or name.exists())

5.7.3.3 Relationship to ISO 11179

One of the sources of the DataElement resource definition was the ISO 11179 Metadata Registries specification. It defines a logical model for the classification, identification, naming and registration of Data Elements, Data Element Concepts and their associated Value and Conceptual Domains.

The DataElement resource can be used to represent both Data_Element and Data_Element_Concept in the ISO logical model. (The ValueSet resource provides the details for Value_Domain and Conceptual_Domain for enumerated elements.) The determination of whether a DataElement resource instance is an ISO 11179 Data_Element or Data_Element_Concept is determined by whether the type property is specified - which corresponds to the ISO property Data_Element.domain.datatype which is required for Data_Elements.

Unlike the 11179 logical specification, the DataElement resource does not require a linkage from data element to a distinct data element concept, though this linkage can be established through an extension if desired. The typical means of identification of the data element concept is expected to instead occur through the mapping of the data element to a particular code or reference model that formally defines the concept. It is possible this reference model could be based on ISO's Object_Class and Property mechanism. However, mappings to other reference models are also possible, for example:

* HL7's Reference Information Model

* Open EHR's reference model

* the BRIDG logical model

* terminologies such as LOINC or SNOMED CT

In the event of multiple codes and/or mappings, the "authoritative" mapping (the one formally defining the concept) is identified by either setting the "primary" element on the code or by specifying the "primary mapping" ISO 11179 extension.

It is possible to create instances that are "conforming", or even "strictly conforming" to the 11179 specification. However, doing so will require the use of extensions to convey certain properties that are not part of the core resource and data types. An initial starter ISO 11179 profile is included in the FHIR specification. It defines extensions that are relevant to the SDC DataElement profile. If there is sufficient interest, the existing starter set may be enhanced to contain a complete set of extensions and a full 11179 in a future release of the specification.

5.7.3.4 Data Elements vs. Codes

Data elements (and their registries) and codes (and their code systems) serve different purposes. Data elements describe the characteristics of a piece of data - identifier, name, definition, data type, length, permitted value set, usage notes, etc. On the other hand, while codes describe a particular concept including a symbol for use when exchanging the concept, display names, definitions and relationships to other concepts. There is a subset of information that is common for both codes and data elements - identifier, name and definition. Because of this, a list of data elements might sometimes be seen as a list of codes.

One of the purposes of data elements is to define variables, observations, or questions that include information about how to collect a variable's value--including information about data types and guidance about the use of about answer lists and/or units of measure. This function is typically done by code systems such as LOINC and SNOMED CT . (Note that LOINC provides considerable detail about allowed answers, while most other code systems such as SNOMED CT only allow identifying the question.) It would be possible to treat a registry of data elements using a common identifier system/namespace as a code system. Similarly, data elements can be mapped to each other (and to codes) in the same manner as codes can be mapped by making use of the ConceptMap resource.

That said, FHIR treats code systems as simple structures that provide identity, names and definitions. Code systems that provide significant additional detail (recommended doses for drugs, data types for data elements, etc.) are treated as a mix of a code system together with a supplementary resource to provide the details. Refer to the Code systems with detailed metadata section of ValueSet resource for additional discussion of the relationship between DataElement and code systems.

5.7.3.5 Standardization through governance and registries

As noted in the introduction, one of the main purposes for the use of data elements is 'standardization' - gaining consistent (and thus comparable) data capture of data in questionnaires, observations and within other resources. However, merely defining data elements is not, in itself, sufficient to improve standardization. To ensure consistency of data, there needs to be several additional things:

* There must be a shared repository of data elements that will be consulted and used by the community where improved interoperability is targeted. Numerous groups will define and register data elements in a wide variety of repositories for a wide variety of purposes. To improve interoperability, everyone in the target community needs to be working from the same set.

* There must be governance over the repository of data elements to ensure consistency of granularity and reference structures - those designing questionnaires, profiles, etc. need to be able to find the appropriate data element easily within the repository and there should ideally only be one data element for a given purpose. If a repository accepts 15 data elements defining different ways of capturing the same piece of information, there will be no improvement in interoperability.

* There must be incentives or governance processes in place that encourage or require the use of data elements from the shared repository when defining forms, profiles and other structures where improved interoperability is desired.

* There must be allowance for (and a transition path for) legacy systems that capture data or have existing data that is not in alignment with the approved data elements.

Data Elements are a tool through which improved standardization can be achieved but without adequate processes, they will not achieve significant benefit.

5.7.3.6 Data Element granularity and hierarchy

Data elements can be defined at a wide variety of granularities. For example:

* Systolic Blood pressure

* Systolic Blood pressure (sitting)

* Systolic Blood pressure (sitting, pressure cuff)

* Systolic Blood pressure (sitting, pressure cuff) mmHg

Coded data elements can be defined without value sets, with suggested value sets, with required but extensible value sets or with completely locked down value sets. Numeric values can be defined with or without allowed ranges. String data can be defined with or without patterns or lengths. All are valid data elements.

When defining data elements, it's important to decide what level of detail/granularity is appropriate for the intended purpose. The tighter the data element is defined, the greater interoperability of the data, however the smaller the set of systems that will be able to satisfy the constraints and the larger the number of data elements that will be required to cover a given domain. In some cases, multiple granularities may be appropriate, though this can introduce challenges in ensuring that the

appropriate granularity is selected for a given use. As a rule, data elements should be defined as loosely as possible while still ensuring that all data captured using the data element will be sufficiently comparable to meet the needs of the group defining and using the data elements. A corollary to this is that data elements defined by one group will not always meet the needs of another group, even if they may be covering the same domain.

5.7.3.7 Comparability and conversion of data elements

Two pieces of data don't necessarily need to be based on the same data element in order to be comparable or aggregatable. As noted above, it's possible for one data element to be a proper subset of another. In addition, it may be possible to convert data from being conformant with one data element to being conformant with another data element. This conversion could be lossless or may involve some loss of semantic precision. For example, a weight measurement captured in pounds can be seamlessly converted to kilograms. Similarly, coded data captured using one value set can be converted to another value set provided mappings are available. The nature of the mappings would determine whether any loss of semantics would occur.

5.7.3.8 Data Element stringency

The stringency element allows identifying the types of usages for which the data element is appropriate. Some uses (direct comparison of data, auto-population of data, etc.) are inappropriate with loose definitions of elements. On the other hand, overly-restrictive data elements can result in a large number of data elements and limited abilities to compare similar but not identical data. For now, the definition of what level of constraints are appropriate/necessary for a given level of stringency is left to implementers, though future versions this specification may assert at least some level of guidance in this area.

5.7.3.9 Definitions

Data elements provide their value by clearly defining the meaning and content of a particular type of data to be exchanged. This value depends on the meaning associated with the element being clear for all potential implementers of the element. I.e. For a data element to be useful, it needs to have a good quality definition. Characteristics of good quality definitions include:

* The definition does not use the name of the element as part of the definition - it should provide additional information above and beyond the name.

* The definition should be more than one or two words.

* Ideally, the definition should be phrased in a manner such that the definition could be substituted for the data element name in a sentence; e.g. "gender" might be defined as "the sexual identity of a person as used for administrative purposes"

* If the data element does not have a value set and does not make use of a simple type such as date, integer or boolean, it should list several example values as part of the definition.

* Information about the rationale for the element or additional usage notes should not be conveyed as part of the definition but should instead make use of the notes or rationale elements.

5.7.3.10 Data Elements and Access Control

Because Data Elements define precise pieces of data that can be conveyed in a variety of places, they may form a useful mechanism for the definition of fine-grained data access controls. The base resource does not include mechanisms for linking access controls directly do data elements, however, an extension could allow a particular data element to be associated with particular Security Labels. Whether this level of control is appropriate may vary by implementation environment.

5.7.3.11 References to and from Data Elements

One of the primary purposes of the DataElement resource is to allow data conveyed in a variety of different forms to be "linked" to the data element and identified as being equivalent/comparable/semantically-aligned. There are a variety of different ways this can occur:

* DataElement.element.code allows a data element to be linked as "semantically equivalent" to the specified code.

* DataElement.element.mapping allows a data element to be linked to a particular data structure such as a FHIR resource, a CDA template, an OpenEHR archetype, etc.

* The Questionnaire.item.definition allows a particular question in a questionnaire to be linked to a data element definition deemed to correspond to that question. This mechanism supports population of questionnaire instances on the basis of known data elements. The intent is that a server will populate anything it recognizes by data elements. However servers may not know all the data elements, and won't populate ones that they don't know.

STU Note: The FHIR Infrastructure work group is considering rolling the DataElement resource into the StructureDefinition resource. If this is done, DataElement will be treated as a type of logical model (whether there will be a distinct 'type' for it is unclear).

Implementers wishing to provide their feedback on the proposed change are invited to make comments on the associated Tracker item .

5.7.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name	Type	Description	Expression	In Common
code	token	A code for the data element (server may choose to do subsumption)	DataElement.element.code	
date	date	The data element publication date	DataElement.date	

description | string | Text search in the description of the data element. This corresponds to the definition of the first DataElement.element. | DataElement.element.definition
identifier | token | External identifier for the data element | DataElement.identifier
jurisdiction | token | Intended jurisdiction for the data element | DataElement.jurisdiction
name | string | Computationally friendly name of the data element | DataElement.name
publisher | string | Name of the publisher of the data element | DataElement.publisher
status | token | The current status of the data element | DataElement.status
stringency | token | The stringency of the data element definition | DataElement.stringency
title | string | The human-friendly name of the data element | DataElement.title
url | uri | The uri that identifies the data element | DataElement.url
version | token | The business version of the data element | DataElement.version

5.7.5 Resource DataElement - Examples

URL: https://hl7.org/fhir/STU3/dataelement-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.7.5 Resource DataElement - Examples

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Patient gender represented as an data element | gender | XML | JSON | Turtle
Prothrombin time for lab test master file | prothrombin | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.7.6 Resource DataElement - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/dataelement-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.7.6 Resource DataElement - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the DataElement resource.

DataElement
Definition | The formal description of a single piece of information that can be gathered and reported.
Control | 1..1
Alternate Names | Master Observation; Data Element; CDE
Comments | Often called a clinical template.
DataElement.url
Definition | An absolute URI that is used to identify this data element when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this data element is (or will be) published. The URL SHOULD include the major version of the data element. For more information see Technical and Business Versions.
Control | 0..1
Type | uri
Requirements | Allows the data element to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.
DataElement.identifier
Definition | A formal identifier that is used to identify this data element when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, e.g., to identify this data element outside of FHIR, where it is not possible to use the logical URI.
LOINC Code | LOINC_NUM (the code is the identifier)
DataElement.version
Definition | The identifier that is used to identify this version of the data element when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the data element author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Control | 0..1
Type | string
Summary | true

Comments | There may be different data element instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the data element with the format [url]|[version].
LOINC Code | N/A
DataElement.status
Definition | The status of this data element. Enables tracking the life-cycle of the content.
Control | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Allows filtering of data elements that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired data element without due consideration.
LOINC Code | STATUS
DataElement.experimental
Definition | A boolean value to indicate that this data element is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.
Control | 0..1
Type | boolean
Is Modifier | true
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level data element.
Summary | true
Comments | Allows filtering of data element that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental data element in production.
DataElement.date
Definition | The date (and optionally time) when the data element was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the data element changes.
Control | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the data element. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
LOINC Code | DATE_LAST_CHANGED
DataElement.publisher
Definition | The name of the individual or organization that published the data element.
Control | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the data element. May also allow for contact.
Summary | true
Comments | Usually an organization, but may be an individual. The publisher (or steward) of the data element is the organization or individual primarily responsible for the maintenance and upkeep of the data element. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the data element. This item SHOULD be populated unless the information is available from context.
LOINC Code | Fixed to Regenstrief or SOURCE
DataElement.name
Definition | A natural language name identifying the data element. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Control | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.
DataElement.title
Definition | A short, descriptive, user-friendly title for the data element.
Control | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc. This may be similar to the question used to prompt someone for a data element value, but the purpose is communicating about the data element, not asking the question. (Refer to the 'question' element for the latter.).
LOINC Code | LONG_COMMON_NAME
DataElement.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
LOINC Code | N/A
DataElement.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate data element instances.
Control | 0..*

Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
DataElement.jurisdiction
Definition | A legal or geographic region in which the data element is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the data element to be used in jurisdictions other than those for which it was originally designed or intended.
DataElement.copyright
Definition | A copyright statement relating to the data element and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the data element.
Control | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the data element and/or its content.
Alternate Names | License; Restrictions
Comments | null The copyright statement does not apply to values for the data element, only to its definition.
DataElement.stringency
Definition | Identifies how precise the data element is in its definition.
Control | 0..1
Terminology Binding | DataElementStringency (Required)
Type | code
Requirements | Allows determining the degree of comparability of data element instances. Less granular data elements result in data that is less comparable (or at least requires more work to compare).
Summary | true
DataElement.mapping
Definition | Identifies a specification (other than a terminology) that the elements which make up the DataElement have some correspondence with.
Control | 0..*
Invariants | Defined on this element dae-3: At least one of name or uri SHALL be present (expression : uri.exists() or name.exists(), xpath: exists(f:uri) or exists(f:name)) Affect this element
DataElement.mapping.identity
Definition | An internal id that is used to identify this mapping set when specific mappings are made on a per-element basis.
Control | 1..1
Type | id
Comments | The specification is described once, with general comments, and then specific mappings are made that reference this declaration.
DataElement.mapping.uri
Definition | An absolute URI that identifies the specification that this mapping is expressed to.
Control | 0..1
Type | uri
Comments | A formal identity for the specification being mapped to helps with identifying maps consistently.
DataElement.mapping.name
Definition | A name for the specification that is being mapped to.
Control | 0..1
Type | string
DataElement.mapping.comment
Definition | Comments about this mapping, including version notes, issues, scope limitations, and other important notes for usage.
Control | 0..1
Type | string
DataElement.element
Definition | Defines the structure, type, allowed values and other constraining characteristics of the data element.
Control | 1..*
Type | ElementDefinition
Summary | true
Comments | For simple data types there will only be one repetition. For complex data types, multiple repetitions will be present defining a nested structure using the "path" element.
Invariants | Defined on this element dae-1: No base allowed (expression : base.empty(), xpath: not(exists(f:base))) dae-2: No slicing allowed (expression : slicing.empty(), xpath: not(exists(f:slicing)))

5.7.7 Resource DataElement - Mappings

URL: https://hl7.org/fhir/STU3/dataelement-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.7.7 Resource DataElement - Mappings

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

LOINC code for the element

HL7 v2 Mapping

ISO 11179

IHE Data Element Exchange (DEX)

Mappings for the dataelement resource.

5.7.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

DataElement | Observation[classCode=OBS, moodCode=DEFN]
url | ./id[use=INSTANCE]
identifier | ./id[use=BUS]
version | ./id[use=VER]
status | ./status (draft = new active = active retired = aborted)
experimental | ./inboundRelationship[typeCode=SUBJ]/source[classCode=OBS,moodCode=EVN,code="experimental"].value[type=BL]
date | ./participation[typeCode=AUT]/time
publisher | ./participation[typeCode=AUT]/role[classCode=ASSIGN]/scoper[classCode=ORG,determinerCode=INSTANCE]/name
name
title | ./title
contact | ./participation[typeCode=AUT]/role[classCode=ASSIGN]/scoper[classCode=ORG,determinerCode=INSTANCE]
useContext | ./inboundRelationship[typeCode=SUBJ]/source[classCode=LIST, moodCode=EVN]/code
jurisdiction
copyright | ./inboundRelationship[typeCode=SUBJ]/source[classCode=CONTRACT,moodCode=EVN].text
stringency | ./target[typeCode=SUBJ]/source[classCode=OBS, moodCode=EVN, code="granularity"]/value
mapping | N/A (MIF)
identity | N/A (MIF)
uri | N/A (MIF)
name | N/A (MIF)
comment | N/A (MIF)
element | N/A (MIF)

5.7.7.2 Mappings for LOINC code for the element (http://loinc.org)

DataElement
url
identifier | LOINC_NUM (the code is the identifier)
version | N/A
status | STATUS
experimental
date | DATE_LAST_CHANGED
publisher | Fixed to Regenstrief or SOURCE
name
title | LONG_COMMON_NAME
contact | N/A
useContext
jurisdiction
copyright
stringency
mapping
identity
uri
name
comment
element

5.7.7.3 Mappings for HL7 v2 Mapping (http://hl7.org/v2)

DataElement | OM1
url
identifier | OM1.2
version | N/A
status | MFE.1
experimental
date | OM1.21
publisher | OM1.16
name
title | OM1.11
contact | OM1.17
useContext | OM1.18, OM1.42, OM1.46, OM7.3
jurisdiction
copyright
stringency | N/A
mapping
identity

uri
name
comment
element

5.7.7.4 Mappings for ISO 11179 (http://metadata-standards.org/11179/)

These mappings are included to indicate where properties of the data elements defined by the FHIR specification relate to similar fields in the ISO 11179 specification.

DataElement | Data_Element
url | (Identified_Item).identifier As assigned by registering authority
identifier | (Identified_Item).identifier
version | (Identified_Item).identifier.version
status | (Administered_item).registration.state new = Incomplete, Candidate, Recorded, Qualified active = Standard, Preferred Standard, Application retired = Retired, Superseded, Historical
experimental
date | (Administered_item).last_change_date
publisher | (Administered_Item).stewardship_record.organization.name
name
title | (Designatable_Item).designation.sign acceptability=preferred in default context
contact | (Administered_Item).stewardship_record.organization.[email_address, phone_number, uri]
useContext | (Classifiable_Item).classifier
jurisdiction
copyright
stringency
mapping
identity
uri
name
comment
element

5.7.7.5 Mappings for IHE Data Element Exchange (DEX) (http://wiki.ihe.net/index.php?title=Data_Element_Exchange)

DataElement | RetrieveMetadata response
url
identifier | id
version | version
status
experimental
date | revisionDate
publisher | registrationAuthority
name
title | displayName
contact
useContext
jurisdiction
copyright
stringency
mapping | 100;0
identity
uri
name
comment
element

5.7.8 Resource DataElement - Design Notes

URL: https://hl7.org/fhir/STU3/dataelement-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.7.8 Resource DataElement - Design Notes

to do

5.7.9 Resource DataElement - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/dataelement-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.7.9 Resource DataElement - Extensions & Profiles

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the DataElement resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

5.7.9.1 Profiles

No Profiles defined for this resource

5.7.9.2 Extensions

11179-de-administrative-status | administrative-status | for ISO 11179 Data Element Profile
11179-de-change-description | change-description | for ISO 11179 Data Element Profile
11179-de-classification-or-context | classification-or-context | for ISO 11179 Data Element Profile
11179-de-contact-address | contact-address | for ISO 11179 Data Element Profile
11179-de-document-reference | document-reference | for ISO 11179 Data Element Profile
11179-de-effective-period | effective-period | for ISO 11179 Data Element Profile
11179-de-is-data-element-concept | is-data-element-concept | for ISO 11179 Data Element Profile
11179-de-registry-org | registry-org | for ISO 11179 Data Element Profile
11179-de-submitter-org | submitter-org | for ISO 11179 Data Element Profile
11179-permitted-value-conceptmap | permitted-value-conceptmap | for ISO 11179 Element Definition Profile
11179-permitted-value-valueset | permitted-value-valueset | for ISO 11179 Element Definition Profile
mapSourcePublisher | mapSourcePublisher | for Standard DataElement Extensions
resource-approvalDate | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | lastReviewDate | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

5.7.9.3 Search Extensions

No Search Extensions defined for this resource

5.7.10 Standard DataElement Extensions

URL: https://hl7.org/fhir/STU3/dataelement-extensions.html

This profile describes common extensions that are used with DataElements

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.7.10 Standard DataElement Extensions

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

5.7.10.1 Content

Extensions:

datadictionary | datadictionary : This extension associates a structure definition with a data dictionary - a collection of data elements that any instance of data that conforms to the structure definition must also conform to. The value of the extension is a uri that defines a query that identifies the collection of relevant data elements This extension can be used on any structure definition that is a constraint on the resource types Observation and Questionnaire. THe profile specifies which elements are mandatory etc., but any elements in a resource that conforms to the structure definition must also conform to one of the defined data elements. For Observation: * Observation.code must be associated with a matching Data Element by DataElement.code. * The structure definition should bind Observation.code to a value set that has the same codes as the collection of data elements (though it can be narrower) * the unit and the quantity value should conform to the rules laid down in the matching Data Element For Questionnaire: * todo.

mapSourcePublisher | mapSourcePublisher : The name of the organization responsible for creating the specification being mapped to.

5.7.11 ISO 11179 Data Element Profile

URL: https://hl7.org/fhir/STU3/dataelement-11179.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.7.11 ISO 11179 Data Element Profile

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

5.7.11.1 Scope and Usage

This profile defines extensions that may be useful in exposing data elements from ISO 11179 -conformant registries. The list of extensions is not currently complete - additional extensions would be required to support all required and optional 11179 elements. The set of define extensions will increase over time based on implementer interest. At some point, a full profile for

11179-conformant systems may be developed.

The list of defined extensions is as follows:

* objectClass - specifies the Object_Class for a Concept_Element or the meaning of a Data_Element

* property - specifies the Object_Class Property for a Concept_Element or the meaning of a Data_Element

5.7.11.2 Content

Extensions:

11179-de-effective-period | effective-period : Identifies the date on which the element became (or is expected to become) active and the date on which the element became (or is expected to become) retired.

11179-de-is-data-element-concept | is-data-element-concept : If true, indicates that the specified coding corresponds to the data element concept for the data element.

11179-de-document-reference | document-reference : Identifies a document that conveys some sort of information related to the data element.

11179-de-classification-or-context | classification-or-context : Identifies whether the useContext is a true context or a classification.

11179-de-administrative-status | administrative-status : Identifies internal processing status for the data element; e.g. "Under review".

11179-de-change-description | change-description : Summarizes changes to the data element since the previous version.

11179-de-submitter-org | submitter-org : Organization responsible for submitting the data element to its registry.

11179-de-contact-address | contact-address : Postal address for a contact for the data element.

11179-de-registry-org | registry-org : The organization that is the custodian for this data element.

5.7.12 Resource DataElement - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/dataelement-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.7.12 Resource DataElement - Detailed Descriptions

Orders and Observations Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for DataElement.

Functional status for this map: 4 tests that all execute ok. 1 fail round-trip testing and 2 r3 resources are invalid (4 errors). (see documentation)

5.7.12.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/DataElement2to3" = "R2 to R3 Conversions for DataElement"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DataElement" alias DataElementR2 as source
uses "http://hl7.org/fhir/StructureDefinition/DataElement" alias DataElement as target

imports "http://hl7.org/fhir/StructureMap/*2to3"

group for type+types  DataElement extends DomainResource
  input src : DataElementR2 as source
  input tgt : DataElement as target

  "DataElement-url" : for src.url make tgt.url
  "DataElement-identifier" : for src.identifier make tgt.identifier
  "DataElement-version" : for src.version make tgt.version
  "DataElement-status" : for src.status make tgt.status
  "DataElement-experimental" : for src.experimental make tgt.experimental
  "DataElement-publisher" : for src.publisher make tgt.publisher
  "DataElement-date" : for src.date make tgt.date
  "DataElement-name" : for src.name make tgt.name
  "DataElement-contact" : for src.contact as vs0 make tgt.contact as vt0 then decontact(vs0, vt0)
  "DataElement-jurisdiction" : for src.useContext where coding.system = 'urn:iso:std:iso:3166' make tgt.jurisdiction
  "DataElement-useContext" : for src.useContext as vs where (coding.system = 'urn:iso:std:iso:3166').not() make
    tgt.useContext as uc,
    uc.code as ucc,
    ucc.system = "http://hl7.org/fhir/usage-context-type",
    ucc.code = "task",
    uc.value = create("CodeableConcept") as vt then CodeableConcept(vs, vt)

  "DataElement-stringency" : for src.stringency make tgt.stringency
  "DataElement-mapping" : for src.mapping as vs0 make tgt.mapping as vt0 then deMapping(vs0, vt0)

  "DataElement-element" : for src.element as vs make tgt.element as vt, create("boolean") as first, first.value =
    "false" then ElementDefinition(vs, vt, first)
endgroup

group decontact extends Element
  input src as source
  input tgt as target

  "ContactDetail-name" : for src.name make tgt.name
  "ContactDetail-telecom" : for src.telecom make tgt.telecom
endgroup

group deMapping extends Element
  input src as source
  input tgt as target
```

```
  "DataElement.mapping-identity" : for src.identity make tgt.identity
  "DataElement.mapping-uri" : for src.uri make tgt.uri
  "DataElement.mapping-name" : for src.name make tgt.name
  "DataElement.mapping-comments" : for src.comments make tgt.comment
endgroup
```

5.7.12.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/DataElement3to2" = "R3 to R2 Conversion for DataElement"

uses "http://hl7.org/fhir/StructureDefinition/DataElement" alias DataElement as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/DataElement" alias DataElementR2 as target

imports "http://hl7.org/fhir/StructureMap/*3to2"

group for type+types  DataElement extends DomainResource
  input src : DataElement as source
  input tgt : DataElementR2 as target

  "DataElement-url" : for src.url make tgt.url
  "DataElement-identifier" : for src.identifier make tgt.identifier
  "DataElement-version" : for src.version make tgt.version
  "DataElement-experimental" : for src.experimental make tgt.experimental
  "DataElement-name" : for src.name make tgt.name
  "DataElement-status" : for src.status make tgt.status
  "DataElement-publisher" : for src.publisher make tgt.publisher
  "DataElement-contact" : for src.contact as vs0 make tgt.contact as vt0 then decontact(vs0, vt0)
  "DataElement-date" : for src.date make tgt.date
  "DataElement-useContext" : for src.jurisdiction make tgt.useContext
  "DataElement-useContext" : for src.useContext as vs then {
    "DataElement-useContext-value" : for vs.value make tgt.useContext
  }

  "DataElement-stringency" : for src.stringency make tgt.stringency
  "DataElement-mapping" : for src.mapping as vs0 make tgt.mapping as vt0 then deMapping(vs0, vt0)

  "DataElement-element" : for src.element as vs make tgt.element as vt then ElementDefinition(src, vs, vt)

endgroup

group decontact extends Element
  input src as source
  input tgt as target

  "ContactDetail-name" : for src.name make tgt.name
  "ContactDetail-telecom" : for src.telecom make tgt.telecom
endgroup

group deMapping extends Element
  input src as source
  input tgt as target

  "DataElement.mapping-identity" : for src.identity make tgt.identity
  "DataElement.mapping-uri" : for src.uri make tgt.uri
  "DataElement.mapping-name" : for src.name make tgt.name
  "DataElement.mapping-comments" : for src.comment make tgt.comments
endgroup
```

5.7.12.3 R3 Validation Errors

gender | WARNING @ DataElement.useContext.valueCodeableConcept.coding[5].display: value should not start or finish with whitespace (src = InstanceValidator)

dataelement-sdc-profile-example-de | ERROR @ DataElement.extension[1].valueReference: Unable to resolve resource '#doc' (src = InstanceValidator) ERROR @ DataElement.extension[1].valueReference: SHALL have a contained resource if a local reference is provided ((url: doc; ids: vm-valueset,pv-valueset,pv-conceptmap,sub-org)) [reference.startsWith('#').not() or (reference.substring(1).trace('url') in %resource.contained.id.trace('ids'))] (src = InstanceValidator) ERROR @ DataElement.element.binding.valueSetReference.extension[2].valueReference: Invalid Resource target type. Found ConceptMap, but expected one of (ValueSet) (src = InstanceValidator)

5.8 Resource ImplementationGuide - Content

URL: https://hl7.org/fhir/STU3/implementationguide.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

A set of rules of how FHIR is used to solve a particular problem. This resource is used to gather all the parts of an implementation guide into a logical whole and to publish a computable definition of all the parts.

5.8.1 Scope and Usage

An implementation guide (IG) is a set of rules about how FHIR resources are used (or should be used) to solve a particular problem, with associated documentation to support and clarify the usage. Classically, FHIR implementation guides are published on the web after they are generated using the FHIR Implementation Guide Publisher .

The ImplementationGuide resource is a single resource that defines the logical content of the IG, along with the important entry pages into the publication, so that the logical package that the IG represents, so that the contents are computable.

In particular, validators are able to use the ImplementationGuide resource to validate content against the implementation guide as a whole. The significant conformance expectation introduced by the ImplementationGuide resource is the idea of Default Profiles. Implementations may conform to multiple implementation guides at once, but this requires that the implementation guides are compatible (see below).

5.8.2 Boundaries and Relationships

Implementation Guides contain two different types of resource references:

* Contents: A set of logical statements which implementations must conform to. These are almost always conformance resources

* Examples: Examples that illustrate the intent of the profiles defined in the implementation guide. These can be any kind of resource

An application's Capability Statement may identify one or more implementation guides that an application conforms to.

5.8.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ImplementationGuide | | | DomainResource | A set of rules about how FHIR is used Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 1..1 | uri | Logical URI to reference this implementation guide (globally unique)
version | ? | 0..1 | string | Business version of the implementation guide
name | ? | 1..1 | string | Name for this implementation guide (computer friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the implementation guide
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for implementation guide (if applicable) Jurisdiction ValueSet (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
fhirVersion | ? | 0..1 | id | FHIR Version this Implementation Guide targets
dependency | ? | 0..* | BackboneElement | Another Implementation guide this depends on
type | ? | 1..1 | code | reference | inclusion GuideDependencyType (Required)
uri | ? | 1..1 | uri | Where to find dependency
package | ? | 0..* | BackboneElement | Group of resources as used in .page.package
name | ? | 1..1 | string | Name used .page.package
description | | 0..1 | string | Human readable text describing the package
resource | ? | 1..* | BackboneElement | Resource in the implementation guide
example | ? | 1..1 | boolean | If not an example, has its normal meaning
name | ? | 0..1 | string | Human Name for the resource
description | | 0..1 | string | Reason why included in guide
acronym | | 0..1 | string | Short code to identify the resource
source[x] | ? | 1..1 | | Location of the resource
sourceUri | | | uri
sourceReference | | | Reference(Any)
exampleFor | | 0..1 | Reference(StructureDefinition) | Resource this is an example of (if applicable)
global | ? | 0..* | BackboneElement | Profiles that apply globally
type | ? | 1..1 | code | Type this profiles applies to ResourceType (Required)
profile | ? | 1..1 | Reference(StructureDefinition) | Profile that all resources must conform to
binary | | 0..* | uri | Image, css, script, etc.
page | ? | 0..1 | BackboneElement | Page/Section in the Guide
source | ? | 1..1 | uri | Where to find that page
title | ? | 1..1 | string | Short title shown for navigational assistance
kind | ? | 1..1 | code | page | example | list | include | directory | dictionary | toc | resource GuidePageKind (Required)
type | | 0..* | code | Kind of resource to include in the list ResourceType (Required)
package | | 0..* | string | Name of package to include
format | | 0..1 | code | Format of the page (e.g. html, markdown, etc.) MimeType (Required)

page | | 0..* | see page | Nested Pages / Sections

Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImplementationGuide xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Logical URI to reference this implementation guide (globally unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the implementation guide -->
 <name value="[string]"/><!-- 1..1 Name for this implementation guide (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the implementation guide -->
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for implementation guide (if applicable)
    --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <fhirVersion value="[id]"/><!-- 0..1 FHIR Version this Implementation Guide targets -->
 <dependency>  <!-- 0..* Another Implementation guide this depends on -->
  <type value="[code]"/><!-- 1..1 reference | inclusion -->
  <uri value="[uri]"/><!-- 1..1 Where to find dependency -->
 </dependency>
 <package>  <!-- 0..* Group of resources as used in .page.package -->
  <name value="[string]"/><!-- 1..1 Name used .page.package -->
  <description value="[string]"/><!-- 0..1 Human readable text describing the package -->
  <resource>  <!-- 1..* Resource in the implementation guide -->
   <example value="[boolean]"/><!-- 1..1 If not an example, has its normal meaning -->
   <name value="[string]"/><!-- 0..1 Human Name for the resource -->
   <description value="[string]"/><!-- 0..1 Reason why included in guide -->
   <acronym value="[string]"/><!-- 0..1 Short code to identify the resource -->
   <source[x]><!-- 1..1 uri|Reference(Any) Location of the resource --></source[x]>
   <exampleFor><!-- 0..1 Reference(StructureDefinition) Resource this is an example of (if applicable) --></exampleFor>
  </resource>
 </package>
 <global>  <!-- 0..* Profiles that apply globally -->
  <type value="[code]"/><!-- 1..1 Type this profiles applies to -->
  <profile><!-- 1..1 Reference(StructureDefinition) Profile that all resources must conform to --></profile>
 </global>
 <binary value="[uri]"/><!-- 0..* Image, css, script, etc. -->
 <page>  <!-- 0..1 Page/Section in the Guide -->
  <source value="[uri]"/><!-- 1..1 Where to find that page -->
  <title value="[string]"/><!-- 1..1 Short title shown for navigational assistance -->
  <kind value="[code]"/><!-- 1..1 page | example | list | include | directory | dictionary | toc | resource -->
  <type value="[code]"/><!-- 0..* Kind of resource to include in the list -->
  <package value="[string]"/><!-- 0..* Name of package to include -->
  <format value="[code]"/><!-- 0..1 Format of the page (e.g. html, markdown, etc.)  -->
  <page><!-- 0..* Content as for ImplementationGuide.page Nested Pages / Sections --></page>
 </page>
</ImplementationGuide>
```

JSON Template

```
{
  "resourceType" : "ImplementationGuide",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Logical URI to reference this implementation guide (globally unique)
  "version" : "<string>", // Business version of the implementation guide
  "name" : "<string>", // R!  Name for this implementation guide (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the implementation guide
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for implementation guide (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "fhirVersion" : "<id>", // FHIR Version this Implementation Guide targets
  "dependency" : [{ // Another Implementation guide this depends on
    "type" : "<code>", // R!  reference | inclusion
    "uri" : "<uri>" // R!  Where to find dependency
  }],
  "package" : [{ // Group of resources as used in .page.package
    "name" : "<string>", // R!  Name used .page.package
    "description" : "<string>", // Human readable text describing the package
    "resource" : [{ // R!  Resource in the implementation guide
      "example" : <boolean>, // R!  If not an example, has its normal meaning
      "name" : "<string>", // Human Name for the resource
      "description" : "<string>", // Reason why included in guide
      "acronym" : "<string>", // Short code to identify the resource
      // source[x]: Location of the resource. One of these 2:
      "sourceUri" : "<uri>",
      "sourceReference" : { Reference(Any) },
      "exampleFor" : { Reference(StructureDefinition) } // Resource this is an example of (if applicable)
```

```
    }]
  }],
  "global" : [{ // Profiles that apply globally
    "type" : "<code>", // R!  Type this profiles applies to
    "profile" : { Reference(StructureDefinition) } // R!  Profile that all resources must conform to
  }],
  "binary" : ["<uri>"], // Image, css, script, etc.
  "page" : { // Page/Section in the Guide
    "source" : "<uri>", // R!  Where to find that page
    "title" : "<string>", // R!  Short title shown for navigational assistance
    "kind" : "<code>", // R!  page | example | list | include | directory | dictionary | toc | resource
    "type" : ["<code>"], // Kind of resource to include in the list
    "package" : ["<string>"], // Name of package to include
    "format" : "<code>", // Format of the page (e.g. html, markdown, etc.)
    "page" : [{ Content as for ImplementationGuide.page }] // Nested Pages / Sections
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImplementationGuide;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImplementationGuide.url [ uri ]; # 1..1 Logical URI to reference this implementation guide (globally unique)
  fhir:ImplementationGuide.version [ string ]; # 0..1 Business version of the implementation guide
  fhir:ImplementationGuide.name [ string ]; # 1..1 Name for this implementation guide (computer friendly)
  fhir:ImplementationGuide.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ImplementationGuide.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ImplementationGuide.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:ImplementationGuide.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ImplementationGuide.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ImplementationGuide.description [ markdown ]; # 0..1 Natural language description of the implementation guide
  fhir:ImplementationGuide.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:ImplementationGuide.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for implementation guide
     (if applicable)
  fhir:ImplementationGuide.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ImplementationGuide.fhirVersion [ id ]; # 0..1 FHIR Version this Implementation Guide targets
  fhir:ImplementationGuide.dependency [ # 0..* Another Implementation guide this depends on
    fhir:ImplementationGuide.dependency.type [ code ]; # 1..1 reference | inclusion
    fhir:ImplementationGuide.dependency.uri [ uri ]; # 1..1 Where to find dependency
  ], ...;
  fhir:ImplementationGuide.package [ # 0..* Group of resources as used in .page.package
    fhir:ImplementationGuide.package.name [ string ]; # 1..1 Name used .page.package
    fhir:ImplementationGuide.package.description [ string ]; # 0..1 Human readable text describing the package
    fhir:ImplementationGuide.package.resource [ # 1..* Resource in the implementation guide
      fhir:ImplementationGuide.package.resource.example [ boolean ]; # 1..1 If not an example, has its normal meaning
      fhir:ImplementationGuide.package.resource.name [ string ]; # 0..1 Human Name for the resource
      fhir:ImplementationGuide.package.resource.description [ string ]; # 0..1 Reason why included in guide
      fhir:ImplementationGuide.package.resource.acronym [ string ]; # 0..1 Short code to identify the resource
      # ImplementationGuide.package.resource.source[x] : 1..1 Location of the resource. One of these 2
        fhir:ImplementationGuide.package.resource.sourceUri [ uri ]
        fhir:ImplementationGuide.package.resource.sourceReference [ Reference(Any) ]
      fhir:ImplementationGuide.package.resource.exampleFor [ Reference(StructureDefinition) ]; # 0..1 Resource this is
    an example of (if applicable)
    ], ...;
  ], ...;
  fhir:ImplementationGuide.global [ # 0..* Profiles that apply globally
    fhir:ImplementationGuide.global.type [ code ]; # 1..1 Type this profiles applies to
    fhir:ImplementationGuide.global.profile [ Reference(StructureDefinition) ]; # 1..1 Profile that all resources must
    conform to
  ], ...;
  fhir:ImplementationGuide.binary [ uri ], ... ; # 0..* Image, css, script, etc.
  fhir:ImplementationGuide.page [ # 0..1 Page/Section in the Guide
    fhir:ImplementationGuide.page.source [ uri ]; # 1..1 Where to find that page
    fhir:ImplementationGuide.page.title [ string ]; # 1..1 Short title shown for navigational assistance
    fhir:ImplementationGuide.page.kind [ code ]; # 1..1 page | example | list | include | directory | dictionary | toc |
     resource
    fhir:ImplementationGuide.page.type [ code ], ... ; # 0..* Kind of resource to include in the list
    fhir:ImplementationGuide.page.package [ string ], ... ; # 0..* Name of package to include
    fhir:ImplementationGuide.page.format [ code ]; # 0..1 Format of the page (e.g. html, markdown, etc.)
    fhir:ImplementationGuide.page.page [ See ImplementationGuide.page ], ... ; # 0..* Nested Pages / Sections
  ];
]
```

Changes since DSTU2

ImplementationGuide

ImplementationGuide.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status

ImplementationGuide.experimental | Now marked as Modifier

ImplementationGuide.contact | Type changed from BackboneElement to ContactDetail

ImplementationGuide.description | Type changed from string to markdown

ImplementationGuide.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)

ImplementationGuide.jurisdiction | Added Element
ImplementationGuide.copyright | Type changed from string to markdown
ImplementationGuide.package | Min Cardinality changed from 1 to 0
ImplementationGuide.package.resource.example | Added Element
ImplementationGuide.page | Min Cardinality changed from 1 to 0
ImplementationGuide.page.title | Renamed from name to title
ImplementationGuide.contact.name | deleted
ImplementationGuide.contact.telecom | deleted
ImplementationGuide.package.resource.purpose | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Structure

Name | Flags | Card. | Type | Description & Constraints
ImplementationGuide | | | DomainResource | A set of rules about how FHIR is used Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 1..1 | uri | Logical URI to reference this implementation guide (globally unique)
version | ? | 0..1 | string | Business version of the implementation guide
name | ? | 1..1 | string | Name for this implementation guide (computer friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the implementation guide
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for implementation guide (if applicable) Jurisdiction ValueSet (Extensible)
copyright | | 0..1 | markdown | Use and/or publishing restrictions
fhirVersion | ? | 0..1 | id | FHIR Version this Implementation Guide targets
dependency | ? | 0..* | BackboneElement | Another Implementation guide this depends on
type | ? | 1..1 | code | reference | inclusion GuideDependencyType (Required)
uri | ? | 1..1 | uri | Where to find dependency
package | ? | 0..* | BackboneElement | Group of resources as used in .page.package
name | ? | 1..1 | string | Name used .page.package
description | | 0..1 | string | Human readable text describing the package
resource | ? | 1..* | BackboneElement | Resource in the implementation guide
example | ? | 1..1 | boolean | If not an example, has its normal meaning
name | ? | 0..1 | string | Human Name for the resource
description | | 0..1 | string | Reason why included in guide
acronym | | 0..1 | string | Short code to identify the resource
source[x] | ? | 1..1 | | Location of the resource
sourceUri | | | uri
sourceReference | | | Reference(Any)
exampleFor | | 0..1 | Reference(StructureDefinition) | Resource this is an example of (if applicable)
global | ? | 0..* | BackboneElement | Profiles that apply globally
type | ? | 1..1 | code | Type this profiles applies to ResourceType (Required)
profile | ? | 1..1 | Reference(StructureDefinition) | Profile that all resources must conform to
binary | | 0..* | uri | Image, css, script, etc.
page | ? | 0..1 | BackboneElement | Page/Section in the Guide
source | ? | 1..1 | uri | Where to find that page
title | ? | 1..1 | string | Short title shown for navigational assistance
kind | ? | 1..1 | code | page | example | list | include | directory | dictionary | toc | resource GuidePageKind (Required)
type | | 0..* | code | Kind of resource to include in the list ResourceType (Required)
package | | 0..* | string | Name of package to include
format | | 0..1 | code | Format of the page (e.g. html, markdown, etc.) MimeType (Required)
page | | 0..* | see page | Nested Pages / Sections
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ImplementationGuide xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 1..1 Logical URI to reference this implementation guide (globally unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the implementation guide -->
 <name value="[string]"/><!-- 1..1 Name for this implementation guide (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
```

```
 <description value="[markdown]"/><!-- 0..1 Natural language description of the implementation guide -->
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for implementation guide (if applicable)
    --></jurisdiction>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <fhirVersion value="[id]"/><!-- 0..1 FHIR Version this Implementation Guide targets -->
 <dependency>  <!-- 0..* Another Implementation guide this depends on -->
  <type value="[code]"/><!-- 1..1 reference | inclusion -->
  <uri value="[uri]"/><!-- 1..1 Where to find dependency -->
 </dependency>
 <package>  <!-- 0..* Group of resources as used in .page.package -->
  <name value="[string]"/><!-- 1..1 Name used .page.package -->
  <description value="[string]"/><!-- 0..1 Human readable text describing the package -->
  <resource>  <!-- 1..* Resource in the implementation guide -->
   <example value="[boolean]"/><!-- 1..1 If not an example, has its normal meaning -->
   <name value="[string]"/><!-- 0..1 Human Name for the resource -->
   <description value="[string]"/><!-- 0..1 Reason why included in guide -->
   <acronym value="[string]"/><!-- 0..1 Short code to identify the resource -->
   <source[x]><!-- 1..1 uri|Reference(Any) Location of the resource --></source[x]>
   <exampleFor><!-- 0..1 Reference(StructureDefinition) Resource this is an example of (if applicable) --></exampleFor>
  </resource>
 </package>
 <global>  <!-- 0..* Profiles that apply globally -->
  <type value="[code]"/><!-- 1..1 Type this profiles applies to -->
  <profile><!-- 1..1 Reference(StructureDefinition) Profile that all resources must conform to --></profile>
 </global>
 <binary value="[uri]"/><!-- 0..* Image, css, script, etc. -->
 <page>  <!-- 0..1 Page/Section in the Guide -->
  <source value="[uri]"/><!-- 1..1 Where to find that page -->
  <title value="[string]"/><!-- 1..1 Short title shown for navigational assistance -->
  <kind value="[code]"/><!-- 1..1 page | example | list | include | directory | dictionary | toc | resource -->
  <type value="[code]"/><!-- 0..* Kind of resource to include in the list -->
  <package value="[string]"/><!-- 0..* Name of package to include -->
  <format value="[code]"/><!-- 0..1 Format of the page (e.g. html, markdown, etc.)  -->
  <page><!-- 0..* Content as for ImplementationGuide.page Nested Pages / Sections --></page>
 </page>
</ImplementationGuide>
```

JSON Template

```
{
  "resourceType" : "ImplementationGuide",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // R!  Logical URI to reference this implementation guide (globally unique)
  "version" : "<string>", // Business version of the implementation guide
  "name" : "<string>", // R!  Name for this implementation guide (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the implementation guide
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for implementation guide (if applicable)
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "fhirVersion" : "<id>", // FHIR Version this Implementation Guide targets
  "dependency" : [{ // Another Implementation guide this depends on
    "type" : "<code>", // R!  reference | inclusion
    "uri" : "<uri>" // R!  Where to find dependency
  }],
  "package" : [{ // Group of resources as used in .page.package
    "name" : "<string>", // R!  Name used .page.package
    "description" : "<string>", // Human readable text describing the package
    "resource" : [{ // R!  Resource in the implementation guide
      "example" : <boolean>, // R!  If not an example, has its normal meaning
      "name" : "<string>", // Human Name for the resource
      "description" : "<string>", // Reason why included in guide
      "acronym" : "<string>", // Short code to identify the resource
      // source[x]: Location of the resource. One of these 2:
      "sourceUri" : "<uri>",
      "sourceReference" : { Reference(Any) },
      "exampleFor" : { Reference(StructureDefinition) } // Resource this is an example of (if applicable)
    }]
  }],
  "global" : [{ // Profiles that apply globally
    "type" : "<code>", // R!  Type this profiles applies to
    "profile" : { Reference(StructureDefinition) } // R!  Profile that all resources must conform to
  }],
  "binary" : ["<uri>"], // Image, css, script, etc.
  "page" : { // Page/Section in the Guide
    "source" : "<uri>", // R!  Where to find that page
    "title" : "<string>", // R!  Short title shown for navigational assistance
    "kind" : "<code>", // R!  page | example | list | include | directory | dictionary | toc | resource
    "type" : ["<code>"], // Kind of resource to include in the list
    "package" : ["<string>"], // Name of package to include
    "format" : "<code>", // Format of the page (e.g. html, markdown, etc.)
    "page" : [{ Content as for ImplementationGuide.page }] // Nested Pages / Sections
  }
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ImplementationGuide;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ImplementationGuide.url [ uri ]; # 1..1 Logical URI to reference this implementation guide (globally unique)
  fhir:ImplementationGuide.version [ string ]; # 0..1 Business version of the implementation guide
  fhir:ImplementationGuide.name [ string ]; # 1..1 Name for this implementation guide (computer friendly)
  fhir:ImplementationGuide.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ImplementationGuide.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ImplementationGuide.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:ImplementationGuide.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ImplementationGuide.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ImplementationGuide.description [ markdown ]; # 0..1 Natural language description of the implementation guide
  fhir:ImplementationGuide.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:ImplementationGuide.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for implementation guide
     (if applicable)
  fhir:ImplementationGuide.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ImplementationGuide.fhirVersion [ id ]; # 0..1 FHIR Version this Implementation Guide targets
  fhir:ImplementationGuide.dependency [ # 0..* Another Implementation guide this depends on
    fhir:ImplementationGuide.dependency.type [ code ]; # 1..1 reference | inclusion
    fhir:ImplementationGuide.dependency.uri [ uri ]; # 1..1 Where to find dependency
  ], ...;
  fhir:ImplementationGuide.package [ # 0..* Group of resources as used in .page.package
    fhir:ImplementationGuide.package.name [ string ]; # 1..1 Name used .page.package
    fhir:ImplementationGuide.package.description [ string ]; # 0..1 Human readable text describing the package
    fhir:ImplementationGuide.package.resource [ # 1..* Resource in the implementation guide
      fhir:ImplementationGuide.package.resource.example [ boolean ]; # 1..1 If not an example, has its normal meaning
      fhir:ImplementationGuide.package.resource.name [ string ]; # 0..1 Human Name for the resource
      fhir:ImplementationGuide.package.resource.description [ string ]; # 0..1 Reason why included in guide
      fhir:ImplementationGuide.package.resource.acronym [ string ]; # 0..1 Short code to identify the resource
      # ImplementationGuide.package.resource.source[x] : 1..1 Location of the resource. One of these 2
        fhir:ImplementationGuide.package.resource.sourceUri [ uri ]
        fhir:ImplementationGuide.package.resource.sourceReference [ Reference(Any) ]
      fhir:ImplementationGuide.package.resource.exampleFor [ Reference(StructureDefinition) ]; # 0..1 Resource this is
    an example of (if applicable)
    ], ...;
  ], ...;
  fhir:ImplementationGuide.global [ # 0..* Profiles that apply globally
    fhir:ImplementationGuide.global.type [ code ]; # 1..1 Type this profiles applies to
    fhir:ImplementationGuide.global.profile [ Reference(StructureDefinition) ]; # 1..1 Profile that all resources must
    conform to
  ], ...;
  fhir:ImplementationGuide.binary [ uri ], ... ; # 0..* Image, css, script, etc.
  fhir:ImplementationGuide.page [ # 0..1 Page/Section in the Guide
    fhir:ImplementationGuide.page.source [ uri ]; # 1..1 Where to find that page
    fhir:ImplementationGuide.page.title [ string ]; # 1..1 Short title shown for navigational assistance
    fhir:ImplementationGuide.page.kind [ code ]; # 1..1 page | example | list | include | directory | dictionary | toc |
     resource
    fhir:ImplementationGuide.page.type [ code ], ... ; # 0..* Kind of resource to include in the list
    fhir:ImplementationGuide.page.package [ string ], ... ; # 0..* Name of package to include
    fhir:ImplementationGuide.page.format [ code ]; # 0..1 Format of the page (e.g. html, markdown, etc.)
    fhir:ImplementationGuide.page.page [ See ImplementationGuide.page ], ... ; # 0..* Nested Pages / Sections
  ];
]
```

Changes since DSTU2

ImplementationGuide

ImplementationGuide.status | Change value set from http://hl7.org/fhir/ValueSet/conformance-resource-status to http://hl7.org/fhir/ValueSet/publication-status

ImplementationGuide.experimental | Now marked as Modifier

ImplementationGuide.contact | Type changed from BackboneElement to ContactDetail

ImplementationGuide.description | Type changed from string to markdown

ImplementationGuide.useContext | Type changed from CodeableConcept to UsageContext Remove Binding http://hl7.org/fhir/ValueSet/use-context (extensible)

ImplementationGuide.jurisdiction | Added Element

ImplementationGuide.copyright | Type changed from string to markdown

ImplementationGuide.package | Min Cardinality changed from 1 to 0

ImplementationGuide.package.resource.example | Added Element

ImplementationGuide.page | Min Cardinality changed from 1 to 0

ImplementationGuide.page.title | Renamed from name to title

ImplementationGuide.contact.name | deleted

ImplementationGuide.contact.telecom | deleted

ImplementationGuide.package.resource.purpose | deleted

See the Full Difference for further information

This analysis is available as XML or JSON.

See R2 <--> R3 Conversion Maps (status = 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid.).

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

5.8.3.1 Terminology Bindings

Path | Definition | Type | Reference
ImplementationGuide.status | The lifecycle status of a Value Set or Concept Map. | Required | PublicationStatus
ImplementationGuide.jurisdiction | Countries and regions within which this artifact is targeted for use | Extensible | Jurisdiction ValueSet
ImplementationGuide.dependency.type | How a dependency is represented when the guide is published. | Required | GuideDependencyType
ImplementationGuide.global.type ImplementationGuide.page.type | One of the resource types defined as part of FHIR. | Required | Resource Types
ImplementationGuide.page.kind | The kind of an included page. | Required | GuidePageKind
ImplementationGuide.page.format | The mime type of an attachment. Any valid mime type is allowed. | Required | BCP 13 (RFCs 2045, 2046, 2047, 4288, 4289 and 2049)

5.8.3.2 Default Profiles

An implementation guide can define default profiles - these are profiles that apply to any resource that does not otherwise have an explicit profile assigned by the implementation guide. Default profiles are always references to profiles (StructureDefinition resources) that are also contained in the resources. By defining default profiles, an implementation guide can save itself from exhaustively defining profiles on every resource type just to profile every reference to a particular resource type.

Note that a resource can conform to the default profile by conforming to any profile derived from it.

5.8.3.3 Compatibility list

This table declares the compatibility between the various resources as determined by the Implementation Guide comparison tool:

Yet to be done

5.8.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
date | date | The implementation guide publication date | ImplementationGuide.date
dependency | uri | Where to find dependency | ImplementationGuide.dependency.uri
description | string | The description of the implementation guide | ImplementationGuide.description
experimental | token | For testing purposes, not real usage | ImplementationGuide.experimental
jurisdiction | token | Intended jurisdiction for the implementation guide | ImplementationGuide.jurisdiction
name | string | Computationally friendly name of the implementation guide | ImplementationGuide.name
publisher | string | Name of the publisher of the implementation guide | ImplementationGuide.publisher
resource | reference | Location of the resource | ImplementationGuide.package.resource.source (Any)
status | token | The current status of the implementation guide | ImplementationGuide.status
url | uri | The uri that identifies the implementation guide | ImplementationGuide.url
version | token | The business version of the implementation guide | ImplementationGuide.version

5.8.5 Resource ImplementationGuide - Examples

URL: https://hl7.org/fhir/STU3/implementationguide-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.8.5 Resource ImplementationGuide - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Example of implementationguide | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.8.6 Resource ImplementationGuide - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/implementationguide-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.8.6 Resource ImplementationGuide - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 1 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ImplementationGuide resource.

ImplementationGuide
Definition | A set of rules of how FHIR is used to solve a particular problem. This resource is used to gather all the parts of an implementation guide into a logical whole and to publish a computable definition of all the parts.

Control | 1..1

Requirements | An implementation guide is able to define default profiles that must apply to any use of a resource, so validation services may need to take one or more implementation guide resources when validating.

ImplementationGuide.url

Definition | An absolute URI that is used to identify this implementation guide when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this implementation guide is (or will be) published. The URL SHOULD include the major version of the implementation guide. For more information see Technical and Business Versions.

Control | 1..1

Type | uri

Requirements | Allows the implementation guide to be referenced by a single globally unique identifier. This is required to allow hosting Implementation Guides on multiple different servers, and to allow for the editorial process.

Alternate Names | url; authoritative-url; destination; identity

Summary | true

Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.

ImplementationGuide.version

Definition | The identifier that is used to identify this version of the implementation guide when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the implementation guide author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.

Note | This is a business versionId, not a resource version id (see discussion)

Control | 0..1

Type | string

Summary | true

Comments | There may be different implementation guide instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the implementation guide with the format [url]|[version].

ImplementationGuide.name

Definition | A natural language name identifying the implementation guide. This name should be usable as an identifier for the module by machine processing applications such as code generation.

Control | 1..1

Type | string

Requirements | Support human navigation and code generation.

Summary | true

Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.

ImplementationGuide.status

Definition | The status of this implementation guide. Enables tracking the life-cycle of the content.

Control | 1..1

Terminology Binding | PublicationStatus (Required)

Type | code

Is Modifier | true

Summary | true

Comments | Allows filtering of implementation guides that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired implementation guide without due consideration.

ImplementationGuide.experimental

Definition | A boolean value to indicate that this implementation guide is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.

Control | 0..1

Type | boolean

Is Modifier | true

Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level implementation guide.

Summary | true

Comments | Allows filtering of implementation guide that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental implementation guide in production.

ImplementationGuide.date

Definition | The date (and optionally time) when the implementation guide was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the implementation guide changes.

Control | 0..1

Type | dateTime

Alternate Names | Revision Date

Summary | true

Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the implementation guide. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.

ImplementationGuide.publisher

Definition | The name of the individual or organization that published the implementation guide.

Control | 0..1

Type | string

Requirements | Helps establish the "authority/credibility" of the implementation guide. May also allow for contact.

Summary | true

Comments | Usually an organization, but may be an individual. The publisher (or steward) of the implementation guide is the organization or individual primarily responsible for the maintenance and upkeep of the implementation guide. This is not necessarily the same individual or

organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the implementation guide. This item SHOULD be populated unless the information is available from context.
ImplementationGuide.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
ImplementationGuide.description
Definition | A free text natural language description of the implementation guide from a consumer's perspective.
Control | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the implementation guide was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the implementation guide as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created).
ImplementationGuide.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate implementation guide instances.
Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
ImplementationGuide.jurisdiction
Definition | A legal or geographic region in which the implementation guide is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the implementation guide to be used in jurisdictions other than those for which it was originally designed or intended.
ImplementationGuide.copyright
Definition | A copyright statement relating to the implementation guide and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the implementation guide.
Control | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the implementation guide and/or its content.
Alternate Names | License; Restrictions
ImplementationGuide.fhirVersion
Definition | The version of the FHIR specification on which this ImplementationGuide is based - this is the formal version of the specification, without the revision number, e.g. [publication].[major].[minor], which is 3.0.2 for this version.
Control | 0..1
Type | id
Summary | true
Comments | An Implementation Guide should specify the target it applies to, as ImplementationGuides are rarely valid across multiple versions of FHIR.
ImplementationGuide.dependency
Definition | Another implementation guide that this implementation depends on. Typically, an implementation guide uses value sets, profiles etc.defined in other implementation guides.
Control | 0..*
Summary | true
ImplementationGuide.dependency.type
Definition | How the dependency is represented when the guide is published.
Control | 1..1
Terminology Binding | GuideDependencyType (Required)
Type | code
Summary | true
Comments | For information about processing included guides, see [Including Guides]{implementationguide.html#include}.
ImplementationGuide.dependency.uri
Definition | Where the dependency is located.
Control | 1..1
Type | uri
Summary | true
Comments | When a guide is published, and the type is 'include', this will be a relative reference within the guide.
ImplementationGuide.package
Definition | A logical group of resources. Logical groups can be used when building pages.
Control | 0..*
Summary | true
ImplementationGuide.package.name
Definition | The name for the group, as used in page.package.
Control | 1..1
Type | string

Summary | true
ImplementationGuide.package.description
Definition | Human readable text describing the package.
Control | 0..1
Type | string
ImplementationGuide.package.resource
Definition | A resource that is part of the implementation guide. Conformance resources (value set, structure definition, capability statements etc.) are obvious candidates for inclusion, but any kind of resource can be included as an example resource.
Control | 1..*
Summary | true
ImplementationGuide.package.resource.example
Definition | Whether a resource is included in the guide as part of the rules defined by the guide, or just as an example of a resource that conforms to the rules and/or help implementers understand the intent of the guide.
Control | 1..1
Type | boolean
Summary | true
Comments | Typically, conformance resources and knowledge resources are directly part of the implementation guide, with their normal meaning, and patient linked resources are usually examples. However this is not always true.
ImplementationGuide.package.resource.name
Definition | A human assigned name for the resource. All resources SHOULD have a name, but the name may be extracted from the resource (e.g. ValueSet.name).
Control | 0..1
Type | string
Summary | true
ImplementationGuide.package.resource.description
Definition | A description of the reason that a resource has been included in the implementation guide.
Control | 0..1
Type | string
Comments | This is mostly used with examples to explain why it is present (though they can have extensive comments in the examples).
ImplementationGuide.package.resource.acronym
Definition | A short code that may be used to identify the resource throughout the implementation guide.
Control | 0..1
Type | string
Comments | This is used when generating schematrons from profiles, for example, but has other uses. Publishing tooling may autogenerate this value, but a human assigned value is more useful. This value must be unique within an implementation guide (across included guides as well).
ImplementationGuide.package.resource.source[x]
Definition | Where this resource is found.
Control | 1..1
Type | uri|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
Summary | true
Comments | Usually this is a relative URL that locates the resource within the implementation guide. If you authoring an implementation guide, and will publish it using the FHIR publication tooling, use a URI that may point to a resource, or to one of various alternative representations (e.g. spreadsheet). The tooling will convert this when it publishes it.
ImplementationGuide.package.resource.exampleFor
Definition | Another resource that this resource is an example for. This is mostly used for resources that are included as examples of StructureDefinitions.
Control | 0..1
Type | Reference(StructureDefinition)
Comments | Examples: * StructureDefinition -> Any * ValueSet -> expansion * OperationDefinition -> Parameters * Questionnaire -> QuestionnaireResponse.
ImplementationGuide.global
Definition | A set of profiles that all resources covered by this implementation guide must conform to.
Control | 0..*
Summary | true
Comments | See Default Profiles for a discussion of which resources are 'covered' by an implementation guide.
ImplementationGuide.global.type
Definition | The type of resource that all instances must conform to.
Control | 1..1
Terminology Binding | Any defined Resource Type name
Type | code
Summary | true
Comments | The type must match that of the profile that is referred to, but is made explicit here as a denormalization so that a system processing the implementation guide resource knows which resources the profile applies to even if the profile itself is not available.
ImplementationGuide.global.profile
Definition | A reference to the profile that all instances must conform to.
Control | 1..1
Type | Reference(StructureDefinition)
Summary | true
ImplementationGuide.binary
Definition | A binary file that is included in the implementation guide when it is published.
Control | 0..*
Type | uri

Comments | This is principally included for the FHIR publishing tool, but may be useful when extracting and reusing content from the implementation guide.
ImplementationGuide.page
Definition | A page / section in the implementation guide. The root page is the implementation guide home page.
Control | 0..1
Summary | true
Comments | Pages automatically become sections if they have sub-pages. By convention, the home page is called index.html.
ImplementationGuide.page.source
Definition | The source address for the page.
Control | 1..1
Type | uri
Summary | true
Comments | The publishing tool will autogenerate source for list (source = n/a), and inject included implementations for include (source = uri of guide to include).
ImplementationGuide.page.title
Definition | A short title used to represent this page in navigational structures such as table of contents, bread crumbs, etc.
Control | 1..1
Type | string
Summary | true
ImplementationGuide.page.kind
Definition | The kind of page that this is. Some pages are autogenerated (list, example), and other kinds are of interest so that tools can navigate the user to the page of interest.
Control | 1..1
Terminology Binding | GuidePageKind (Required)
Type | code
Summary | true
ImplementationGuide.page.type
Definition | For constructed pages, what kind of resources to include in the list.
Control | 0..*
Terminology Binding | Any defined Resource Type name
Type | code
ImplementationGuide.page.package
Definition | For constructed pages, a list of packages to include in the page (or else empty for everything).
Control | 0..*
Type | string
ImplementationGuide.page.format
Definition | The format of the page.
Control | 0..1
Terminology Binding | MimeType :
Type | code
Comments | This is primarily for the publishing tool, that will convert all pages to html format. HTML pages are pre-processed, see xxxx.
ImplementationGuide.page.page
Definition | Nested Pages/Sections under this page.
Control | 0..*
Type | See ImplementationGuide.page
Comments | The implementation guide breadcrumbs are generated from this structure.

5.8.7 Resource ImplementationGuide - Mappings

URL: https://hl7.org/fhir/STU3/implementationguide-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.8.7 Resource ImplementationGuide - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Mappings for the implementationguide resource.

5.8.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

ImplementationGuide | n/a

5.8.8 Resource ImplementationGuide - Design Notes

URL: https://hl7.org/fhir/STU3/implementationguide-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.8.8 Resource ImplementationGuide - Design Notes

to do

5.8.9 Resource ImplementationGuide - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/implementationguide-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.8.9 Resource ImplementationGuide - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ImplementationGuide resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

5.8.9.1 Profiles

No Profiles defined for this resource

5.8.9.2 Extensions

implementationguide-page | page | for FHIR Core ImplementationGuide Profile
resource-approvalDate | approvalDate | for Resource HL7 Extensions
resource-effectivePeriod | effectivePeriod | for Resource HL7 Extensions
resource-lastReviewDate | lastReviewDate | for Resource HL7 Extensions

Extensions that reference this resource:

No Extensions refer to this resource

5.8.9.3 Search Extensions

No Search Extensions defined for this resource

5.8.10 FHIR Core ImplementationGuide Profile

URL: https://hl7.org/fhir/STU3/implementationguide-extensions.html

This profile holds core ImplementationGuide extensions

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.8.10 FHIR Core ImplementationGuide Profile

(No assigned work group) | Maturity Level: N/A | Ballot Status: Informative

5.8.10.1 Content

Extensions:
implementationguide-page | page : The home page location for this resource (e.g. for users to view this resource).

5.8.11 Resource ImplementationGuide - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/implementationguide-version-maps.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.8.11 Resource ImplementationGuide - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

R2 : R3 Converson maps for ImplementationGuide.

Functional status for this map: 1 test that all execute ok. All tests pass round-trip testing and all r3 resources are valid. (see documentation)

5.8.11.1 R2 to R3

```
map "http://hl7.org/fhir/StructureMap/ImplementationGuide2to3" = "R2 to R3 Conversions for ImplementationGuide"

uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ImplementationGuide" alias ImplementationGuideR2 as source
uses "http://hl7.org/fhir/StructureDefinition/ImplementationGuide" alias ImplementationGuide as target

imports "http://hl7.org/fhir/StructureMap/primitives2to3"
imports "http://hl7.org/fhir/StructureMap/BackboneElement2to3"
imports "http://hl7.org/fhir/StructureMap/DomainResource2to3"
imports "http://hl7.org/fhir/StructureMap/ContactPoint2to3"
imports "http://hl7.org/fhir/StructureMap/CodeableConcept2to3"
imports "http://hl7.org/fhir/StructureMap/Reference2to3"

group for type+types  ImplementationGuide extends DomainResource
```

```
  input src : ImplementationGuideR2 as source
  input tgt : ImplementationGuide as target

  "ImplementationGuide-url" : for src.url make tgt.url
  "ImplementationGuide-version" : for src.version make tgt.version
  "ImplementationGuide-name" : for src.name make tgt.name
  "ImplementationGuide-status" : for src.status make tgt.status
  "ImplementationGuide-experimental" : for src.experimental make tgt.experimental
  "ImplementationGuide-publisher" : for src.publisher make tgt.publisher
  "ImplementationGuide-contact" : for src.contact as vs0 make tgt.contact as vt0 then {
    "ContactDetail-name" : for vs0.name make vt0.name
    "ContactDetail-telecom" : for vs0.telecom make vt0.telecom
  }

  "ImplementationGuide-description" : for src.description make tgt.description
  "ImplementationGuide-copyright" : for src.copyright make tgt.copyright
  "ImplementationGuide-jurisdiction" : for src.useContext where coding.system = 'urn:iso:std:iso:3166' make
    tgt.jurisdiction
  "ImplementationGuide-useContext" : for src.useContext as vs where (coding.system = 'urn:iso:std:iso:3166').not() make
      tgt.useContext as uc,  uc.code as ucc,  ucc.system = "http://hl7.org/fhir/usage-context-type",  ucc.code = "task",
      uc.value = create("CodeableConcept") as vt then CodeableConcept(vs, vt)
  "ImplementationGuide-date" : for src.date make tgt.date
  "ImplementationGuide-fhirVersion" : for src.fhirVersion make tgt.fhirVersion
  "ImplementationGuide-dependency" : for src.dependency as vs0 make tgt.dependency as vt0 then {
    "ImplementationGuide.dependency-type" : for vs0.type make vt0.type
    "ImplementationGuide.dependency-uri" : for vs0.uri make vt0.uri
  }

  "ImplementationGuide-package" : for src.package as vs0 make tgt.package as vt0 then {
    "ImplementationGuide.package-name" : for vs0.name make vt0.name
    "ImplementationGuide.package-description" : for vs0.description make vt0.description
    "ImplementationGuide.package-resource" : for vs0.resource as vs make vt0.resource as vt then resource(vs, vt)
  }

  "ImplementationGuide-global" : for src.global as vs0 make tgt.global as vt0 then {
    "ImplementationGuide.global-type" : for vs0.type make vt0.type
    "ImplementationGuide.global-profile" : for vs0.profile make vt0.profile
  }

  "ImplementationGuide-binary" : for src.binary make tgt.binary
  "ImplementationGuide-page" : for src.page as vs0 make tgt.page as vt0 then page(vs0, vt0)
endgroup

group resource extends BackboneElement
  input src as source
  input tgt as target

  "ImplementationGuide.resource-purpose" : for src where src.purpose = 'example' make tgt.example = true
  "ImplementationGuide.resource-name" : for src.name make tgt.name
  "ImplementationGuide.resource-description" : for src.description make tgt.description
  "ImplementationGuide.resource-acronym" : for src.acronym make tgt.acronym
  "ImplementationGuide.resource-source" : for src.source make tgt.source
  "ImplementationGuide.resource-exampleFor" : for src.exampleFor make tgt.exampleFor
endgroup

group page extends BackboneElement
  input src as source
  input tgt as target

  "ImplementationGuide.page-source" : for src.source make tgt.source
  "ImplementationGuide.page-title" : for src.name make tgt.title
  "ImplementationGuide.page-kind" : for src.kind make tgt.kind
  "ImplementationGuide.page-type" : for src.type make tgt.type
  "ImplementationGuide.page-package" : for src.package make tgt.package
  "ImplementationGuide.page-format" : for src.format make tgt.format
  "ImplementationGuide.page-page" : for src.page as vs0 make tgt.page as vt0 then page(vs0, vt0)
endgroup
```

5.8.11.2 R3 to R2

```
map "http://hl7.org/fhir/StructureMap/ImplementationGuide3to2" = "R3 to R2 Conversion for ImplementationGuide"

uses "http://hl7.org/fhir/StructureDefinition/ImplementationGuide" alias ImplementationGuide as source
uses "http://hl7.org/fhir/DSTU2/StructureDefinition/ImplementationGuide" alias ImplementationGuideR2 as target

imports "http://hl7.org/fhir/StructureMap/primitives3to2"
imports "http://hl7.org/fhir/StructureMap/BackboneElement3to2"
imports "http://hl7.org/fhir/StructureMap/DomainResource3to2"
imports "http://hl7.org/fhir/StructureMap/ContactPoint3to2"
imports "http://hl7.org/fhir/StructureMap/Reference3to2"
imports "http://hl7.org/fhir/StructureMap/CodeableConcept3to2"

group for type+types  ImplementationGuide extends DomainResource
  input src : ImplementationGuide as source
  input tgt : ImplementationGuideR2 as target

  "ImplementationGuide-url" : for src.url make tgt.url
  "ImplementationGuide-version" : for src.version make tgt.version
  "ImplementationGuide-name" : for src.name make tgt.name
  "ImplementationGuide-status" : for src.status make tgt.status
```

```
  "ImplementationGuide-experimental" : for src.experimental make tgt.experimental
  "ImplementationGuide-publisher" : for src.publisher make tgt.publisher
  "ImplementationGuide-contact" : for src.contact as vs0 make tgt.contact as vt0 then {
    "ImplementationGuide.contact-name" : for vs0.name make vt0.name
    "ImplementationGuide.contact-telecom" : for vs0.telecom make vt0.telecom
  }

  "ImplementationGuide-date" : for src.date make tgt.date
  "ImplementationGuide-description" : for src.description make tgt.description
  "ImplementationGuide-useContext" : for src.jurisdiction make tgt.useContext
  "ImplementationGuide-copyright" : for src.copyright make tgt.copyright
  "ImplementationGuide-fhirVersion" : for src.fhirVersion make tgt.fhirVersion
  "ImplementationGuide-dependency" : for src.dependency as vs0 make tgt.dependency as vt0 then {
    "ImplementationGuide.dependency-type" : for vs0.type make vt0.type
    "ImplementationGuide.dependency-uri" : for vs0.uri make vt0.uri
  }

  "ImplementationGuide-package" : for src.package as vs0 make tgt.package as vt0 then {
    "ImplementationGuide.package-name" : for vs0.name make vt0.name
    "ImplementationGuide.package-description" : for vs0.description make vt0.description
    "ImplementationGuide.package-resource" : for vs0.resource as vs make vt0.resource as vt then resource(vs, vt)
  }

  "ImplementationGuide-global" : for src.global as vs0 make tgt.global as vt0 then {
    "ImplementationGuide.global-type" : for vs0.type make vt0.type
    "ImplementationGuide.global-profile" : for vs0.profile make vt0.profile
  }

  "ImplementationGuide-binary" : for src.binary make tgt.binary
  "ImplementationGuide-page" : for src.page as vs0 make tgt.page as vt0 then page(vs0, vt0)
endgroup

group resource extends BackboneElement
  input src as source
  input tgt as target

  "ImplementationGuide.resource-purpose" : for src where example = true make tgt.purpose = 'example'
  "ImplementationGuide.resource-name" : for src.name make tgt.name
  "ImplementationGuide.resource-description" : for src.description make tgt.description
  "ImplementationGuide.resource-acronym" : for src.acronym make tgt.acronym
  "ImplementationGuide.resource-source" : for src.source make tgt.source
  "ImplementationGuide.resource-exampleFor" : for src.exampleFor make tgt.exampleFor
endgroup

group page extends BackboneElement
  input src as source
  input tgt as target

  "ImplementationGuide.page-source" : for src.source make tgt.source
  "ImplementationGuide.page-name" : for src.title make tgt.name
  "ImplementationGuide.page-kind" : for src.kind make tgt.kind
  "ImplementationGuide.page-type" : for src.type make tgt.type
  "ImplementationGuide.page-package" : for src.package make tgt.package
  "ImplementationGuide.page-format" : for src.format make tgt.format
  "ImplementationGuide.page-page" : for src.page as vs0 make tgt.page as vt0 then page(vs0, vt0)
endgroup
```

5.8.11.3 R3 Validation Errors

5.9 Resource GraphDefinition - Content

URL: https://hl7.org/fhir/STU3/graphdefinition.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

FHIR Infrastructure Work Group | Maturity Level: 0 | Draft | Compartments: Not linked to any defined compartments

A formal computable definition of a graph of resources - that is, a coherent set of resources that form a graph by following references. The Graph Definition resource defines a set and makes rules about the set.

5.9.1 Scope and Usage

The GraphDefinition resource provides a formal computable definition of a graph of resources - that is, a coherent set of resources that form a graph by following references. The Graph Definition resource defines a set and makes rules about the set. The GraphDefinition resource can be used to:

* Summarize a set of profiles on resources

* Define a graph of resources to return in a query

* Define a graph of resources to include in a document

* Document rules about the relationship between a set of resources e.g. must all resources concern the same patient?

5.9.2 Boundaries and Relationships

There is a close relationship between Profiles and GraphDefinitions:

* A StructureDefinition defines a profile, and profiles can make rules about the relationships between resources. A carefully defined set of profiles implies part of what is in a GraphDefinition

* A GraphDefinition defines rules about the relationships between resources, and in so doing, implies some constraints that would need to be represented in their profiles

Profiles and Graph Definitions can be used together, or separately. When used together, they should be consistent. Note, though, that a graph definition may contain a subset or a superset of the relationships explicitly described in the profiles it refers to.

It is possible that in some circumstances, a GraphRelationship makes incompatible rules with the Profiles it refers to - in this case, no graph if resources will meet the constraints expressed. Applications should - but are not required - detect when such incompatibilities arise.

5.9.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
GraphDefinition | | | DomainResource | Definition of an graph of resources Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this graph definition (globally unique)
version | ? | 0..1 | string | Business version of the graph definition
name | ? | 1..1 | string | Name for this graph definition (computer friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the graph definition
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for graph definition (if applicable) Jurisdiction ValueSet (Extensible)
purpose | | 0..1 | markdown | Why this graph definition is defined
start | | 1..1 | code | Type of resource at which the graph starts ResourceType (Required)
profile | | 0..1 | uri | Profile on base resource
link | | 0..* | BackboneElement | Links this graph makes rules about
path | | 1..1 | string | Path in the resource that contains the link
sliceName | | 0..1 | string | Which slice (if profiled)
min | | 0..1 | integer | Minimum occurrences for this link
max | | 0..1 | string | Maximum occurrences for this link
description | | 0..1 | string | Why this link is specified
target | | 1..* | BackboneElement | Potential target for the link
type | | 1..1 | code | Type of resource this link refers to ResourceType (Required)
profile | | 0..1 | uri | Profile for the target resource
compartment | | 0..* | BackboneElement | Compartment Consistency Rules
code | | 1..1 | code | Identifies the compartment CompartmentType (Required)
rule | | 1..1 | code | identical | matching | different | custom GraphCompartmentRule (Required)
expression | | 0..1 | string | Custom rule, as a FHIRPath expression
description | | 0..1 | string | Documentation for FHIRPath expression
link | | 0..* | see link | Additional links from target resource
Documentation for this format

UML Diagram (Legend)

XML Template

```
<GraphDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this graph definition (globally unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the graph definition -->
 <name value="[string]"/><!-- 1..1 Name for this graph definition (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the graph definition -->
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
```

```
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for graph definition (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this graph definition is defined -->
 <start value="[code]"/><!-- 1..1 Type of resource at which the graph starts -->
 <profile value="[uri]"/><!-- 0..1 Profile on base resource -->
 <link>  <!-- 0..* Links this graph makes rules about -->
  <path value="[string]"/><!-- 1..1 Path in the resource that contains the link -->
  <sliceName value="[string]"/><!-- 0..1 Which slice (if profiled) -->
  <min value="[integer]"/><!-- 0..1 Minimum occurrences for this link -->
  <max value="[string]"/><!-- 0..1 Maximum occurrences for this link -->
  <description value="[string]"/><!-- 0..1 Why this link is specified -->
  <target>  <!-- 1..* Potential target for the link -->
   <type value="[code]"/><!-- 1..1 Type of resource this link refers to -->
   <profile value="[uri]"/><!-- 0..1 Profile for the target resource -->
   <compartment>  <!-- 0..* Compartment Consistency Rules -->
    <code value="[code]"/><!-- 1..1 Identifies the compartment -->
    <rule value="[code]"/><!-- 1..1 identical | matching | different | custom -->
    <expression value="[string]"/><!-- 0..1 Custom rule, as a FHIRPath expression -->
    <description value="[string]"/><!-- 0..1 Documentation for FHIRPath expression -->
   </compartment>
   <link><!-- 0..* Content as for GraphDefinition.link Additional links from target resource --></link>
  </target>
 </link>
</GraphDefinition>
```

JSON Template

```
{
  "resourceType" : "GraphDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this graph definition (globally unique)
  "version" : "<string>", // Business version of the graph definition
  "name" : "<string>", // R!  Name for this graph definition (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the graph definition
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for graph definition (if applicable)
  "purpose" : "<markdown>", // Why this graph definition is defined
  "start" : "<code>", // R!  Type of resource at which the graph starts
  "profile" : "<uri>", // Profile on base resource
  "link" : [{ // Links this graph makes rules about
    "path" : "<string>", // R!  Path in the resource that contains the link
    "sliceName" : "<string>", // Which slice (if profiled)
    "min" : <integer>, // Minimum occurrences for this link
    "max" : "<string>", // Maximum occurrences for this link
    "description" : "<string>", // Why this link is specified
    "target" : [{ // R!  Potential target for the link
      "type" : "<code>", // R!  Type of resource this link refers to
      "profile" : "<uri>", // Profile for the target resource
      "compartment" : [{ // Compartment Consistency Rules
        "code" : "<code>", // R!  Identifies the compartment
        "rule" : "<code>", // R!  identical | matching | different | custom
        "expression" : "<string>", // Custom rule, as a FHIRPath expression
        "description" : "<string>" // Documentation for FHIRPath expression
      }],
      "link" : [{ Content as for GraphDefinition.link }] // Additional links from target resource
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:GraphDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:GraphDefinition.url [ uri ]; # 0..1 Logical URI to reference this graph definition (globally unique)
  fhir:GraphDefinition.version [ string ]; # 0..1 Business version of the graph definition
  fhir:GraphDefinition.name [ string ]; # 1..1 Name for this graph definition (computer friendly)
  fhir:GraphDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:GraphDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:GraphDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:GraphDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:GraphDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:GraphDefinition.description [ markdown ]; # 0..1 Natural language description of the graph definition
  fhir:GraphDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:GraphDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for graph definition (if
    applicable)
  fhir:GraphDefinition.purpose [ markdown ]; # 0..1 Why this graph definition is defined
  fhir:GraphDefinition.start [ code ]; # 1..1 Type of resource at which the graph starts
  fhir:GraphDefinition.profile [ uri ]; # 0..1 Profile on base resource
  fhir:GraphDefinition.link [ # 0..* Links this graph makes rules about
    fhir:GraphDefinition.link.path [ string ]; # 1..1 Path in the resource that contains the link
```

```
    fhir:GraphDefinition.link.sliceName [ string ]; # 0..1 Which slice (if profiled)
    fhir:GraphDefinition.link.min [ integer ]; # 0..1 Minimum occurrences for this link
    fhir:GraphDefinition.link.max [ string ]; # 0..1 Maximum occurrences for this link
    fhir:GraphDefinition.link.description [ string ]; # 0..1 Why this link is specified
    fhir:GraphDefinition.link.target [ # 1..* Potential target for the link
      fhir:GraphDefinition.link.target.type [ code ]; # 1..1 Type of resource this link refers to
      fhir:GraphDefinition.link.target.profile [ uri ]; # 0..1 Profile for the target resource
      fhir:GraphDefinition.link.target.compartment [ # 0..* Compartment Consistency Rules
        fhir:GraphDefinition.link.target.compartment.code [ code ]; # 1..1 Identifies the compartment
        fhir:GraphDefinition.link.target.compartment.rule [ code ]; # 1..1 identical | matching | different | custom
        fhir:GraphDefinition.link.target.compartment.expression [ string ]; # 0..1 Custom rule, as a FHIRPath expression
        fhir:GraphDefinition.link.target.compartment.description [ string ]; # 0..1 Documentation for FHIRPath
    expression
      ], ...;
      fhir:GraphDefinition.link.target.link [ See GraphDefinition.link ], ... ; # 0..* Additional links from target
    resource
    ], ...;
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
GraphDefinition | | | DomainResource | Definition of an graph of resources Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this graph definition (globally unique)
version | ? | 0..1 | string | Business version of the graph definition
name | ? | 1..1 | string | Name for this graph definition (computer friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
contact | ? | 0..* | ContactDetail | Contact details for the publisher
description | | 0..1 | markdown | Natural language description of the graph definition
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for graph definition (if applicable) Jurisdiction ValueSet (Extensible)
purpose | | 0..1 | markdown | Why this graph definition is defined
start | | 1..1 | code | Type of resource at which the graph starts ResourceType (Required)
profile | | 0..1 | uri | Profile on base resource
link | | 0..* | BackboneElement | Links this graph makes rules about
path | | 1..1 | string | Path in the resource that contains the link
sliceName | | 0..1 | string | Which slice (if profiled)
min | | 0..1 | integer | Minimum occurrences for this link
max | | 0..1 | string | Maximum occurrences for this link
description | | 0..1 | string | Why this link is specified
target | | 1..* | BackboneElement | Potential target for the link
type | | 1..1 | code | Type of resource this link refers to ResourceType (Required)
profile | | 0..1 | uri | Profile for the target resource
compartment | | 0..* | BackboneElement | Compartment Consistency Rules
code | | 1..1 | code | Identifies the compartment CompartmentType (Required)
rule | | 1..1 | code | identical | matching | different | custom GraphCompartmentRule (Required)
expression | | 0..1 | string | Custom rule, as a FHIRPath expression
description | | 0..1 | string | Documentation for FHIRPath expression
link | | 0..* | see link | Additional links from target resource
Documentation for this format

UML Diagram (Legend)

XML Template

```
<GraphDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this graph definition (globally unique) -->
 <version value="[string]"/><!-- 0..1 Business version of the graph definition -->
 <name value="[string]"/><!-- 1..1 Name for this graph definition (computer friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <description value="[markdown]"/><!-- 0..1 Natural language description of the graph definition -->
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for graph definition (if applicable) --></jurisdiction>
 <purpose value="[markdown]"/><!-- 0..1 Why this graph definition is defined -->
 <start value="[code]"/><!-- 1..1 Type of resource at which the graph starts -->
 <profile value="[uri]"/><!-- 0..1 Profile on base resource -->
 <link>  <!-- 0..* Links this graph makes rules about -->
```

```
  <path value="[string]"/><!-- 1..1 Path in the resource that contains the link -->
  <sliceName value="[string]"/><!-- 0..1 Which slice (if profiled) -->
  <min value="[integer]"/><!-- 0..1 Minimum occurrences for this link -->
  <max value="[string]"/><!-- 0..1 Maximum occurrences for this link -->
  <description value="[string]"/><!-- 0..1 Why this link is specified -->
  <target>  <!-- 1..* Potential target for the link -->
   <type value="[code]"/><!-- 1..1 Type of resource this link refers to -->
   <profile value="[uri]"/><!-- 0..1 Profile for the target resource -->
   <compartment>  <!-- 0..* Compartment Consistency Rules -->
    <code value="[code]"/><!-- 1..1 Identifies the compartment -->
    <rule value="[code]"/><!-- 1..1 identical | matching | different | custom -->
    <expression value="[string]"/><!-- 0..1 Custom rule, as a FHIRPath expression -->
    <description value="[string]"/><!-- 0..1 Documentation for FHIRPath expression -->
   </compartment>
   <link><!-- 0..* Content as for GraphDefinition.link Additional links from target resource --></link>
  </target>
 </link>
</GraphDefinition>
```

JSON Template

```
{
  "resourceType" : "GraphDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this graph definition (globally unique)
  "version" : "<string>", // Business version of the graph definition
  "name" : "<string>", // R!  Name for this graph definition (computer friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "description" : "<markdown>", // Natural language description of the graph definition
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for graph definition (if applicable)
  "purpose" : "<markdown>", // Why this graph definition is defined
  "start" : "<code>", // R!  Type of resource at which the graph starts
  "profile" : "<uri>", // Profile on base resource
  "link" : [{ // Links this graph makes rules about
    "path" : "<string>", // R!  Path in the resource that contains the link
    "sliceName" : "<string>", // Which slice (if profiled)
    "min" : <integer>, // Minimum occurrences for this link
    "max" : "<string>", // Maximum occurrences for this link
    "description" : "<string>", // Why this link is specified
    "target" : [{ // R!  Potential target for the link
      "type" : "<code>", // R!  Type of resource this link refers to
      "profile" : "<uri>", // Profile for the target resource
      "compartment" : [{ // Compartment Consistency Rules
        "code" : "<code>", // R!  Identifies the compartment
        "rule" : "<code>", // R!  identical | matching | different | custom
        "expression" : "<string>", // Custom rule, as a FHIRPath expression
        "description" : "<string>" // Documentation for FHIRPath expression
      }],
      "link" : [{ Content as for GraphDefinition.link }] // Additional links from target resource
    }]
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:GraphDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:GraphDefinition.url [ uri ]; # 0..1 Logical URI to reference this graph definition (globally unique)
  fhir:GraphDefinition.version [ string ]; # 0..1 Business version of the graph definition
  fhir:GraphDefinition.name [ string ]; # 1..1 Name for this graph definition (computer friendly)
  fhir:GraphDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:GraphDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:GraphDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:GraphDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:GraphDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:GraphDefinition.description [ markdown ]; # 0..1 Natural language description of the graph definition
  fhir:GraphDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:GraphDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for graph definition (if
    applicable)
  fhir:GraphDefinition.purpose [ markdown ]; # 0..1 Why this graph definition is defined
  fhir:GraphDefinition.start [ code ]; # 1..1 Type of resource at which the graph starts
  fhir:GraphDefinition.profile [ uri ]; # 0..1 Profile on base resource
  fhir:GraphDefinition.link [ # 0..* Links this graph makes rules about
    fhir:GraphDefinition.link.path [ string ]; # 1..1 Path in the resource that contains the link
    fhir:GraphDefinition.link.sliceName [ string ]; # 0..1 Which slice (if profiled)
    fhir:GraphDefinition.link.min [ integer ]; # 0..1 Minimum occurrences for this link
    fhir:GraphDefinition.link.max [ string ]; # 0..1 Maximum occurrences for this link
    fhir:GraphDefinition.link.description [ string ]; # 0..1 Why this link is specified
    fhir:GraphDefinition.link.target [ # 1..* Potential target for the link
```

```
      fhir:GraphDefinition.link.target.type [ code ]; # 1..1 Type of resource this link refers to
      fhir:GraphDefinition.link.target.profile [ uri ]; # 0..1 Profile for the target resource
      fhir:GraphDefinition.link.target.compartment [ # 0..* Compartment Consistency Rules
        fhir:GraphDefinition.link.target.compartment.code [ code ]; # 1..1 Identifies the compartment
        fhir:GraphDefinition.link.target.compartment.rule [ code ]; # 1..1 identical | matching | different | custom
        fhir:GraphDefinition.link.target.compartment.expression [ string ]; # 0..1 Custom rule, as a FHIRPath expression
        fhir:GraphDefinition.link.target.compartment.description [ string ]; # 0..1 Documentation for FHIRPath
    expression
      ], ...;
      fhir:GraphDefinition.link.target.link [ See GraphDefinition.link ], ... ; # 0..* Additional links from target
    resource
    ], ...;
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

5.9.3.1 Terminology Bindings

Path	Definition	Type	Reference
GraphDefinition.status	The lifecycle status of a Value Set or Concept Map.	Required	PublicationStatus
GraphDefinition.jurisdiction	Countries and regions within which this artifact is targeted for use	Extensible	Jurisdiction ValueSet
GraphDefinition.start GraphDefinition.link.target.type	One of the resource types defined as part of FHIR.	Required	Resource Types
GraphDefinition.link.target.compartment.code	Identifies a compartment	Required	CompartmentType
GraphDefinition.link.target.compartment.rule	How a compartment must be linked	Required	GraphCompartmentRule

5.9.3.2 Using GraphDefinitions

The GraphDefinition resource can be used to:

* Summarize a set of profiles on resources

* Define a graph of resources to return in a query

* Define a graph of resources to include in a document

* Document rules about the relationship between a set of resources e.g. must all resources concern the same patient?

5.9.3.2.1 Summarize a set of profiles on resources

FHIR resources are relatively granular. In many/most cases, many resources are needed to handle any particular task. A typical example of this is a complex diagnostic report: it will start with a DiagnosticReport, which will link to a set of panels (Observation resources), each of which link to a set of Observation resources for atomic data items.

One way to represent this is to profile each of the resources, creating hundreds of profiles, and then leave it to the user to infer the overall pattern of the report from the detailed profiles for each observation in the report. But it's not easy to see the forest for the trees. A GraphDefinition can summarise the overall picture and present a summary to the user.

Here's an example of the kind of summary this represents. (Todo: make this an actual graph definition, and clone into the main spec)

5.9.3.2.2 Fetching a graph of resources

As another example of using many resources, to completely represent a medication dispense, an application needs not only the MedicationDispense resource, but also resources to represent the patient, provider, organizations, and the associated prescription.

A client can retrieve a single resource:

```
GET [base]/MedicationDispense/example
```

Then, when it reads the returned resource, it can fetch the referenced resources:

```
GET [base]/Patient/example
GET [base]/Practitioner/example
GET [base]/MedicationRequest/example
... etc
```

This is a very inefficient way to retrieve all the required resources. An alternative approach is to do a search, and _include the required resources:

```
GET [base]/MedicationDispense?_id=example
  &_include=MedicationDispense:authorizingPrescription
  &_include=MedicationDispense:subject
```

But scaling this approach to fetch a full package with its dependencies becomes increasingly difficult as the package gets deeper. A graph definition can be used instead to inform the server what to return as part of the search:

```
GET [base]/MedicationDispense?_id=example&_graph=med-package
```

This is a reference to the local graph package 'med-package', with the intent that the server returns the graph as outlined by the definition. In this case, the graph definition would look approximately like this:

```
MedicationDispense
  .subject
  .context
```

```
.performer.actor
.authorizingPrescription
   .requester.agent
.substitution.responsibleParty
```

5.9.3.2.3 Building a document

A very similar issue applies when building a document using the $document operation. A document must include all the resources linked directly from the composition, but whether to include additional linked resources is at the discretion of the document author. How does the user inform the $document operation which linked resources to include? One option is a boolean flag for including all linked data, but this may be extensive - up to an entire patient record - and may include resources that are not desired.

An operation can use a graph definition as a parameter to the $document operation:

```
GET [base]/Composition/example/$document?_graph=example
```

This tells the server to include the graph of resources defined in the example GraphDefinition - in this case, any resources referred to from lists, when the section content is a list.

5.9.3.2.4 Document Relationship rules

One important question about the use of resources is cross-resource consistency. For example, if an Observation refers to both a Patient and Encounter, does the Encounter have to refer to the same patient?

In general, the answer to this is that it usually should - the record needs to be consistent. However there are edge cases where the references may differ. For example, with regard to patient references, they may differ for:

* Health Records concerning mother and baby

* Organ Transplants

* Counselling, particularly family counselling

Other reasons for the references to differ - mixing records about the same patient from different servers, or specific records about patients mixed with records about groups of patients (particularly common in veterinarian care).

The GraphDefinition resource allows for compartment consistency rules to be made regarding the links between resources. For each link in the graph, the graph definition can make a rule about the compartment consistency. The rule can specify one of the following consistencies:

Code | Meaning
identical | The compartment must be identical (the same literal reference)
matching | The compartment must be the same - the record must be about the same patient, but the reference may be different
different | The compartment must be different
custom | The compartment rule is defined in the accompanying FHIRPath expression

Todo: how would this be validated? - where is the graph referred to?

5.9.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
date | date | The graph definition publication date | GraphDefinition.date
description | string | The description of the graph definition | GraphDefinition.description
jurisdiction | token | Intended jurisdiction for the graph definition | GraphDefinition.jurisdiction
name | string | Computationally friendly name of the graph definition | GraphDefinition.name
publisher | string | Name of the publisher of the graph definition | GraphDefinition.publisher
start | token | Type of resource at which the graph starts | GraphDefinition.start
status | token | The current status of the graph definition | GraphDefinition.status
url | uri | The uri that identifies the graph definition | GraphDefinition.url
version | token | The business version of the graph definition | GraphDefinition.version

5.9.5 Resource GraphDefinition - Examples

URL: https://hl7.org/fhir/STU3/graphdefinition-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.9.5 Resource GraphDefinition - Examples

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

GraphDefinition example | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

5.9.6 Resource GraphDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/graphdefinition-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.9.6 Resource GraphDefinition - Detailed Descriptions

FHIR Infrastructure Work Group | Maturity Level: 0 | Draft | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the GraphDefinition resource.

GraphDefinition

Definition | A formal computable definition of a graph of resources - that is, a coherent set of resources that form a graph by following references. The Graph Definition resource defines a set and makes rules about the set.

Control | 1..1

GraphDefinition.url

Definition | An absolute URI that is used to identify this graph definition when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this graph definition is (or will be) published. The URL SHOULD include the major version of the graph definition. For more information see Technical and Business Versions.

Control | 0..1

Type | uri

Requirements | Allows the graph definition to be referenced by a single globally unique identifier.

Summary | true

Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.

GraphDefinition.version

Definition | The identifier that is used to identify this version of the graph definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the graph definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.

Note | This is a business versionId, not a resource version id (see discussion)

Control | 0..1

Type | string

Summary | true

Comments | There may be different graph definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the graph definition with the format [url]|[version].

GraphDefinition.name

Definition | A natural language name identifying the graph definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.

Control | 1..1

Type | string

Requirements | Support human navigation and code generation.

Summary | true

Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.

GraphDefinition.status

Definition | The status of this graph definition. Enables tracking the life-cycle of the content.

Control | 1..1

Terminology Binding | PublicationStatus (Required)

Type | code

Is Modifier | true

Summary | true

Comments | Allows filtering of graph definitions that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired graph definition without due consideration.

GraphDefinition.experimental

Definition | A boolean value to indicate that this graph definition is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.

Control | 0..1

Type | boolean

Is Modifier | true

Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level graph definition.

Summary | true

Comments | Allows filtering of graph definition that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental graph definition in production.

GraphDefinition.date

Definition | The date (and optionally time) when the graph definition was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the graph definition changes.

Control | 0..1

Type | dateTime

Alternate Names | Revision Date

Summary | true

Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the graph definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.

GraphDefinition.publisher

Definition | The name of the individual or organization that published the graph definition.

Control | 0..1

Type | string
Requirements | Helps establish the "authority/credibility" of the graph definition. May also allow for contact.
Summary | true
Comments | Usually an organization, but may be an individual. The publisher (or steward) of the graph definition is the organization or individual primarily responsible for the maintenance and upkeep of the graph definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the graph definition. This item SHOULD be populated unless the information is available from context.
GraphDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
GraphDefinition.description
Definition | A free text natural language description of the graph definition from a consumer's perspective.
Control | 0..1
Type | markdown
Comments | This description can be used to capture details such as why the graph definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the graph definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created).
GraphDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate graph definition instances.
Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
GraphDefinition.jurisdiction
Definition | A legal or geographic region in which the graph definition is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the graph definition to be used in jurisdictions other than those for which it was originally designed or intended.
GraphDefinition.purpose
Definition | Explaination of why this graph definition is needed and why it has been designed as it has.
Control | 0..1
Type | markdown
Comments | This element does not describe the usage of the graph definition Instead it provides traceability of "why" the resource is either needed or "why" it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this graph definition.
GraphDefinition.start
Definition | The type of FHIR resource at which instances of this graph start.
Control | 1..1
Terminology Binding | Any defined Resource Type name
Type | code
GraphDefinition.profile
Definition | The profile that describes the use of the base resource.
Control | 0..1
Type | uri
Comments | The code does not include the '$' prefix that is always included in the URL when the operation is invoked.
GraphDefinition.link
Definition | Links this graph makes rules about.
Control | 0..*
GraphDefinition.link.path
Definition | Path in the resource that contains the link.
Control | 1..1
Type | string
Comments | The path can be a literal path (e.g. Observation.subject), or it can be a query for reverse links (e.g. Provenance?target=[id]).
GraphDefinition.link.sliceName
Definition | Which slice (if profiled).
Control | 0..1
Type | string
GraphDefinition.link.min
Definition | Minimum occurrences for this link.
Control | 0..1
Type | integer
GraphDefinition.link.max
Definition | Maximum occurrences for this link.
Control | 0..1
Type | string
GraphDefinition.link.description

Definition | Information about why this link is of interest in this graph definition.
Control | 0..1
Type | string
GraphDefinition.link.target
Definition | Potential target for the link.
Control | 1..*
GraphDefinition.link.target.type
Definition | Type of resource this link refers to.
Control | 1..1
Terminology Binding | Any defined Resource Type name
Type | code
GraphDefinition.link.target.profile
Definition | Profile for the target resource.
Control | 0..1
Type | uri
GraphDefinition.link.target.compartment
Definition | Compartment Consistency Rules.
Control | 0..*
GraphDefinition.link.target.compartment.code
Definition | Identifies the compartment.
Control | 1..1
Terminology Binding | CompartmentType (Required)
Type | code
GraphDefinition.link.target.compartment.rule
Definition | identical | matching | different | no-rule | custom.
Control | 1..1
Terminology Binding | GraphCompartmentRule (Required)
Type | code
GraphDefinition.link.target.compartment.expression
Definition | Custom rule, as a FHIRPath expression.
Control | 0..1
Type | string
GraphDefinition.link.target.compartment.description
Definition | Documentation for FHIRPath expression.
Control | 0..1
Type | string
GraphDefinition.link.target.link
Definition | Additional links from target resource.
Control | 0..*
Type | See GraphDefinition.link

5.9.7 Resource GraphDefinition - Mappings

URL: https://hl7.org/fhir/STU3/graphdefinition-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.9.7 Resource GraphDefinition - Mappings

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Workflow Mapping

Mappings for the graphdefinition resource.

5.9.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

GraphDefinition | n/a

5.9.7.2 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

GraphDefinition | Definition
url | Definition.url
version | Definition.version
name | Definition.title
status | Definition.status (different ValueSet)
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
contact | Definition.contact
description | Definition.description
useContext | Definition.useContext

jurisdiction | Definition.jurisdiction
purpose | Definition.purpose
start
profile
link
path
sliceName
min
max
description
target
type
profile
compartment
code
rule
expression
description
link

5.9.8 Resource GraphDefinition - Design Notes

URL: https://hl7.org/fhir/STU3/graphdefinition-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.9.8 Resource GraphDefinition - Design Notes

to do

5.9.9 Resource GraphDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/graphdefinition-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

5.9.9 Resource GraphDefinition - Extensions & Profiles

FHIR Infrastructure Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the GraphDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

5.9.9.1 Profiles

No Profiles defined for this resource

5.9.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

5.9.9.3 Search Extensions

No Search Extensions defined for this resource